

SEVENTH EDITION



EVALUATING RESEARCH IN ACADEMIC JOURNALS

A PRACTICAL GUIDE TO REALISTIC EVALUATION

Fred Pyrczak and
Maria Tcherni-Buzzeo

Evaluating Research in Academic Journals

Evaluating Research in Academic Journals is a guide for students who are learning how to evaluate reports of empirical research published in academic journals. It breaks down the process of evaluating a journal article into easy-to-understand steps, and emphasizes the practical aspects of evaluating research – not just how to apply a list of technical terms from textbooks.

The book avoids oversimplification in the evaluation process by describing the nuances that may make an article publishable even when it has serious methodological flaws. Students learn when and why certain types of flaws may be tolerated, and why evaluation should not be performed mechanically.

Each chapter is organized around evaluation questions. For each question, there is a concise explanation of how to apply it in the evaluation of research reports. Numerous examples from journals in the social and behavioral sciences illustrate the application of the evaluation questions, and demonstrate actual examples of strong and weak features of published reports. Common-sense models for evaluation combined with a lack of jargon make it possible for students to start evaluating research articles the first week of class.

New to this edition

- New chapters on:
 - Evaluating mixed methods research
 - Evaluating systematic reviews and meta-analyses
 - Program evaluation research
- Updated chapters and appendices that provide more comprehensive information and recent examples
- Full new online resources: test bank questions and PowerPoint slides for instructors, and self-test chapter quizzes, further readings, and additional journal examples for students.

Maria Tcherni-Buzzeo is an Associate Professor of Criminal Justice at the University of New Haven. She received her PhD in Criminal Justice from the University at Albany (SUNY), and her research has been published in the *Journal of Quantitative Criminology*, *Justice Quarterly*, and *Deviant Behavior*.



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Evaluating Research in Academic Journals

A Practical Guide to
Realistic Evaluation

Seventh Edition

Fred Pyrczak and
Maria Tcherni-Buzzeo

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Introduction to the Seventh Edition

When students in the social and behavioral sciences take advanced courses in their major field of study, they are often required to read and evaluate original research reports published as articles in academic journals. This book is designed as a guide for students who are first learning how to engage in this process.

Major Assumptions

First, it is assumed that the students using this book have limited knowledge of research methods, even though they may have taken a course in introductory research methods (or may be using this book while taking such a course). Because of this assumption, technical terms and jargon such as *true experiment* are defined when they are first used in this book.

Second, it is assumed that students have only a limited grasp of elementary statistics. Thus, the chapters on evaluating statistical reporting in research reports are confined to criteria that such students can easily comprehend.

Finally, and perhaps most important, it is assumed that students with limited backgrounds in research methods and statistics can produce adequate evaluations of research reports – evaluations that get to the heart of important issues and allow students to draw sound conclusions from published research.

This Book Is *Not* Written for . . .

This book is not written for journal editors or members of their editorial review boards. Such professionals usually have had first-hand experience in conducting research and have taken advanced courses in research methods and statistics. Published evaluation criteria for use by these professionals are often terse, full of jargon, and composed of many elements that cannot be fully comprehended without advanced training and experience. This book is aimed at a completely different audience: students who are just beginning to learn how to evaluate original reports of research published in journals.

Applying the Evaluation Questions in This Book

Chapters 2 through 15 are organized around evaluation questions that may be answered with a simple “yes” or “no,” where a “yes” indicates that students judge a characteristic to be satisfactory. However, for evaluation questions that deal with complex issues, students may also want to rate each one using a scale from 1 to 5, where 5 is the highest rating. In addition, N/A (not applicable) may be used when students believe a characteristic does not apply, and I/I (insufficient information) may be used if the research report does not contain sufficient information for an informed judgment to be made.

Evaluating Quantitative and Qualitative Research

Quantitative and qualitative research differ in purpose as well as methodology. Students who are not familiar with the distinctions between the two approaches are advised to read Appendix A, which presents a very brief overview of the differences, and also explains what mixed methods research is. Students are also encouraged to check the online resources for Chapter 11 that include an overview of important issues in the evaluation of qualitative research.

Note from the Authors

I have taken over the updating of this text for its current, 7th edition, due to Fred Pyrczak’s untimely departure from this earth in 2014. His writing in this book is amazing: structured, clear, and concise. It is no surprise that the text has been highly regarded by multiple generations of students who used it in their studies. In fact, many students in my Methods classes have commented how much they like this text and how well written and helpful it is.

I have truly enjoyed updating this edition for the new generation of students, and tried my best to retain all the strengths of Fred’s original writing. I am also grateful to my colleague Anne Li Kringen, who is an expert on mixed methods research, for contributing a new Chapter 12 (on evaluating mixed methods research) to the current edition. Also, new in the current edition are Chapter 14 (on evaluating meta-analyses and systematic reviews) and Appendix B (on evaluating programs and policies). The remainder of the chapters and appendices have been updated throughout with new information and examples. I hope this text will serve you well in your adventures of reading research articles!

Maria Tcherni-Buzzeo
New Haven, 2018

My best wishes are with you as you master the art and science of evaluating research. With the aid of this book, you should find the process both undaunting and fascinating as you seek defensible conclusions regarding research on topics that interest you.

Fred Pyrczak
Los Angeles, 2014

Background for Evaluating Research Reports

The vast majority of research reports are initially published in academic journals. In these reports, or *empirical* journal articles,¹ researchers describe how they have identified a research problem, made relevant observations or measurements to gather data, and analyzed the data they collected. The articles usually conclude with a discussion of the results in view of the study limitations, as well as the implications of these results. This chapter provides an overview of some general characteristics of such research. Subsequent chapters present specific questions that should be applied in the evaluation of empirical research articles.

✓ **Guideline 1: Researchers Often Examine Narrowly Defined Problems**

Comment: While researchers usually are interested in broad problem areas, they very often examine only narrow aspects of the problems because of limited resources and the desire to keep the research manageable by limiting its focus. Furthermore, they often examine problems in such a way that the results can be easily reduced to statistics, further limiting the breadth of their research.²

Example 1.1.1 briefly describes a study on two correlates of prosocial behavior (i.e., helping behavior). To make the study of this issue manageable, the researchers greatly limited its scope. Specifically, they examined only one very narrow type of prosocial behavior (making donations to homeless men who were begging in public).

- 1 Note that empirical research articles are different from other types of articles published in peer-reviewed journals in that they specifically include an original analysis of empirical data (data could be qualitative or quantitative, which is explained in more detail in Appendix A). Other types of articles include book reviews or overview articles that summarize the state of knowledge and empirical research on a specific topic or propose agenda for future research. Such articles do not include original data analyses and thus are not suitable for being evaluated using the criteria in this text.
- 2 Qualitative researchers (see Appendix A) generally take a broader view when defining a problem to be explored in research and are not constrained by the need to reduce the results to numbers and statistics. More information about examining the validity of qualitative research can be found in the online resources for Chapter 11 of this text.

Example 1.1.1

A STUDY ON PROSOCIAL BEHAVIOR, NARROWLY DEFINED

In order to study the relationship between prosocial behavior and gender as well as age, researchers located five men who appeared to be homeless and were soliciting money on street corners using cardboard signs. Without approaching the men, the researchers observed them from a short distance for two hours each. For each pedestrian who walked within ten feet of the men, the researchers recorded whether the pedestrian made a donation. The researchers also recorded the gender and approximate age of each pedestrian.

Because researchers often conduct their research on narrowly defined problems, an important task in the evaluation of research is to judge whether a researcher has defined the problem so narrowly that it fails to make an important contribution to the advancement of knowledge.

✓ Guideline 2: Researchers Often Conduct Studies in Artificial Settings

Comment: Laboratories on university campuses are often the settings for research. To study the effects of alcohol consumption on driving behavior, a group of participants might be asked to drink carefully measured amounts of alcohol in a laboratory and then “drive” using virtual-reality simulators. Example 1.2.1 describes the preparation of the cocktails in a study of this type.

Example 1.2.1³

ALCOHOLIC BEVERAGES PREPARED FOR CONSUMPTION IN A LABORATORY SETTING

The preparation of the cocktail was done in a separate area out of view of the participant. All cocktails were a 16-oz mixture of orange juice, cranberry juice, and grapefruit juice (ratio 4:2:1, respectively). For the cocktails containing alcohol, we added 2 oz of 190-proof grain alcohol mixed thoroughly. For the placebo cocktail, we lightly sprayed the surface of the juice cocktail with alcohol using an atomizer placed slightly above the juice surface to impart an aroma of alcohol to the glass and beverage surface. This placebo cocktail was then immediately given to the participant to consume. This procedure results in the same alcohol aroma being imparted to the placebo cocktail as the alcohol cocktail . . .

Such a study might have limited generalizability to drinking in out-of-laboratory settings, such as nightclubs, the home, picnics, and other places where those who are consuming alcohol may

3 Barkley, R. A., Murphy, K. R., O’Connell, T., Anderson, D., & Connor, D. F. (2006). Effects of two doses of alcohol on simulator driving performance in adults with attention-deficit/hyperactivity disorder. *Neuropsychology*, 20(1), 77–87.

be drinking different amounts at different rates while consuming (or not consuming) various foods. Nevertheless, conducting such research in a laboratory allows researchers to simplify, isolate, and control variables such as the amount of alcohol consumed, the types of food being consumed, the type of distractions during the “car ride”, and so on. In short, researchers very often opt against studying variables in complex, real-life settings for the more interpretable research results typically obtained in a laboratory.

✓ **Guideline 3: Researchers use Less-than-perfect Methods of Measurement**

Comment: In research, *measurement* can take many forms—from online multiple-choice achievement tests to essay examinations, from administering a paper-and-pencil attitude scale with choices from “strongly agree” to “strongly disagree” to conducting unstructured interviews to identify interviewees’ attitudes.⁴ *Observation* is a type of measurement that includes direct observation of individuals interacting in either their natural environments or laboratory settings.

It is safe to assume that all methods of observation or measurement are flawed to some extent. To see why this is so, consider a professor/researcher who is interested in studying racial relations in society in general. Because of limited resources, the researcher decides to make direct observations of White and African American students interacting (and/or not interacting) in the college cafeteria. The observations will necessarily be limited to the types of behaviors typically exhibited in cafeteria settings – a weakness in the researcher’s method of observation. In addition, observations will be limited to certain overt behaviors because, for instance, it will be difficult for the researcher to hear most of what is being said without intruding on the privacy of the students.

On the other hand, suppose that another researcher decides to measure racial attitudes by having students respond anonymously to racial statements by circling “agree” or “disagree” for each one. This researcher has an entirely different set of weaknesses in the method of measurement. First is the matter of whether students will reveal their real attitudes on such a scale – even if the response is anonymous – because most college students are aware that negative racial attitudes are severely frowned on in academic communities. Thus, some students might indicate what they believe to be socially desirable (i.e., socially or politically “correct”) rather than reveal their true attitudes. Moreover, people may often be unaware of their own implicit racial biases.⁵

In short, *there is no perfect way to measure complex variables*. Instead of expecting perfection, a consumer of research should consider this question: *Is the method sufficiently valid and reliable to provide potentially useful information?*

4 Researchers sometimes refer to measurement tools as *instruments*, especially in older research literature.

5 For more information, check Project Implicit hosted by Harvard University and run by an international collaboration of researchers (see the link in the online resources for this chapter).

Examples 1.3.1 and 1.3.2 show statements from research articles in which the researchers acknowledge limitations in their methods of measurement.

Example 1.3.1⁶

RESEARCHERS' ACKNOWLEDGMENT OF A LIMITATION OF THEIR MEASURES

In addition, the assessment of marital religious discord was limited to one item. Future research should include a multiple-items scale of marital religious discord and additional types of measures, such as interviews or observational coding, as well as multiple informants.

Example 1.3.2⁷

RESEARCHERS' ACKNOWLEDGMENT OF LIMITATIONS OF SELF-REPORTS

Despite these strengths, this study is not without limitations. First, the small sample size decreases the likelihood of finding statistically significant interaction effects. [. . .] Fourth, neighborhood danger was measured from mothers' self-reports of the events which had occurred in the neighborhood during the past year. Adding other family member reports of the dangerous events and official police reports would clearly strengthen our measure of neighborhood danger.

Chapter 8 provides more information on evaluating observational methods and measures typically used in empirical studies. Generally, it is important to look for whether the researchers themselves properly acknowledge in the article some key limitations of their measurement strategies.

✓ Guideline 4: Researchers use Less-than-perfect Samples

Comment: Arguably, the most common sampling flaw in research reported in academic journals is the use of *convenience samples* (i.e., samples that are readily accessible to the researchers). Most researchers are professors, and professors often use samples of college students – obviously as a matter of convenience. Another common flaw is relying on voluntary responses to mailed surveys, which are often quite low, with some researchers arguing that a response rate of about 40–60% or more is acceptable. For online surveys, it may be even more difficult to evaluate the response rate unless we know how many people saw the survey solicitation. (Problems related to the use of online versus mailed surveys are discussed in Chapter 6.)

6 Kor, A., Mikulincer, M., & Pirutinsky, S. (2012). Family functioning among returnees to Orthodox Judaism in Israel. *Journal of Family Psychology*, 26(1), 149–158.

7 Callahan, K. L., Scaramella, L. V., Laird, R. D., & Sohr-Preston, S. L. (2011). Neighborhood disadvantage as a moderator of the association between harsh parenting and toddler-aged children's internalizing and externalizing problems. *Journal of Family Psychology*, 25(1), 68–76.

Other samples are flawed because researchers cannot identify and locate all members of a population (e.g., injection drug users). Without being able to do this, it is impossible to draw a sample that a researcher can reasonably defend as being representative of the population.⁸ In addition, researchers often have limited resources, which forces them to use small samples and which in turn might produce unreliable results.

Researchers sometimes explicitly acknowledge the limitations of their samples. Examples 1.4.1 through 1.4.3 show portions of such statements from research articles.

Example 1.4.1⁹

RESEARCHERS' ACKNOWLEDGMENT OF LIMITATION OF SAMPLING (CONVENIENCE SAMPLE)

The present study suffered from several limitations. First of all, the samples were confined to university undergraduate students and only Chinese and American students. For broader generalizations, further studies could recruit people of various ages and educational and occupational characteristics.

Example 1.4.2¹⁰

RESEARCHERS' ACKNOWLEDGMENT OF LIMITATION OF SAMPLING (LOW RATE OF PARTICIPATION)

Data were collected using a random sample of e-mail addresses obtained from the university's registrar's office. The response rate (23%) was lower than desired; however, it is unknown what percentage of the e-mail addresses were valid or were being monitored by the targeted student.

Example 1.4.3¹¹

RESEARCHER'S ACKNOWLEDGMENT OF LIMITATION OF SAMPLING (LIMITED DIVERSITY)

There are a number of limitations to this study. The most significant of them relates to the fact that the study was located within one school and the children studied were primarily from a White, working-class community. There is a need to identify how socially and ethnically diverse groups of children use online virtual worlds.

- 8 Qualitative researchers emphasize selecting a *purposive* sample—one that focuses on people with specific characteristics and is likely to yield useful information – rather than a *representative* sample.
- 9 Jiang, F., Yue, X. D., & Lu, S. (2011). Different attitudes toward humor between Chinese and American students: Evidence from the Implicit Association Test. *Psychological Reports*, 109(1), 99–107.
- 10 Cox, J. M., & Bates, S. C. (2011). Referent group proximity, social norms, and context: Alcohol use in a low-use environment. *Journal of American College Health*, 59(4), 252–259.
- 11 Marsh, J. (2011). Young children's literacy practices in a virtual world: Establishing an online interaction order. *Reading Research Quarterly*, 46(2), 101–118.

In Chapters 6 and 7, specific criteria for evaluating samples are explored in detail. Again, it is important to look for statements in which researchers honestly acknowledge limitations of sampling in their study. It does not mitigate the resulting problems but can help researchers properly recognize some likely biases and problems with the generalizability of their results.

✓ **Guideline 5: Even a Straightforward Analysis of Data can Produce Misleading Results**

Comment: Obviously, data-input errors and computational errors are possible sources of errors in results. Some commercial research firms have the data they collect entered independently by two or more data-entry clerks. A computer program checks to see whether the two sets of entries match perfectly – if they do not, the errors must be identified before the analysis can proceed. Unfortunately, taking such care in checking for mechanical errors in entering data is hardly ever mentioned in research reports published in academic journals.

In addition, there are alternative statistical methods for most problems, and different methods can yield different results. (See Chapter 10 for specific examples regarding the selection of statistics.)

Finally, even a non-statistical analysis can be problematic. For instance, if two or more researchers review extensive transcripts of unstructured interviews, they might differ in their interpretations of the interviewees' responses. Discrepancies such as these suggest that the results may be flawed or at least subject to different interpretations.

Chapter 10 provides evaluation criteria for quantitative Analysis and Results sections of research reports, while Chapter 11 does the same for qualitative Analysis and Results sections, and Chapter 12 for mixed methods research.

✓ **Guideline 6: Even a Single, Isolated Flaw in Research Methods can lead to Seriously Misleading Results**

Comment: A seemingly minor flaw such as a poorly worded question on attitudes in a survey questionnaire might lead to results that are incorrect. Likewise, a treatment that has been misapplied in an experiment might lead to misleading conclusions regarding the effectiveness of the treatment. Or a sample that only had volunteers eager to participate in a specific study can lead to skewed results. (This type of situation can lead to *self-selection bias*, which is discussed in more detail in Chapter 6.) For these reasons, empirical research articles should be detailed, so that consumers of research can have enough information to judge whether the research methods were flawed. This leads to the next guideline.

✓ **Guideline 7: Research Reports Often Contain many Details, Which can be Very Important when Evaluating a Report**

Comment: The old saying “The devil is in the details” applies here. Students who have relied exclusively on secondary sources for information about their major field of study may be surprised at the level of detail in many research reports, which is typically much greater than

is implied in sources such as textbooks and classroom lectures. Example 1.7.1 illustrates the level of detail that can be expected in many empirical research articles published in academic journals. It describes part of an intervention for postal service letter carriers.

Example 1.7.1¹²

AN EXCERPT FROM AN ARTICLE ILLUSTRATING THE LEVEL OF DETAIL OFTEN INCLUDED IN RESEARCH REPORTS IN ACADEMIC JOURNALS

Within 2 weeks of the baseline measurement, Project SUNWISE health educators visited intervention stations to give out hats, install and dispense sunscreen, distribute materials that prompted use of solar protective strategies, and deliver the initial educational presentation. [. . .] The machine-washable dark blue hat was made of Cordura nylon, it had a brim that was 4 inches wide in the front and back and 3 inches wide on the sides, and it had an adjustable cord chin strap. In addition to the initial free hat provided by Project SUNWISE, letter carriers at intervention stations were given discounts on replacement hats by the vendor (Watership Trading Company, Bellingham, WA).

Locker rooms at intervention stations were stocked with large pump bottles of sunscreen (Coppertone Sport, SPF 30, Schering-Plough HealthCare Products, Inc., Memphis, TN) that were refilled regularly by the research staff. Additionally, letter carriers were given free 12 ounce bottles of the sunscreen, which they could refill with sunscreen from the pump bottles. The decision about which sunscreen to use was made on the basis of formative work that identified a product with a high SPF that had an acceptable fragrance and consistency and minimal rub-off from newsprint onto skin. [. . .]

Finally, Project SUNWISE health educators delivered 6 brief onsite educational presentations over 2 years. The 5- to 10-minute presentations were modeled after the “stand-up talks” letter carriers regularly participated in; the educators used large flip charts with colorful graphics that were tailored to letter carriers. Key points of the introductory presentation included the amount of UVR carriers are exposed to and UVR as a skin cancer risk factor, a case example of a former carrier who recently had a precancerous growth removed, feasible protection strategies, and specific information about the hats and sunscreen. The themes of subsequent presentations were (1) importance of sun safety, even in winter; (2) sun safety for the eyes; (3) sharing sun safety tips with loved ones; (4) relevance of sun safety to letter carriers of all races/ethnicities; and (5) recap and encouragement to continue practicing sun safety behaviors.

Note the level of detail, such as (a) the color and size of the hats and (b) the specific brand of sunscreen that was distributed. Such details are useful for helping consumers of research understand exactly the nature of the intervention examined in the study. Knowing what was said and done to participants as well as how the participants were observed makes it possible

12 Mayer, J. A., Slymen, D. J., Clapp, E. J., Pichon, L. C., Eckhardt, L., Eichenfield, L. F., . . . Oh, S. S. (2007). Promoting sun safety among U.S. Postal Service letter carriers: Impact of a 2-year intervention. *American Journal of Public Health*, 97, 559–565.

to render informed evaluations of research. Having detailed descriptions is also helpful for other researchers who might want to replicate the study in order to confirm the findings.

✓ **Guideline 8: Many Research Articles Provide Precise Definitions of Key Terms to Help Guide the Measurement of the Associated Concepts**

Comment: Often, students complain that research articles are dry and boring and “Why do they include all those definitions anyway?” To the credit of researchers writing these articles, they include definitions to help rather than annoy the reader. Consider some of the complex concepts that need to be measured in a typical study. For example, researchers are interested in how prevalent domestic violence (DV) is. What is domestic violence? Do we only consider physical acts as domestic violence or psychological and verbal abuse as well? What about financial abuse? What about threats? These questions can be answered by using a careful and precisely worded definition of domestic violence. This can also help the reader figure out what the researchers may be missing if they use police reports rather than a survey of self-reported victimization. Example 1.8.1 illustrates some of the issues:

Example 1.8.1¹³

AN EXCERPT FROM AN ARTICLE ILLUSTRATING HOW DOMESTIC VIOLENCE
DEFINITION IS RELATED TO ITS MEASUREMENT

By using different definitions and ways of operationalizing DV, other forms of family violence may be omitted from the analysis. Pinchevsky and Wright (2012) note that researchers should expand their definitions of abuse in future research to be broader and more inclusive of different types of abuse. The current research uses a broader definition of DV by examining all domestic offenses that were reported in Chicago and each of the counties in Illinois and aims to capture a more accurate representation of the different forms of DV.

Thus, precise definitions for key terms help guide the most appropriate strategy to measure these terms, and help translate the concept into a variable. More information about conceptual and operational definitions of key terms in a study is provided in Chapter 4.

✓ **Guideline 9: Many Research Reports lack Information on Matters that are Potentially Important for Evaluating the Quality of Research**

Comment: In most journals, research reports of more than 15 pages are rare. Journal space is limited by economics: journals have limited readership and thus a limited paid circulation, and they seldom have advertisers. Even with electronic-only versions, there is a consideration of

13 Morgan, R. E., & Jasinski, J. L. (2017). Tracking violence: Using structural-level characteristics in the analysis of domestic violence in Chicago and the state of Illinois. *Crime & Delinquency*, 63(4), 391–411.

curbing the editorial/peer-review workload, and thus a requirement to describe the study as concisely as possible.¹⁴ Given this situation, researchers must judiciously choose the details to include into the report. Sometimes, they may omit information that readers deem important.

Omitted details can cause problems during research evaluation. For instance, it is common for researchers to describe in general terms the questionnaires and attitude scales they used without reporting the exact wording of the questions.¹⁵ Yet there is considerable research indicating that how items are worded can affect the results of a study.

Another important source of information about a study is descriptive statistics for the main variables included into subsequent analyses. This information is often crucial in judging the sample, as well as the appropriateness of analytical and statistical methods used in the study. The fact that full descriptive statistics are provided can also serve as an important proxy for the authors' diligence, professionalism, and integrity. Chapter 10 provides more information on how to evaluate some of the statistical information often presented in research articles.

As students apply the evaluation criteria in the remaining chapters of this book while evaluating research, they may often find that they must answer "insufficient information to make a judgment" and thus put I/I (insufficient information) instead of grading the evaluation criterion on a scale from 1 (very unsatisfactory) to 5 (very satisfactory).

✓ **Guideline 10: Some Published Research is Obviously Flawed**

Comment: With many hundreds of editors of and contributors to academic journals, it is understandable that published empirical articles vary in quality, with some being very obviously weak in terms of their research methodology.¹⁶

Undoubtedly, some weak articles simply slip past less-skilled editors. More likely, an editor may make a deliberate decision to publish a weak study report because the problem it explores is of current interest to the journal's readers. This is especially true when there is a new topic of interest, such as a new educational reform, a newly recognized disease, or a new government initiative. The editorial board of a journal might reasonably conclude that publishing studies on such new topics is important, even if the initial studies are weak.

14 Also consider the fact that our culture is generally moving towards a more fast-paced, quick-read (140-characters?) environment, which makes long(ish) pieces often untenable.

15 This statement appears in each issue of *The Gallup Poll Monthly*: "In addition to sampling error, readers should bear in mind that question wording [. . .] can introduce additional systematic error or 'bias' into the results of opinion polls." Accordingly, *The Gallup Poll Monthly* reports the exact wording of the questions it uses in its polls. Other researchers cannot always do this because the measures they use may be too long to include in a journal article or may be copyrighted by publishers prohibiting the release of the items to the public.

16 Many journals are refereed, or peer-reviewed. This means that the editor has experts who act as referees by evaluating each paper submitted for possible publication. These experts make their judgments without knowing the identity of the researcher who submitted the paper (that is why the process is also called 'blind peer review'), and the editor uses their input in deciding which papers to publish as journal articles. The author then receives the editor's decision, which includes anonymous peer reviews of the author's manuscript.

Sometimes, studies with very serious methodological problems are labeled as *pilot studies*, in either their titles or introductions to the articles. A pilot study is a preliminary study that allows a researcher to try out new methods and procedures for conducting research, often with small samples. Pilot studies may be refined in subsequent, more definitive, larger studies. Publication of pilot studies, despite their limited samples and other potential weaknesses, is justified on the basis that they may point other researchers in the direction of promising new leads and methods for further research.

✓ **Guideline 11: Many Researchers Acknowledge Obvious Flaws in Their Research**

Comment: Many researchers very briefly point out the most obvious flaws in their research. They typically do this in the last section of their reports, which is the Discussion section. While they tend to be brief and deal with only the most obvious problems, these acknowledgments can be a good starting point in the evaluation of a research report.

Example 1.11.1 shows the researchers' description of the limitations of their research on Mexican American men's college persistence intentions.

Example 1.11.1¹⁷

RESEARCHERS' DESCRIPTION OF THE LIMITATIONS OF THEIR RESEARCH

Despite the contributions of this study in expanding our understanding of Mexican American men's college persistence intentions, there also are some clear limitations that should be noted. First, several factors limit our ability to generalize this study's findings to other populations of Mexican American male undergraduates. The participants attended a Hispanic-serving 4-year university in a predominantly Mexican American midsize southern Texas town located near the U.S.-México border. While the majority of U.S. Latinos live in the Southwest region, Latinos are represented in communities across the U.S. (U.S. Census Bureau, 2008c). Additionally, the study's generalizability is limited by the use of nonrandom sampling methods (e.g., self-selection bias) and its cross-sectional approach (Heppner, Wampold, & Kivlighan, 2007).

✓ **Guideline 12: No Research Report Provides "Proof"**

Comment: Conducting research is fraught with pitfalls, any one study may have very misleading results, and all studies can be presumed to be flawed to some extent. In light of this, individual empirical research articles should be evaluated carefully to identify those that are most likely to provide sound results. In addition, a consumer of research should consider the entire body of research on a given problem. If different researchers using different research

17 Ojeda, L., Navarro, R. L., & Morales, A. (2011). The role of la familia on Mexican American men's college persistence intentions. *Psychology of Men & Masculinity*, 12(3), 216–229.

methods with different types of strengths and weaknesses all reach similar conclusions, consumers of research may say that they have *considerable confidence* in the conclusions of the body of research.

The process of conducting repeated studies on the same topic using different methods or target populations is called *replication*. It is one of the most important ways in science to check whether the findings of previous studies hold water or are a result of random chance. To the extent that the body of research on a topic yields mixed results, consumers of research should lower their degree of confidence. For instance, if the studies with a more scientifically rigorous methodology point in one direction while weaker ones point in a different direction, consumers of research might say that they have *some confidence* in the conclusion suggested by the stronger studies but that the evidence is not conclusive yet.

✓ **Guideline 13: Other Things Being Equal, Research Related to Theories is more Important than Non-Theoretical Research**

Comment: A given theory helps explain interrelationships among a number of variables and often has implications for understanding human behavior in a variety of settings.¹⁸ Theories provide major causal explanations to help us “see the forest for the trees”, to make sense of the world around us. Why do people commit crimes? What causes autism? How do children learn a language? Why are people reluctant to consider evidence contradicting their worldview? Why are lower-class voters less likely to participate in elections? These and many other questions can be best answered with a logical big-picture explanation, or theory.

Studies that have results consistent with a theory lend support to the theory. Those with inconsistent results argue against the theory. (Remember that no one study ever provides proof.) After a number of studies relating to the theory have been conducted, their results provide accumulated evidence that argues for or against the theory, as well as lend evidence that can assist in modifying the theory. Often, researchers explicitly discuss theories that are relevant to their research, as illustrated in Example 1.13.1.

Example 1.13.1¹⁹

PORTIONS OF RESEARCHERS' DISCUSSION OF A THEORY RELATED TO THEIR RESEARCH

One of the most influential theories regarding women's intentions to stay in or leave abusive relationships is social exchange theory, which suggests that these kinds of relational decisions follow from an analysis of the relative cost-benefit ratio of remaining in a relationship (Kelley & Thibaut, 1978). On the basis of this theory, many researchers have

18 Notice that the word theory has a similar meaning when used in everyday language: for example, “I have a theory on why their relationship did not work out.”

19 Gordon, K. C., Burton, S., & Porter, L. (2004). Predicting the intentions of women in domestic violence shelters to return to partners: Does forgiveness play a role? *Journal of Family Psychology*, 18(2), 331–338.

posited that whereas escaping the abuse may appear to be a clear benefit, the costs associated with leaving the relationship may create insurmountable barriers for many abused women.

The role of theoretical considerations in the evaluation of research is discussed in greater detail in Chapter 4.

✓ **Guideline 14: As a Rule, the Quality of a Research Article is Correlated with the Quality of a Journal in Which the Article is Published**

Comment: It is no surprise that most authors want their research published in the best, highest-ranked journals. Thus, the top journals in each field of science get the most article submissions and, as a result, can be very selective in choosing which research reports to publish (basically, the best ones). Those authors whose paper got rejected from the top journal then usually move down the list and submit the article (or its revised version) to the next best one. If rejected from that one as well, the article then gets submitted to a second-tier journal, and so on. This typical process is another reason why the quality/ranking of a journal is usually a good proxy for the quality of articles published there.

Generally, the *journal impact factor* is a metric that provides a good idea of the journal quality. Impact factor for a journal is calculated based on how often the studies recently published in the journal are cited by other researchers. A quick Google search of journal rankings by discipline can provide an easy way to see how journals stack up against one another in your field of study.²⁰

✓ **Guideline 15: To Become an Expert on a Topic, one must Become an Expert at Evaluating Original Reports of Research**

Comment: An expert is someone who knows not only broad generalizations about a topic but also the nuances of the research that underlie them. In other words, he or she knows the particular strengths and weaknesses of the major studies used to arrive at the generalizations. Put another way, an expert on a topic knows the *quality of the evidence* regarding that topic and bases generalizations from the research literature on that knowledge.

20 The reader should also be very cautious of any journal that has no impact factor metric. See more information about predatory journals and publishers in the online resources for this chapter.

Chapter 1 Exercises

Part A

Directions: The 15 guidelines discussed in this chapter are repeated below. For each one, indicate the extent to which you were already familiar with it before reading this chapter. Use a scale from 1 (not at all familiar) to 5 (very familiar).

Guideline 1: Researchers often examine narrowly defined problems.

Familiarity rating: 1 2 3 4 5

Guideline 2: Researchers often conduct studies in artificial settings.

Familiarity rating: 1 2 3 4 5

Guideline 3: Researchers use less-than-perfect methods of measurement.

Familiarity rating: 1 2 3 4 5

Guideline 4: Researchers use less-than-perfect samples.

Familiarity rating: 1 2 3 4 5

Guideline 5: Even a straightforward analysis of data can produce misleading results.

Familiarity rating: 1 2 3 4 5

Guideline 6: Even a single, isolated flaw in research methods can lead to seriously misleading results.

Familiarity rating: 1 2 3 4 5

Guideline 7: Research reports often contain many details, which can be very important when evaluating a report.

Familiarity rating: 1 2 3 4 5

Guideline 8: Many research articles provide precise definitions of key terms to help guide the measurement of the associated concepts.

Familiarity rating: 1 2 3 4 5

Guideline 9: Many research reports lack information on matters that are potentially important for evaluating a research article.

Familiarity rating: 1 2 3 4 5

Guideline 10: Some published research is obviously flawed.

Familiarity rating: 1 2 3 4 5

Guideline 11: Many researchers acknowledge obvious flaws in their research.

Familiarity rating: 1 2 3 4 5

Guideline 12: No research report provides “proof.”

Familiarity rating: 1 2 3 4 5

Guideline 13: Other things being equal, research related to theories is more important than non-theoretical research.

Familiarity rating: 1 2 3 4 5

Guideline 14: As a rule, the quality of research articles is correlated with the quality of the journal the article is published in.

Familiarity rating: 1 2 3 4 5

Guideline 15: To become an expert on a topic, one must become an expert at evaluating original reports of research.

Familiarity rating: 1 2 3 4 5

Part B: Application

Directions: Read an empirical research article published in an academic, peer-reviewed journal, and respond to the following questions. The article may be one that you select or one that is assigned by your instructor. If you are using this book without any prior training in research methods, do the best you can in answering the questions at this point. As you work through this book, your evaluations will become increasingly sophisticated.

1. How narrowly is the research problem defined? In your opinion, is it too narrow? Is it too broad? Explain.
2. Was the research setting artificial (e.g., a laboratory setting)? If yes, do you think that the gain in the control of extraneous variables offsets the potential loss of information that would be obtained in a study in a more real-life setting? Explain.
3. Are there any obvious flaws or weaknesses in the researcher's methods of measurement or observation? Explain. (Note: This aspect of research is usually described under the subheading Measures.)
4. Are there any obvious sampling flaws? Explain.
5. Was the analysis statistical or non-statistical? Was the description of the results easy to understand? Explain.
6. Are definitions of the key terms provided? Is the measurement strategy for the associated variables aligned with the provided definitions? Explain.
7. Were the descriptions of procedures and methods sufficiently detailed? Were any important details missing? Explain.
8. Does the report lack information on matters that are potentially important for evaluating it?
9. Do the researchers include a discussion of the limitations of their study?
10. Does the researcher imply that his or her research proves something? Do you believe that it proves something? Explain.

11. Does the researcher describe related theories?
12. Can you assess the quality of the journal the article is published in? Can you find information online about the journal's ranking or impact factor?
13. Overall, was the research obviously very weak? If yes, briefly describe its weaknesses and speculate on why it was published despite them.
14. Do you think that as a result of reading this chapter and evaluating a research report you are becoming more expert at evaluating research reports? Explain.

Evaluating Titles

Titles help consumers of research to identify journal articles of interest to them. A preliminary evaluation of a title should be made when it is first encountered. After the article is read, the title should be re-evaluated to ensure that it accurately reflects the contents of the article.

Apply the questions that follow while evaluating titles. The questions are stated as ‘yes–no’ questions, where a “yes” indicates that you judge the characteristic to be satisfactory. You may also want to rate each characteristic using a scale from 1 to 5, where 5 is the highest rating. N/A (not applicable) and I/I (insufficient information to make a judgment) may also be used when necessary.

___ **1. Is the Title Sufficiently Specific?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: On any major topic in the social and behavioral sciences, there are likely to be many hundreds of research articles published in academic journals. In order to help potential readers locate those that are most relevant to their needs, researchers should use titles that are sufficiently specific so that each article can be differentiated from the other research articles on the same topic.

Consider the topic of depression, which has been extensively investigated. The title in Example 2.1.1 is insufficiently specific. Contrast it with the titles in Example 2.1.2, each of which contains information that differentiates it from the others.

Example 2.1.1

A TITLE THAT IS INSUFFICIENTLY SPECIFIC

— An Investigation of Adolescent Depression and Its Implications

Example 2.1.2

THREE TITLES THAT ARE MORE SPECIFIC THAN THE ONE IN EXAMPLE 2.1.1

- Gender Differences in the Expression of Depression by Early Adolescent Children of Alcoholics
- The Impact of Social Support on the Severity of Postpartum Depression Among Adolescent Mothers
- The Effectiveness of Cognitive Therapy in the Treatment of Adolescent Students with Severe Clinical Depression

___ 2. Is the Title Reasonably Concise?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: While a title should be specific (see the previous evaluation question), it should be fairly concise. Titles of research articles in academic journals typically are 15 words or fewer. When a title contains more than 20 words, it is likely that the researcher is providing more information than is needed by consumers of research who want to locate articles.¹

___ 3. Are the Primary Variables Mentioned in the Title?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Variables are the characteristics that vary from one participant to another. In Example 2.3.1, the variables are (1) television viewing habits, (2) mathematics achievement, and (3) reading achievement. For instance, the children *vary* (or differ) in their reading achievement, with some children achieving more than others. Likewise, they vary in terms of their mathematics achievement and their television viewing habits.

Example 2.3.1

A TITLE THAT MENTIONS THREE VARIABLES

- The Relationship Between Young Children’s Television Viewing Habits and Their Achievement in Mathematics and Reading

Note that “young children” is *not* a variable because the title clearly suggests that only young children were studied. In other words, being a young child does not vary in this study. Instead, it is a *common trait* of all the participants in the study, or a characteristic of the study sample.

¹ Titles of theses and dissertations tend to be longer than those of journal articles.

___ 4. When There are Many Variables, are the *Types of Variables* Referred to?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: When researchers examine many specific variables in a given study, they may refer to the *types* of variables in their titles rather than naming each one individually. For instance, suppose a researcher administered a standardized achievement test that measured spelling ability, reading comprehension, vocabulary knowledge, mathematical problem-solving skills, and so on. Naming all these variables would create a title that is too long. Instead, the researcher could refer to this collection of variables measured by the test as *academic achievement*, which is done in Example 2.4.1.

Example 2.4.1

A TITLE IN WHICH TYPES OF VARIABLES (ACHIEVEMENT VARIABLES) ARE IDENTIFIED WITHOUT BEING NAMED SPECIFICALLY

— The Relationship Between Parental Involvement in Schooling and Academic Achievement in the Middle Grades

___ 5. Does the Title Identify the Types of Individuals who Participated or the Types of Aggregate Units in the Sample?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: It is often desirable to include names of populations in the title. From the title in Example 2.5.1, it is reasonable to infer that the population of interest consists of graduate students who are taking a statistics class. This would be of interest to a consumer of research who is searching through a list of the many hundreds of published articles on cooperative learning. For instance, knowing that the research report deals with this particular population might help a consumer rule it out as an article of interest if he or she is trying to locate research on cooperative learning in elementary school mathematics.

Example 2.5.1

A TITLE IN WHICH THE TYPE OF PARTICIPANTS IS MENTIONED

— Effects of Cooperative Learning in a Graduate-Level Statistics Class

Example 2.5.2 also names an important characteristic of the research participants – the fact that they are registered nurses employed by public hospitals.

Example 2.5.2

A TITLE IN WHICH THE TYPE OF PARTICIPANTS IS MENTIONED

- Administrative Management Styles and Job Satisfaction Among Registered Nurses Employed by Public Hospitals

Sometimes, instead of using individuals in a sample, studies use aggregate-level sampling units (such as cities, states, or countries) and compare them to one another. For titles of such research reports, it is important to mention the type of units in the study sample as well. In Example 2.5.3, neighborhoods are such sampling units.

Example 2.5.3

A TITLE IN WHICH THE TYPE OF UNITS IN THE SAMPLE IS NOT ADEQUATELY MENTIONED

- Domestic Violence and Socioeconomic Status: Does the Type of Neighborhood Matter?

Take a closer look at the title in Example 2.5.3 – does it give sufficiently specific information about where the study was conducted? In fact, it is an inadequate title because it fails to mention the key characteristic of the neighborhoods in the study – that they are all located in the city of Sao Paulo, Brazil. Thus, a researcher who is looking, say, for studies conducted in South American countries may not even realize that this article should be checked. A more appropriate title for the study would be: “Domestic Violence and Socioeconomic Status in Sao Paolo, Brazil: Does the Type of Neighborhood Matter?”

Often, researchers use a particular group of units or participants only because they are readily available, such as college students enrolled in an introductory psychology class who are required to participate in research projects. Researchers might use such individuals even though they are conducting research that might apply to all types of individuals. For instance, a researcher might conduct research to test a social relations theory that might apply to all types of individuals. In such a case, the researcher might omit mentioning the types of individuals (e.g., college students) in the title because the research is not specifically directed at that population.

____ **6. If a Study is Strongly Tied to a Theory, is the Name of the Specific Theory Mentioned in the Title?**

Very							Very		
unsatisfactory	1	2	3	4	5		satisfactory	or N/A	I/I

Comment: Theories help to advance science because they are propositions regarding relationships that have applications in many diverse, specific situations. For instance, a particular learning theory might have applications for teaching kindergarten children as well as for training astronauts. A useful theory leads to predictions about human behavior that can be tested through

research. Many consumers of research are seeking information on specific theories, and mention of them in titles helps these consumers to identify reports of relevant research. Thus, when research is closely tied to a theory, the theory should be mentioned. Example 2.6.1 shows two titles in which specific theories are mentioned.

Example 2.6.1

TWO TITLES THAT MENTION SPECIFIC THEORIES (DESIRABLE)

- Application of Terror Management Theory to Treatment of Rural Battered Women
- Achievement in Science-Oriented Charter Schools for Girls: A Critical Test of the Social Learning Theory

Note that simply using the term *theory* in a title without mentioning the name of the specific theory is not useful to consumers of research. Example 2.6.2 has this undesirable characteristic.

Example 2.6.2

A TITLE THAT REFERS TO THEORY WITHOUT NAMING THE SPECIFIC THEORY (UNDESIRABLE)

- An Examination of Voting Patterns and Social Class in a Rural Southern Community: A Study Based on Theory

7. Has the Author Avoided Describing Results in the Title?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: It is usually inappropriate for a title to describe the results of a research project. Research often raises more questions than it answers. In addition, the results of research are often subject to more than one interpretation. Given that titles need to be concise, attempting to state results in a title is likely to lead to oversimplification.

Consider the title in Example 2.7.1, which undoubtedly oversimplifies the results of the study. A meaningful accounting of the results should address issues such as the following: What type of social support (e.g., parental support, peer support, and so on) is effective? How strong does it need to be to lessen the depression? By how much is depression lessened by strong social support? Because it is almost always impossible to state results accurately and unambiguously in a short title, results ordinarily should *not* be stated at all, as illustrated in the Improved Version of Example 2.7.1.

Example 2.7.1

A TITLE THAT INAPPROPRIATELY DESCRIBES RESULTS

- Strong Social Support Lessens Depression in Delinquent Young Adolescents

Improved Version of Example 2.7.1

A TITLE THAT APPROPRIATELY DOES NOT DESCRIBE RESULTS

- The Relationship Between Social Support and Depression in Delinquent Young Adolescents

8. Has the Author Avoided Using a “Yes–No” Question as a Title?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Because research rarely yields simple, definitive answers, it is seldom appropriate to use a title that poses a simple “yes–no” question. For instance, the title in Example 2.8.1 implies that there is a simple answer to the question it poses. However, a study on this topic undoubtedly explores *the extent to which men and women differ in their opinions on social justice issues* – a much more interesting topic than the one suggested by the title. The Improved Version is cast as a statement and is more appropriate as the title of a research report for publication in an academic journal.

Example 2.8.1

A TITLE THAT INAPPROPRIATELY POSES A “YES–NO” QUESTION

- Do Men and Women Differ in Their Opinions on Social Justice Issues?

Improved Version of Example 2.8.1

A TITLE AS A STATEMENT

- Gender Differences in Opinions on Social Justice Issues

9. If There are a Main Title and a Subtitle, do both Provide Important Information About the Research?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Failure on this evaluation question often results from an author’s use of a ‘clever’ main title that is vague or catchy², followed by a subtitle that identifies the specific content of the research report. Example 2.9.1 illustrates this problem. In this example, the main title fails to impart specific information. In fact, it could apply to many thousands of studies in hundreds of fields, as diverse as psychology and physics, in which researchers find that various combinations of variables (the parts) contribute to our understanding of a complex whole.

2 For additional information about amusing or humorous titles in research literature, see the online resources for this chapter.

Example 2.9.1

A TWO-PART TITLE WITH A VAGUE MAIN TITLE (INAPPROPRIATE)

- The Whole Is Greater Than the Sum of Its Parts: The Relationship Between Playing with Pets and Longevity Among the Elderly

Example 2.9.2 is also deficient because the main title is catchy but does not carry any information about the study.

Example 2.9.2

A TWO-PART TITLE WITH A CATCHY BUT VAGUE MAIN TITLE (INAPPROPRIATE)

- The “Best of the Best”: The Upper-Class Mothers’ Involvement in Their Children’s Schooling

In contrast to the previous two examples, Example 2.9.3 has a main title and a subtitle that both refer to specific variables examined in a research study. The first part names two major variables (“attachment” and “well-being”), while the second part names the two groups that were compared in terms of these variables.

Example 2.9.3

A TWO-PART TITLE IN WHICH BOTH PARTS PROVIDE IMPORTANT INFORMATION

- Attachment to Parents and Emotional Well-Being: A Comparison of African American and White Adolescents

The title in Example 2.9.3 could also be rewritten as a single statement without a subtitle, as illustrated in Example 2.9.4.

Example 2.9.4

A REWRITTEN VERSION OF EXAMPLE 2.9.3

- A Comparison of the Emotional Well-Being and Attachment to Parents in African American and White Adolescents

Examples 2.9.3 and 2.9.4 are equally good. The evaluation question being considered here is neutral on whether a title should be broken into a main title and subtitle. Rather, it suggests that if it is broken into two parts, both parts should provide important information specific to the research being reported.

10. If the Title Implies Causality, does the Method of Research Justify it?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Example 2.10.1 implies that causal relationships (i.e., cause-and-effect relationships) have been examined because the title contains the word *effects*. This is a keyword frequently used by researchers in their titles to indicate that they have explored causality in their studies.

Example 2.10.1

A TITLE IN WHICH CAUSALITY IS IMPLIED BY THE WORD *EFFECTS*

— The Effects of Computer-Assisted Instruction in Mathematics on Students' Computational Skills

A common method for examining causal relationships is conducting an *experiment*. An experiment is a study in which researchers give treatments to participants to determine whether the treatments *cause* changes in outcomes.³

In a traditional experiment, different groups of participants are given different treatments (e.g., one group receives computer-assisted instruction while a more traditional method is used to teach another group). The researcher then compares the outcomes obtained through the application of the various treatments.⁴ When such a study is conducted, the use of the word *effects* in the title is justified.⁵

The title in Example 2.10.2 also suggests that the researcher examined a causal relationship because of the inclusion of the word *effects*. Note that in this case, however, the researcher probably did *not* investigate the relationship using an experiment because it would be unethical to manipulate breakfast as an independent variable (i.e., researchers would not want to assign some students to receive breakfast while denying it to others for the purposes of an experiment).

Example 2.10.2

A TITLE IN WHICH CAUSALITY IS IMPLIED BY THE WORD *EFFECTS*

— The Effects of Breakfast on Student Achievement in the Primary Grades

- 3 Notice that the word *experiment* is used in a similar way in everyday language: for example, "I don't know if using local honey would actually relieve my allergy symptoms but I will try it as an experiment."
- 4 Experiments can also be conducted by treating a given person or group differently *at different points in time*. For instance, a researcher might praise a child for staying in his or her seat in the classroom on some days and not praise him or her on others and then compare the child's seat-staying behavior under the two conditions.
- 5 The evaluation of experiments is considered in Chapter 9. Note that this evaluation question merely asks whether there is a basis for suggesting causality in the title. This evaluation question does not ask for an evaluation of the quality of the experiment or quasi-experiment.

When it is not possible to conduct an experiment on a causal issue, researchers often conduct what are called *ex post facto studies* (also called *causal-comparative* or *quasi-experimental studies*). In these studies, researchers identify students who differ on some outcome (such as students who are high and low in achievement in the primary grades) but who are the same on demographics and other potentially influential variables (such as parents' highest level of education, parental income, quality of the schools the children attend, and so on). Comparing the breakfast-eating habits of the two groups (i.e., high- and low-achievement groups) might yield some useful information on whether eating breakfast *affects*⁶ students' achievement because the two groups are similar on other variables that might account for differences in achievement (e.g., their parents' level of education is similar). If a researcher has conducted such a study, the use of the word *effects* in the title is justified.

Note that simply examining a relationship without controlling for potentially confounding variables does *not* justify a reference to causality in the title. For instance, if a researcher merely compared the achievement of children who regularly eat breakfast with those who do not, without controlling for other explanatory variables, a causal conclusion (and, hence, a title suggesting it) usually cannot be justified.

Also note that synonyms for *effect* are *influence* and *impact*. They should usually be reserved for use in the titles of studies that are either experiments or quasi-experiments (like *ex post facto* studies).

11. Is the Title Free of Jargon and Acronyms that Might be Unknown to the Audience for the Research Report?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Professionals in all fields use jargon and acronyms (i.e., shorthand for phrases, usually in capital letters) for efficient and accurate communication with their peers. However, their use in titles of research reports is inappropriate unless the researchers are writing exclusively for such peers. Consider Example 2.11.1. If ACOA⁷ is likely to be well known to all the readers of the journal in which this title appears, its use is probably appropriate. Otherwise, it should be spelled out or have its meaning paraphrased. As you can see, it can be difficult to make this judgment without being familiar with the journal and its audience.

Example 2.11.1

A TITLE WITH AN ACRONYM THAT IS NOT SPELLED OUT (MAY BE INAPPROPRIATE IF NOT WELL-KNOWN BY THE READING AUDIENCE)

— Job Satisfaction and Motivation to Succeed Among ACOA in Managerial Positions

⁶ Note that in reference to an outcome caused by some treatment, the word is spelled *effect* (i.e., it is a noun). As a verb meaning “to influence”, the word is spelled *affect*.

⁷ ACOA stands for Adult Children of Alcoholics.

___ 12. Are any Highly Unique or Very Important Characteristics of the Study Referred to in the Title or Subtitle?

Very 1 2 3 4 5 Very or N/A I/I
unsatisfactory satisfactory

Comment: On many topics in the social and behavioral sciences, there may be hundreds of studies. To help readers identify those with highly unusual or very important characteristics, reference to these should be made in the title. For instance, in Example 2.12.1, the mention of a “nationally representative sample” may help distinguish that study from many others employing only local convenience samples.

Example 2.12.1

A TITLE THAT POINTS OUT AN IMPORTANT STRENGTH IN SAMPLING

— The Relationship Between Teachers’ Job Satisfaction and Compensation in a Nationally Representative Sample

___ 13. Overall, is the Title Effective and Appropriate?

Very 1 2 3 4 5 Very or N/A I/I
unsatisfactory satisfactory

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter, and taking into account any additional considerations and concerns you may have after reading the entire research article.

Chapter 2 Exercises

Part A

Directions: Evaluate each of the following titles to the extent that it is possible to do so without reading the complete research reports. The references for the titles are given below. All are from journals that are widely available in large academic libraries. More definitive application of the evaluation criteria for titles is made possible by reading the articles in their entirety and then evaluating their titles. Keep in mind that there can be considerable subjectivity in determining whether a title is adequate.

1. Sugar and Spice and All Things Nice: The Role of Gender Stereotypes in Jurors’ Perceptions of Criminal Defendants⁸

8 Strub, T., & McKimmie, B. M. (2016). *Psychiatry, Psychology and Law*, 23, 487–498.

Titles

2. Being a Sibling⁹
3. Estimating the Potential Health Impact and Costs of Implementing a Local Policy for Food Procurement to Reduce the Consumption of Sodium in the County of Los Angeles¹⁰
4. More Than Numbers Matter: The Effect of Social Factors on Behaviour and Welfare of Laboratory Rodents and Non-Human Primates¹¹
5. Social Support Provides Motivation and Ability to Participate in Occupation¹²
6. Cognitive Abilities of Musicians¹³
7. Social Exclusion Decreases Prosocial Behavior¹⁴
8. ICTs, Social Thinking and Subjective Well-Being: The Internet and Its Representations in Everyday Life¹⁵
9. Child Care and Mothers' Mental Health: Is High-Quality Care Associated with Fewer Depressive Symptoms?¹⁶
- 10 Education: Theory, Practice, and the Road Less Followed¹⁷
11. Wake Me Up When There's a Crisis: Progress on State Pandemic Influenza Ethics Preparedness¹⁸
12. Teachers' Perceptions of Integrating Information and Communication Technologies into Literacy Instruction: A National Survey in the United States¹⁹
13. Provincial Laws on the Protection of Women in China: A Partial Test of Black's Theory²⁰

Part B

Directions: Examine several academic journals that publish on topics of interest to you. Identify two empirical articles with titles you think are especially strong in terms of the evaluation questions presented in this chapter. Also, identify two titles that you believe have clear weaknesses. Bring the four titles to class for discussion.

- 9 Baumann, S. L., Dyches, T. T., & Braddick, M. (2005). *Nursing Science Quarterly*, 18, 51.
- 10 Gase, L. N., Kuo, T., Dunet, D., Schmidt, S. M., Simon, P. A., & Fielding, J. E. (2011). *American Journal of Public Health*, 101, 1501.
- 11 Olsson, I. A. S., & Westlund, K. (2007). *Applied Animal Behaviour Science*, 103, 229.
- 12 Isaksson, G., Lexell, J., & Skär, L. (2007). *OTJR: Occupation, Participation and Health*, 27, 23.
- 13 Giovagnoli, A. R., & Raglio, A. (2011). *Perceptual and Motor Skills*, 113, 563.
- 14 Twenge, J. M., Baumeister, R. F., DeWall, C. N., Ciarocco, N. J., & Bartels, J. M. (2007). *Journal of Personality and Social Psychology*, 92, 56.
- 15 Contarello, A., & Sarrica, M. (2007). *Computers in Human Behavior*, 23, 1016.
- 16 Gordon, R., Usdansky, M. L., Wang, X., & Gluzman, A. (2011). *Family Relations*, 60, 446.
- 17 Klaczynski, P. A. (2007). *Journal of Applied Developmental Psychology*, 28, 80.
- 18 Thomas, J. C., & Young, S. (2011). *American Journal of Public Health*, 101, 2080.
- 19 Hutchison, A., & Reinking, D. (2011). *Reading Research Quarterly*, 46, 312.
- 20 Lu, H., & Miethe, T. D. (2007). *International Journal of Offender Therapy and Comparative Criminology*, 51, 25.

Evaluating Abstracts

An abstract is a summary of a research report that appears below its title. Like the title, it helps consumers of research identify articles of interest. This function of abstracts is so important that the major computerized databases in the social and behavioral sciences provide the abstracts as well as the titles of the articles they index.

Many journals have a policy on the maximum length of abstracts. It is common to allow a maximum of 100 to 250 words.¹ When evaluating abstracts, you will need to make subjective decisions about how much weight to give to the various elements included within them, given that their length typically is severely restricted.

Make a preliminary evaluation of an abstract when you first encounter it. After reading the associated article, re-evaluate the abstract. The evaluation questions that follow are stated as ‘yes–no’ questions, where a “yes” indicates that you judge the characteristic being considered as satisfactory. You may also want to rate each characteristic using a scale from 1 to 5, where 5 is the highest rating. N/A (not applicable) and I/I (insufficient information to make a judgment) may also be used when necessary.

____ 1. Is the Purpose of the Study Referred to or at Least Clearly Implied?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Many writers begin their abstracts with a brief statement of the purpose of their research. Examples 3.1.1 and 3.1.2 show the first sentences of abstracts in which this was done.

¹ The *Publication Manual of the American Psychological Association* (APA) suggests that an abstract should not exceed 150 words.

Example 3.1.1²

FIRST SENTENCE OF AN ABSTRACT THAT SPECIFICALLY STATES THE PURPOSE OF THE STUDY (ACCEPTABLE)

The purpose of the current investigation is to examine the characteristics of college students with attention-deficit hyperactivity disorder symptoms who misuse their prescribed psychostimulant medications.

Example 3.1.2³

FIRST SENTENCE OF AN ABSTRACT THAT IMPLIES THE PURPOSE OF THE STUDY (ALSO ACCEPTABLE)

This is a pioneering study examining the effect of different types of social support on the mental health of the physically disabled in mainland China.

Note that even though the word *purpose* is not used in Example 3.1.2, the purpose of the study is clearly implied: to examine the effects of social support on mental health in a particular population.

___ 2. Does the Abstract Mention Highlights of the Research Methodology?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Given the shortness of an abstract, researchers usually can provide only limited information on their research methodology. However, even brief highlights can be helpful to consumers of research who are looking for research reports of interest. Consider Example 3.2.1, which is taken from an abstract. The fact that the researchers used qualitative methodology employing interviews with small samples is an important methodological characteristic that might set this study apart from others on the same topic.

Example 3.2.1⁴

EXCERPT FROM AN ABSTRACT THAT DESCRIBES HIGHLIGHTS OF RESEARCH METHODOLOGY (DESIRABLE)

- 2 Jardin, B., Looby, A., & Earleywine, M. (2011). Characteristics of college students with attention-deficit hyperactivity disorder symptoms who misuse their medications. *Journal of American College Health*, 59(5), 373–377.
- 3 Wu, Q., & Mok, B. (2007). Mental health and social support: A pioneering study on the physically disabled in Southern China. *International Journal of Social Welfare*, 16(1), 41–54.
- 4 Saint-Jacques, M.-C., Robitaille, C., Godbout, É., Parent, C., Drapeau, S., & Gagne, M.-H. (2011). The process distinguishing stable from unstable stepfamily couples: A qualitative analysis. *Family Relations*, 60(5), 545–561.

Second marriages are known to be more fragile than first marriages. To better understand the factors that contribute to this fragility, this qualitative study compared stepfamilies that stayed together with those that separated by collecting interview data from one adult in each of the former ($n = 31$) and latter ($n = 26$) stepfamilies.

Likewise, Example 3.2.2 provides important information about research methodology (the fact that a telephone survey was used).

Example 3.2.2⁵

EXCERPT FROM AN ABSTRACT THAT DESCRIBES HIGHLIGHTS OF RESEARCH
METHODOLOGY (DESIRABLE)

Data were collected via telephone survey with the use of a 42-item survey instrument.

3. Has the Researcher Omitted the Titles of Measures (Except when These are the Focus of the Research)?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Including the full, formal titles of published measures such as tests, questionnaires, and scales in an abstract is *usually* inappropriate (see the exception below) because their names take up space that could be used to convey more important information. Note that consumers of research who are interested in the topic will be able to find the full names of the measures in the body of the article, where space is less limited than in an abstract. A comparison of Examples 3.3.1 and 3.3.2 shows how much space can be saved by omitting the names of the measures while conveying the same essential information.

Example 3.3.1

AN EXCERPT FROM AN ABSTRACT THAT NAMES THE TITLES OF MEASURES
(INAPPROPRIATE DUE TO SPACE LIMITATIONS IN ABSTRACTS)

A sample of 483 college males completed the Attitudes Toward Alcohol Scale (Fourth Edition, Revised), the Alcohol Use Questionnaire, and the Manns–Herschfield Quantitative Inventory of Alcohol Dependence (Brief Form).

5 Miller, L. M. (2011). Emergency contraceptive pill (ECP) use and experiences at college health centers in the mid-Atlantic United States: Changes since ECP went over-the-counter. *Journal of American College Health*, 59(8), 683–689.

Example 3.3.2

AN IMPROVED VERSION OF EXAMPLE 3.3.1

A sample of 483 college males completed measures of their attitudes toward alcohol, their alcohol use, and their dependence on alcohol.

The exception: If the primary purpose of the research is to evaluate the reliability and validity of one or more specific measures, it is appropriate to name them in the abstract as well as in the title. This will help readers who are interested in locating research on the characteristics of specific measures. In Example 3.3.3, mentioning the name of a specific measure is appropriate because the purpose of the study is to determine a characteristic of the measure (its reliability).

Example 3.3.3EXCERPT FROM AN ABSTRACT THAT PROVIDES THE TITLE OF A MEASURE
(APPROPRIATE BECAUSE THE PURPOSE OF THE RESEARCH IS TO INVESTIGATE
THE MEASURE)

Test-retest reliability of the Test of Variables of Attention (T.O.V.A.) was investigated in two studies using two different time intervals: 90 min and 1 week (7 days). To investigate the 90-min reliability, 31 school-age children ($M = 10$ years, $SD = 2.66$) were administered the T.O.V.A., then re-administered the test.

___ 4. Are the Highlights of the Results Described?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Example 3.4.1 shows the last three sentences of an abstract, which describe the highlights of the results of a study. Notice that the researchers make general statements about their results, such as “working-class women, in particular, view marriage less favorably,” without stating precisely how much less favorably. General statements of this type are acceptable given the need for brevity in an abstract. In other words, it is acceptable to point out highlights of the results in general terms.

Example 3.4.1⁶

LAST THREE SENTENCES OF ABSTRACT (HIGHLIGHTS OF RESULTS REPORTED)

More than two thirds of respondents mentioned concerns with divorce. Working-class women, in particular, view marriage less favorably than do their male and middle-class

6 Miller, A. J., Sassler, S., & Kusi-Appouh, D. (2011). The specter of divorce: Views from working- and middle-class cohabitators. *Family Relations*, 60(5), 602–616.

counterparts, in part because they see marriage as hard to exit and are reluctant to assume restrictive gender roles. Middle-class cohabitators are more likely to have concrete wedding plans and believe that marriage signifies a greater commitment than does cohabitation.

Note that there is nothing inherently wrong with providing specific statistical results in an abstract if space permits and the statistics are understandable within the limited context of an abstract. Example 3.4.2 illustrates how this might be done.

Example 3.4.2⁷

PART OF AN ABSTRACT WITH SOME SPECIFIC RESULTS REPORTED
AS HIGHLIGHTS

Results suggest that increasing the proportion of peers who engage in criminal activities by 5% will increase the likelihood an individual engages in criminal activities by 3 percentage points.

___ 5. If the Study is Strongly Tied to a Theory, is the Theory Mentioned in the Abstract?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: As indicated in the previous chapter, a theory that is central to a study might be mentioned in the title. If such a theory is not mentioned in the title, it should be mentioned in the abstract, as illustrated in Example 3.5.1. It is also acceptable to mention it in both the title and abstract, as illustrated in Example 3.5.2. (Note that italics have been used in these examples for emphasis.)

Example 3.5.1⁸

TITLE AND ABSTRACT IN WHICH A SPECIFIC THEORY IS NAMED IN THE ABSTRACT
BUT NOT IN THE TITLE (ACCEPTABLE TO DE-EMPHASIZE THEORY)

Title: Self-Efficacy Program to Prevent Osteoporosis Among Chinese Immigrants

Objectives: The aim of this study was to evaluate the preliminary effectiveness of an educational intervention based on the *self-efficacy theory* aimed at increasing the knowledge of osteoporosis and adoption of preventive behaviors, including regular exercise and osteoporosis medication adherence, designed for Chinese immigrants, aged 45 years or above, living in the United States.

7 Kim, J., & Fletcher, J. M. (2018). The influence of classmates on adolescent criminal activities in the United States. *Deviant Behavior*, 39(3), 275–292.

8 Qi, B.-B., Resnick, B., Smeltzer, S. C., & Bausell, B. (2011). Self-efficacy program to prevent osteoporosis among Chinese immigrants. *Nursing Research*, 60(6), 393–404.

Example 3.5.2⁹

TITLE AND ABSTRACT IN WHICH A SPECIFIC THEORY IS MENTIONED IN THE TITLE AND ABSTRACT (ACCEPTABLE TO EMPHASIZE THEORY)

Title: An Exploration of Female Offenders' Memorable Messages from Probation and Parole Officers on the Self-Assessment of Behavior from a *Control Theory* Perspective

Abstract (first half): Guided by *control theory*, this study examines memorable messages that women on probation and parole receive from their probation and parole agents. Women interviewed for the study were asked to report a memorable message they received from an agent, and to describe situations if/when the message came to mind in three contexts likely to emerge from a *control theory* perspective: when they did something of which they were proud, when they stopped themselves from doing something they would later regret, and when they did something of which they were not proud.

___ 6. Has the Researcher Avoided Making Vague References to Implications and Future Research Directions?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Most researchers discuss the implications of their research and directions for future research near the end of their articles. The limited amount of space allotted to abstracts usually should not be used to make vague references to these matters. Example 3.6.1 is the closing sentence from an abstract. It contains vague references to implications and future research.

Example 3.6.1

LAST SENTENCE OF AN ABSTRACT WITH VAGUE REFERENCES TO IMPLICATIONS AND FUTURE RESEARCH (INAPPROPRIATE)

This article concludes with a discussion of both the implications of the results and directions for future research.

The phrase in Example 3.6.1 could safely be omitted from the abstract without causing a loss of important information, because most readers will correctly assume that most research reports discuss these elements. An alternative is to state something specific about these matters, as illustrated in Example 3.6.2. Notice that in this example, the researcher does not describe the implications but indicates that the implications will be of special interest to a particular group

9 Cornacchione, J., Smith, S. W., Morash, M., Bohmert, M. N., Cobbina, J. E., & Kashy, D. A. (2016). An exploration of female offenders' memorable messages from probation and parole officers on the self-assessment of behavior from a control theory perspective. *Journal of Applied Communication Research*, 44(1), 60–77.

of professionals – school counselors. This will alert school counselors that this article (among the many hundreds of others on drug abuse) might be of special interest to them. If space does not permit such a long closing sentence in the abstract, it could be shortened to “Implications for school counselors are discussed.”

Example 3.6.2

IMPROVED VERSION OF EXAMPLE 3.6.1 (LAST SENTENCE OF AN ABSTRACT)

While these results have implications for all professionals who work with adolescents who abuse drugs, special attention is given to the implications for school counselors.

In short, implications and future research do not necessarily need to be mentioned in abstracts. If they are mentioned, however, something specific should be said about them.

___ 7. Does the Abstract Include Purpose/Objectives, Methods, and Results of the Study?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: A recent trend in some academic journals is to require the tri-partitioning of abstracts into Objective–Methods–Results or quad-partitioning them into Purpose–Methods–Results–Conclusions. This is a convenient way to make sure that the key pieces of information are included in the abstract with an explicit subheading. Examples 3.7.1 and 3.7.2 provide an illustration of such partitioned abstracts.

Example 3.7.1¹⁰

THE TRI-PARTITIONING OF AN ABSTRACT INTO OBJECTIVE-METHODS-RESULTS
(VERY HELPFUL)

Objective: The purpose of this study was to examine challenges and recommendations (identified by college administrators) to enforcing alcohol policies implemented at colleges in the southeastern United States. *Methods:* Telephone interviews were conducted with 71 individuals at 21 institutions. *Results:* Common challenges included inconsistent enforcement, mixed messages received by students, and students’ attitudes toward alcohol use. The most common recommendations were ensuring a comprehensive approach, collaboration with members of the community, and enhanced alcohol education.

10 Cremeens, J. L., Usdan, S. L., Umstattd, M. R., Talbott, L. L., Turner, L., & Perko, M. (2011). Challenges and recommendations to enforcement of alcohol policies on college campuses: An administrator’s perspective. *Journal of American College Health, 59*(5), 427–430.

Example 3.7.2¹¹

THE QUAD-PARTITIONING OF AN ABSTRACT INTO PURPOSE-METHODS-RESULTS-CONCLUSIONS (VERY HELPFUL)

Purpose: The present study examines whether experiences of household food insecurity during childhood are predictive of low self-control and early involvement in delinquency.

Methods: In order to examine these associations, we employ data from the Fragile Families and Child Wellbeing Study (FFCWS) – a national study that follows a large group of children born in the U.S. between 1998 and 2000.

Results: Children raised in food *insecure* households exhibit significantly lower levels of self-control during early childhood and higher levels of delinquency during late childhood than children raised in food *secure* households, net of covariates. Both transient and persistent food insecurity are significantly and positively associated with low self-control and early delinquency, although persistent food insecurity is associated with larger increases in the risk of low self-control and early delinquency. Ancillary analyses reveal that low self-control partly explains the association between food insecurity and early delinquency.

Conclusions: The general theory of crime may need to be expanded to account for the role of early life stressors linked to a tenuous supply of healthy household foods in the development of self-control. Future research should seek to further elucidate the process by which household food insecurity influences childhood self-control and early delinquency.

However, even if a particular journal does not require the partitioning of abstracts, it is still a good rule of thumb to look for these key pieces of information when evaluating an abstract.

___ 8. Overall, is the Abstract Effective and Appropriate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter, while taking into account any additional considerations and concerns you may have.

When answering this evaluation question, pay special attention to whether all three major elements described in the previous section (objectives, methods, and results) are included in the abstract.

11 Jackson, D. B., Newsome, J., Vaughn, M. G., & Johnson, K. R. (2018). Considering the role of food insecurity in low self-control and early delinquency. *Journal of Criminal Justice*, 56, 127–139.

Chapter 3 Exercises

Part A

Directions: Evaluate each of the following abstracts (to the extent that it is possible to do so without reading the associated articles) by answering Evaluation Question 8 (“Overall, is the abstract effective and appropriate?”) using a scale from 1 (very unsatisfactory) to 5 (very satisfactory). In the explanations for your ratings, refer to the other evaluation questions in this chapter. Point out *both* strengths and weaknesses, if any, of the abstracts.

1. *Title:* Effect of an Aerobic Training Program as Complementary Therapy in Patients with Moderate Depression¹²

Abstract: The aim of this study was to assess the effects of an aerobic training program as complementary therapy in patients suffering from moderate depression. Eighty-two female patients were divided into a group that received traditional pharmacotherapy (Fluoxetine 20 mg) and a group that received pharmacotherapy plus an aerobic training program. This program was carried out for eight consecutive weeks, three days per week, and included gymnastics, dancing, and walking. Depressive symptoms were measured with the Beck Depression Inventory and the ICD-10 Guide for Depression Diagnosis, both administered before and after treatments. The results confirm the effectiveness of the aerobic training program as a complementary therapy to diminish depressive symptoms in patients suffering from moderate depression.

Overall, is the abstract effective and appropriate?

1 2 3 4 5

Explain your rating.

2. *Title:* What’s the Problem? A Look at Men in Marital Therapy¹³

Abstract: This study examined the premise that men’s lack of awareness of relational problems contributes to their reluctance to consider, seek, and benefit from couples therapy. Ninety-two couples reported on couple and family problem areas using the Dyadic Adjustment Scale and the Family Assessment Device. No gender differences were found in either the frequency or the pattern of initial problem reports or improvement rates during ten sessions of couples therapy at a university training outpatient clinic. Implications for treatment and recommendations for future research are discussed.

- 12 de la Cerda, P., Cervelló, E., Cocca, A., & Vicianá, J. (2011). Effect of an aerobic training program as complementary therapy in patients with moderate depression. *Perceptual and Motor Skills*, 112(3), 761–769.
- 13 Moynihan, J., & Adams, J. (2007). What’s the problem? A look at men in marital therapy. *American Journal of Family Therapy*, 35(1), 41–51.

Overall, is the abstract effective and appropriate?

1 2 3 4 5

Explain your rating.

3. *Title: Middle School Drinking: Who, Where, and When*¹⁴

Abstract: The goal of this research was to describe the most common drinking situations for young adolescents ($N = 1171$; 46.6% girls), as well as determine predictors of their drinking in the seventh and eighth grades. Middle school students most frequently drank at parties with three to four teens, in their home or at a friend's home, and reported alcohol-related problems including conflicts with friends or parents, memory loss, nausea, and doing things they would not normally do. Differences emerged in predicting higher levels of drinking on the basis of sex, race, grade, positive alcohol expectancies, impulsivity, and peer drinking. These findings suggest both specific and general factors are implicated in drinking for middle school students. Contextual factors, including drinking alone, in public places, and at or near school, are characteristic of the most problematic alcohol involvement in middle school and may have utility in prevention and early intervention.

Overall, is the abstract effective and appropriate?

1 2 3 4 5

Explain your rating.

4. *Title: The Multifaceted Nature of Poverty and Differential Trajectories of Health Among Children*¹⁵

Abstract: The relationships between poverty and children's health have been well documented, but the diverse and dynamic nature of poverty has not been thoroughly explored. Drawing on cumulative disadvantage and human capital theory, we examined to what extent the duration and depth of poverty, as well as the level of material hardship, affected changes in physical health among children over time. Data came from eight waves of the Korea Welfare Panel Study between 2006 and 2013. Using children who were under age 10 at baseline ($N = 1657$, Observations = 13,256), we conducted random coefficient regression in a multilevel growth curve framework to examine poverty group differences in intra-individual change in health status. Results showed that

14 Anderson, K. G., & Brown, S. A. (2011). Middle school drinking: Who, where, and when. *Journal of Child & Adolescent Substance Abuse*, 20(1), 48–62.

15 Kwon, E., Kim, B., & Park, S. (2017). The multifaceted nature of poverty and differential trajectories of health among children. *Journal of Children and Poverty*, 23(2), 141–160.

chronically poor children were most likely to have poor health. Children in households located far below the poverty line were most likely to be in poor health at baseline, while near-poor children's health got significantly worse over time. Material hardship also had a significant impact on child health.

Overall, is the abstract effective and appropriate?

1 2 3 4 5

Explain your rating:

5. *Title:* Prevention of Child Sexual Abuse by Targeting Pre-Offenders Before First Offense¹⁶

Abstract: The population of potential child abuse offenders has largely been unstudied. In the current study, we examine whether a six-component model used for primary diabetes prevention could be adapted to child sexual abuse pre-offenders, whereby individuals who are prone to sexual abuse but have not yet committed an offense can be prevented from committing a first offense. The six components include: define and track the magnitude of the problem; delineate a well-established risk factor profile so that at-risk persons can be identified; define valid screening tests to correctly rule in those with the disease and rule out those without disease; test effectiveness of interventions – the Dunkelfeld Project is an example; produce and disseminate reliable outcome data so that widespread application can be justified; and establish a system for continuous improvement. By using the diabetes primary prevention model as a model, the number of victims of child sexual abuse might be diminished.

Overall, is the abstract effective and appropriate?

1 2 3 4 5

Explain your rating:

Part B

Directions: Examine several academic journals that publish on topics of interest to you. Identify two with abstracts that you think are especially strong in terms of the evaluation questions presented in this chapter. Also, identify two abstracts that you believe have clear weaknesses. Bring the four abstracts to class for discussion.

16 Levine, J. A., & Dandamudi, K. (2016). Prevention of child sexual abuse by targeting pre-offenders before first offense. *Journal of Child Sexual Abuse*, 25(7), 719–737.

Evaluating Introductions and Literature Reviews

Research reports in academic journals usually begin with an introduction in which literature is cited.¹ An introduction with an integrated literature review has the following five purposes: (a) introduce the problem area, (b) establish its importance, (c) provide an overview of the relevant literature, (d) show how the current study will advance knowledge in the area, and (e) describe the researcher’s specific research questions, purposes, or hypotheses, which usually are stated in the last paragraph of the introduction.

This chapter presents evaluation questions regarding the introductory material in a research report. In the next chapter, the evaluation of the literature review portion is considered.

1. Does the Researcher Begin by Identifying a Specific Problem Area?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ²
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: Some researchers start their introductions with statements that are so broad they fail to identify the specific area of investigation. As the beginning of an introduction to a study on the effects of a tobacco control program for military troops, Example 4.1.1 is deficient. Notice that it fails to identify the specific area (tobacco control) to be explored in the research.

Example 4.1.1

BEGINNING OF AN INAPPROPRIATELY BROAD INTRODUCTION

The federal government expends considerable resources for research on public health issues, especially as they relate to individuals serving in the military. The findings of this research are used to formulate policies that regulate health-related activities in military settings. In addition to helping establish regulations, agencies develop educational programs so that

1 In theses and dissertations, the first chapter usually is the introduction, with relatively few references to the literature. This is followed by a chapter that provides a comprehensive literature review.
2 Continuing with the same scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement”.

individuals have appropriate information when making individual lifestyle decisions that may affect their health.

Example 4.1.2 illustrates a more appropriate beginning for a research report on a tobacco control program for the military.

Example 4.1.2³

A SPECIFIC BEGINNING (COMPARE WITH EXAMPLE 4.1.1)

Given the negative health consequences associated with tobacco use and their impact on physical fitness and readiness, the Department of Defense (DoD) has identified the reduction of tobacco use as a priority for improving the health of U.S. military forces (Department of Defense, 1977, 1986, 1994a, 1994b, 1999). Under these directives, tobacco use in official buildings and vehicles is prohibited; information regarding the health consequences of tobacco use is provided at entry into the military; and health care providers are encouraged to inquire about their patients' tobacco use. Recently, the DoD (1999) developed the *Tobacco Use Prevention Strategic Plan* that established DoD-wide goals. These goals include promoting a tobacco-free lifestyle and culture in the military, reducing the rates of cigarette and smokeless tobacco use, decreasing the availability of tobacco products, and providing targeted interventions to identified tobacco users.

Despite DoD directives and programs that focus on tobacco use reduction, the 2002 DoD worldwide survey indicated that past-month cigarette use in all branches of the military increased from 1998 to 2002 (from 29.9% to 33.8%; Bray et al., 2003).

Deciding whether a researcher has started the introduction by being reasonably specific often involves some subjectivity. As a general rule, the researcher should get to the point quickly, without using valuable journal space to outline a very broad problem area rather than the specific one(s) that he or she has directly studied.

___ 2. Does the Researcher Establish the Importance of the Problem Area?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Researchers select research problems they believe are important, and they should specifically address this belief early in their introductions. Often, this is done by citing previously published statistics that indicate how widespread a problem is, how many individuals are affected by it, and so on. Example 4.2.1 illustrates how researchers did this in the first paragraph of a study of a program intended to reduce school bullying.

3 Klesges, R. C., DeBon, M., Vander Weg, M. W., Haddock, C. K., Lando, H. A., Relyea, G. E., . . . Talcott, G. W. (2006). Efficacy of a tailored tobacco control program on long-term use in a population of U.S. military troops. *Journal of Consulting and Clinical Psychology*, 74(2), 295–306.

Example 4.2.1⁴

FIRST PARAGRAPH OF AN INTRODUCTION THAT INCLUDES STATISTICS TO ESTABLISH THE IMPORTANCE OF A PROBLEM AREA

Bullying in schools is a pervasive and ongoing threat to the mental health and school success of students. A meta-analysis of 21 U.S. studies showed that on average 18% of youth were involved in bullying perpetration, 21% of youth were involved in bullying victimization, and 8% of youth were involved in both perpetration and victimization (Cook, Williams, Guerra, & Kim, 2010). In addition, the Youth Risk Behavior Survey, which started measuring bullying victimization in 2009, has shown that the prevalence rate has remained at 20% since that time (Centers for Disease Control and Prevention [CDC], 2016).

Example 4.2.2 also uses statistical information to justify the importance of a study on alcohol abuse among active-duty military personnel.

Example 4.2.2⁵

BEGINNING OF AN INTRODUCTION THAT INCLUDES STATISTICAL INFORMATION TO ESTABLISH THE IMPORTANCE OF A PROBLEM AREA

Despite reductions in tobacco and illicit substance use in U.S. military personnel, alcohol misuse remains a significant problem (Bray et al., 2010). Data from the 2011 Department of Defense Health Related Behavior Survey suggests that across all military branches (Army, Navy, Marine Corps, Air Force, and Coast Guard), 84.5% of those on active duty report using alcohol, and over 25% report moderate to heavy use (Department of Defense, 2013). In addition, there are financial costs of alcohol use. A survey of TRICARE Prime beneficiaries in 2006 estimated that alcohol use cost the Department of Defense an estimated \$1.2 billion (Harwood, Zhang, Dall, Olaiya, & Fagan, 2009). Alcohol use problems also appear to be on the rise; trends across the years 1998 to 2008 show significant increases in the percentage of individuals who have engaged in recent binge drinking among those on active duty (Bray, Brown, & Williams, 2013), suggesting that alcohol issues remain a serious problem in the Department of Defense.

Instead of providing statistics on the prevalence of problems, researchers sometimes use other strategies to convince readers of the importance of the research problems they have studied. One approach is to show that prominent individuals or influential authors have considered and addressed the issue that is being researched. Another approach is to show that a topic is of current interest because of actions taken by governments (such as legislative actions), major

4 Hall, W. J., & Chapman, M. V. (2018). Fidelity of implementation of a state antibullying policy with a focus on protected social classes. *Journal of School Violence*, 17(1), 58–73.

5 Derefinko, K. J., Linde, B. D., Klesges, R. C., Boothe, T., Colvin, L., Leroy, K., . . . & Bursac, Z. (2018). Dissemination of the Brief Alcohol Intervention in the United States Air Force: Study rationale, design, and methods. *Military Behavioral Health*, 6(1), 108–117.

corporations, and professional associations. Example 4.2.3 illustrates the latter technique, in which the actions of both a prominent professional association and state legislatures are cited.

Example 4.2.3⁶

BEGINNING OF AN INTRODUCTION THAT USES A NONSTATISTICAL ARGUMENT TO ESTABLISH THE IMPORTANCE OF A PROBLEM

Less than 10 years after the American Psychological Association (APA) Council officially endorsed prescriptive authority for psychologists and outlined recommended training (APA, 1996), psychologists are prescribing in New Mexico and Louisiana. In both 2005 and again in 2006 seven states and territories introduced prescriptive authority legislation and RxP Task Forces were active in many more states (Sullivan, 2005; Baker, 2006). Commenting on this dramatic maturing of the prescriptive authority agenda, DeLeon (2003, p. XIII) notes it is “fundamentally a social policy agenda ensuring that all Americans have access to the highest possible quality of care . . . wherein psychotropics are prescribed in the context of an overarching psychologically based treatment paradigm.” The agenda for psychologists prescribing is inspired by the premise that psychologists so trained will play central roles in primary health care delivery.

Finally, a researcher may attempt to establish the nature and importance of a problem by citing anecdotal evidence or personal experience. While this is arguably the weakest way to establish the importance of a problem, a unique and interesting anecdote might convince readers that the problem is important enough to investigate.

A caveat: When you apply Evaluation Question 2 to the introduction of a research report, do *not* confuse the importance of a problem with your personal interest in the problem. It is possible to have little personal interest in a problem yet still recognize that a researcher has established its importance. On the other hand, it is possible to have a strong personal interest in a problem but judge that the researcher has failed to make a strong argument (or has failed to present convincing evidence) to establish its importance.

___ 3. Are any Underlying Theories Adequately Described?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: If a theory is named in the introduction to a research article, the theory should be adequately described. As a general rule, even a well-known theory should be described in at least a short paragraph (along with one or more references where additional information can be found). Lesser-known theories and new theories should be described in more detail.

6 LeVine, E. S. (2007). Experiences from the frontline: Prescribing in New Mexico. *Psychological Services*, 4(1), 59–71.

Example 4.3.1 briefly but clearly summarizes a key aspect of general strain theory, which underlies the author's research.⁷

Example 4.3.1⁸

EXCERPT FROM THE INTRODUCTION TO A RESEARCH ARTICLE THAT DESCRIBES A THEORY UNDERLYING THE RESEARCH

This study applies general strain theory to contribute to literature that explores factors associated with engagement in cyberbullying. General strain theory posits that individuals develop negative emotions as a result of experiencing strain (e.g., anger and stress), and are susceptible to engaging in criminal or deviant behavior (Agnew, 1992). In contrast with other studies on cyberbullying, this study applies general strain theory to test the impact that individual and social factors of adolescents have on engagement in cyberbullying.

Note that much useful research is *non-theoretical*.⁹ Sometimes, the purpose of a study is only to collect and interpret data in order to make a practical decision. For instance, a researcher might poll parents to determine what percentage favors a proposed regulation that would require students to wear uniforms when attending school. Non-theoretical information on parents' attitudes toward requiring uniforms might be an important consideration when a school board is making a decision on the issue.

Another major reason for conducting non-theoretical research is to determine whether there is a problem and/or the incidence of a problem (*descriptive* research). For instance, without regard to theory, a researcher might collect data on the percentage of pregnant women attending a county medical clinic who use tobacco products during pregnancy. The resulting data will help decision makers determine the prevalence of this problem within the clinic's population.

Another common type of study – again, mostly non-theoretical – evaluates the effectiveness of a policy or program (*evaluation* research). For example, researchers are wondering whether boot camps reduce juvenile delinquency compared to a traditional community service approach. Thus, the researchers secure the judge's agreement to randomly assign half of the youth adjudicated for minor offenses to boot camps and the other half to community service. Then the researchers compare the rates of recidivism between the two groups of juveniles a year later. Evaluation research is covered in Appendix B: *A Special Case of Program or Policy Evaluation*.

7 Notice that this is a very brief description of a theory in the introduction of a research article. Further in the article, discussion of the theory is expanded considerably.

8 Paez, G. R. (2018). Cyberbullying among adolescents: A general strain theory perspective. *Journal of School Violence*, 17(1), 74–85.

9 Traditionally, empirical studies in social sciences are divided into 4 types: exploratory, descriptive, explanatory, and evaluation (of a policy or program's effectiveness). Among these, only the explanatory type is often related to a theory (tests a theoretical explanation). Studies of the other 3 types are often non-theoretical.

Evaluation studies are very important in assessing the effectiveness of various interventions and treatments but are unlikely to involve a theoretical basis. In Chapter 14, you can find out more information about evidence-based programs and research aimed at creating such evidence base.

When applying Evaluation Question 3 to non-theoretical research, “not applicable” (N/A) will usually be the fitting answer.

A special note for evaluating qualitative research: Often, qualitative researchers explore problem areas without initial reference to theories and hypotheses (this type of research is often called *exploratory*). Sometimes, they develop new theories (and models and other generalizations) as they collect and analyze data¹⁰. The data often take the form of transcripts from open-ended interviews, notes on direct observation and involvement in activities with participants, and so on. Thus, in a research article reporting on qualitative research, a theory might not be described until the Results and Discussion sections (instead of the Introduction). When this is the case, apply Evaluation Question 3 to the point at which theory is discussed.

___ 4. Does the Introduction Move from Topic to Topic Instead of from Citation to Citation?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Introductions that typically fail on this evaluation question are organized around citations rather than topics. For instance, a researcher might *inappropriately* first summarize Smith’s study, then Jones’s study, then Miller’s study, and so on. The result is a series of annotations that are merely strung together. This fails to show readers how the various sources relate to each other and what they mean as a whole.

In contrast, an introduction should be organized around topics and subtopics, with references cited as needed, often in groups of two or more citations per source. For instance, if four empirical studies support a certain point, the point usually should be stated with all four references cited together (as opposed to citing them in separate statements or paragraphs that summarize each of the four sources).

In Example 4.4.1, there are three citations for each of the points made in two separate sentences.

Example 4.4.1¹¹

AN EXCERPT FROM A LITERATURE REVIEW WITH SOURCES CITED IN GROUPS

For most individuals facing the end of life, having control over their final days, dying in a place of their choosing, and being treated with dignity and respect are central concerns (Chochinov et al., 2002; Steinhauser et al., 2000; Vig, Davenport, & Pearlman, 2002).

10 Such theories developed in qualitative research or by summarizing data/observations are called *grounded*.

11 Thompson, G. N., McClement, S. E., & Chochinov, H. M. (2011). How respect and kindness are experienced at the end of life by nursing home residents. *Canadian Journal of Nursing Research*, 43(3), 96–118.

However, research suggests that quality end-of-life care is often lacking in [nursing homes], resulting in residents dying with their symptoms poorly managed, their psychological or spiritual needs neglected, and their families feeling dissatisfied with the care provided (Teno, Kabumoto, Wetle, Roy, & Mor, 2004; Thompson, Menec, Chochinov, & McClement, 2008; Wetle, Shield, Teno, Miller, & Welch, 2005).

When a researcher is discussing a particular source that is crucial to a point being made, that source should be discussed in more detail than in Example 4.4.1. However, because research reports in academic journals are expected to be relatively brief, detailed discussions of individual sources should be presented sparingly and only for the most important related literature.

5. Are Very Long Introductions Broken into Subsections, Each with its Own Subheading?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: When there are a number of issues to be covered in a long introduction, there may be several sub-essays, each with its own subheading. The subheadings help to guide readers through long introductions, visually and substantively breaking them down into more easily ‘digestible’ parts. For instance, Example 4.5.1 shows the five subheadings used within the introduction to a study of risk and protective factors for alcohol and marijuana use among urban and rural adolescents.

Example 4.5.1¹²

FIVE SUBHEADINGS USED WITHIN AN INTRODUCTION

- Individual Factors
- Family Factors
- Peer Factors
- Community Factors
- Risk and Protective Factors among Urban and Rural Youths

6. Has the Researcher Provided Adequate Conceptual Definitions of Key Terms?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

12 Clark, T. T., Nguyen, A. B., & Belgrave, F. Z. (2011). Risk and protective factors for alcohol and marijuana use among African American rural and urban adolescents. *Journal of Child & Adolescent Substance Abuse*, 20(3), 205–220.

Comment: Often, researchers will pause at appropriate points in their introductions to offer formal conceptual definitions, such as the one shown in Example 4.6.1. We have discussed in Chapter 1 why definitions are important in research reports. A *conceptual definition* explains what the term means or includes while an *operational definition* explains how the term is measured in the study.¹³ Note that it is acceptable for a researcher to cite a previously published definition, which is done in Example 4.6.1. Also, note that the researchers contrast the term being defined (i.e., academic self-concept) with a term with which it might be confused (i.e., academic engagement).

Example 4.6.1¹⁴

A CONCEPTUAL DEFINITION PROVIDED IN AN ARTICLE'S INTRODUCTION

Academic self-concept refers to perceptions of one's own academic competence, and develops out of past experiences, evaluative feedback from important others, and social comparisons (Dweck, 2002; Harter, 1998). *Academic engagement* refers to enthusiastic and focused involvement in academic activities and manifests in behaviors such as effort and active class participation (Kindermann, 2007; Ryan, 2001). Youths' academic self-concepts and engagement are interrelated: Academic self-concept predicts expectations for success and the value placed on academic achievement, which, in turn, affects levels of academic engagement (e.g., Wigfield & Eccles, 2002).

Conceptual definitions do not need to be lengthy as long as their meaning is clear. The first sentence in Example 4.6.2 shows a brief conceptual definition.

Example 4.6.2¹⁵

A BRIEF CONCEPTUAL DEFINITION PROVIDED IN AN ARTICLE'S INTRODUCTION

In Cobb's (1976) classic disquisition, *social support* is defined as the perception that one is loved, valued and esteemed, and able to count on others should the need arise. The desire and need for social support have evolved as an adaptive tool for survival, and our perceptions of the world around us as being supportive emerge from our interactions and attachments experienced early in the life course (Bowlby, 1969, 1973; Simpson, & Belsky, 2008). Consequent to the seminal review articles of Cobb (1976) and Cassel (1976),

- 13 A *conceptual definition* identifies a term using only general concepts but with enough specificity that the term is not confused with other related terms or concepts. As such, it resembles a dictionary definition. In contrast, an *operational definition* describes the physical process used to create the corresponding variable. For instance, an operational definition for "psychological control" by parents includes the use of a particular observation checklist, which would be described under the heading Measures later in a research report (see Chapter 8).
- 14 Molloy, L. E., Gest, S. D., & Rulison, K. L. (2011). Peer influences on academic motivation: Exploring multiple methods of assessing youths' most "influential" peer relationships. *Journal of Early Adolescence*, 31(1), 13–40.
- 15 Gayman, M. D., Turner, R. J., Cislo, A. M., & Eliassen, A. H. (2011). Early adolescent family experiences and perceived social support in young adulthood. *Journal of Early Adolescence*, 31(6), 880–908.

a vast and consistent body of evidence has accumulated suggesting that social support from family and friends is protective against a variety of adverse health outcomes.

At times, researchers may not provide formal conceptual definitions because the terms have widespread commonly held definitions. For instance, in a report of research on various methods of teaching handwriting, a researcher may not offer a formal definition of handwriting, which might be acceptable.

In sum, this evaluation question should not be applied mechanically by looking to see whether there is a specific statement of a definition. The mere absence of one does not necessarily mean that a researcher has failed on this evaluation question, because a conceptual definition is not needed for some variables. When this is the case, you may give the article a rating of N/A (“not applicable”) for this evaluation question.

7. Has the Researcher Cited Sources for “Factual” Statements?

Very unsatisfactory 1 2 3 4 5 Very satisfactory or N/A I/I

Comment: Researchers should avoid making statements that sound like facts without referring to their source. Example 4.7.1 is deficient in this respect. Compare it with its Improved Version, in which sources are cited for various assertions.

Example 4.7.1

UNREFERENCED FACTUAL CLAIMS (UNDESIRABLE)

Nursing is widely recognized as a high-stress occupation, which is highly demanding yet has limited resources to support nurses with their occupational stress. Providing palliative care to patients with fatal diseases is especially stressful, causing both emotional and professional challenges for nurses.

Improved Version of Example 4.7.1¹⁶

SOURCES CITED FOR FACTUAL CLAIMS (COMPARE WITH EXAMPLE 4.7.1)

Considering the large number of demands and the limited resources available to support them, nurses represent a high-risk group for experiencing occupational stress (Bourbonnais, Comeau, & Vézina, 1999; Demerouti, Bakker, Nachreiner, & Schaufeli, 2000). Numerous studies suggest that those offering palliative care could be particularly at risk (Twycross, 2002; Wilkes et al., 1998). Palliative care provides comfort, support, and quality of life to patients living with fatal diseases, such as cancer (Ferris et al., 2002). Nurses involved

16 Fillion, L., Tremblay, I., Truchon, M., Côté, D., Struthers, C. W., & Dupuis, R. (2007). Job satisfaction and emotional distress among nurses providing palliative care: Empirical evidence for an integrative occupational stress-model. *International Journal of Stress Management*, 14(1), 1–25.

in the provision of this type of care meet several recurrent professional, emotional, and organizational challenges (Fillion, Saint-Laurent, & Rousseau, 2003; Lu, While, & Barriball, 2005; Newton & Waters, 2001; Plante & Bouchard, 1995; Vachon, 1995, 1999).

At the same time, not every factual statement should be provided with a reference. Some factual statements reflect common knowledge and thus do not need any references to a specific source of such knowledge. For example, an assertion like “Violent crime has devastating consequences not only for the victims but also for the victims’ families” is fairly self-evident and reflects a common understanding about the direct and indirect effects of violent crime.

8. Do the Specific Research Purposes, Questions, or Hypotheses Logically Flow from the Introductory Material?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Typically, the specific research purposes, questions, or hypotheses on which a study is based are stated in the final paragraphs of the introduction.¹⁷ The material preceding them should set the stage and logically lead to them. For instance, if a researcher argues that research methods used by previous researchers are not well suited for answering certain research questions, it would not be surprising to learn that his or her research purpose is to re-examine the research questions using alternative research methods. In Example 4.8.1, which includes the last paragraphs in the introduction to a research report, the researchers refer to the literature that they reviewed in the introduction. This sets the stage for the specific research questions, which are stated in the last sentence of the example.

Example 4.8.1¹⁸

LAST PARAGRAPHS OF AN INTRODUCTION (BEGINNING WITH A SUMMARY OF THE RESEARCH THAT WAS REVIEWED AND ENDING WITH A STATEMENT OF THE PURPOSES OF THE CURRENT STUDY)

These somewhat conflicting results [of studies reviewed above] point to a need of further research into how persistence of victimization and variation in experiences of bullying relate to different aspects of children’s lives. [. . .]

The goal for this study is to examine patterns, including gender differences, of stability or persistence of bullying victimization, and how experiences of being bullied relate to children’s general well-being, including somatic and emotional symptomology.

¹⁷ Some researchers state their research purpose and research questions or hypotheses in general terms near the beginning of their introductions, and then restate them more specifically at the end of introduction.

¹⁸ Hellfeldt, K., Gill, P. E., & Johansson, B. (2018). Longitudinal analysis of links between bullying victimization and psychosomatic maladjustment in Swedish schoolchildren. *Journal of School Violence*, 17(1), 86–98.

___ 9. Overall, is the Introduction Effective and Appropriate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	<i>or</i> N/A	I/I
------------------------	---	---	---	---	---	----------------------	---------------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and taking into account any additional considerations and concerns you may have. Be prepared to explain your overall evaluation.

Chapter 4 Exercises

Part A

Directions: Following are the beginning paragraphs of introductions to research articles. Answer the questions that follow each one.

1. Longitudinal and experimental studies on children in orphanages, children's homes, and foster families have confirmed the adverse effects of long-term institutional care on children's personality development (American Academy of Child and Adolescent Psychiatry, 2005; Castle et al., 1999; Chisholm, 1998; Marcovitch et al., 1997; O'Connor, Marvin, Rutter, Olrick, & Britner, 2003; Roy, Rutter, & Pickles, 2000; Tizard & Hodges, 1978; Tizard & Rees, 1975; Vorria, Rutter, Pickles, Wolkind, & Hobsbaum, 1998; Wolkind, 1974; Zeanah, 2000; Zeanah, Smyke, & Dumitrescu, 2002). Consistently reported effects on children's behavior include hyperactivity, inability to concentrate, poor school performance, ineffective coping skills, conduct disorder (CD) symptoms, disruptive attention-seeking, difficulties with peers, few close relationships, emotional withdrawal, and indiscriminate relationships with adults. Similar effects have been observed in adolescents (Hodges & Tizard, 1989a, b), together with an early report of schizoid personality traits (Goldfarb, 1943). Institutional rearing conveys a greater risk of hyperactivity and inattention, compared to foster home rearing (Roy et al., 2000; Vorria et al., 1998).

Providing subsequent family care and improving the quality of caregivers' parenting skills both reduce the risk of problem behavior (Webster-Stratton, 1998) and improve cognitive development (Loeb, Fuller, Kagan, & Carrol, 2004). These consistent findings have influenced policymakers for child welfare in different countries (Broad, 2001; Department for Education and Skills, 1989; Maunders, 1994; NSW Community Services Commission, 1996) to prioritize foster home or kinships over children's home care and to increase investment to raise standards within care systems.¹⁹

19 Yang, M., Ullrich, S., Roberts, A., & Coid, J. (2007). Childhood institutional care and personality disorder traits in adulthood: Findings from the British National Surveys of Psychiatric Morbidity. *American Journal of Orthopsychiatry*, 77(1), 67–75.

- a. How well have the researchers established the importance of the problem area? Explain.
 - b. Does the material move from topic to topic instead of from citation to citation? Explain.
 - c. Have the researchers cited sources for factual statements? Explain.
2. "This man is just not cooperating and just doesn't want to be in therapy." A doctoral student working with a 26-year-old white man in counseling was frustrated at her inability to get her client to reveal what she regarded to be his true feelings. She believed that he was resistant to therapy because of his reticence to show emotions. However, her supervisor, someone trained in the psychology of men, explained to her the difficulty some men have in expressing emotions: that, in fact, some men are unaware of their emotional states. Working with the supervisor, the trainee focused part of the therapy on helping the client identify and normalize his emotions and providing some psycho-education on the effects of his masculine socialization process. This critical incident could be repeated in psychology training programs around the country. As men come to therapy, the issue for many psychologists becomes, How do psychologists become competent to work with men? This question may seem paradoxical given the sentiment that most if not all of psychology is premised on men's, especially white men's, worldviews and experiences (Sue, Arredondo, & McDavis, 1992; Sue & Sue, 2003). But several authors have suggested that working with men in therapy is a clinical competency and just as complex and difficult as working with women and other multicultural communities (Addis & Mahalik, 2003; Liu, 2005).²⁰
 - a. How well have the researchers established the importance of the problem area? Explain.

Part B

Directions: Following are excerpts from various sections of introductions. Answer the questions that follow each one.

3. The current article focuses on one such intermediate perspective: the dialect theory of communicating emotion. Dialect theory proposes the presence of cultural differences in the use of cues for emotional expression that are subtle enough to allow accurate communication across cultural boundaries in general, yet substantive enough to result in a potential for miscommunication (Elfenbein & Ambady, 2002b, 2003).²¹
 - a. Is the theory adequately described? Explain.

20 Mellinger, T. N., & Liu, W. M. (2006). Men's issues in doctoral training: A survey of counseling psychology programs. *Professional Psychology: Research and Practice*, 37(2), 196–204.

21 Elfenbein, H. A., Beaupré, M., Lévesque, M., & Hess, U. (2007). Toward a dialect theory: Cultural differences in the expression and recognition of posed facial expressions. *Emotion*, 7(1), 131–146.

4. Terror management theory (see Greenberg et al., 1997, for a complete presentation) is based on the premise that humans are in a precarious position due to the conflict between biological motives to survive and the cognitive capacity to realize life will ultimately end. This generally unconscious awareness that death is inevitable, coupled with proclivities for survival, creates potentially paralyzing anxiety that people manage by investing in a meaningful conception of the world (cultural worldview) that provides prescriptions for valued behavior and thus a way to also maintain self-esteem. For instance, support for the theory has been provided by numerous findings that reminding people of their own eventual death (mortality salience) results in an attitudinal and behavioral defense of their cultural worldview (*worldview defense*, e.g., Greenberg et al., 1990) and a striving to attain self-esteem (e.g., Routledge, Arndt, & Goldenberg, 2004; see Pyszczynski, Greenberg, Solomon, Arndt, & Schimel, 2004, for a review). Although terror management theory has traditionally focused on the effects of unconscious concerns with mortality on these symbolic or indirect distal defenses, recent research has led to the conceptualization of a dual defense model that also explicates responses provoked by conscious death-related thoughts (Arndt, Cook, & Routledge, 2004; Pyszczynski, Greenberg, & Solomon, 1999).²²
 - a. Is the theory adequately described? Explain.
5. An emergency medical condition is defined as a medical condition manifesting itself by acute symptoms of sufficient severity (including severe pain, psychiatric disturbances and/or symptoms of substance abuse) such that the absence of immediate medical attention could reasonably be expected to result in placing the health of the individual (or, with respect to a pregnant woman, the health of the woman or her unborn child) in serious jeopardy.²³
 - a. Is the conceptual definition adequate? Explain.

Part C

Directions: Read two empirical articles in academic journals on a topic of interest to you. Apply the evaluation questions in this chapter to their introductions, and select the one to which you have given the highest ratings. Bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

22 Arndt, J., Cook, A., Goldenberg, J. L., & Cox, C. R. (2007). Cancer and the threat of death: The cognitive dynamics of death-thought suppression and its impact on behavioral health intentions. *Journal of Personality and Social Psychology*, 92(1), 12–29.

23 Kunen, S., Niederhauser, R., Smith P. O., Morris, J. A., & Marx, B. D. (2005). Race disparities in psychiatric rates in emergency departments. *Journal of Consulting and Clinical Psychology*, 73(1), 116–126.

A Closer Look at Evaluating Literature Reviews

As indicated in the previous chapter, literature reviews usually are integrated into the researcher's introductory statements. In that chapter, the emphasis was on the functions of the introduction and the most salient characteristics of a literature review. This chapter explores the quality of literature reviews in more detail.

1. Has the Researcher Avoided Citing a Large Number of Sources for a Single Point?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: As a rough rule, citing more than six sources for a single point is often inappropriate.

When there are many sources for a single point, three things can be done. First, the researcher can break them into two or more subgroups. For instance, those sources dealing with one population (such as children) might be cited in one group, while those sources dealing with another population (such as adolescents) might be cited in another group.

Second, the researcher can cite only the most salient (or methodologically strong) sources as examples of the sources that support a point, which is illustrated in Example 5.1.1. Notice that the researchers make reference to “vast empirical literature,” indicating that there are many sources that support the point. Then they use *e.g.*, (meaning “for example,”) to cite two selected sources.

Example 5.1.1²

USING *E.G.*, TO CITE SELECTED SOURCES (ITALICS USED FOR EMPHASIS)

A vast empirical literature has substantiated the existence of a link between symptoms of depression and marital conflict. Although this relationship is undoubtedly bidirectional and

1 Continuing with the same scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement”.

2 Marshall, A. D., Jones, D. E., & Feinberg, M. E. (2011). Enduring vulnerabilities, relationship attributions, and couple conflict: An integrative model of the occurrence and frequency of intimate partner violence. *Journal of Family Psychology*, 25(5), 709–718.

reciprocal (*e.g.*, Whisman, Uebelacker, & Weinstock, 2004), data suggest that the effect may be more strongly in the direction of depression leading to marital conflict (*e.g.*, Atkins, Dimidjian, Bedics, & Christensen, 2009).

Third, to avoid citing a long string of references for a single point, researchers may refer the reader to the most recent comprehensive review of the relevant literature, as illustrated in Example 5.1.2.

Example 5.1.2³

REFERRING TO A SINGLE COMPREHENSIVE RECENT SOURCE THAT SUMMARIZES
OTHER RELEVANT RESEARCH SOURCES (ITALIC FONT ADDED FOR EMPHASIS)

Thus, individual victimizations only represent the tip of the iceberg in terms of financial losses. Different methodologies of calculating losses and different definitions of online crime (identity theft, credit/debit card fraud, etc.) lead to different estimates of per person and overall losses. Moreover, surveys of individuals can bias estimates of losses upwards, if the percentage of population affected is small and may not be represented well, even in fairly large samples (*see Florencio & Herley, 2013, for an excellent discussion of this issue*).

2. Is the Literature Review Critical?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: A researcher should consider the strengths and weaknesses of previously published studies.⁴ Note that criticism can be positive (as in Example 5.2.1, in which the authors refer to “well-designed” studies).

Example 5.2.1⁵

POSITIVE CRITICISM IN A LITERATURE REVIEW

In the past 20 years, well-designed studies (*e.g.*, those using more representative samples, clear exclusion criteria, subjects blind to study purpose, and standardized instruments) have challenged the view that children of alcoholics necessarily have poor psychosocial outcomes

- 3 Tcherni, M., Davies, A., Lopes, G., & Lizotte, A. (2016). The dark figure of online property crime: Is cyberspace hiding a crime wave? *Justice Quarterly*, 33(5), 890–911.
- 4 Articles based on reasonably strong methodology may be cited without comments on their strengths. However, researchers have an obligation to point out which studies are exceptionally weak. This might be done with comments such as “A small pilot study suggested . . .” or “Even though the authors were not able to test other likely alternative explanations of their results . . .”.
- 5 Amodeo, M., Griffin, M., & Paris, R. (2011). Women’s reports of negative, neutral, and positive effects of growing up with alcoholic parents. *Families in Society: The Journal of Contemporary Social Services*, 92(1), 69–76.

in adulthood. For example, Werner’s (1986) pioneering longitudinal study of children born in Hawaii found differential effects of parental alcoholism in offspring, with 59% of the offspring reaching age 18 with no measureable adjustment problems. Clair and Genest (1987) showed that . . .

Of course, negative criticisms are often warranted. An instance of this is shown in Example 5.2.2.

Example 5.2.2⁶

NEGATIVE CRITICISM IN A LITERATURE REVIEW

Nevertheless, several methodological limitations occurred in prior studies of parental involvement. An important limitation in these studies is one that is related to methodology; these studies focused on parental involvement as a whole but did not distinguish between mother’s involvement and father’s involvement (Flouri & Buchanan, 2004; Shek, 2007). The mother’s influence on her child may be different from father’s influence, as there are clear differences in how mother and father treat their children (McBride & Mills, 1993). [. . .]

A second limitation of research in this area is that, although a limited amount of research has been done to compare the different effects of father’s involvement and mother’s involvement, these studies used only infant or preschool children and did not include adolescents. [. . .]

Another limitation of prior research is that parental involvement and the effect of such involvement on adolescent academic achievement is often confounded by ethnic and cultural factors . . .

Sometimes, the authors are very subtle in the way they assess previous research: highlighting its strengths while still mentioning the weaknesses in a very balanced way, as shown in 5.2.3.

Example 5.2.3⁷

BALANCED CRITICISM IN A LITERATURE REVIEW

Previous research on police–community interactions has relied on citizens’ recollection of past interactions (10)⁸ or researcher observation of officer behavior (17–20) to assess procedural fairness. Although these methods are invaluable, they offer an indirect view

6 Hsu, H.-Y., Zhang, D., Kwok, O.-M., Li, Y., & Ju, S. (2011). Distinguishing the influences of father’s and mother’s involvement on adolescent academic achievement: Analyses of Taiwan Educational Panel Survey Data. *Journal of Early Adolescence*, 31(5), 694–713.

7 Voigt, R., Camp, N. P., Prabhakaran, V., Hamilton, W. L., Hetey, R. C., Griffiths, C. M., . . . & Eberhardt, J. L. (2017). Language from police body camera footage shows racial disparities in officer respect. *Proceedings of the National Academy of Sciences*, 114(25), 6521–6526.

8 Notice that the citation format in this example is different from the standard APA-style in-text citations. In this case, references are numbered as they appear in the text, which is more typical of journals in exact sciences like engineering and in some social sciences like public health.

of officer behavior and are limited to a small number of interactions. Furthermore, the very presence of researchers may influence the police behavior those researchers seek to measure (21).

___ 3. Is Current Research Cited?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: The currency of the literature can be checked by noting whether research published in recent years has been cited. Keep in mind, however, that relevance to the research topic is more important than currency. A 15-year-old study that is highly relevant and has superior research methodology may deserve more attention than a less relevant, methodologically weaker one that was recently published. When this is the case, the researcher should explicitly state why an older research article is being discussed in more detail than newer ones.

Also, note that a researcher may want to cite older sources to establish the historical context for the study. In Example 5.3.1, the researchers link a particular finding to Ferster and Skinner's work in 1957. Skinner is the best known of the early behavior analysts. References to more current literature follow.

Example 5.3.1⁹

AN EXCERPT FROM A LITERATURE REVIEW SHOWING HISTORICAL LINKS

Behavior analysts often allude to the imperviousness of schedule effects to particular reinforcement histories (e.g., Ferster & Skinner, 1957), but rarely is evidence adduced to substantiate that point. There is currently a small body of mixed evidence for reinforcement history effects on FI [fixed-interval] performance (Baron & Leinenweber, 1995; Cole, 2001 [. . .]). For example, Wanchisen et al. (1989) found . . .

___ 4. Has the Author Cited any Contradictory Research Findings?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Researchers cannot be reviewing only research literature that supports their case while ignoring any studies that contradict (or do not support) their hypothesis or expected results. An important goal of maintaining impartiality and objectivity in science requires that the authors of an empirical study include both the studies that support their view as well as those that produced opposite or inconclusive results. It may be that such unfavorable results came from studies that are methodologically weaker than the studies with supportive findings – then

9 Ludvig, E. A., & Staddon, J. E. R. (2004). The conditions for temporal tracking under interval schedules of reinforcement. *Journal of Experimental Psychology: Animal Behavior Processes*, 30(4), 299–316.

the researchers can discuss the limitations and draw comparisons. But if the authors only cite those studies that are in line with their thinking, while omitting any mention of ‘inconvenient’ contradictory findings, this is a problem and a serious flaw of the literature review.

In Example 5.4.1, contradictory findings regarding the success of job training programs for former prisoners are cited and explained.

Example 5.4.1¹⁰

CONTRADICTIONARY FINDINGS ARE INCLUDED IN THE LITERATURE REVIEW (BOTH SUPPORTIVE RESULTS AND UNFAVORABLE ONES)

The main findings were quite discouraging. SVORI [Serious and Violent Offender Reentry Initiative] provided modest enhancements in services to offenders before and after release, and appears to have had some effect on intermediate outcomes like self-reported employment, drug use, housing, and criminal involvement. However, there was no reduction in recidivism as measured by administrative data on arrest and conviction (Lattimore et al. 2010). [. . .]

The most prominent experiment of the decade of the 1970s was the National Supported Work Demonstration program, which provided recently released prisoners and other high-risk groups with employment opportunities on an experimental basis. [. . .] A re-analysis by Christopher Uggen (2000) which combined the ex-offenders with illicit-drug abusers and youthful dropouts found some reduction in arrests for older participants (over age 26), but not for the younger group. He has speculated that older offenders are more amenable to employment-oriented interventions (Uggen and Staff 2001), perhaps because they are more motivated. [. . .]

In sum, the evidence on whether temporary programs that improve employment opportunities have any effect on recidivism is mixed. There have been both null findings and somewhat encouraging findings.

5. Has the Researcher Distinguished between Opinions and Research Findings?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Researchers should use wording that helps readers understand whether the cited literature presents opinions or research results.

For indicating that a citation is research-based, there is a variety of options, several of which are shown in Example 5.5.1.

10 Cook, P. J., Kang, S., Braga, A. A., Ludwig, J., & O’Brien, M. E. (2015). An experimental evaluation of a comprehensive employment-oriented prisoner re-entry program. *Journal of Quantitative Criminology*, 31(3), 355–382.

Example 5.5.1

EXAMPLES OF KEY TERMS AND EXPRESSIONS INDICATING THAT INFORMATION IS RESEARCH-BASED

- Recent data suggest that . . .
- In laboratory experiments . . .
- Recent test scores show . . .
- Group A has outperformed its counterparts on measures of . . .
- Research on XYZ has established . . .
- Data from surveys comparing . . .
- Doe (2017) has found that the rate of . . .
- These studies have greatly increased knowledge of . . .
- The mean scores for women exceed . . .
- The percentage of men who have performed . . .

Note that if a researcher cites a specific statistic from the literature (e.g., “Approximately 22% of Canadian teenagers between 15 and 19 years currently smoke cigarettes [*Health Canada*, 2003].”),¹¹ it is safe to assume that factual information is being cited.

Sometimes, researchers cite the opinions of others. When they do this, they should word their statements in such a way that readers are made aware that opinions (and not data-based research findings) are being cited. Example 5.5.2 shows some examples of key words and phrases that researchers sometimes use to do this.

Example 5.5.2

EXAMPLES OF KEY TERMS AND EXPRESSIONS INDICATING THAT AN OPINION IS BEING CITED

- Jones (2016) has argued that . . .
- These kinds of assumptions were . . .
- Despite this speculation . . .
- These arguments predict . . .
- This logical suggestion . . .

11 As cited in Golmier, I., Chebat, J.-C., & Gelinas-Chebat, C. (2007). Can cigarette warnings counterbalance effects of smoking scenes in movies? *Psychological Reports*, 100(1), 3–18.

- Smith has strongly advocated the use of . . .
- Based on the theory, Miller (2018) predicted that . . .

___ 6. Has the Researcher Noted any Gaps in the Literature?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Gaps in the literature on a topic (areas not fully explored in previous studies) can be as important as areas already explored by researchers. The gaps point to areas needing research in the future. In Example 5.6.1, the researchers point out a gap.

Example 5.6.1¹²

EXCERPT POINTING OUT A GAP IN THE LITERATURE

Although the importance of fathers has been established, the majority of research on fathering is based on data from middle-class European American families, and research on ethnic minority fathers, especially Latino fathers, has lagged significantly behind (Cabrera & Garcia-Coll, 2004). This is a shortcoming of the literature . . .

Note that the presence of a gap in the literature can then be used to justify a study when the purpose of the study is to fill the gap.

___ 7. Has the Researcher Interpreted Research Literature in Light of the Inherent Limits of Empirical Research?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: As indicated in Chapter 1, empirical research has inherent limitations. As a result, no research study offers definitive proof. Instead, research results offer *degrees of evidence*, which are sometimes extremely strong (such as the relationship between cigarette smoking and health), and much more often, are only modest or weak (such as the relationship between mental illness and crime).

Terms that researchers might use to indicate that the results of research offer strong evidence are shown in Example 5.7.1.

12 Cruz, R. A., King, K. M., Widaman, K. F., Leu, J., Cauce, A. M., & Conger, R. D. (2011). Cultural influences on positive father involvement in two-parent Mexican-origin families. *Journal of Family Psychology*, 25(5), 731–740.

Example 5.7.1

EXAMPLES OF TERMINOLOGY (IN ITALICS) THAT CAN BE USED TO INDICATE STRONG EVIDENCE

- Results of three recent studies *strongly suggest* that X and Y are . . .
- Most studies of X and Y *clearly indicate the possibility* that X and Y are . . .
- This type of evidence *has led most researchers to conclude* that X and Y . . .

Terms that researchers can use to indicate that the results of research offer moderate to weak evidence are shown in Example 5.7.2.

Example 5.7.2

EXAMPLES OF TERMINOLOGY (IN ITALICS) THAT CAN BE USED TO INDICATE MODERATE TO WEAK EVIDENCE

- The results of a recent pilot study *suggest* that X and Y are . . .
- To date, there is *only limited evidence* that X and Y are . . .
- Although empirical evidence *is inconclusive*, X and Y seem to be . . .
- Recent research *implies* that X and Y may be . . .
- The relationship between X and Y has been examined, with results *pointing towards* . . .

It is not necessary for a researcher to indicate the degree of confidence that should be accorded every finding discussed in a literature review. However, if a researcher merely states what the results of research indicate without qualifying terms, readers will assume that the research being cited is reasonably strong.

___ 8. Has the Researcher Avoided Overuse of Direct Quotations from the Literature?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Direct quotations should be rarely used in literature reviews for two reasons. First, they often take up more journal space, which is very limited, than a paraphrase would take. Second, they often interrupt the flow of the text because of the differences in writing styles of the reviewer and the author of the literature.

An occasional quotation may be used if it expresses an idea or concept that would lose its impact in a paraphrase. When something is written so beautifully or in such a perfect way that it would enhance the narrative of the article citing it, then it is a good idea to include such a direct quote. This may be the case with a quotation shown in Example 5.8.1, which appeared in the first paragraph of a research report on drug abuse and its association with loneliness.

Example 5.8.1¹³

A DIRECT QUOTATION IN A LITERATURE REVIEW (ACCEPTABLE IF DONE
VERY SPARINGLY)

Recent studies suggest that a large proportion of the population are frequently lonely (Rokach & Brock, 1997). Ornish (1998) stated at the very beginning of his book *Love & Survival*: “Our survival depends on the healing power of love, intimacy, and relationships. Physically. Emotionally. Spiritually. As individuals. As communities. As a culture. Perhaps even as a species.” (p. 1.) Indeed, loneliness has been linked to depression, anxiety and . . .

9. After Reading the Literature Review, Does a Clear Picture Emerge of What the Previous Research has Accomplished and Which Questions Still Remain Unresolved?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: A good literature review is supposed to educate the reader on the state of research about the issue the study sets out to investigate. The key findings and highlights from the literature should be clearly synthesized in the introduction and literature review. The following questions are useful to ask after you have read the literature review portion of an empirical article:

- Does it provide enough information on the state of research about the problem the study sets out to investigate?
- Are the key findings and highlights from the literature clearly synthesized in the review?
- Do you feel that you understand the state of research related to the main research question asked (usually, it is in the title of the article)?

If, after reading the literature review, you are still confused about what the previous studies have found and what still remains to be discovered about the narrow topic the study is focused on, give a low mark on this evaluation question.

10. Overall, is the Literature Review Portion of the Introduction Appropriate and Effective?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and taking into account any additional considerations and concerns you may have. Be prepared to explain your overall evaluation.

13 Orzeck, T., & Rokach, A. (2004). Men who abuse drugs and their experience of loneliness. *European Psychologist*, 9(3), 163–169.

Chapter 5 Exercises

Part A

Directions: Answer the following questions.

1. Consider Statement A and Statement B. They both contain the same citations. In your opinion, which statement is superior? Explain.

Statement A: The overall positive association between nonverbal decoding skills and workplace effectiveness has been replicated with adults in a variety of settings (Campbell, Kagan, & Krathwohl, 1971; Costanzo & Philpott, 1986; DiMatteo, Friedman, & Taranta, 1979; Halberstadt & Hall, 1980; Izard, 1971; Izard et al., 2001; Nowicki & Duke, 1994; Schag, Loo, & Levin, 1978; Tickle-Degnen, 1998).

Statement B: The overall positive association between nonverbal decoding skills and workplace effectiveness has been replicated with adults in counseling settings (Campbell, Kagan, & Krathwohl, 1971; Costanzo & Philpott, 1986; Schag, Loo, & Levin, 1978) and medical settings (DiMatteo, Friedman, & Taranta, 1979; Tickle-Degnen, 1998), and with children in academic settings (Halberstadt & Hall, 1980; Izard, 1971; Izard et al., 2001; Nowicki & Duke, 1994).¹⁴

2. Consider Statement C. This statement could have been used as an example for which evaluation question in this chapter?

Statement C: In contrast to the somewhat sizable body of research informing secular program practice to reduce relapse and recidivism, the literature on faith-based religious programming has produced very few outcome-based studies. With regard to community-based corrections-related programming, evaluations are almost nonexistent.¹⁵

3. Consider Statement D. This statement could have been used as an example for which evaluation question in this chapter?

Statement D: Research on happiness and subjective well-being has generated many intriguing findings, among which is that happiness is context dependent and relative (e.g., Brickman & Campbell, 1971; Easterlin, 1974, 2001; Parducci, 1995; Ubel, Loewenstein, & Jepson, 2005; see Diener et al., 2006; Hsee & Hastie, 2006, for reviews). For example, paraplegics can be nearly as happy as lottery winners (Brickman et al., 1978).¹⁶

14 Effenbein, H. A., & Ambady, N. (2002). Predicting workplace outcomes from the ability to eavesdrop on feelings. *Journal of Applied Psychology*, 87(5), 963–971.

15 Roman, C. G., Wolff, A., Correa, V., & Buck, J. (2007). Assessing intermediate outcomes of a faith-based residential prisoner reentry program. *Research on Social Work Practice*, 17(2), 199–215.

16 Hsee, C. K., & Tang, J. N. (2007). Sun and water: On a modulus-based measurement of happiness. *Emotion*, 7(1), 213–218.

4. Consider Statement E. This statement could have been used as an example for which evaluation question in this chapter?

Statement E: When speaking of “help-seeking” behaviors or patterns, Rogler and Cortes (1993) proposed that “from the beginning, psychosocial and cultural factors impinge upon the severity and type of mental health problems; these factors [thus] interactively shape the [help-seeking] pathways’ direction and duration” (p. 556).¹⁷

5. Consider Statement F. This statement could have been used as an example for which evaluation question in this chapter?

Statement F: In the majority of studies referred to above, the findings have been correlational in nature, with the result that it has not been possible to draw causal inferences between low cortisol concentrations and antisocial behavior.¹⁸

Part B

Directions: Read the introductions to three empirical articles in academic journals on a topic of interest to you. Apply the evaluation questions in this chapter to the literature reviews in their introductions, and select the one to which you gave the highest ratings. Bring it to class for discussion. Be prepared to discuss its specific strengths and weaknesses.

17 Akutsu, P. D., Castillo, E. D., & Snowden, L. R. (2007). Differential referral patterns to ethnic-specific and mainstream mental health programs for four Asian American groups. *American Journal of Orthopsychiatry*, 77(1), 95–103.

18 van Goozen, S. H. M., Fairchild, G., Snoek, H., & Harold, G. T. (2007). The evidence for a neurobiological model of childhood antisocial behavior. *Psychological Bulletin*, 133(1), 149–182.

Evaluating Samples when Researchers Generalize

Immediately after the Introduction, which includes a literature review, most researchers insert the main heading of *Method* or *Methods* (or *Data and Methods*). In the *Method* section, researchers almost always begin by describing the individuals they studied. This description is usually prefaced with one of these subheadings: *Data*, or *Sample*, or *Subjects*, or *Participants*.¹

A *population* is any group in which a researcher is ultimately interested. It might be large, such as all registered voters in Pennsylvania, or it might be small, such as all members of a local teachers' association. Researchers often study only *samples* (i.e., a subset of a population) for the sake of efficiency, then *generalize* their results to the population of interest. In other words, they infer that the data they collected by studying a sample are similar to the data they would have obtained by studying the entire population. Such generalizability only makes sense if the sample is *representative* of the population. In this chapter, we will discuss some of the criteria that can help you figure out whether a study sample is representative, and thus, whether the study results can be generalized to a wider population.

Because many researchers do not explicitly state whether they are attempting to generalize, consumers of research often need to make a judgment on this matter in order to decide whether to apply the evaluation questions in this chapter to the empirical research article being evaluated. To make this decision, consider these questions:

- Does the researcher *imply* that the results apply to some larger population?
- Does the researcher discuss the implications of his or her research for a larger group of individuals than the one directly studied?

If the answers are clearly “yes”, apply the evaluation questions in this chapter to the article being evaluated. Note that the evaluation of samples when researchers are clearly *not* attempting to generalize to populations (a much less likely scenario for social science research) is considered in the next chapter.

1 In older research literature, the term *participants* would indicate that the individuals being studied had consented to participate after being informed of the nature of the research project, its potential benefits, and its potential harm; while the use of the term *subjects* would be preferred when there was no consent – such as in animal studies.

1. Was Random Sampling Used?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ²
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: Using random, or probability, sampling (like drawing names out of a hat³) yields an *unbiased* sample (i.e., a sample that does not systematically favor any particular type of individual or group in the selection process). If a sample is unbiased and reasonably large, researchers are likely to make sound generalizations. (Sample size will be discussed later in this chapter.)

The desirability of using random sampling as the basis for making generalizations is so widely recognized among researchers that they are almost certain to mention its use if it was employed in selecting their sample. Examples 6.1.1 and 6.1.2 show two instances of how this has recently been expressed in published research.

Example 6.1.1⁴

DESCRIPTION OF THE USE OF RANDOM SAMPLING (A NATIONALLY REPRESENTATIVE SAMPLE OF ADOLESCENTS IN THE UNITED STATES)

Data for this study came from the National Longitudinal Study of Adolescent Health (Add Health; Harris, 2009). The Add Health is a longitudinal and nationally representative sample of adolescents enrolled in grades 7 through 12 for the 1994–1995 academic year. The general focus of the Add Health study was to assess the health and development of American adolescents. In order to do so, a sample of high schools was first selected by employing stratified random sampling techniques. During this step, 132 schools were selected for participation and all students attending these schools were asked to complete a self-report questionnaire (N ~ 90,000).

Beginning in April 1995 and continuing through December 1995, the Add Health research team collected more detailed information from a subsample of the students who completed the in-school surveys. Not all 90,000 students who completed in-school surveys also completed the follow-up interview (i.e. wave 1). Instead, students listed on each school's roster provided a sample frame from which respondents were chosen. In all, wave 1 in-home interviews were conducted with 20,745 adolescents. Respondents ranged between 11 and 21 years of age at wave 1.

- 2 Continuing with the same scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”
- 3 For a more modern version of this procedure, see the online resources for this chapter (a link to a random number generator).
- 4 Barnes, J. C., Golden, K., Mancini, C., Boutwell, B. B., Beaver, K. M., & Diamond, B. (2014). Marriage and involvement in crime: A consideration of reciprocal effects in a nationally representative sample. *Justice Quarterly*, 31(2), 229–256.

Example 6.1.2⁵

DESCRIPTION OF THE USE OF RANDOM SAMPLING (A REPRESENTATIVE SAMPLE OF COURT CASES WHERE SCHIZOPHRENIA IS SUSPECTED OR CONFIRMED)

The litigated cases are a 10% random sample of 3543 cases litigated in all courts during the period 2010 to 2012 in which one of the keywords is “schizophrenia.” The cases were retrieved from the Lexis Nexis database of court cases at all court levels. Only cases in which the person with schizophrenia was a litigant were included. This reduced the total number of usable cases to 299.

2. If Random Sampling was Used, was it Stratified?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Researchers use *stratified random sampling* by drawing individuals separately at random from different strata (i.e., subgroups) within a population. In Example 6.1.1 above, the sample of schools selected for the National Longitudinal Study of Adolescent Health (Add Health) was stratified by region of the country, urbanicity, and school size and type, to make sure that schools from various parts of the country were represented, that rural, urban, and suburban schools were included in the sample, that small as well as large schools were represented, and so on.

Stratifying will improve a sample only if the stratification variable (e.g., geography) is related to the variables to be studied. For instance, if a researcher is planning to study how psychologists work with illicit substance abusers in New York State, stratifying on geography will improve the sample if the various areas of the state (for example, rural upstate New York areas versus areas in and around New York City) tend to have different types of drug problems, which may require different treatment modalities.

Note that geography is often an excellent variable on which to stratify, because people tend to cluster geographically on the basis of many variables that are important in the social and behavioral sciences. For instance, they often cluster according to race/ethnicity, income/personal wealth, language preference, religion, and so on. Thus, a geographically representative sample is likely to be representative in terms of these other variables as well. Other common stratification variables are gender, age, occupation, highest educational level attained, and political affiliation.

In Example 6.2.1, geography was used as a stratification variable.

5 LaVan, M., LaVan, H., & Martin, W. M. M. (2017). Antecedents, behaviours, and court case characteristics and their effects on case outcomes in litigation for persons with schizophrenia. *Psychiatry, Psychology and Law*, 24(6), 866–887.

Example 6.2.1⁶

DESCRIPTION OF THE USE OF STRATIFIED RANDOM SAMPLING

The data for our investigation came from a survey of 3,690 seventh-grade students from 65 middle schools in randomly selected counties in the state of Kentucky. Four strata were used: (1) counties with a minimum population of 150,000, (2) counties with population sizes between 40,000 and 150,000, (3) counties with population sizes between 15,000 and 40,000, and (4) counties with population sizes below 15,000.

If random sampling without stratification is used (like in Example 6.1.2 in the previous section, where 10% of all relevant cases were randomly selected), the technique is called *simple random sampling*. In contrast, if stratification is used to form subgroups from which random samples are drawn, the technique is called *stratified random sampling*.

Despite the almost universal acceptance that an unbiased sample obtained through simple or stratified random sampling is highly desirable for making generalizations, the vast majority of research from which researchers want to make generalizations is based on studies in which nonrandom (biased) samples were used. There are three major reasons for this:

- a) Even though a random selection of names might have been drawn, a researcher often cannot convince all those selected to participate in the research project. This problem is addressed in the next three evaluation questions.
- b) Many researchers have limited resources with which to conduct research: limited time, money, and assistance. Often, they will reach out to individuals who are readily accessible or convenient to use as participants. For instance, college professors conducting research often find that the most convenient samples consist of students enrolled in their classes, which are not even random samples of students on their campuses. This is called *convenience sampling*, which is a highly suspect method for drawing samples from which to generalize.
- c) For some populations, it is difficult to identify all members. If a researcher cannot do this, he or she obviously cannot draw a random sample of the entire population.⁷ Examples of populations whose members are difficult to identify are the homeless in a large city, successful burglars (i.e., those who have never been caught), and illicit drug users.

Because so many researchers study nonrandom samples, it is unrealistic to count failures on the first two evaluation questions in this chapter as fatal flaws in research methodology. If journal editors routinely refused to publish research reports with this type of deficiency, there would be very little published research on many of the most important problems in the social and behavioral sciences. Thus, when researchers use nonrandom samples when attempting to generalize, the additional evaluation questions raised in this chapter should be applied in order

6 This example is loosely based on the work of Ousey, G. C., & Wilcox, P. (2005). Subcultural values and violent delinquency: A multilevel analysis in middle schools. *Youth Violence and Juvenile Justice*, 3(1), 3–22.

7 You might have already figured out that the only way for researchers to draw a simple or stratified random sample is if the researchers have a list of all population members they would be choosing from.

to distinguish between studies from which it is reasonable to make tentative, very cautious generalizations and those that are hopelessly flawed with respect to their sampling.

3. If Some Potential Participants Refused to Participate, Was the Rate of Participation Reasonably High?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Defining *reasonably high* is problematic. For instance, a professional survey organization, with trained personnel and substantial resources, would be concerned if it had a response rate of less than 80% when conducting a national survey. On the other hand, researchers with limited resources using mailed questionnaires often are satisfied with a return rate as low as 50%, especially because rates of returns to mailed surveys are notoriously poor. As a very rough rule of thumb, then, response rates of substantially less than 50% raise serious concerns about the generalizability of the findings.

Example 6.3.1 reports a reasonable response rate for a mailed survey.

Example 6.3.1⁸

REASONABLE RESPONSE RATES FOR A MAILED SURVEY

Surveys returned without forwarding addresses, for deceased respondents, or those with incomplete responses were eliminated from the sample. The response rates were 56.7% psychologists ($n = 603$), 45.8% psychiatrists ($n = 483$), and 58.2% social workers ($n = 454$), resulting in a 53% overall survey response rate and a total sample ($N = 1,540$).

The situation becomes even murkier when electronic or online surveys are solicited through email, text message, or an ad placed at a website. The pace of technological advances is so high, and changes in the use of phones, tablets, email, and specific social media platforms are so unpredictable, that it is difficult to make any specific judgments or draw even tentative thresholds about the “typical” response rates for online surveys. There is also paucity of research and knowledge on this topic exactly because of the fast pace of changes.

For example, a study published in 2008 (that used teachers in Ohio and South Carolina as survey participants) suggests that web-based surveys solicited through email yield a lower rate of response than mailed surveys⁹, while another similar study published a year later (that used evaluators from the American Evaluation Association as survey participants) suggests online surveys yield a higher response than traditional mailed ones.¹⁰ And it is likely that the situation has changed in the several years since these studies were conducted.

- 8 Pottick, K. J., Kirk, S. A., Hsieh, D. K., & Tian, X. (2007). Judging mental disorder in youths: Effects of client, clinical, and contextual differences. *Journal of Consulting and Clinical Psychology*, 75, 1–8.
- 9 Converse, P. D., Wolfe, E. W., Huang, X., & Oswald, F. L. (2008). Response rates for mixed-mode surveys using mail and e-mail/web. *American Journal of Evaluation*, 29(1), 99–107.
- 10 Greenlaw, C., & Brown-Welty, S. (2009). A comparison of web-based and paper-based survey methods: testing assumptions of survey mode and response cost. *Evaluation Review*, 33(5), 464–480.

Moreover, any comparisons between mailed and emailed/online surveys can only be investigated using specific categories of people as survey participants (for example, federal employees,¹¹ Illinois public school guidance counselors,¹² doctors in Australia,¹³ or PhD holders from Spanish universities¹⁴), and thus any findings obtained are likely not generalizable to other populations.

The percentages mentioned above regarding response rates to surveys should not be applied mechanically during research evaluation because exceptions may be made for cases in which participation in the research is burdensome or invasive or raises sensitive issues that might make it understandable to obtain a lower rate of participation. For instance, if a researcher needed to draw samples of blood from students on campus to estimate the incidence of a certain type of infection, or needed to put a sample of students through a series of rigorous physical fitness tests that spanned several days for a study in sports psychology, a consumer of research might judge a participation rate of substantially less than 50% to be reasonable in light of the demanding nature of research participation in the study, keeping in mind that any generalizations to wider populations would be highly tenuous.

Overall, lower rates of participation have a high potential for introducing a *selection bias* (or *self-selection bias*), which means that those who have agreed to participate are different in some fundamental ways from those who refused to participate, and thus the study results will not correctly reflect the total population.

4. If the Response Rate Was Low, Did the Researcher Make Multiple Attempts to Contact Potential Participants?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Researchers often make multiple attempts to contact potential participants. For instance, a researcher might contact potential participants several times (e.g., by several mailings and by phone) and still achieve a response rate of less than 50%. In this case, a consumer of research might reach the conclusion that this is the highest rate of return that might be expected for the researcher's particular research problem and population. In effect, the consumer might judge that this is the best that can be done, keeping in mind that generalizations from such a sample are exceedingly risky because nonparticipants might be fundamentally different from those who agreed to participate (self-selection bias).

- 11 Lewis, T., & Hess, K. (2017). The effect of alternative e-mail contact timing strategies on response rates in a self-administered web survey. *Field Methods*, 29(4), 351–364.
- 12 Mackety, D. M. (2007). *Mail and web surveys: A comparison of demographic characteristics and response quality when respondents self-select the survey administration mode*. Ann Arbor, MI: ProQuest Information and Learning Company.
- 13 Scott, A., Jeon, S. H., Joyce, C. M., Humphreys, J. S., Kalb, G., Witt, J., & Leahy, A. (2011). A randomised trial and economic evaluation of the effect of response mode on response rate, response bias, and item non-response in a survey of doctors. *BMC Medical Research Methodology*, 11(1), 126.
- 14 Barrios, M., Villarroja, A., Borrego, Á., & Ollé, C. (2011). Response rates and data quality in web and mail surveys administered to PhD holders. *Social Science Computer Review*, 29(2), 208–220.

Example 6.4.1 describes multiple contacts made by researchers in an effort to achieve a high response rate.

Example 6.4.1¹⁵

MULTIPLE ATTEMPTS TO OBTAIN A SAMPLE

Potential participants were first contacted with an e-mail invitation that included a link to complete the online survey. This was followed by up to 5 reminder e-mails sent by the survey center and up to 10 attempted follow-up telephone contacts as needed. The telephone calls served as a reminder to complete the survey online and an opportunity to complete the survey over the phone. Only 3 of our respondents chose to complete the survey over the phone versus online.

5. Is There Reason to Believe that the Participants and Nonparticipants are Similar on Relevant Variables?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: In some instances, researchers have information about those who do not participate, which allows for a comparison of nonparticipants with participants. For instance, a researcher might note the zip codes on the envelopes in which returned questionnaires were posted. This might allow a researcher to determine whether those in affluent neighborhoods were more responsive than those in less affluent ones.¹⁶

In institutional settings such as schools, hospitals, and prisons, it is often possible to determine whether participants and nonparticipants differ in important respects. For instance, in a survey regarding political attitudes held by college students, participants might be asked for background information such as major, GPA, and age. These background characteristics are usually known for the population of students on the campus, allowing for a comparison of participants and the entire student body. If there are substantial differences, the results will need to be interpreted in light of them. For instance, if political science majors were a much larger percentage of the participants than exists in the whole student body, the researcher should be highly cautious in generalizing the results to all students.

In the evaluation of a new component for the Head Start program in rural areas of Oregon, only 56% agreed to participate. However, the researchers noted the similarities of these participants with the general population in Example 6.5.1. This provides some assurance that those who chose to participate in the research were not substantially different from nonparticipants in terms of important background characteristics (i.e., demographics).

15 Winters, K. C., Toomey, T., Nelson, T. F., Erickson, D., Lenk, K., & Miazga, M. (2011). Screening for alcohol problems among 4-year colleges and universities. *Journal of American College Health*, 59(5), 350–357.

16 If such a bias were detected, statistical adjustments might be made to correct for it by mathematically giving more weight to the respondents from the underrepresented zip codes.

Example 6.5.1¹⁷

COMPARISON OF A FLAWED SAMPLE TO THE LARGER POPULATION

Forty-five percent of children [were] living in families including both biological parents. Sixty percent of the children and families received public assistance. Eighty-three percent were Caucasian, and 13% were other ethnic groups, primarily Hispanic. These demographics are representative of the rural population in Oregon.

It is also important to consider what is called *attrition*, or selective dropout of participants from the study,¹⁸ for those studies that are conducted over a period of time (such studies are called *longitudinal* if done over longer periods of time¹⁹). If out of 120 participants who signed up for the study and completed the first round of interviews, only 70 are left by the third round of interviews one year later, it is important to compare the characteristics of those who dropped out of the study with those who stayed. If the two groups differ on some important study variables or demographic characteristics, the possibility of self-selection bias should be discussed by the researchers. It is very likely that by the third wave, the remaining participants are not as representative of the larger population as were the original 120, and thus the study results could be misleading or hard to generalize.

6. If a Sample is Not Random, Was it at Least Drawn from the Target Group for the Generalization?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: There are many instances in the published literature in which a researcher studied one type of participant (e.g., college freshmen) and used the data to make generalizations to a different target group (e.g., young adults in general).²⁰ If a researcher does not have the wherewithal to at least tap into the target group of interest, it might be better to leave the research to other researchers who have the resources and contacts that give them access to members of the target group. Alternatively, the researcher should be honest about generalizing the results to the actual population the sample was drawn from.

Example 6.6.1 describes the convenience sample (nonrandom) used in a study on the provision of mental health services to college students. The researchers wanted to apply the results

17 Kaminski, R. A., Stormshak, E. A., Good, R. H. III, & Goodman, M. R. (2002). Prevention of substance abuse with rural Head Start children and families: Results of Project STAR. *Psychology of Addictive Behaviors*, 16(4S), S11–S26.

18 *Attrition* is especially important to consider for studies that involve experiments. These issues are discussed in more detail in Chapter 9.

19 In contrast, studies conducted “in one shot” are called *cross-sectional*.

20 In this context, it is interesting to note that the editor of the *Journal of Adolescent Research* pointed out that “Many articles currently published in journals on adolescence are based on American middle-class samples but draw conclusions about adolescents in general.” (p. 5). Arnett, J. J. (2005). The vitality criterion: A new standard of publication for *Journal of Adolescent Research*. *Journal of Adolescent Research*, 20(1), 3–7.

only to college students. Thus, the sample is adequate in terms of this evaluation question because the sample was drawn from the target group.

Example 6.6.1²¹

NONRANDOM SAMPLE FROM THE TARGET GROUP (COLLEGE STUDENTS)

Three hundred students (201 women, 98 men, 1 not indicating gender) enrolled in introductory college courses served as participants. Students were at least age 18, attending a medium-sized state university in the Midwestern United States. Participants were recruited from their university’s multidepartment research pool ($n = 546$) for research or extra credit through a password-protected Website listing available university-specific studies for electronic sign-up.

___ 7. If a Sample is Not Random, Was it Drawn from Diverse Sources?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Did a researcher generalize to all college students after studying only students attending a small religious college in which 99% of the students have the same ethnic/racial background? Did a researcher generalize to men and women regarding the relationship between exercise and health after studying only men attending a cardiac unit’s exercise program? An answer of “yes” to these types of questions might lead to a low rating for this evaluation question.

When a researcher wants to generalize to a larger population in the absence of random sampling, consider whether the researcher sought participants from several sources, which increases the odds of representativeness. For instance, much educational research is conducted in just one school. Using students from several schools within the district would increase the odds that the resulting sample will reflect the diversity of the district.

In Example 6.7.1, the researchers used three methods for drawing a sample for a study of parents with disabilities. This is vastly superior to using just one method for locating participants in a hard-to-reach population.

Example 6.7.1²²

DIVERSE SOURCES FOR A SAMPLE (HELPS INCREASE REPRESENTATIVENESS)

We used three avenues for recruitment of parents with disabilities. The first was to distribute survey packets to many disability organizations and service agencies and to ask

21 Elhai, J. D., & Simons, J. S. (2007). Trauma exposure and posttraumatic stress disorder predictors of mental health treatment use in college students. *Psychological Services*, 4(1), 38–45.

22 Olkin, R., Abrams, K., Preston, P., & Kirshbaum, M. (2006). Comparison of parents with and without disabilities raising teens: Information from the NHIS and two national surveys. *Rehabilitation Psychology*, 51(1), 43–49.

them to distribute the survey packets. There are drawbacks to this method. [. . .] This distribution method solicits responses only from families connected to a disability or service agency in some way. Such families may differ from those with no connections to such agencies.

The second method was to solicit participants directly by placing announcements and ads in many different venues and having interested parents call us for a survey. This was our primary recruitment method. Contact was made with 548 agencies, resulting in announcements or ads in newsletters or other publications associated with those agencies.

The third method of outreach was through the Internet. E-mail and Website postings went to agencies serving people with disabilities, parents, and/or children, as well as bulletin boards, and were updated frequently. Approximately 650 websites were visited and requested to help distribute information about this survey. Additionally, we investigated 65 electronic mailing lists and subscribed to 27. Last, we purchased a list of addresses, phone numbers, and e-mail addresses of various disability-related agencies, magazines, and newsletters. We contacted these sites by phone and followed up with an informational e-mail.

8. If a Sample is Not Random, Does the Researcher Explicitly Discuss This Limitation and How it May Have Affected the Generalizability of the Study Findings?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: A common way for researchers to recruit people into their samples is to ask for volunteers. But do those who volunteer for study participation differ in some important ways from those who never responded to the study recruitment ads? Could this selective volunteering (or self-selection) have affected the study results and conclusions?

Some researchers²³ think this is exactly what happened in the famous Stanford Prison Experiment (SPE): that its results would have been very different if a different way of recruiting participants had been used. The ad looking for SPE volunteers mentioned “a psychological study of prison life”, which might have attracted students with more psychopathic personalities than the general student population on the campus. As a result, such volunteers might have been more prone to using emotionally abusive tactics in the “prison guard” role.²⁴

While researchers may discuss the limitations of their methodology (including sampling) in any part of their reports, many explicitly discuss limitations in the Discussion section at the end of their articles. Example 6.8.1 appeared near the end of a research report.

23 Carnahan, T., & McFarland, S. (2007). Revisiting the Stanford Prison Experiment: Could participant self-selection have led to the cruelty? *Personality and Social Psychology Bulletin*, 33(5), 603–614.

24 For more information about the Stanford Prison Experiment and possible interpretations of its results, see the online resources for this chapter.

Example 6.8.1²⁵

STATEMENT OF A LIMITATION IN SAMPLING

The findings of the current study should be considered in light of its limitations. [. . .]
[Our] sample consisted of higher risk adjudicated delinquents from a single southeastern state in the United States, thus limiting its generalizability.

Example 6.8.2 is an acknowledgment of a sampling limitation that appeared as the last few sentences in a research report. While such an acknowledgement does not remedy the flaws in the sampling procedure, it is important for the researchers to point out how it limits the generalizability of the study findings.

Example 6.8.2²⁶

STATEMENT OF A LIMITATION IN SAMPLING

Finally, the fact that patients with a lifetime history of psychotic disorder, or alcohol or drug addiction, were not included in the study may have biased the sample, limiting the generalizability of the findings. The results should be treated with caution, and replication, preferably including a larger sample size, is recommended.

Such acknowledgments of limitations do not improve researchers’ ability to generalize. However, they do perform two important functions: (a) they serve as warnings to naïve readers regarding the problem of generalizing, and (b) they reassure all readers that the researchers are aware of a serious flaw in their methodology.

___ 9. Has the Author Described Relevant Characteristics (Demographics) of the Sample?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: A researcher should describe the relevant background characteristics of the sample. For instance, when studying physicians’ attitudes toward assisted suicide, it would be relevant to know their religious affiliations. For studying consumers’ preferences, it would be helpful to know their economic status.

In addition to the participants’ characteristics that are directly relevant to the variables being studied, it usually is desirable to provide an overall demographic profile, including variables such as age, gender, race/ethnicity, and highest level of education. This is especially

25 Craig, J. M., Intravia, J., Wolff, K. T., & Baglivio, M. T. (2017). What can help? Examining levels of substance (non)use as a protective factor in the effect of ACEs on crime. *Youth Violence and Juvenile Justice* [Online first].

26 Chioqueta, A. P., & Stiles, T. C. (2004). Suicide risk in patients with somatization disorder. *Crisis: The Journal of Crisis Intervention and Suicide*, 25(1), 3–7.

important when a nonrandom sample of convenience has been used because readers will want to visualize the particular participants who were part of such a sample.

Example 6.9.1 is from a study on how religious functioning is related to mental health outcomes in military veterans.

Example 6.9.1²⁷

DESCRIPTION OF RELEVANT DEMOGRAPHICS

Military veterans (N = 90) completed an online survey for the current study. The sample was primarily male (80%) and Caucasian (79%). The mean age of the sample was 39.46 (SD = 15.10). Deployments were primarily related to Operation Iraqi Freedom/Operation Enduring Freedom (OIF/OEF) (n = 62), with other reported deployments to Vietnam (n = 12), the Balkan conflict (n = 4), and other conflicts (n = 3). Nine participants did not report the location of their deployments. The mean number of deployments was 1.47, and the mean time since last deployment was 13.10 years (SD = 13.56; Median = 8.00).

When information on a large number of demographic characteristics has been collected, researchers often present these in statistical tables instead of in the narrative of the report.

10. Is the Overall Size of the Sample Adequate?

Very	1	2	3	4	5	Very		
unsatisfactory						satisfactory	or N/A	I/I

Comment: Students who are new to research methods are sometimes surprised to learn that there often is no simple answer to the question of how large a sample should be. First, it depends in part on how much error a researcher is willing to tolerate. For public opinion polls, a stratified random sample of about 1,500 produces a margin of error of about one to three percentage points. A sample size of 400 produces a margin of error of about four to six percentage points.²⁸ If a researcher is trying to predict the outcome of a close election, clearly a sample size of 400 would be inadequate.²⁹

Responding to a public opinion poll usually takes little time and may be of interest to many participants, thus making it easier for the researchers to reach a large sample size. Other types of studies, however, may be of less interest to potential participants and/or may require extensive effort on the part of participants. In addition, certain data collection methods

27 Boals, A., & Lancaster, S. (2018). Religious coping and mental health outcomes: The mediating roles of event centrality, negative affect, and social support for military veterans. *Military Behavioral Health*, 6(1), 22–29.

28 The exact size of the margin of error depends on whether the sample was stratified and on other sampling issues that are beyond the scope of this book.

29 With a sample of only 400 individuals, there would need to be an 8–12 percentage-point *difference* (twice the four- to six-point margin of error) between the two candidates for a reliable prediction to be made (i.e., a statistically significant prediction).

(such as individual interviews) may require expenditure of considerable resources by researchers. Under such circumstances, it may be unrealistic to expect a researcher to use large samples. Thus, a consumer of research should ask whether the researchers used a reasonable number given the particular circumstances of their study. Would it have been an unreasonable burden to use substantially more participants? Is the number of participants so low that there is little hope of making sound generalizations? Would it be reasonable to base an important decision on the results of the study given the number of participants used? Subjective answers to these types of questions will guide consumers of research on this evaluation question.³⁰

It is important to keep in mind that a large sample size does not compensate for a bias in sampling due to the failure to use random sampling. Using large numbers of unrepresentative participants does not get around the problem of their unrepresentativeness.

____ **11. Is the Number of Participants in Each Subgroup Sufficiently Large?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: When several groups of people are compared, for example, in the context of an experiment where one group has received “treatment” while the other is a comparison group, the rule of thumb is to have at least 30 participants per group (or subgroup) if the groups are fairly homogenous.³¹ A larger number of participants per group is needed if the groups are heterogeneous, i.e., if there is a lot of variation among the participants on sociodemographic characteristics like race, gender, age, income, or any other relevant variable.

Consider the hypothetical information in Example 6.11.1, where the numbers of participants in each subgroup are indicated by n , and the mean (average) scores are indicated by m .

Example 6.11.1

A SAMPLE IN WHICH SOME SUBGROUPS ARE VERY SMALL

A random sample of 100 college freshmen was surveyed on its knowledge of alcoholism. The mean (m) scores out of a maximum of 25 were as follows: White ($m = 18.5$, $n = 78$), African American ($m = 20.1$, $n = 11$), Hispanic/Latino ($m = 19.9$, $n = 9$), and Chinese American ($m = 17.9$, $n = 2$). Thus, for each of the four ethnic/racial groups, there was a reasonably high average knowledge of alcoholism.

Although the total number in the sample is 100 (a number that might be acceptable for some research purposes), the numbers of participants in the last three subgroups in Example 6.11.1

30 There are statistical methods for estimating optimum sample sizes under various assumptions. While these methods are beyond the scope of this book, note that they do not take into account the practical matters raised here.

31 There is nothing magic about the number 30 – the reasons are purely statistical and have a lot to do with statistical significance testing (see more on this topic in Appendix C: *The Limitations of Significance Testing*).

are so small that it would be highly inappropriate to generalize from them to their respective populations. The researcher should either obtain larger numbers of them or refrain from reporting separately on the individual subgroups. Notice that there is nothing wrong with indicating ethnic/racial backgrounds (such as the fact that there were two Chinese American participants) in describing the demographics of the sample. Instead, the problem is that the number of individuals in some of the subgroups used for comparison is too small to justify calculating a mean and making any valid comparisons or inferences about them. For instance, a mean of 17.9 for the Chinese Americans is meaningless for the purpose of generalizing because there are only two individuals in this subgroup. Here, at least 30 people per subgroup would be needed.

___ 12. Has Informed Consent Been Obtained?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: It is almost always highly desirable to obtain written, informed consent from the participants in a study. Participants should be informed of the nature of the study and, at least in general terms, the nature of their involvement. They should also be informed of their right to withdraw from the study at any time without penalty. Typically, researchers report only very briefly on this matter, as illustrated in Example 6.12.1, which presents a statement similar to many found in research reports in academic journals. It is unrealistic to expect much more detail than shown here because, by convention, the discussion of this issue is typically brief.

Example 6.12.1

A BRIEF DESCRIPTION OF INFORMED CONSENT

Students from the departmental subject pool volunteered to participate in this study for course credit. Prior to participating in the study, students were given an informed consent form that had been approved by the university's institutional review board. The form described the experiment as "a study of social interactions between male and female students" and informed them that if they consented, they were free to withdraw from the study at any time without penalty.

___ 13. Has the Study Been Approved by an Ethics Review Agency (Institutional Review Board, or IRB, if in the United States or a Similar Agency if in Another Country)?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: For any study that involves human subjects, even if indirectly, the researchers planning the study must undergo a research ethics review process. In the United States, committees

responsible for such ethics reviews are called Institutional Review Boards (IRBs). In Canada, similar agencies are called Research Ethics Boards (REBs). In the United Kingdom, there is a system of Research Ethics Committees (RECs). Such an ethics committee checks that the study meets required ethical standards and does not present any undue danger of harm to the participants (usually, three types of harm are considered: physical, psychological, and legal harm). Only after the approval of a study by the relevant ethics committee has been granted, the study can commence. It is not required to mention the IRB's or an analogous agency's approval in the research report but is often a good idea to do so. Example 6.13.1 shows how such an approval can be stated in an article (though a separate subheading is uncommon).

Example 6.13.1³²

A BRIEF MENTION OF APPROVAL BY THE RELEVANT ETHICS REVIEW COMMITTEE,
UNDER A SEPARATE SUBHEADING

Ethics Approval

The Ethics Committee of the Institut de la statistique du Québec and the Research Ethics Board of the CHU Sainte-Justine Research Center approved each phase of the study, and informed consent was obtained.

There may be times when a consumer of research judges that the study is so innocuous that informed consent might not be needed. An example is an observational study in which individuals are observed in public places, such as a public park or shopping mall, while the observers are in plain view. Because public behaviors are being observed by researchers in such instances, privacy would not normally be expected and informed consent may not be required. Even for such studies, however, approval from an ethics review committee *is required*.

Example 6.13.2 shows a typical way the ethical review committee's approval of a study is mentioned in the article, even though this study did not involve any direct contact with its subjects.

Example 6.13.2³³

A DESCRIPTION OF APPROVAL BY THE RELEVANT ETHICS REVIEW COMMITTEE
AND THE NATURE OF THE STUDY

The researcher applied for and received ethics approval from the Department of Community Health (DCH) Institutional Review Board (IRB). All data were kept confidential and

32 Geoffroy, M. C., Boivin, M., Arseneault, L., Renaud, J., Perret, L. C., Turecki, G., . . . & Tremblay, R. E. (2018). Childhood trajectories of peer victimization and prediction of mental health outcomes in mid-adolescence: a longitudinal population-based study. *Canadian Medical Association Journal*, 190(2), E37–E43.

33 Gay, J. G., Ragatz, L., & Vitacco, M. (2015). Mental health symptoms and their relationship to specific deficits in competency to proceed to trial evaluations. *Psychiatry, Psychology and Law*, 22(5), 780–791.

presented in an anonymous format such that individual defendants were unidentifiable. This study was archival in nature and did not involve any direct contact with the study subjects. A list of competency assessments conducted through a state psychiatric facility in the southeastern United States was generated for the years 2010 to 2013, including both inpatient and outpatient evaluations. Four independent raters, consisting of a forensic psychologist, a doctoral student, and two undergraduate students were assigned to read the competency evaluations and complete a coding template developed by one of the study's authors.

___ 14. Overall, is the Sample Appropriate for Generalizing?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and taking into account any additional considerations and concerns you may have. Be prepared to discuss your response to this evaluation question.

Concluding Comment

Although a primary goal of much research in all the sciences is to make sound generalizations from samples to populations, researchers in the social and behavioral sciences face special problems regarding access to and cooperation from samples of humans. Unlike other published lists of criteria for evaluating samples, the criteria discussed in this chapter urge consumers of research to be pragmatic when making these evaluations. A researcher may exhibit some relatively serious flaws in sampling, yet a consumer may conclude that the researcher did a reasonable job under the circumstances.

However, this does not preclude the need to be exceedingly cautious in making generalizations from studies with weak, non-representative samples. Confidence in certain generalizations based on weak samples can be increased, however, if various researchers with different patterns of weaknesses in their sampling methods arrive at similar conclusions when studying the same research problems (this important process, already mentioned in Chapter 1, is called *replication*).

In the next chapter, the evaluation of samples when researchers do *not* attempt to generalize is considered.

Chapter 6 Exercises

Part A

Directions: Answer the following questions.

1. Suppose a researcher conducted a survey on a college campus by interviewing students that she or he approached while they were having dinner in the campus cafeteria one evening. In your opinion, is this a random sample of all students enrolled in the college? Could the method be improved? How?
2. Briefly explain why *geography* is often an excellent variable on which to stratify when sampling.
3. According to this chapter, the vast majority of research is based on biased samples. Cite one reason that is given in this chapter for this circumstance.
4. If multiple attempts have been made to contact potential participants, and yet the response rate is low, would you be willing to give the report a reasonably high rating for sampling? Explain.
5. Is it important to know whether participants and nonparticipants are similar on relevant variables? Explain.
6. Does the use of a large sample compensate for a bias in sampling? Explain.

Part B

Directions: Locate several research reports in academic journals in which the researchers are concerned with generalizing from a sample to a population, and apply the evaluation questions in this chapter. Select the one to which you gave the highest overall rating and bring it to class for discussion. Be prepared to discuss the strengths and weaknesses of the sampling method used.

Evaluating Samples when Researchers Do *Not* Generalize

As indicated in the previous chapter, researchers often study samples in order to make inferences about the populations from which the samples were drawn. This process is known as generalizing.

Not all research is aimed at generalizing. Here are the major reasons why:

1. Researchers often conduct *pilot studies*. These are designed to determine the feasibility of methods for studying specific research problems. For instance, a novice researcher who wants to conduct an interview study of the social dynamics of marijuana use among high school students might conduct a pilot study to determine, among other things, how much cooperation can be obtained from school personnel for such a study, what percentage of the parents give permission for their children to participate in interviews on this topic, whether students have difficulty understanding the interview questions and whether they are willing to answer them, the optimum length of the interviews, and so on. After the research techniques are refined in a pilot study with a sample of convenience, a more definitive study with a more appropriate sample for generalizing might be conducted.

Note that it is not uncommon for journals to publish reports of pilot studies, especially if they yield interesting results and point to promising directions for future research. Also note that while many researchers will explicitly identify their pilot studies as such (by using the term *pilot study*), at other times consumers of research will need to infer that a study is a pilot study from statements such as “The findings from this preliminary investigation suggest that . . .”

2. Some researchers focus on *developing and testing theories*. A theory is a proposition or set of propositions that provides a cohesive explanation of the underlying dynamics of certain aspects of behavior. For instance, self-verification theory indicates that people attempt to maintain stable self-concepts. On the basis of this theory, researchers can make a number of predictions. For instance, if the theory is correct, a researcher might predict that people with poor self-concepts will seek out negative social reinforcement (e.g., seek out people who give them negative feedback about themselves) while avoiding or rejecting positive reinforcement. They do not do this because they enjoy negative reinforcement. Instead, according to the theory,

it is an attempt to validate their perceptions of themselves.¹ Such predictions can be tested with empirical research, which sheds light on the validity of a theory, as well as data that may be used to further develop and refine it.

In addition to testing whether the predictions made on the basis of a theory are supported by data, researchers conduct studies to determine under what circumstances the elements of a theory hold up (e.g., in intimate relationships only? with mildly as well as severely depressed patients?). One researcher might test one aspect of the theory with a convenience sample of adolescent boys who are being treated for depression, another might test a different aspect with a convenience sample of high-achieving women, and so on. Note that they are focusing on the theory as an evolving concept rather than as a static explanation that needs to be tested with a random sample for generalization to a population. These studies may be viewed as *developmental tests* of a theory. For *preliminary* developmental work of this type, rigorous and expensive sampling from a large population usually is not justified.

3. Some researchers prefer to study *purposive samples* rather than random samples. A purposive sample is one in which a researcher has a special interest because the individuals in a sample have characteristics that make them especially rich sources of information. For instance, an anthropologist who is interested in studying tribal religious practices might purposively select a tribe that has remained isolated and, hence, may have been less influenced by outside religions than other tribes that are less isolated. Note that the tribe is not selected at random but is selected deliberately (i.e., purposively). The use of purposive samples is a tradition in *qualitative* research. (See Appendix A for a brief overview of the differences between qualitative and quantitative research, as well as mixed methods research.)

4. Some researchers study entire populations – not samples. This is especially true in institutional settings such as schools, where all the seniors in a school district (the population) might be tested. Nevertheless, when researchers write research reports on population studies, they should describe their populations in some detail.

Also, it is important to realize that in some studies, a sample may look like an entire population but the inferences from the study are supposed to extend beyond the specific time or “snapshot” of the population’s characteristics. For example, if a researcher is interested in describing the relationship between income inequality and violent crime rates in the United States during the 1990s, she may use all U.S. states as her entire population. At the same time, she may also intend to generalize her findings about the relationship between inequality and violent crime to other time periods, beyond the decade included in the study.

1 For more information on this theory and its potential application to a particular behavioral issue, see Trouilloud, D., Sarrazin, P., Bressoux, P., & Bois, J. (2006). Relation between teachers’ early expectations and students’ later perceived competence in physical education classes: Autonomy-supportive climate as a moderator. *Journal of Educational Psychology*, 98(1), 75–86.

___ **1. Has the Researcher Described the Sample/Population in Sufficient Detail?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ²
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Comment: As indicated in the previous chapter, researchers should describe relevant demographics (i.e., background characteristics) of the participants when conducting studies in which they are generalizing from a sample to a population. This is also true when researchers are not attempting to generalize.

Example 7.1.1 shows a description of demographics from a qualitative research report in which the researchers are seeking in-depth information about a group of women living in a shelter because of domestic violence. The description of the demographics helps consumers of research “see” the participants, which makes the results of the study more meaningful.

Example 7.1.1³

DETAILED DESCRIPTION OF THE DEMOGRAPHICS OF PARTICIPANTS

Ten participants were recruited from the local domestic violence shelter. They ranged in age from 20 to 47 years ($M = 35.4$, $SD = 7.5$). All 10 participants were women. Of the participants, 5 (50%) were Native American, 4 (40%) were European American, and 1 (10%) was Latina. Two (20%) participants were married, 2 (20%) were divorced, 2 (20%) were single, and 4 (40%) were separated from their spouses. Nine of the 10 (90%) participants had children, and the children’s ages ranged from under 1 year to over 27 years. Educational levels included 5 (50%) participants who had taken some college or technical courses, 2 (20%) participants with a high school diploma or general equivalency diploma (GED), 1 participant (10%) with a 10th-grade education, 1 participant (10%) with a technical school degree, and 1 participant (10%) who was a doctoral candidate. Four participants were unemployed, 2 worked as secretaries, 1 worked as a waitress, 1 worked as a housekeeper, 1 worked in a local retail store, and 1 worked in a factory. Each participant listed a series of short-term, low-pay positions such as convenience store clerk.

___ **2. For a Pilot Study or Developmental Test of a Theory, Has the Researcher Used a Sample with Relevant Demographics?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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- 2 Continuing with the same scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”
- 3 Wettersten, K. B., Rudolph, S. E., Faul, K., Gallagher, K., Trangsrud, H. B., Adams, K., . . . Terrance, C. (2004). Freedom through self-sufficiency: A qualitative examination of the impact of domestic violence on the working lives of women in shelter. *Journal of Counseling Psychology*, 51(4), 447–462.

Comment: Studies that often fail on this evaluation question are those in which college students are used as participants (for convenience in sampling). For instance, some researchers have stretched the limits of credulity by conducting studies in which college students are asked to respond to questions that are unrelated to their life experiences, such as asking unmarried, childless college women what disciplinary measures they would take if they discovered that their hypothetical teenage sons were using illicit drugs. Obviously, posing such hypothetical questions to an inappropriate sample might yield little relevant information even in a pilot study.

Less extreme examples are frequently found in published research literature. For instance, using college students in tests of learning theories when the theories were constructed to explain the learning behavior of children would be inappropriate. When applying this evaluation question to such studies, make some allowance for minor “misfits” between the sample used in a pilot study (or developmental test of a theory) and the population of ultimate interest. Keep in mind that pilot studies are not designed to provide definitive data – only preliminary information that will assist in refining future research.

___ **3. Even if the Purpose is Not to Generalize to a Population,
Has the Researcher Used a Sample of Adequate Size?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Very preliminary studies might be conducted using exceedingly small samples. While such studies might be useful to the researcher who is testing new methodology, the results frequently are not publishable. Because there are no scientific standards for what constitutes a reasonable sample size for a pilot study to be publishable, consumers of research need to make subjective judgments when answering this evaluation question. Likewise, there are no standards for sample sizes for developmental tests of a theory.

For purposive samples, which are common in qualitative research, the sample size may be determined by the availability of participants who fit the sampling profile for the purposive sample. For instance, to study the career paths of high-achieving women in education, a researcher might decide to use female directors of statewide education agencies. If there are only a handful of such women, the sample will necessarily be limited in number.

On the other hand, when there are many potential participants who meet the standards for a purposive sample, a researcher might continue contacting additional participants until the point of *saturation*, that is, the point at which additional participants are adding little new information to the picture that is emerging from the data the researchers are collecting. In other words, saturation occurs when new participants are revealing the same types of information as those who have already participated. Example 7.3.1 illustrates how this was described in the report of a qualitative study. Note the use of the term *saturation*, which has been italicized for emphasis. Using the criterion of data saturation sometimes results in the use of small samples.

Example 7.3.1⁴

A STATEMENT USING SATURATION TO JUSTIFY THE USE OF A SMALL PURPOSEIVE SAMPLE IN A QUALITATIVE STUDY (ITALICS ADDED FOR EMPHASIS)

Saturation, as described by Lincoln and Guba (1985), was achieved upon interviewing nine dyads, as there was no new or different information emerging; however, a total of 12 dyads were interviewed to confirm redundancy and maintain rigor.

Note that those who conduct qualitative research often have extended contact with their participants as a result of using techniques such as in-depth personal interviews or prolonged observational periods. With limited resources, their samples might necessarily be small. On the other hand, quantitative researchers often have more limited contact due to using techniques such as written tests or questionnaires, which can be administered to many participants at little cost. As a result, consumers of research usually should expect quantitative researchers to use larger samples than qualitative researchers.

4. Is the Sample Size Adequate in Terms of its Orientation (Quantitative Versus Qualitative)?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Traditionally, qualitative researchers use smaller samples than quantitative researchers. For instance, using fewer than 15 participants is quite common and is usually considered acceptable in qualitative research (for reasons, see the discussion under the previous evaluation question). Using such a small number of participants in quantitative research would usually be considered a serious flaw.⁵

5. If a Purposive Sample Has Been Used, Has the Researcher Indicated the Basis for Selecting Participants?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: When using purposive sampling, researchers should indicate the basis or criteria for the selection of the participants. Example 7.5.1 is taken from a qualitative study on gender differences in stress among professional managers. Notice that the researchers did not simply

4 Cummings, J. (2011). Sharing a traumatic event: The experience of the listener and the storyteller within the dyad. *Nursing Research*, 60(6), 386–392.

5 Quantitative researchers usually conduct significance tests. Sample size is an important determinant of significance. If the size is very small, a significance test may fail to identify a “true” difference as statistically significant.

rely on managers they happened to know to serve as participants. Instead, they selected a purposive sample of managers that met specific criteria.

Example 7.5.1⁶

A DESCRIPTION OF THE CRITERIA FOR SELECTING A PURPOSIVE SAMPLE FOR
A QUALITATIVE STUDY

Participants were selected based on purposive criterion sampling from a list, purchased by the research team, which consisted of professionals who had managerial positions in business, governmental, or nongovernmental organizations in a western Canadian city. The criteria for participation included the following: (a) individuals were responsible for making decisions that affected the direction of their business or organization on a regular basis and (b) individuals had to score 3, 4, or 5 on at least three of four questions that asked about level of stress in their work, family, personal life, and overall life situations using a 5 point scale (1 = *not stressful at all* to 5 = *extremely stressful*). The first criterion verified that each individual held a managerial position, whereas the second criterion ensured that the participant generally felt stressed in his or her life. A research assistant randomly called listings from the database to describe the purpose of the study, make sure these individuals met the criteria for being participants, explain the tasks of each participant, and find out whether they were interested in being involved in the study. Attention was also paid to ensuring that both women and men were recruited to participate.

Note that even if a researcher calls his or her sample purposive, usually it should be regarded as merely a sample of convenience unless the specific basis for selection is described.

6. If a Population Has Been Studied, Has it Been Clearly Identified and Described?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Researchers who conduct population studies often disguise the true identity of their populations (for ethical and legal reasons), especially if the results reflect negatively on the population. Nevertheless, information should be given that helps the reader visualize the population, as illustrated in Example 7.6.1. Notice that the specific city is not mentioned, which helps protect the identity of the participants. Also, note that “prospective interviewees and survey respondents from substance use treatment and child welfare” agencies constitutes the population.

6 Iwasaki, Y., MacKay, K. J., & Ristock, J. (2004). Gender-based analyses of stress among professional managers: An exploratory qualitative study. *International Journal of Stress Management*, 11(1), 56–79.

Example 7.6.1⁷**DESCRIPTION OF A POPULATION THAT WAS STUDIED**

First, a purposive sample of prospective interviewees and survey respondents from substance use treatment and child welfare were developed with key contacts at the British Columbia Center for Excellence in Women's Health and the Ministry of Children and Family Development. Prospective interviewees were identified based on the following criteria: (a) experience in working across systems in direct service, consultant, supervisory, or management roles; and (b) representation of different regions in the province. Because a majority of parents who are concurrently involved in child welfare systems are women, special efforts were made to recruit interviewees from agencies whose services include specialized treatment for women with addiction problems. Prospective interviewees were contacted by e-mail to inform them of the purpose of the study and to invite participation. Prospective interviewees who did not respond to initial contacts received follow-up e-mails and phone calls to invite their participation in the study. Out of 36 prospective interviewees identified for the study, 12 did not respond to preliminary e-mail invitations (66% response rate).

With information such as that provided in Example 7.6.1, readers can make educated judgments as to whether the results are likely to apply to other populations of social workers.

___ 7. Has Informed Consent Been Obtained?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: This evaluation question was raised in the previous chapter on evaluating samples when researchers generalize (see Evaluation Question 12 in Chapter 6). It is being raised again in this chapter because it is an important question that applies whether or not researchers are attempting to generalize.

___ 8. Has the Study Been Approved by an Ethics Review Committee?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Again, like in the previous chapter (see Evaluation, Question 13 in Chapter 6), the ethics review process is relevant for any study involving human subjects, regardless of whether or not the study involves direct interaction with the participants. And the ethical considerations of avoiding or minimizing any potential harm to the subjects are just as relevant, even if researchers are not interested in generalizing from their sample to the population.

7 Drabble, L., & Poole, N. (2011). Collaboration between addiction treatment and child welfare fields: Opportunities in a Canadian context. *Journal of Social Work Practice in the Addictions*, 11(2), 124–149.

___ 9. Overall, is the Description of the Sample Adequate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and taking into account any additional considerations and concerns you may have.

Chapter 7 Exercises

Part A

Directions: Answer the following questions.

1. Very briefly explain in your own words how theory development might impact the selection of a sample.
2. The use of *purposive* samples is a tradition in which type of research?
A. Qualitative B. Quantitative.
3. Suppose you were evaluating a pilot study on college students' voting behavior. What are some demographics that you think should be described for such a study?
4. Very briefly describe in your own words the meaning of *data saturation*. Is this concept more closely affiliated with quantitative *or* qualitative research?
5. Small samples are more common in which type of research?
A. Qualitative B. Quantitative.
6. Which evaluation questions were regarded as so important that they were posed in both Chapter 6 and this chapter?

Part B

Directions: Locate three research reports of interest to you in academic journals, in which the researchers are not directly concerned with generalizing from a sample to a population, and apply the evaluation questions in this chapter. Select the one to which you gave the highest overall rating and bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

Evaluating Measures

Immediately after describing the sample or population, researchers typically describe their measurement procedures. A *measure* is any tool or method for measuring a trait or characteristic. The description of measures in research reports is usually identified with the subheading *Measures*.¹

Often, researchers use published measures. About equally as often, researchers use measures that they devise specifically for their particular research purposes. As a general rule, researchers should provide more information about such newly developed measures than on previously published ones that have been described in detail in other publications, such as test manuals and other research reports.

While a consumer of research would need to take several sequential courses in measurement to become an expert, he or she will be able to make preliminary evaluations of researchers' measurement procedures by applying the evaluation questions discussed in this chapter.

1. Have the Actual Items and Questions (or at Least a Sample of Them) Been Provided?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ²
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Comment: Providing sample items and questions is highly desirable because they help to operationalize what was measured. Note that researchers *operationalize* when they specify the aspects and properties of the concepts on which they are reporting.

In Example 8.1.1, the researchers provide sample items for two areas measured (alcohol and drug use). Note that by being given the actual words used in the questions, consumers of research can evaluate whether the wording is appropriate and unambiguous.

- 1 As indicated in Chapter 1, *observation* is one of the ways of measurement. The term *measures* refers to the materials, scales, and tests that are used to make the observations or obtain the measurements. Participants (or Sample) and Measures are typical subheadings under the main heading Method in a research report.
- 2 Continuing with the same grading scheme as in the previous chapters, N/A stands for "Not applicable" and I/I stands for "Insufficient information to make a judgement".

Example 8.1.1³

SAMPLE INTERVIEW QUESTIONS

The poly-substance intoxication index asks youth seven questions about their alcohol and drug use (e.g., “Have you ever smoked a cigarette?” “Have you ever drunk more than just a few sips of alcohol?”), which are answered with 0 (*no*) or 1 (*yes*). The questions ask whether the youth has ever drunk alcohol; smoked cigarettes, marijuana, or hashish; sniffed glue or paint; used ecstasy; used prescription hard drugs or medication; and whether the youth has ever used Vicodin, Percocet, or Oxycontin. The index ranges from 0 to 7, with 7 indicating the use of all substances.

Example 8.1.2 also illustrates this guideline. The questions were asked in a qualitative study in which the questions were open-ended.

Example 8.1.2⁴

OPEN-ENDED QUESTIONS USED IN A QUALITATIVE STUDY

Respondents were asked, via an anonymous online survey, to provide comments about the former colleague’s strengths and weaknesses as a leader. For the comment focusing on strengths, the instructions read, “We’d like to hear your views about this person’s strengths as a colleague and as a leader. Please write a few brief thoughts below.” For the comment focusing on weaknesses, the instructions read, “Consider areas where you think this person could improve as a colleague and leader. What do you wish they would do differently . . . what do you wish they would change? Please be honest and constructive.” To minimize contrived or meaningless responses, we informed raters that the comments were optional: “These comments are important, but if nothing constructive comes to mind, click below to continue.”

Many achievement tests have items that vary in difficulty. When this is the case, including sample items that show the range of difficulty is desirable. The researchers who wrote Example 8.1.3 did this.

Example 8.1.3⁵

SAMPLE ACHIEVEMENT ITEMS THAT SHOW THEIR RANGE OF DIFFICULTY

This task [mental computation of word problems] was taken from the arithmetic subtest of the WISC-III (Wechsler, 1991). Each word problem was orally presented and was solved

- 3 Oelsner, J., Lippold, M. A., & Greenberg, M. T. (2011). Factors influencing the development of school bonding among middle school students. *Journal of Early Adolescence*, 31(3), 463–487.
- 4 Ames, D. R., & Flynn, F. J. (2007). What breaks a leader: The curvilinear relation between assertiveness and leadership. *Journal of Personality and Social Psychology*, 92(2), 307–324.
- 5 Swanson, H. L., & Beebe-Frankenberger, M. (2004). The relationship between working memory and mathematical problem solving in children at risk and not at risk for serious math difficulties. *Journal of Educational Psychology*, 96(3), 471–491.

without paper or pencil. Questions ranged from simple addition (e.g., If I cut an apple in half, how many pieces will I have?) to more complex calculations (e.g., If three children buy tickets to the show for \$6.00 each, how much change do they get back from \$20.00?).

Keep in mind that many measures are copyrighted, and their copyright holders might insist on keeping the actual items secure from public exposure. Obviously, a researcher should not be faulted for failing to provide sample questions when this is the case.

___ 2. Are any Specialized Response Formats, Settings, and/or Restrictions Described in Detail?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: It is desirable for researchers to indicate the response format (e.g., multiple-choice, responses on a scale from Strongly Agree to Strongly Disagree, and so on).

Examples of settings that should be mentioned are the place where the measures were used (such as in the participants' homes), whether other individuals were present (such as whether parents were present while their children were interviewed), whether a laptop was handed to the participants for the sensitive-topic portion of interview.

Examples of restrictions that should be mentioned are time limits and tools that participants are permitted (or are not permitted) to use, such as not allowing the use of calculators during a mathematics test.

Qualitative researchers also should provide details on these matters. This is illustrated in Example 8.2.1, in which the person who conducted the qualitative interviews is indicated as well as the length of the interviews, the languages used, and the incentive to participate.

Example 8.2.1⁶

DESCRIPTION OF DATA COLLECTION IN A QUALITATIVE STUDY

After informed consent was obtained, the first author interviewed adolescents twice and nonparental adults once. Each interview lasted 30–90 minutes and was conducted in English or Spanish, as per participants' choice. Participants were paid \$10 per interview session. Interviews were audiotaped and transcribed verbatim. Transcripts were verified against audiotapes by the research team. All names were removed from the transcripts to ensure confidentiality.

6 Sanchez, B., Reyes, O., & Singh, J. (2006). A qualitative examination of the relationships that serve a mentoring function for Mexican American older adolescents. *Cultural Diversity and Ethnic Minority Psychology, 12*(4), 615–631.

3. When Appropriate, Were Multiple Methods or Sources Used to Collect Data/Information on Key Variables?

Very unsatisfactory 1 2 3 4 5 Very satisfactory or N/A I/I

Comment: As indicated in Chapter 1, it is safe to assume that all methods of measurement (e.g., testing, interviewing, making observations) are flawed. Thus, the results of a study can be more definitive if more than one method for collecting data or more than one source of data is used for key variables.

In quantitative research, researchers emphasize the development of objective measures that meet statistical standards for reliability⁷ and validity⁸, which are discussed later in this chapter. When researchers use these highly developed measures, they often do not believe that it is important to use multiple measures. For instance, they might use a well-established multiple-choice reading comprehension test that was extensively investigated (with regard to its validity and reliability) prior to publication of the test. A quantitative researcher would be *unlikely* to supplement such highly developed measures with other measures such as teachers' ratings of students' reading comprehension.

At the same time, some quantitative researchers may use multiple sources of data on the same variable (for example, violence rates) to compensate for the weaknesses of each data source. Using the example of violence rates, according to official crime statistics based on police reports, the rates of violent crimes committed by women have not been decreasing since the early 1990s as drastically as the rates of violent crimes committed by men, even though men generally commit much more violent offenses than women (a so-called "gender gap"). Do police reports reflect the reality, i.e. the decreasing gender gap in violence, correctly?

To investigate this issue, it would be useful to compare these official crime statistics with data on violence from victimization surveys (which ask people, for example, if they have been attacked or beaten and, if so, what the characteristics of the attacker and the incident were). We could also use a third source of data broken down by gender – rates of imprisonment, which reflect only the most serious forms of violence. Example 8.3.1 provides an excerpt from such a research study that investigates whether the gender gap in violence has been decreasing in recent decades. The study uses the process of *triangulation*, or comparing several measures of the same variable from different sources.

7 *Reliability* of a measure refers to how well its results are reproduced in repeated measurements, or how consistent the results are when they are measured the same way (and the characteristic being measured has not changed). For example, if we administer the Stanford–Binet IQ test again a week later, will its results be the same if there has been no change in intellectual abilities of the children (and no training has been administered in between the two measurements)? If the answer is yes, the test is reliable.

8 *Validity* refers to whether the instrument measures what it is designed to measure. For example, if the Stanford–Binet IQ test is designed to measure innate intelligence while it actually measures a combination of innate intelligence and the quality of education received by the child, the test is not a valid measure of innate intelligence, even if the test is a reliable measure.

Example 8.3.1⁹

DESCRIPTION OF MEASURES OF VIOLENT CRIMES USING SEVERAL SOURCES OF INFORMATION, TO COMPARE MALE AND FEMALE VIOLENCE TRENDS

Data triangulation involves combining data sources and methodologies to strengthen internal and external validity¹⁰ and reliability and increase confidence in conclusions by lessening the influence of biases in any one source or violence measure. [. . .]

Arrest statistics. The FBI publishes annually the *Uniform Crime Reports* (UCR) (FBI 1979–2003). Each UCR includes aggregated *arrest counts* based on a compilation of thousands of local police precinct reports broken out by crime type and by demographic characteristics of the offender (e.g., age, sex). [. . .]

Prison admission counts. The National Corrections Reporting Program (NCRP) (U.S. Bureau of Justice Statistics 1996), an annual national census of state and federal prison admissions and releases, has collected data continuously since 1986. The information gathered from prisoner records of those entering or leaving the correctional system includes conviction offense, sentence length, and defendant characteristics like age and gender. Admissions include court commitments, parole revocations, and transfers. We use new *court commitments* to derive female and male imprisonment rates for violent offending.

Victim-based reports. The National Crime Victimization Survey (NCVS) (U.S. Bureau of Justice Statistics 1992–2005), conducted annually by the Census Bureau since 1973, gathers information from a national sample of approximately 50,000 household respondents age 12 or older about any violent crime they experienced, excepting homicide. The NCVS provides trend data on violent crimes that did not necessarily come to the attention of the police or result in a recorded arrest. For personal crimes, the survey asks about the perpetrator(s), including age and gender. We generate offending estimates based on victim identification of perpetrator sex and age [. . .].

In qualitative studies, researchers are also likely to use *triangulation of data sources*, or multiple measures of a single phenomenon, for several reasons. First, qualitative researchers strive to conduct research that is intensive and yields highly detailed results (often in the form of themes supported by verbal descriptions – as opposed to numbers). The use of multiple measures helps qualitative researchers probe more intensively from different points of view. In addition, qualitative researchers tend to view their research as *exploratory*¹¹. When conducting exploratory research, it is difficult to know which type of measure for a particular variable is likely to be

9 Schwartz, J., Steffensmeier, D. J., & Feldmeyer, B. (2009). Assessing trends in women's violence via data triangulation: Arrests, convictions, incarcerations, and victim reports. *Social Problems*, 56(3), 494–525.

10 *Internal validity* refers to how well the cause-and-effect relationship has been established in a study (usually, in an experiment), and these issues will be discussed in detail in the next chapter (Chapter 9). *External validity* is often used as another term for generalizability (of the study's findings).

11 We have discussed the types of research (descriptive, exploratory, explanatory, and explanation) in Chapter 4 (Evaluation Question #3).

most useful, and thus it would make sense to use several ways of measuring or observing the same phenomenon, if possible. Finally, qualitative researchers see the use of multiple measures as a way to check the validity of their results. In other words, if different measures of the same phenomenon yield highly consistent results, the measures (including the interpretation of the data) might be more highly regarded as being valid than if only one data source was used.

Sometimes, it is not realistic to expect researchers to use multiple measures of all key variables. Measurement of some variables is so straightforward that it would be a poor use of a researcher's time to measure them in several ways. For instance, when assessing the age of students participating in a study, most of the time it is sufficient to ask them to indicate it. If this variable is more important (for example, to ensure that nobody under the age of 18 is included), the researcher may use information about the students' birth dates collected from the Registrar's Office of the university. But in either case, it is unnecessary to use several sources of data on the participants' age (unless the study specifically focuses on a research question such as: Which personality characteristics are associated with lying about one's age?).

4. For Published Measures, Have Sources Been Cited Where Additional Information can be Obtained?

Very unsatisfactory 1 2 3 4 5 Very satisfactory or N/A I/I

Comment: Researchers should provide references to sources of additional information on the published measures used in their research.

Some measures are published, or previously reproduced in full, in journal articles. Such articles typically describe the development and statistical properties of the measures. Other measures are published by commercial publishers as separate publications (e.g., test booklets) that usually have accompanying manuals that describe technical information on the measures.

In Example 8.4.1, the researchers briefly describe the nature of one of the measures they used, following it with a statement that the validity and reliability of the measure have been established. It is important to note that they provide a reference (shown in italics) where more information on the measure's reliability and validity may be found.

Example 8.4.1¹²

BRIEF DESCRIPTION OF A MEASURE IN WHICH A REFERENCE FOR MORE INFORMATION ON RELIABILITY AND VALIDITY IS PROVIDED (ITALICS ADDED FOR EMPHASIS)

Motivations for drinking alcohol were assessed using the 20-item Drinking Motives Questionnaire (DMQ-R; Cooper, 1994), encompassing the 4 subscales of Coping ($\alpha = .87$), Conformity ($\alpha = .79$), Enhancement ($\alpha = .92$), and Social Motives ($\alpha = .94$). *The DMQ-R*

12 LaBrie, J. W., Kenney, S. R., Migliuri, S., & Lac, A. (2011). Sexual experience and risky alcohol consumption among incoming first-year college females. *Journal of Child & Adolescent Substance Abuse*, 20(1), 15–33.

has proven to be the most rigorously tested and validated measurement of drinking motives (Maclean & Lecci, 2000; Stewart, Loughlin, & Rhyno, 2001). Respondents were prompted with, “Thinking of the time you drank in the past 30 days, how often would you say that you drank for the following reasons?” Participants rated each reason (e.g., “because it makes social gatherings more fun” and “to fit in”) on a 1 (*almost never/never*) to 5 (*almost always/always*).

In Example 8.4.2, the researchers also briefly describe the nature of one of the measures they used, following it with a statement that describes its technical and statistical properties, including reliability and validity.

Example 8.4.2¹³

BRIEF DESCRIPTION OF A MEASURE IN WHICH A REFERENCE FOR MORE INFORMATION IS PROVIDED (ITALICS ADDED FOR EMPHASIS)

Youths completed the RSE (Rosenberg, 1979), a 10-item scale assessing the degree to which respondents are satisfied with their lives and feel good about themselves. Children respond on a 4-point scale, ranging from 1 (*strongly agree*) to 4 (*strongly disagree*); higher scores indicate more positive self-esteem. *Studies across a wide range of ages yield adequate internal consistency (α between .77 to .88), temporal stability (test-retest correlations between .82 and .88), and construct validity (i.e., moderate correlations with other measures of self-concept and depression symptoms) (Blascovich & Tomeka, 1993).*

If a study does not include previously published measures, the most fitting answer to this evaluation question would be N/A (not applicable).

5. When Delving into Sensitive Matters, is There Reason to Believe that Accurate Data Were Obtained?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Some issues are sensitive because they deal with illegal matters such as illicit substance use, gang violence, and so on. Others are sensitive because of societal taboos such as those regarding certain forms of sexual behavior. Still others may be sensitive because of idiosyncratic personal views on privacy. For instance, sexual orientation and income are sensitive issues for many individuals. Participants often decline to answer these questions or may not answer honestly. Thus, self-reports by participants may sometimes lack validity. The authors of Example 8.5.1 discuss the limitations of self-reports and how they might have affected the results of their research.

13 Goodman, S. H., Tully, E., Connell, A. M., Hartman, C. L., & Huh, M. (2011). Measuring children's perceptions of their mother's depression: The Children's Perceptions of Others' Depression Scale–Mother Version. *Journal of Family Psychology*, 25(2), 163–173.

Example 8.5.1¹⁴

DISCUSSION OF LIMITATIONS OF SELF-REPORTS IN RELATION TO A PARTICULAR STUDY

Our data are based exclusively on self-reported assessments of psychological distress, and, thus, our ability to draw conclusions is limited by the validity and reliability of this methodology. In general, self-report data are subject to threats to validity such as social desirability and response-style biases.¹⁵ Thus, as suggested above, it may be that the veterans in the treatment groups were hesitant to acknowledge much change in the status of their distress as they may fear that to do so would impact their service connection or their identity associated with being a traumatized veteran.

A common technique for encouraging honest answers to sensitive questions is to collect the responses *anonymously*. For instance, participants may be asked to mail in questionnaires with the assurance that they are not coded in any way that would reveal their identity. In group settings, participants who respond in writing may also be assured that their responses are anonymous. However, if a group is small, such as a class of 20 students, some participants might be hesitant to be perfectly honest regarding highly sensitive matters because a small group does not provide much “cover” for hiding the identity of a participant who engages in illegal or taboo behaviors.

With techniques such as interviewing or direct physical observation, or in longitudinal studies where researchers need to connect each person’s responses among several waves of data collection, it is usually not possible to provide anonymity. The most a researcher might be able to do is assure *confidentiality*. Such an assurance is likely to work best if the participants already know and trust the interviewer (such as a school counselor) or if the researcher has spent enough time with the participants to develop rapport and trust. The latter is more likely to occur in qualitative research than quantitative research because qualitative researchers often spend substantial amounts of time interacting with their participants in an effort to bond with them.

Another technique of increasing the likelihood of honest answers about sensitive matters in a questionnaire, for example, when measuring the involvement in illegal activities like shoplifting or drug use, is to include questions asking how often the person thinks his or her peers do “so and so” before asking the respondent how often he or she does “so and so”.¹⁶

14 Bolton, E. E. et al. (2004). Evaluating a cognitive-behavioral group treatment program for veterans with posttraumatic stress disorder. *Psychological Services*, 1(2), 140–146.

15 *Social desirability* refers to the tendency of some respondents to provide answers that are considered socially desirable, i.e. making the respondent look good. *Response-style bias* refers to the tendency of some participants to respond in certain ways (such as tending to select the middle category on a scale) regardless of the content of the question.

16 In fact, research shows that when a person is asked about the illegal activities of his or her peers (especially the type of activities about which direct knowledge is limited), the respondent often projects his own behavior in assigning it to his peers. For more, see Haynie, D. L., & Osgood, D. W. (2005). Reconsidering peers and delinquency: How do peers matter? *Social Forces*, 84(2), 1109–1130.

___ 6. Have Steps Been Taken to Keep the Measures from Influencing Any Overt Behaviors that Were Observed?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: If participants know they are being directly observed, they may temporarily change their behavior.¹⁷ Clearly, this is likely to happen in the study of highly sensitive behaviors, but it can also affect data collection on other matters. For instance, some students may show their best behavior if they come to class to find a newly installed video camera scanning the classroom (to gather research data). Other students may show off by acting up in the presence of the camera.

One solution would be to make surreptitious observations, such as with a hidden video camera or a one-way mirror. In most circumstances, such techniques raise serious ethical and legal problems.

Another solution is to make the observational procedures a routine part of the research setting. For instance, if it is routine for a classroom to be visited frequently by outsiders (e.g., parents, school staff, and university observers), the presence of a researcher may be unlikely to obtrude on the behavior of the students.

___ 7. If the Collection and Coding of Observations Involves Subjectivity, is There Evidence of Inter-rater (or Inter-observer) Reliability?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Suppose a researcher observes groups of adolescent females interacting in various public settings, such as shopping malls, in order to collect data on aggressive behavior. Identifying some aggressive behaviors may require considerable subjectivity. If an adolescent puffs out her chest, is this a threatening behavior or merely a manifestation of a big sigh of relief? Is a scowl a sign of aggression or merely an expression of unhappiness? Answering such questions sometimes requires considerable subjectivity.

An important technique for addressing this issue is to have two or more independent observers make observations of the same participants at the same time. If the *rate of agreement* on the identification and classification of the behavior is reasonably high (say, 80% or more), a consumer of research will be assured that the resulting data are not idiosyncratic to one particular observer and his or her powers of observation and possible biases.

In Example 8.7.1, the researchers reported rates of agreement of 90% and 96%. Note that to achieve such high rates of agreement, the researchers first trained the raters by instructing them to rate the behavior with groups that were not part of the main study.

17 This is referred to as the *Hawthorne effect*. For more information, check the online resources for this chapter.

Example 8.7.1¹⁸

DISCUSSION OF OBSERVER TRAINING AND INTER-OBSERVER RELIABILITY

Two independent raters first practiced the categorization of self-disclosure on five group sessions that were not part of this study and discussed each category until full agreement was reached. Next, each rater identified the “predominant behavior” (Hill & O’Brien, 1999) – that is, the speech turn that contained the disclosure – on which they reached agreement on 90%. Finally, each rater classified the participants into the three levels of self-disclosure. Interrater agreement was high (96%).

The rate of agreement often is referred to as *inter-rater reliability*, or *inter-observer reliability*. When the observations are reduced to scores for each participant (such as a total score for nonverbal aggressiveness), the scores based on two independent raters’ observations can be expressed as an *inter-rater reliability coefficient*. In reliability studies, these can range from 0.00 to 1.00, with coefficients of about 0.70 or higher indicating adequate inter-observer reliability.¹⁹

8. If a Measure is Designed to Measure a Single Unitary Trait, Does it Have Adequate Internal Consistency?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: A test of computational skills in mathematics at the primary grade levels measures a relatively homogeneous trait. However, a mathematics battery that measures verbal problem solving and mathematical reasoning in addition to computational skills measures a more heterogeneous trait. Likewise, a self-report measure of depression measures a much more homogeneous trait than does a measure of overall mental health.

For measures designed to measure homogeneous traits, it is important to ask whether they are *internally consistent* (i.e., to what extent do the items or questions within the measure yield results that are consistent with each other?). While it is beyond the scope of this book to explain how and why it works, a statistic known as Cronbach’s alpha (whose symbol is α) provides a statistical measure of internal consistency.²⁰ As a special type of correlation coefficient, it ranges from 0.00 to 1.00, with values of about 0.70 or above indicating adequate internal consistency and values above 0.90 indicating excellence on this characteristic.

18 Shechtman, Z., & Rybko, J. (2004). Attachment style and observed initial self-disclosure as explanatory variables of group functioning. *Group Dynamics: Theory, Research, and Practice*, 8(3), 207–220.

19 Mathematically, these coefficients are the same as *correlation coefficients*, which are covered in all standard introductory statistics courses. Correlation coefficients can range from –1.00 to 1.00, with a value of 0.00 indicating no relationship. In practice, however, negatives are not found in reliability studies. Values near 1.00 indicate a high rate of agreement.

20 *Split-half reliability* also measures internal consistency, but Cronbach’s alpha is widely considered a superior measure. Hence, split-half reliability is seldom reported.

Values below 0.70 suggest that more than one trait is being measured by the measure, which is undesirable when a researcher wants to measure only one homogeneous trait.

In Example 8.8.1, the value of Cronbach's alpha is above the cutoff point of 0.70.

Example 8.8.1²¹

STATEMENT REGARDING INTERNAL CONSISTENCY USING CRONBACH'S ALPHA

We employed the widely used Grasmick et al. (1993) scale to measure self-control attitudinally. Respondents answered 24 questions addressing the six characteristics of self-control (i.e. impulsive, risk seeking, physical, present oriented, self-centered, and simple minded). Response categories were adjusted so that higher values represent higher levels of self-control. The items were averaged and then standardized. Consistent with the behavioral measure of self-control, sample respondents reported a slightly higher than average level of attitudinal self-control (3.3 on the unstandardized scale ranging from 1.3 to 4.6). The scale exhibits good internal reliability ($\alpha = .82$).

Internal consistency (sometimes also called *internal reliability*) usually is regarded as an issue only when a measure is designed to measure a single homogeneous trait, *and* yields numerical scores (as opposed to qualitative measures used to identify patterns that are described in words).

If a measure does not meet these two criteria, "not applicable" is an appropriate answer to this evaluation question.

___ 9. For Stable Traits, is There Evidence of Temporal Stability?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Suppose a researcher wants to measure aptitude (i.e., potential) for learning algebra. Such an aptitude is widely regarded as being stable. In other words, it is unlikely to fluctuate much from one week to another. Hence, a test of such an aptitude should yield results that are stable across at least short periods of time. For instance, if a student's score on such a test administered this week indicates that he or she has very little aptitude for learning algebra, this test should yield a similar assessment if administered to the same student next week.

Likewise, in the area of personality measurement, most measures also should yield results that have temporal stability (i.e., are stable over time). For instance, a researcher would expect that a student who scored very highly on a measure of self-control one week would also score very highly the following week because self-control is unlikely to fluctuate much over short periods of time.

The most straightforward approach to assessing temporal stability (e.g., stability of the measurements over time) is to administer a measure to a group of participants twice at different

21 Zimmerman, G. M., Botchkovar, E. V., Antonaccio, O., & Hughes, L. A. (2015). Low self-control in "bad" neighborhoods: Assessing the role of context on the relationship between self-control and crime. *Justice Quarterly*, 32(1), 56–84.

points in time, typically with a couple of weeks between administrations. The two sets of scores can be correlated, and if a coefficient (whose symbol is r) of about 0.70 or more (on a scale from 0.00 to 1.00) is obtained, there is evidence of temporal stability. This type of reliability is commonly known as *test–retest reliability*. It is usually examined only for tests or scales that yield scores (as opposed to open-ended interviews, which yield meanings and ideas derived from responses).

In Example 8.9.1, researchers describe how they established the test–retest reliability of a measure. Note that they report values above the suggested cutoff point of 0.70 for middle-aged adults and the less optimal range of r values for older adults. The authors also use the symbol r when discussing their results.

Example 8.9.1²²

STATEMENT REGARDING TEMPORAL STABILITY (TEST-RETEST RELIABILITY) ESTABLISHED BY THE RESEARCHERS

To conduct another survey for test–retest reliability purposes, the company again emailed those who participated in survey 1 with an invitation to and link for the web survey two weeks after the Survey 1 (Survey 2). All told, 794 participants responded to the second round of the survey (re-response proportion: 90.0%). [. . .]

The correlation coefficients between TIPI-J [Ten-Item Personality Inventory, Japanese version] scores at the two time points were 0.74–0.84 (middle-aged individuals) and 0.67–0.79 (older individuals). [. . .]

These results are consistent with previous studies: Oshio et al. (2012) reported test–retest reliability of the TIPI-J among undergraduates as ranging from $r = 0.64$ (Conscientiousness) to $r = 0.86$ (Extraversion), and Gosling et al. (2003) reported values ranging from 0.62 to 0.77. As a whole, these findings indicate the almost acceptable reliability of the TIPI-J.

In Example 8.9.2, the researchers report on the range of test–retest reliability coefficients for the Perceived Racism Scale that were reported earlier by other researchers (i.e., McNeilly et al., 1996). All of them were above the suggested 0.70 cutoff point for acceptability.

Example 8.9.2²³

STATEMENT REGARDING TEMPORAL STABILITY (TEST-RETEST RELIABILITY)

The PRS [Perceived Racism Scale] is a 32-item instrument that measures emotional reactions to racism [in four domains]. [. . .] McNeilly et al. (1996) reported . . . test-retest reliability coefficients ranging from .71 to .80 for the four domains.

22 Iwasa, H., & Yoshida, Y. (2018). Psychometric evaluation of the Japanese version of Ten Item Personality Inventory (TIPI-J) among middle-aged and elderly adults: Concurrent validity, internal consistency and test-retest reliability. *Cogent Psychology*, 5(1), 1–10.

23 Liang, C. T. H., Li, L. C., & Kim, B. S. K. (2004). The Asian American Racism-Related Stress Inventory: Development, factor analysis, reliability, and validity. *Journal of Counseling Psychology*, 51(1), 103–114.

10. When Appropriate, is There Evidence of Content Validity?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: An important issue in the evaluation of achievement tests is the extent to which the contents of the tests (i.e., the stimulus materials and skills) are suitable in light of the research purpose. For instance, if a researcher has used an achievement test to study the extent to which the second graders in a school district have achieved the skills expected of them at this grade level, a consumer of the research will want to know whether the contents of the test are aligned with (or match) the contents of the second-grade curriculum.

While *content validity* is most closely associated with measurement of achievement, it also is sometimes used as a construct for evaluating other types of measures. For instance, in Example 8.10.1, the researchers had the contents of the measure of depression evaluated by experts.

Example 8.10.1²⁴

A MEASURE SUBJECTED TO CONTENT VALIDATION BY EXPERTS

To test content validity, the C-PDSS [Chinese Version of the Postpartum Depression Screening Scale] was submitted to a panel consisting of six experts from different fields, including a psychology professor, a clinician from a psychiatric clinic, a senior nurse in psychiatric and mental health nursing, a university professor in obstetric nursing, and two obstetricians from two regional public hospitals. The rating of each item was based on two criteria: (a) the applicability of the content (applicability of expression and content to the local culture and the research object) and (b) the clarity of phrasing.

11. When Appropriate, is There Evidence of Empirical Validity?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Empirical validity refers to validity established by collecting data using a measure in order to determine the extent to which the data make sense.²⁵ For instance, a depression scale might be empirically validated by being administered to an institutionalized, clinically depressed group of adult patients as well as to a random sample of adult patients visiting family physicians for annual checkups. A researcher would expect the scores of the two groups to differ substantially in a predicted direction (i.e., the institutionalized sample should have higher depression scores). If they do not, the validity of the scale would be questionable.

Sometimes, the empirical validity of a test is expressed with a correlation coefficient. For instance, a test maker might correlate scores on the College Board's SATs with freshman grades

24 Li, L., Liu, F., Zhang, H., Wang, L., & Chen, X. (2011). Chinese version of the Postpartum Depression Screening Scale: Translation and validation. *Nursing Research*, 60(4), 231–239.

25 In contrast, *face validity* is a subjective assessment of whether the measure seems like it measures what it is supposed to measure, based on one's understanding of the underlying concept and logic.

in college. A correlation of 0.40 or more might be interpreted as indicating that the test has validity as a modest predictor of college grades.

Empirical validity comes in many forms, and a full exploration of it is beyond the scope of this book. Some key terms that suggest that empirical validity has been explored are *predictive validity*, *concurrent validity*, *criterion-related validity*, *convergent validity*, *discriminant validity*, *construct validity*, and *factor analysis*.

When researchers describe empirical validity, they usually briefly summarize the information, and these summaries are typically fairly comprehensible to individuals with limited training in tests and measurements.

In Example 8.11.1, the researchers briefly describe the empirical validity of a measure they used in their research. Notice that sources where additional information may be obtained are cited.

Example 8.11.1²⁶

STATEMENT REGARDING EMPIRICAL VALIDITY OF A MEASURE WITH A REFERENCE TO SOURCES WHERE MORE INFORMATION MAY BE OBTAINED

Supporting the convergent validity of the measure, PGIS [Personal Growth Initiative Scale] scores correlated positively with assertiveness, internal locus of control, and instrumentality among both European American (Robitschek, 1998) and Mexican American college students (Robitschek, 2003).

Often, information on validity is exceptionally brief. For instance, in Example 8.11.2, the researchers refer to the validity of a questionnaire as “excellent.” The source that is cited (McDowell & Newell, 1996) would need to be consulted to determine whether this refers to empirical validity.

Example 8.11.2²⁷

STATEMENT REGARDING EMPIRICAL VALIDITY OF A MEASURE WITH A REFERENCE TO WHERE MORE INFORMATION MAY BE OBTAINED

We assessed general psychological distress using the 12-item version of the General Health Questionnaire (GHQ-12; Goldberg & Huxley, 1992; McDowell & Newell, 1996). This scale, based on a 4-point Likert scale, was designed to be a broad screening instrument for psychological problems in a general population and has excellent validity and reliability (McDowell & Newell, 1996).

Note that it is traditional for researchers to address empirical validity only for measures that yield scores, as opposed to measures such as semi-structured, open-ended interviews.

26 Hardin, E. E., Weigold, I. K., Robitschek, C., & Nixon, A. E. (2007). Self-discrepancy and distress: The role of a personal growth initiative. *Journal of Counseling Psychology*, 54(1), 86–92.

27 Adams, R. E., Boscarino, J. A., & Figley, C. R. (2006). Compassion fatigue and psychological distress among social workers: A validation study. *American Journal of Orthopsychiatry*, 76(1), 103–108.

Comment: By discussing limitations of their measures, researchers help consumers of research to understand the extent to which the data presented in the results can be trusted. In Example 8.5.1 (see Evaluation Question 5 above), the researchers discussed how the limitations of using self-reports might have affected the outcomes of their study. In Example 8.12.1 that follows, the researchers discuss other possible limitations.

STATEMENT ACKNOWLEDGING A WEAKNESS IN MEASURES

With regard to measurement, it should be noted that the history of victimization measure was limited by a one-year historical time frame. This time frame might have excluded youths who were still experiencing the traumatic effects of victimizing events that occurred over a year before their completion of the survey. The victimization measure was also limited in that it did not include a measure of sexual victimization for male youths.

If, in your judgment, there are no obvious limitations to the measures described in a research report, a rating of N/A (not applicable) should be made for this evaluation question.

13. Overall, are the Measures Adequate?

Comment: The amount of information about measures used in research that is reported in academic journals is often quite limited. The provision of references for obtaining additional information helps to overcome this problem.

Typically, if a researcher provides too little information for a consumer of research to make an informed judgment about the measures used in the study and/or does not provide references where additional information can be obtained, the consumer should give it a low rating on this evaluation question or respond that there is insufficient information (I/I).

Even if enough information or additional references about the measures are provided, rate this evaluation question, taking into account your answers to the previous questions in this chapter, as well as any additional considerations you may have about the measures.

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Chapter 8 Exercises

Part A

Directions: Answer the following questions.

1. Name two or three issues that some participants might regard as sensitive and, hence, are difficult to measure. Answer this question with examples that are *not* mentioned in this chapter. (See the discussion of Evaluation Question 5.)
2. Have you ever changed your behavior because you knew (or thought) you were being observed? If yes, briefly describe how or why you were being observed and what behavior(s) you changed. (See Evaluation Question 6 and online resources for this chapter.)
3. According to this chapter, what is a reasonably high rate of agreement when two or more independent observers classify behavior (i.e., of inter-rater reliability)?
4. For which of the following would it be more important to consider internal consistency using Cronbach's alpha? Explain your answer.
 - A. For a single test of mathematics ability for first graders that yields a single score.
 - B. For a single test of reading and mathematics abilities for first graders that yields a single score.
5. Suppose a researcher obtained a test-retest reliability coefficient of 0.86. According to this chapter, does this indicate adequate temporal stability? Explain.
6. Which type of validity is mentioned in this chapter as being an especially important issue in the evaluation of achievement tests?

Part B

Directions: Locate two research reports of interest to you in academic journals. Evaluate the descriptions of the measures in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Select the one to which you gave the highest overall rating, and bring it to class for discussion. Be prepared to discuss both its strengths and weaknesses.

Evaluating Experimental Procedures

An experiment is a study in which treatments are given in order to determine their effects. For instance, one group of students might be trained to use conflict-resolution techniques (the experimental group) while a control group is not given any training. Then, the students in both groups could be observed on the playground to determine whether the experimental group uses more conflict-resolution techniques than the control group.

The treatments (i.e., training versus no training) constitute what are known as the *independent variables*, which are sometimes called the *stimuli* or *input variables*. The resulting behavior on the playground constitutes the *dependent variable*, which is sometimes called the *output* (or *outcome*) or *response variable*.

Any study in which even a single treatment is given to just a single participant is an experiment as long as the purpose of the study is to determine the effects of the treatment on another variable (some sort of outcome). A study that does not meet this minimal condition is *not* an experiment. Thus, for instance, a political poll in which questions are asked but no treatments are given is *not* an experiment and should *not* be referred to as such.

The following evaluation questions cover basic guidelines for the evaluation of experiments.

____ **1. If Two or More Groups Were Compared, Were the Participants Assigned at Random to the Groups?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: Assigning participants at random to groups guarantees that there is no bias in the assignment, so the groups are comparable (similar on average). For instance, random assignment to two groups in the experiment on conflict-resolution training (mentioned previously) would provide assurance that there is no bias, such as systematically assigning the less-aggressive

1 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”

children to the experimental group. Random assignment is a key feature of a *true experiment*, also called a *randomized controlled trial*.

Note that it is *not* safe to assume the assignment was random unless a researcher explicitly states that it was.² Example 9.1.1 illustrates how this was stated in reports on three different experiments.

Example 9.1.1

EXCERPTS FROM THREE EXPERIMENTS WITH RANDOM ASSIGNMENT EXPLICITLY MENTIONED

Experiment 1: Participants aged 14 to 18 years were randomly assigned to each flicker-type condition.³

Experiment 2: Using an experimental design, 150 college football players were randomly assigned to four conditions in which they read different vignettes about Jack, a football player who cries after a football game.⁴

Experiment 3: Socially anxious students were randomly assigned to either the experimental group, which received video and social feedback ($n = 12$), or the control group ($n = 13$).⁵

Note that assigning *individuals* to treatments at random is vastly superior to assigning previously existing *groups* to treatments at random. For instance, in educational research, it is not uncommon to assign one class to an experimental treatment and have another class serve as the control group. Because students are not ordinarily randomly assigned to classes, there may be systematic differences between the students in the two classes. For instance, one class might have more highly motivated students, another might have more parental involvement, and so on. Thus, a consumer of research should *not* answer “yes” to this evaluation question unless *individuals* (or a large number of *aggregate units*) were assigned at random.

What do we mean by *aggregate units*? This term can refer to police beats, neighborhoods, or courthouses. Some (rare) examples of studies where a large number of such aggregate units is randomly assigned to treatment and control conditions can be found in criminal justice research. Example 9.1.2 describes the random assignment of so-called crime ‘hot spots’ (areas with high frequency of violent crime) to different police intervention strategies.

- 2 Since true experiments (the ones with random assignment) are the strongest research design to establish a cause-and-effect relationship, researchers would never fail to mention this crucial feature of their study.
- 3 Huang, K.-C., Lin, R.-T., & Wu, C.-F. (2011). Effects of flicker rate, complexity, and color combinations of Chinese characters and backgrounds on visual search performance with varying flicker types. *Perceptual and Motor Skills*, 113(1), 201–214.
- 4 Wong, Y. J., Steinfeldt, J. A., LaFollette, J. R., & Tsao, S.-C. (2011). Men’s tears: Football players’ evaluations of crying behavior. *Psychology of Men & Masculinity*, 12(4), 297–310.
- 5 Kanai, Y., Sasagawa, S., Chen, J., & Sakano, Y. (2011). The effects of video and nonnegative social feedback on distorted appraisals of bodily sensations and social anxiety. *Psychological Reports*, 109(2), 411–427.

Example 9.1.2⁶

AN EXAMPLE OF A TRUE EXPERIMENT WHERE A LARGE NUMBER OF AGGREGATE UNITS IS RANDOMLY ASSIGNED TO THE EXPERIMENTAL AND CONTROL GROUPS

Jacksonville is the largest city in Florida. [. . .] Like many large cities, Jacksonville has a violent crime problem. The number of violent crimes in Jacksonville has gone up from 2003 to 2008. [. . .] For this project, . . . JSO [Jacksonville Sheriff's Office] experimented with a more geographically focused approach to violence reduction that involved concentrating patrol and problem-solving efforts on well-defined "micro" hot spots of violence.

As discussed below, we took 83 violent hot spots and randomly assigned them to one of three conditions: 40 control hot spots, 21 saturation/directed patrol hot spots (we use this hybrid term to capture the fact that officers were directed to specific hot spots and that their extended presence at these small locations, which typically lasted for several hours at a time, amounted to a saturation of the areas), or 22 problem-oriented policing (POP) hot spots. Each of these three conditions was maintained for a 90-day period. [. . .] Yet while the intervention period was short, the intensity of the intervention was high, particularly in the POP areas. As described below, POP officers conducted problem-solving activities full-time, 7 days a week and were able to complete many POP responses at each location. Further, our analysis examines changes in crime during the 90 days following the intervention to allow for the possibilities that the effects of POP would take more than 90 days to materialize and/or that the effects of either or both interventions would decay quickly.

Again, if the answer to this evaluation question is "yes," the experiment being evaluated is known as a *true experiment*. Note that this term does not imply that the experiment is perfect in all respects. Instead, it indicates only that participants were assigned at random to comparison groups to make the groups approximately similar. There are other important features that should be considered, including the size of the groups, which is discussed next.

___ 2. If Two or More Groups Were Compared, Were There Enough Participants (or Aggregate Units) per Group?

Very	1	2	3	4	5	Very		
unsatisfactory						satisfactory	or N/A	I/I

Comment: Remember, in Chapter 6, we mentioned an important rule of thumb for studies that involve comparisons among subgroups or generalizations from a sample to a population: each group or subgroup should have at least 30 individuals or aggregate units (if the groups are fairly homogenous). The same rule applies to groups compared in an experiment.

6 Taylor, B., Koper, C. S., & Woods, D. J. (2011). A randomized controlled trial of different policing strategies at hot spots of violent crime. *Journal of Experimental Criminology*, 7(2), 149–181.

For example, if an antidepressant drug treatment was administered to one group of depressed patients, and exercise therapy – to another group of similar patients (ideally, the patients were randomly assigned to these two groups), researchers would want to see which group had a reduced incidence of depression following the treatments. Moreover, they would want to be able to make meaningful inferences about whether the difference in outcomes between the groups is statistically significant (and not just a fluke). For that, each group would ideally need to have 30 or more patients.

If we apply this rule to aggregate units in the experiment described in Example 9.1.2 above, we would need 30 or more hot spots for each of the three conditions: saturation/direct patrol, problem-oriented policing, and control group. Unfortunately, the experiment had 21, 22, and 40 hot spots, correspondingly, so it would not be able to get the highest rating on this evaluation question but it comes fairly close to the required 30+ units per group. Obviously, it is much harder to get 30+ aggregate units compared to many experiments where 30+ individuals per group would be needed.

This rule about group size applies both to true experiments (with random assignment to groups) and to *quasi-experiments*, or experiments with no random assignment to groups.

Besides random assignment and group size, some other important considerations for experiments are mentioned in Evaluation Questions 5 through 15 below (with Questions 3 and 4 referring specifically to quasi-experiments).

3. If Two or More Comparison Groups Were Not Formed at Random, is there Evidence that They Were Initially Equal in Important Ways?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Suppose a researcher wants to study the impact of a new third-grade reading program being used with all third-graders in a school (the experimental group). For a control group, the researcher will have to use third-graders in another school. The use of two intact groups (groups that were already formed), with both a pretest and a post-test to ensure that the groups are similar enough, is known as a *quasi-experiment*⁷ – as opposed to a *true experiment*. Because students are not randomly assigned to schools, this experiment will get low marks on Evaluation Question 1. However, if the researcher selects a control school in which the first-graders have standardized reading test scores similar to those in the experimental school and are similar in other important respects such as parents' socioeconomic status, the experiment may yield useful experimental evidence.

Note, however, that similarity between groups is not as satisfactory as assigning participants at random to groups. For instance, the children in the two schools in the example being considered may be different in some important respect that the researcher has overlooked or

7 There are other types of quasi-experiments, besides the non-equivalent group design (NEGD). Some of the most popular among them are ex post facto designs, before-and-after and time-series designs, and a recently popular statistical approach of propensity score matching.

on which the researcher has no information. Perhaps the children's teachers in the experimental school are more experienced. Their experience in teaching, rather than the new reading program, might be the cause of any differences in reading achievement between the two groups.

When using two intact groups (such as classrooms), it is important to give both a pre-test and a post-test to measure the dependent variable before and after the treatment. For instance, to evaluate the reading program, a researcher should give a pretest in reading in order to establish the baseline reading scores and to check whether the two intact groups are initially similar on the dependent variable. Of course, if the two groups are highly dissimilar, the results of the experiment will be difficult to interpret.⁸

Notice that some pre-existing groups could have been formed at random: for example, if court cases get assigned to different judges at random, then the groups of cases ruled on by each judge can be expected to be approximately equal on average. That is, even if there is a lot of variation among such cases, each judge is supposed to get a group with a similar range of variations (if there is a sufficiently large number of cases in each group). Then researchers could wait a few years and compare the groups to examine whether offenders are more likely to commit new crimes when their cases had been decided by more punitive judges or by more lenient ones.⁹ Thus, even though it was not the researchers who formed the groups using random assignment, this example represents a true experiment.

___ 4. If Only a Single Participant or a Single Group is Used, Have the Treatments been Alternated?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Not all quasi-experiments involve the comparison of two or more groups. Consider, for instance, a teacher who wants to try using increased praise for appropriate behaviors in the classroom to see whether it reduces behaviors such as inappropriate out-of-seat behavior (IOSB). To conduct an experiment on this, the teacher could count the number of IOSBs for a week or two before administering the increased praise. This would yield what are called the *baseline data*. Suppose the teacher then introduces the extra praise and finds a decrease in the IOSBs. This might suggest that the extra praise *caused* the improvement.¹⁰ However, such a conclusion

- 8 If the groups are initially dissimilar, a researcher should consider locating another group that is more similar to serve as the control. If this is not possible, a statistical technique known as analysis of covariance can be used to adjust the post-test scores in light of the initial differences in pretest scores. Such a statistical adjustment can be risky if the assumptions underlying the test have been violated, a topic beyond the scope of this book.
- 9 In fact, the study that inspired this example has found that there is no statistically significant difference among the groups, even though there is a tendency of offenders to recidivate more if their cases happen to be assigned to more punitive judges: Green, D. P., & Winik, D. (2010). Using random judge assignments to estimate the effects of incarceration and probation on recidivism among drug offenders. *Criminology*, 48(2), 357–387.
- 10 If the teacher stopped the experiment at that point, it would represent what is called a *before-and-after design* (one of the simplest quasi-experimental designs).

would be highly tenuous because children’s environments are constantly changing in many ways, and some other environmental influence (such as the school principal scolding the students on the playground without the teacher’s knowledge) might be the real cause of the change. A more definitive test would be for the teacher to reverse the treatment and go back to giving less praise, then revert to the higher-praise condition again. If the data form the expected pattern, the teacher would have reasonable evidence that increased praise reduces IOSB.

Notice that in the example being considered, the single group serves as the control group during the baseline, serves as the experimental group when the extra praise is initially given, serves as the control group again when the condition is reversed, and finally serves as the experimental group again when the extra praise is reintroduced. Such a design has this strength: The same children with the same backgrounds are both the experimental and control groups. (In a two-group experiment, the children in one group may be different from the children in the other group in some important way that affects the outcome of the experiment.) The major drawback of a single-group design is that the same children are being exposed to multiple treatments, which may lead to unnatural reactions. How does a child feel when some weeks he or she gets extra praise for appropriate behaviors but other weeks does not? Such reactions might confound the results of the experiment.¹¹

If two preexisting classes were available for the type of experiment being considered, a teacher could use what is called a *multiple baseline design*, in which the initial extra-praise condition is started on a different week for each group. If the pattern of decreased IOSB under the extra-praise condition holds up across both groups, the causal conclusion would be even stronger than when only one group was used.

The type of experimentation being discussed under this evaluation question is often referred to as *single-subject research* or *behavior analysis*. When a researcher has only a single participant or one intact group that cannot be divided at random into two or more groups, such a design can provide useful information about causality.

___ **5. Are the Treatments Described in Sufficient Detail?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Because the purpose of an experiment is to estimate the effects of treatment on dependent variables, researchers should provide detailed descriptions of the treatment that was administered. If the treatments are complex, such as two types of therapy in clinical psychology, researchers should provide references to additional publications where detailed accounts can be found, if possible. The same rule applies to treatments that have been used in previous studies – references to the previous research should be provided.

In Example 9.5.1, the researchers begin by giving references for the experimental task and then they describe how it was used in their study. Only a portion of their detailed description of the treatment is shown in the example.

11 Researchers refer to this problem as *multiple-treatment interference*.

Example 9.5.1¹²

EXCERPT SHOWING REFERENCES FOR MORE INFORMATION ON EXPERIMENTAL TREATMENT FOLLOWED BY A DETAILED DESCRIPTION (PARTIAL DESCRIPTION SHOWN HERE)

The 6.3 min video was titled *Bullying or Not?* (available online at www.youtube.com) because it was designed to help students distinguish bullying from other forms of peer conflict. In the opening scene of the video, two student commentators (boy and girl) reviewed the definition of bullying, emphasizing the power imbalance concept. Next, three pairs of scenes illustrated the difference between bullying and ordinary peer conflict that is not bullying. In each pair, the first scene demonstrated a clear instance of bullying, and in the companion scene, the same actors enacted a similar peer conflict that was not bullying. For example, two scenes illustrated the difference between verbal bullying and a verbal argument between two peers of comparable size and status. Similarly, two scenes distinguished social bullying from an argument between friends, and two scenes distinguished physical bullying from a physical struggle between two boys of comparable size and strength. The student commentators explained the power imbalance present in each of the bullying scenes. At the end of the video, the student commentators emphasized the importance of preventing bullying and encouraged students to answer survey questions correctly when asked about bullying.

— **6. If the Treatments Were Administered by Individuals Other than the Researcher, Were Those Individuals Properly Trained and Monitored?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Researchers often rely on other individuals, such as graduate assistants, teachers, and psychologists, to administer the treatments they use in experiments. When this is the case, it is desirable for the researcher to assure consumers of research that there was proper training. Otherwise, it is possible that the treatments were modified in some unknown way. Example 9.6.1 shows a statement regarding the training of student therapists who administered treatments in an experiment. Note that such statements are typically brief.

Example 9.6.1¹³

EXCERPT ON TRAINING THOSE WHO ADMINISTERED THE TREATMENTS

Student therapists received 54 h of training in EFT–AS [emotion-focused therapy for adult survivors of child abuse]. This consisted of reviewing the treatment manual and videotapes

12 Baly, M. W., & Cornell, D. G. (2011). Effects of an educational video on the measurement of bullying by self-report. *Journal of School Violence*, 10(3), 221–238.

13 Paivio, S. C., Holowaty, K. A. M., & Hall, I. E. (2004). The influence of therapist adherence and competence on client reprocessing of child abuse memories. *Psychotherapy: Theory, Research, Practice, Training*, 41(1), 56–68.

of therapy sessions with expert therapists, as well as supervised peer skills practice and three sessions of therapy with volunteer “practice” clients.

Even if those who administered the treatments were trained, they normally should be monitored. This is especially true for long and complex treatment cycles. For instance, if psychologists will be trying out new techniques with clients over a period of several months, the psychologists should be monitored by spot-checking their efforts to determine whether they are applying the techniques they learned in their training. This can be done by directly observing them or by questioning them.

___ **7. If Each Treatment Group Had a Different Person Administering a Treatment, Did the Researcher Try to Eliminate the Personal Effect?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Suppose that the purpose of an experiment is to compare the effectiveness of three methods for teaching decoding skills in first-grade reading instruction. If each method is used by a different teacher, differences in the teachers (such as ability to build rapport with students, level of enthusiasm, ability to build effective relationships with parents) may cause any observed differences in achievement. That is, the teachers’ personal characteristics rather than their respective teaching method may have had an impact on the outcome.

One solution to this problem is to have each of the three methods used by a large number of teachers, with the teachers assigned at random to the methods. If such a large-scale study is not possible, another solution is to have each teacher use all three methods. In other words, Teacher A could use Method X, Method Y, and Method Z at different points in time with different children. The other two teachers would do likewise. When the results are averaged, the personal effect of each teacher will have contributed to the average scores earned under each of the three methods.

If this issue is not applicable to the experiment you are evaluating, give it ‘N/A’ on this evaluation question.

___ **8. If Treatments Were Self-administered, Did the Researcher Check on Treatment Compliance?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Some treatments are self-administered, out of view of the researcher. For instance, an experimental group might be given a new antidepressant drug to self-administer over a period of months. The researcher could check treatment compliance by asking participants how faithful they are being in taking the drug. More elaborate checks would include instructing participants to keep a diary of their drug-taking schedule, or even conducting tests that detect the presence of the drug.

One of the most famous experiments where the participants did not comply with the treatment assignments as designed was the iconic Minneapolis Domestic Violence Experiment. Police officers responding to a dispute involving domestic violence were instructed to follow a randomly assigned action of either making an arrest or administering one of the two non-arrest options: counseling the parties on the scene or sending the offending party away for 8 hours. In about a quarter of the cases where a non-arrest action was assigned, the officers arrested the perpetrator (for various reasons, some of which might have been largely outside of the officers' control). Example 9.9.1 discusses how this treatment non-compliance may have affected the results of this *natural*¹⁴ experiment.

Example 9.8.1¹⁵

EXCERPT ON HOW VIOLATIONS OF THE ASSIGNED TREATMENTS HAVE LIKELY AFFECTED THE GROUP COMPOSITION AND THUS THE RESULTS OF THE EXPERIMENT

Table 1 [in the original article] shows the degree to which the treatments were delivered as designed. Ninety-nine percent of the suspects targeted for arrest actually were arrested, while only 78 percent of those to receive advice did, and only 73 percent of those to be sent out of the residence for eight hours were actually sent. One explanation for this pattern, consistent with the experimental guidelines, is that mediating and sending were more difficult ways for police to control the situation, with a greater likelihood that officers might resort to arrest as a fallback position. When the assigned treatment is arrest, there is no need for a fallback position. For example, some offenders may have refused to comply with an order to leave the premises.

Such differential attrition would potentially bias estimates of the relative effectiveness of arrest by removing uncooperative and difficult offenders from the mediation and separation treatments. Any deterrent effect could be underestimated and, in the extreme, artefactual support for deviance amplification could be found. That is, the arrest group would have too many “bad guys” *relative* to the other treatments. [*Italics in the original*]

9. Except for Differences in the Treatments, Were All Other Conditions the Same in the Experimental and Control Groups?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The results of an experiment can be influenced by many variables other than the independent variable. For instance, if experimental and control groups are treated at different

14 *Natural* refers to the fact that the experiment was conducted not in a lab but *in the field*, as part of police officers' daily jobs. See Evaluation Question 11 further in this chapter for more information on experiments in natural versus artificial settings.

15 Sherman, L. W., & Berk, R. A. (1984). The specific deterrent effects of arrest for domestic assault. *American Sociological Review*, 49(2), 261–272.

times of the day or in different rooms in a building (where one room is noisy and the other is not), these factors might influence the outcome of an experiment. Researchers refer to variables such as these as *confounding variables*¹⁶ because they confound the interpretation of the results.

One especially striking illustration of such confounding comes from experiments testing the effects of surgeries for a specific health condition. For example, is surgery the best treatment for osteoarthritis of the knee?¹⁷ It turns out that if the patients are just randomly assigned to either undergo surgery or to complete a round of physical therapy, the results would be confounded by the patients' knowledge of which treatment they have received. Thus, it would be hard to say whether it is the surgery or the knowledge that one had the surgery that made him or her feel better. To remove this confounding variable, the researchers went to a pretty extreme extent of equalizing the experimental and control group conditions: patients were randomly assigned to either real or *placebo* surgeries (sometimes also called *sham surgeries*¹⁸). That is, each patient participating in the study had a surgery, they just did not know whether they got the real procedure (with cartilage removal) or a simulated one (they got the same anesthesia and a scalpel cut on their knee but the cut was then just stitched back up, with no additional surgical procedures taking place). Admittedly, this is a much more involved experiment than randomizing patients into a drug pill versus a placebo pill, but it dramatically reduces the important confounding difference by essentially equalizing the subjective experiences of participants in the experimental and control groups.

The Minneapolis Domestic Violence Experiment (MDVE) used in Example 9.8.1 above can also serve as an illustration of confounding. When we consider what led to the lower likelihood of repeat offending by those who had been arrested for domestic violence, it is possible that it was the police officers' decisions about whom to arrest rather than the actual arrests that produced the effect. In this case, the police officers' discretion is likely a confounding variable that impacted both the treatment (the independent variable: arrest or no arrest) and the outcome (the dependent variable: recidivism).

In fact, when a decision was made to replicate the MDVE in other cities, the procedures needed to be tweaked to limit the confounding influence of police officers' discretion, by making it much harder for the officers to change the assigned treatment. The necessary funding was obtained

- 16 In quasi-experimental designs, it is even harder to rule out confounders than in true experiments. For example, consider a study where a group of subjects who experienced abuse or neglect as children has been matched *ex post facto* (after the fact) with a control group of adults of the same age, gender, race, and socioeconomic status who grew up in the same neighborhoods as the group of child maltreatment survivors. Then the researchers compare the two groups in terms of outcomes like engaging in violence as adults. Let's say the study has found that the control group of adults has far fewer arrests for violence than maltreatment survivors. How can we be sure that this difference in outcomes is a result of child maltreatment experiences? It is very likely that other important variables confound the intergenerational transmission of violence found in such a hypothetical study.
- 17 Moseley, J. B., O'Malley, K., Petersen, N. J., Menke, T. J., Brody, B. A., Kuykendall, D. H., . . . & Wray, N. P. (2002). A controlled trial of arthroscopic surgery for osteoarthritis of the knee. *New England Journal of Medicine*, 347(2), 81–88.
- 18 For another example, see the following article: Frank, S., Kiebertz, K., Holloway, R., & Kim, S. Y. (2005). What is the risk of sham surgery in Parkinson disease clinical trials? A review of published reports. *Neurology*, 65(7), 1101–1103.

and, most importantly, the cooperation of law enforcement authorities in several other cities across the United States was secured, and the replications of MDVE were completed in five cities.¹⁹ When the results came in, they were confusing, to say the least: in some cities, arrests for domestic violence reduced recidivism among the arrested offenders, in other cities arrests increased recidivism, and in still others there were no differences in repeat offending between the “arrest” and “no-arrest” groups of offenders.

10. Were the Effects or Outcomes of Treatment Evaluated by Individuals Who Were *Not* Aware of the Group Assignment Status?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: If a researcher is also the individual administering the treatment (or one of the treatment providers), it is very important that the outcomes are assessed by somebody else – importantly, by a person who is *not* aware of the treatment group assignment. Even if the effects of the treatment or intervention are evaluated using fairly objective procedures and tests, the assessor’s knowledge of group assignment status can inadvertently impact the assessments and bias the results.

It is considered to be a gold standard of experimentation to use what is called a *double-blind* procedure: (1) the participants are not aware of whether they are in the treatment or control group, and (2) the individuals assessing the outcomes are not aware of the participants’ group assignment either.

For example, in the placebo surgery experiment described in the previous section, nurses would assess the changes in the patients’ knee function after the surgery using subjective measures of pain (something like: “On a scale of 1 to 10 . . .”), as well as using objective measures like the number of seconds it takes a patient to climb up and down a flight of stairs as quickly as possible. As illustrated in Example 9.10.1 below, neither the nurses assessing these outcomes nor the patients themselves were aware of what type of surgery they had received.

Example 9.10.1²⁰

ASSESSMENT OF THE RESULTS OF EXPERIMENT USING DOUBLE-BLIND PROCEDURES

Study personnel who were unaware of the treatment-group assignments performed all postoperative outcome assessments; the operating surgeon did not participate in any way. Data on end points were collected 2 weeks, 6 weeks, 3 months, 6 months, 12 months, 18 months, and 24 months after the procedure. To assess whether patients remained

¹⁹ For more information, see the online resources for this chapter.

²⁰ Moseley, J. B., O’Malley, K., Petersen, N. J., Menke, T. J., Brody, B. A., Kuykendall, D. H., . . . & Wray, N. P. (2002). A controlled trial of arthroscopic surgery for osteoarthritis of the knee. *New England Journal of Medicine*, 347(2), 81–88.

unaware of their treatment-group assignment, they were asked at each follow-up visit to guess which procedure they had undergone. Patients in the placebo group were no more likely than patients in the other two groups to guess that they had undergone a placebo procedure.

___ 11. When Appropriate, Have the Researchers Considered Possible Demand Characteristics?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: If participants have no knowledge of whether they are in the experimental or control group, such experiments are called *blind*, or *blinded* (double-blind if the assessors of outcomes also don't know the participants' group assignments). However, it is not always possible to conduct blinded experiments. If participants know (or suspect) the purpose of an experiment, this knowledge may influence their responses. For instance, in a study on the effects of a film showing negative consequences of drinking alcohol, the experimental-group participants might report more negative attitudes toward alcohol only because they suspect the researcher has hypothesized that this will happen. In other words, sometimes participants try to give researchers what they think the researchers expect. This is known as a *demand characteristic*. It is called this because the phenomenon operates as though a researcher is subtly demanding a certain outcome.

Certain types of measures are more prone to the effects of demand characteristics than others. Self-report measures (such as self-reported attitudes toward alcohol) are especially sensitive to them. When interpreting the results obtained with such measures, researchers should consider whether any differences are due to the demands of the experiment. One way to overcome this difficulty is to supplement self-report measures with other measures, such as reports by friends or significant others.

On the other hand, an achievement test is less sensitive to the demands of an experiment because students who do not have the skills being tested will not be successful on a test even if they want to please the researcher by producing the desired behavior. Likewise, many physical or biological measures are insensitive to this type of influence. In an experiment on methods for reducing cocaine use, for instance, a participant will not be able to alter the results of a blood test for the presence of cocaine.

___ 12. Is the Setting for the Experiment Natural?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Sometimes, researchers conduct experiments in artificial settings. When they do this, they limit their study's *external validity*, that is, what is found in the artificial environment of an experiment may not be found in more natural settings (i.e., the finding may not be valid outside of the laboratory where the study took place). External validity is often used synonymously with *generalizability*.

Experiments conducted in laboratory settings are likely to have poor external validity. Notice the unnatural aspects of Example 9.12.1 below. First, the amount and type of alcoholic beverages were assigned (rather than being selected by the participants as they would be in a natural setting). Second, the female was an accomplice of the experimenters (not someone the males were actually dating). Third, the setting was a laboratory, where the males would be likely to suspect that their behavior was being monitored in some way. While the researchers have achieved a high degree of physical control over the experimental setting, they have sacrificed external validity in the process.

Example 9.12.1

EXPERIMENT WITH POOR EXTERNAL VALIDITY

A research team was interested in the effects of alcohol consumption on aggressiveness in males when dating. In the experiment, some of the males were given moderate amounts of beer to consume, while controls were given nonalcoholic beer. Then all males were observed interacting with a female cohort of the experimenters. The interactions took place in a laboratory on a college campus, and observations were made through a one-way mirror.

At the same time, experiments conducted *in the field*, like the Minneapolis Domestic Violence Experiment (MDVE) discussed earlier in this chapter, present the opposite problem: it is often impossible for researchers to control all the important aspects of the experiment. MDVE was a *natural experiment*, with actual arrests for domestic violence (and thus good external validity), but the trade-off was the researchers' inability to ensure that every randomly assigned condition was actually carried out as planned.

Thus, experiments in natural settings (or *field experiments*) often present problems with *internal validity*. Internal validity of an experiment refers to whether the experiment can help clearly determine a cause-and-effect relationship, to rule out confounding variables (or alternative explanations for the results). In case of MDVE, is it the arrest that was the true cause of subsequent reductions in reoffending or is it the discretion of a police officer about whom to arrest (even when the experiment called for no arrest) that made a difference in recidivism outcomes?

13. Has the Researcher Distinguished between Random Selection and Random Assignment?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The desirability of using *random selection* to obtain samples from which researchers can generalize with confidence to larger populations was discussed in Chapter 6. Such selection is highly desirable in most studies – whether they are experiments or not. *Random assignment*, on the other hand, refers to the process of assigning participants to the various treatment conditions (i.e., to the treatments, including any control condition).

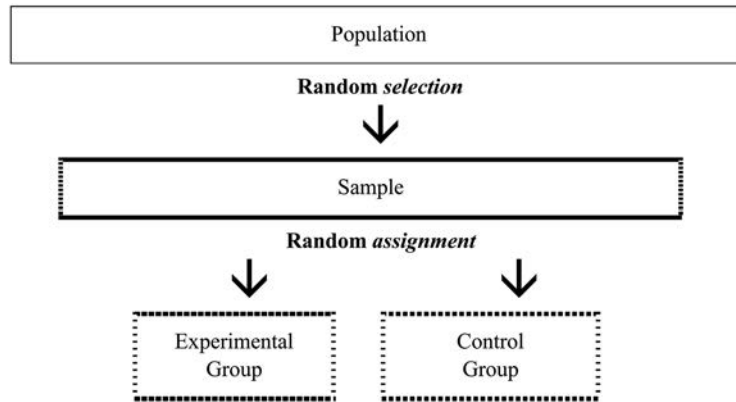


Figure 9.13.1 Ideal combination of random selection and random assignment.

Note that in any given experiment, *selection* may or may not be random. Likewise, *assignment* may or may not be random. Figure 9.13.1 illustrates the ideal situation, where first there is random selection from a population of interest to obtain a sample. This is followed by random assignment to treatment conditions.

When discussing the generalizability of the results of an experiment, a researcher should do so in light of the type of *selection* used. In other words, a properly selected sample (ideally, one selected at random) allows for more confidence in generalizing the results to a population.²¹ On the other hand, when discussing the comparability of the two groups, a researcher should consider the type of *assignment* used. In other words, proper assignment to a group (ideally, assignment at random) increases researchers' confidence that the two groups were initially equal – permitting a valid comparison of the outcomes of treatment and control conditions and thus ensuring the *internal validity* of the experiment.

14. Has the Researcher Considered Attrition?

Very	1	2	3	4	5	Very	or N/A	I/I
unsatisfactory						satisfactory		

Comment: The phenomenon of individuals dropping out of a study is referred to as *attrition* (sometimes called *experimental mortality*). In Chapter 6, we have already mentioned this with regard to longitudinal studies. It can be especially problematic for experiments. The longer and more complex the experimental treatment, the more likely it is that some participants will drop out. This can affect the generalizability of the results because they will apply only to the types of individuals who continued to participate.

For researchers who conduct experiments, *differential attrition* can be an important source of confounding (also referred to as *attrition bias*). Differential attrition refers to the possibility that those who drop out of an experimental condition are of a type different from those who

21 Recall the discussion about the Stanford Prison Experiment in Chapter 6 – it could be that its results are not generalizable due to the way the sample was selected (asking for volunteers for a “psychological study of prison life”), even though random assignment to ‘guards’ and ‘prisoners’ was used in the experiment.

drop out of a control condition. For instance, in an experiment on a weight-loss program, those in the experimental group who get discouraged by failing to lose weight may drop out. Thus, those who remain in the experimental condition are those who are more successful in losing weight, leading to an overestimate of the beneficial effects of the weight-loss program.

Researchers usually cannot physically prevent attrition (participants should be free to withdraw from a study, and it should be mentioned in the informed consent form). However, often the researchers can compare those who dropped out with those who remained in the study in an effort to determine whether those who remained and those who dropped out are similar in important ways. Example 9.14.1 shows a portion of a statement dealing with this matter.

Example 9.14.1²²

DESCRIPTION OF AN ATTRITION ANALYSIS

The participant attrition rate in this study raised the concern that the participants successfully completing the procedure were different from those who did not in some important way that would render the results less generalizable. Thus, an attrition analysis was undertaken to determine which, if any, of a variety of participant variables could account for participant attrition. Participant variables analyzed included ages of the participants and the parents, birth order and weight, socioeconomic status, duration of prenatal health care, prenatal risk factor exposure, hours spent weekly in day care, parental ratings of quality of infant's previous night's sleep, and time elapsed since last feeding and diaper change. This analysis revealed two effects: On average, participants who completed the procedure had been fed more recently than those who did not complete the procedure [. . .], and those who completed the procedure were slightly younger (153.5 days) than those who did not (156 days).

An alternative approach to dealing with this issue is an *intent-to-treat (ITT) analysis* when treatment dropouts are included into the calculations along with the participants who have completed the treatment. This is a very conservative approach that makes it less likely to find statistically significant effects of treatment (since dropouts are unlikely to exhibit any positive treatment outcomes). Thus, if the treatment is found to have a statistically significant impact with the intent-to-treat analysis, we can be much more confident in the actual effectiveness of the treatment.

____ 15. Has the Researcher Used Ethical and Politically Acceptable Treatments?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: This evaluation question is applicable primarily to experiments in applied areas such as criminal justice, education, clinical psychology, social work, and nursing. For instance,

22 Moore, D. S., & Cocas, L. A. (2006). Perception precedes computation: Can familiarity preferences explain apparent calculation by human babies? *Developmental Psychology*, 42(4), 666–678.

has the researcher used treatments to promote classroom discipline that will be acceptable to parents, teachers, and the community? Has the researcher used methods such as moderate corporal punishment by teachers, which may be unacceptable in typical classroom settings?

A low mark on this question means that the experiment is unlikely to have an impact in the applied area in which it was conducted.

At the same time, it is important to remember that if the proposed treatments are non-ethical, they are usually ruled out at the ethics board or IRB review stage, before the experiment even takes place, so this guideline might be more relevant when evaluating older studies²³ or studies that were not subjected to review by an IRB or ethics board.

____ 16. Overall, Was the Experiment Properly Conducted?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Rate the overall quality of the experimental procedures on the basis of the answers to the evaluation questions in this chapter and taking into account any other concerns you may have.

Chapter 9 Exercises

Part A

Directions: Answer the following questions.

1. In an experiment, a treatment constitutes what is known as
 - A. an independent variable.
 - B. a dependent variable.
2. Which of the following is described in this chapter as being vastly superior to the other?
 - A. Assigning a small number of previously existing groups to treatments at random.
 - B. Assigning individuals to treatments at random.

23 For example, in the United States, human subject research regulations have been tightened considerably at the end of the 1970s–early 1980s, with the publication of the Belmont Report (1979) and the adoption of the Code of Federal Regulations (1981).

3. Suppose a psychology professor conducted an experiment in which one of her sections of Introduction to Social Psychology was assigned to be the experimental group and the other section served as the control group during a given semester. The experimental group used computer-assisted instruction while the control group received instruction via a traditional lecture/discussion method. Although both groups are taking a course in social psychology during the same semester, the two groups might be initially different in other ways. Speculate on what some of the differences might be. (See Evaluation Question 3.)
4. In this chapter, what is described as a strength of an experimental design in which one group serves as both the treatment group and its own control group? What is the weakness of this experimental design?
5. Very briefly describe how the personal effect might confound an experiment.
6. What is the difference between a *blind* and a *double-blind* experiment?
7. What is the name of the phenomenon in which participants may be influenced by knowledge of the purpose of an experiment?
8. What are the main advantages and drawbacks of *natural* experiments? What about lab experiments?
9. Briefly explain how *random selection* differs from *random assignment*.
10. Is it possible to have *nonrandom selection* yet still have *random assignment* in an experiment? Explain.

Part B

Directions: Locate empirical articles on two experiments on topics of interest to you. Evaluate them in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Select the one to which you gave the highest overall rating, and bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

Evaluating Analysis and Results Sections: Quantitative Research

This chapter discusses the evaluation of Analysis and Results sections in *quantitative research* reports. These almost always contain statistics that summarize the data that were collected, such as means, medians, and standard deviations. These types of statistics are known as *descriptive statistics*. The Results sections of quantitative research reports also usually contain *inferential statistics* (like various regression analyses), which help in making inferences from the sample that was actually studied to the population from which the sample was drawn. It is assumed that the reader has a basic knowledge of elementary statistical methods.

Note that the evaluation of Analysis and Results sections of *qualitative research* reports is covered in the next chapter. The guidelines for evaluating Analysis and Results sections of *mixed methods research* are explained in Chapter 12.

1. When Percentages are Reported, are the Underlying Numbers of Cases also Reported?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
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Comment: Percentages are very widely reported in empirical articles published in academic journals. When reporting percentages, it is important for researchers to also report the underlying number of cases for each percentage. Otherwise, the results can be misleading. Consider Example 10.1.1, which contains only percentages. The percentage decrease in this example seems dramatic. However, when the underlying numbers of cases (whose symbol is *n*) are shown, as in Example 10.1.2, it becomes clear that the percentage represents only a very small decrease in absolute terms (i.e., a decrease from 4 students to 2 students).

1 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”

Example 10.1.1

PERCENTAGE REPORTED WITHOUT UNDERLYING NUMBER OF CASES
(POTENTIALLY MISLEADING)

Since the end of the Cold War, interest in Russian language studies has decreased dramatically. For instance, at Zaneville Language Institute, the number of students majoring in Russian has decreased by 50% from a decade earlier.

Example 10.1.2

PERCENTAGE REPORTED WITH UNDERLYING NUMBER OF CASES (NOT MISLEADING)

Since the end of the Cold War, interest in Russian language studies has decreased dramatically. For instance, at Zaneville Language Institute, the number of students majoring in Russian has decreased by 50% from a decade earlier ($n = 4$ in 2002, $n = 2$ in 2012).

___ 2. Are Means Reported Only for Approximately Symmetrical Distributions?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The *mean*, which is the most commonly reported average, should be used only when a distribution is *not* highly skewed. In other words, it should be used only when a distribution is approximately symmetrical.

A skewed distribution is one in which there are some extreme scores on one side of the distribution (such as some very high scores without some very low scores to counterbalance them). Example 10.2.1 shows a skewed distribution. It is skewed because there is a very high score of 310, which is not balanced out by a very low score at the lower end of the distribution of scores. This is known as a distribution that is *skewed to the right*.² The mean, which is supposed to represent the central tendency of the entire group of scores, has been pulled up by a single very high score, resulting in a mean of 82.45, which is higher than all of the scores except the highest one (the score of 310).

Example 10.2.1

A SKEWED DISTRIBUTION (SKEWED TO THE RIGHT) AND A MISLEADING MEAN

Scores: 55, 55, 56, 57, 58, 60, 61, 63, 66, 66, 310

Mean = 82.45, standard deviation = 75.57

The raw scores for which a mean was calculated are very seldom included in research reports, which makes it impossible to inspect for skewedness. However, a couple of simple computations using

2 A distribution that is skewed to the right is also said to have a *positive skew*.

only the mean and standard deviation (which are usually reported) can reveal whether the mean was misapplied to a distribution that is highly skewed to the right. These are the calculations:

1. Round the mean and standard deviation to whole numbers (to keep the computations simple). Thus, the rounded mean is 82, and the rounded standard deviation is 76 for Example 10.2.1.
2. Multiply the standard deviation by 2 (i.e., $76 \times 2 = 152$).
- 3a. *SUBTRACT* the result of Step 2 from the mean (i.e., $82 - 152 = -70$).
- 3b. *ADD* the result of Step 2 to the mean (i.e., $82 + 152 = 234$).

Steps 3a and 3b show the lower and upper bounds of a distribution that would be fittingly described by the mean. If the result of Step 3a is lower than the lowest possible score, which is usually zero, the distribution is highly skewed to the right.³ (In this example, -70 is much lower than zero.) This indicates that the mean was applied to a skewed distribution, resulting in a misleading value for an average (i.e., an average that is misleadingly high).⁴ If the result of Step 3b is higher than the highest score, the distribution is highly *skewed to the left*.⁵ In such a case (which is not the case here because $234 < 310$), the mean would be a misleadingly low value for an average.

This type of inappropriate selection of an average is rather common, perhaps because researchers often compute the mean and standard deviation for a set of scores without first considering whether the distribution of scores is skewed. A more appropriate measure of central tendency for skewed distributions would be the *median* (the mid-point of the distribution if the raw scores are listed from lowest to highest) or the *mode*⁶ (the most common raw score in the distribution).

If a consumer of research detects that a mean has been computed for a highly skewed distribution by performing the set of calculations described above, there is little that can be done to correct it short of contacting the researcher to request the raw scores. If this is not feasible, and if the alternative measures of central tendency (the median or mode) are not provided in the research report, the mean should be interpreted with great caution, and the article should be given a low mark on this evaluation question.

- 3 In a normal, symmetrical distribution, there are 3 standard deviation units on each side of the mean. Thus, there should be 3 standard deviation units on both sides of the mean in a distribution that is not skewed. In this example, there are not even 2 standard deviation units to the left of the mean (because the standard deviation was multiplied by 2). Even without understanding this theory, a consumer of research can still apply the simple steps described here to identify the misapplication of the mean. Note that there are precise statistical methods for detecting a skew. However, for their use to be possible, the original scores would be needed, and those are almost never available to consumers of research.
- 4 This procedure will not detect all highly skewed distributions. If the result of Step 3a is lower than the lowest score obtained by any participant, the distribution is also skewed. However, researchers seldom report the lowest score obtained by participants.
- 5 A distribution that is skewed to the left is said to have a *negative skew*.
- 6 A mode is also the only measure of central tendency that can be used for describing non-numerical data but it is much more common to present the distribution of non-numerical data as percentages (for example, “65% of the sample was White, 23% African American, 4% Asian, and 8% were other or mixed race”).

___ 3. If any Differences are Statistically Significant but Substantively Small, Have the Researchers Noted that They are Small?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Statistically significant differences are sometimes very small, especially when researchers are using large samples. (See Appendix C for an explanation of this point.) When this is the case, it is a good idea for a researcher to point this out. Obviously, a small but statistically significant difference will be interpreted differently from a large and statistically significant difference.

Example 10.3.1 illustrates how a significant but substantively small difference might be pointed out.⁷

Example 10.3.1

DESCRIPTION OF A SMALL BUT STATISTICALLY SIGNIFICANT DIFFERENCE

Although the difference between the means of the experimental group ($M = 24.55$) and control group ($M = 23.65$) was statistically significant ($t = 2.075, p < .05$), the small size of the difference, in absolute terms, suggests that the effects of the experimental treatment were weak.

This evaluation question is important in that researchers sometimes incorrectly imply that because a difference is statistically significant, it is necessarily large and important. More details about the limitations of significance testing are provided in Appendix C.

___ 4. Is the Results Section a Cohesive Essay?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: The Results section should be an essay – not just a collection of statistics. In other words, researchers should describe results in paragraphs, each of which outlines some aspect of the results. These paragraphs generally will contain statistics. The essay usually should be organized around the research hypotheses, research questions, or research purposes. See the example under the next guideline.

⁷ An increasingly popular statistic, *effect size*, is designed to draw readers' attention to the size of any significant difference. In general terms, it indicates by how many standard deviations two groups differ from each other. However, the effect size measures are mostly used in meta-analyses (see Chapter 14 for more details).

5. Does the Researcher Refer Back to the Research Hypotheses, Purposes, or Questions Originally Stated in the Introduction?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: This guideline may not be applicable to a very short research report with a single hypothesis, question, or purpose. Most empirical articles, however, contain several research hypotheses, so readers should be shown how different elements of the results relate to the specific hypotheses, questions, or purposes, as illustrated in Example 10.5.1. The example refers to three research purposes, which are briefly restated in the narrative. The tables referred to in the example are not shown here.

Example 10.5.1

RESULTS DISCUSSED IN TERMS OF RESEARCH PURPOSES STATED EARLIER IN THE REPORT

The first purpose was to determine adolescent students' estimates of the frequency of use of illicit drugs by students-at-large in their high schools. Table 1 shows the percentages for each . . .

Regarding the second purpose (estimates of illicit drug use by close friends), the percentages in Table 2 clearly indicate . . .

Finally, results relating to the third purpose are shown in Table 3. Since the purpose was to determine the differences between . . .

6. When There are Several Related Statistics, Have They Been Presented in a Table?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Even when there are only a small number of related statistics, a table can be helpful. For instance, consider Example 10.6.1, in which percentages and numbers of cases (*n*) are presented in a paragraph. Compare it with Example 10.6.2, in which the same statistics are reported in tabular form. Clearly, the tabular form is easier to follow.

Example 10.6.1⁸

TOO MANY STATISTICS PRESENTED IN A PARAGRAPH (COMPARE WITH EXAMPLE 10.6.2, WHICH PRESENTS THE SAME STATISTICS IN A TABLE)

⁸ Adapted from Erling, A., & Hwang, C. P. (2004). Body-esteem in Swedish 10-year-old children. *Perceptual and Motor Skills*, 99(2), 437–444. In the research report, the statistics are reported in tabular form, as recommended here.

Two percent of the girls ($n = 8$) and 2% of the boys ($n = 8$) reported that they were “Far too skinny.” Boys and girls were also identical in response to the choice “A little skinny” (8%, $n = 41$ for girls and 8%, $n = 34$ for boys). For “Just right,” a larger percentage of boys (76%, $n = 337$) than girls (70%, $n = 358$) responded. For “A little fat,” the responses were 18% ($n = 92$) and 13% ($n = 60$) for girls and boys, respectively. Also, a slightly higher percentage of girls than boys reported being “Far too fat” with 2% ($n = 12$) for girls and 1% ($n = 6$) for boys.

Example 10.6.2

USING A TABLE TO PRESENT RESULTS IN AN EASY-TO-UNDERSTAND FORMAT

Table 1 Answers to the Research Question on Self-perceived Weight

<i>Answer</i>	<i>Girls</i>		<i>Boys</i>	
	<i>%</i>	<i>n</i>	<i>%</i>	<i>n</i>
Far too skinny	2	8	2	8
A little skinny	8	41	8	34
Just right	70	358	76	337
A little fat	18	92	13	60
Far too fat	2	12	1	6

7. If There are Tables, are Their Highlights Discussed in the Narrative of the Results Section?

Very
unsatisfactory

1 2 3 4 5

Very
satisfactory

or N/A I/I

Comment: Researchers should point out important highlights of statistical tables, as illustrated in Example 10.7.1, which shows part of the discussion of the statistics in Example 10.6.2. Note that only *highlights* of the statistics should be presented. To repeat them all in paragraph form would be redundant.

When there are large tables, pointing out the highlights can be especially helpful for consumers of the research.

Example 10.7.1⁹

HIGHLIGHTS OF EXAMPLE 10.6.2 POINTED OUT

The same percentage of boys as girls (10%) perceived themselves as a little or far too skinny, while 20% of the girls and 14% of the boys perceived themselves as a little or far too fat (see Table 1). Of the 104 girls who perceived themselves as fat (a little fat or far too fat), only . . .

9 Ibid.

8. Have the Researchers Presented Descriptive Statistics Before Presenting the Results of Inferential Tests?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Descriptive statistics include frequencies, percentages, averages (usually the mean or median), and measures of variability (usually the standard deviation or inter-quartile range). Descriptive statistics are called this way because they *describe* the sample.

Inferential statistics, such as tests of differences between the means, correlation coefficients (usually the Pearson's r), regression analyses, etc., allow researchers to *infer*, or generalize, from the sample statistics to the population. In technical terms, inferential statistics determine the probability that any differences among descriptive statistics are due to chance (random sampling error). Obviously, it makes no sense to discuss the results of a test performed on descriptive statistics unless those descriptive statistics have first been presented. Failure on this evaluation question is very rare (and represents a serious flaw in a research report).¹⁰

9. Overall, is the Presentation of the Results Comprehensible?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	<i>or</i> N/A	I/I
------------------------	---	---	---	---	---	----------------------	---------------	-----

Comment: Even when the analysis is complex and advanced statistical methods have been applied, the essay that describes the results should be comprehensible to any intelligent layperson. Specifically, the essay should describe the results conceptually, using everyday language while presenting the statistical results for consumers of research who wish to consider them. Example 10.9.1 illustrates such a description in a study of seasonal crime rate fluctuations.

Example 10.9.1¹¹

THE RESULTS OF COMPLEX STATISTICAL ANALYSES ARE EXPLAINED IN AN EASY-TO-UNDERSTAND LANGUAGE

Results for the basic homicide and assault models appear in Table 1 [in the original article]. To avoid unnecessary detail and to simplify the presentation, the table does not include the coefficients for the 87 cross-sectional fixed effects. [. . .]

At a broad level of comparison, homicide and assault have similar seasonal cycles. Both offenses peak in July, and both are lowest in January. Assault nevertheless displays considerably more variability than homicide . . . For homicide, the seasonal fluctuations are less extreme, and none of the months between June and November significantly differ

10 If articles that omit descriptive statistics but go straight to presenting the results of, say, regression analyses, are published regularly in a journal, this speaks volumes to the low quality of the journal and its editorial process.

11 McDowall, D., & Curtis, K. M. (2015). Seasonal variation in homicide and assault across large US cities. *Homicide Studies*, 19(4), 303–325.

from December. Both assault and homicide rates are seasonal overall and both follow generally comparable patterns. Still, homicide is flatter over its yearly cycle than is assault, and the impact of seasonality is much smaller.

___ 10. Overall, is the Presentation of the Results Adequate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and taking into account any additional considerations and concerns you may have.

Chapter 10 Exercises

Part A

Directions: Answer the following questions.

1. When reporting percentages, what else is it important for researchers to present?
2. Should the mean be used to report the average of a highly skewed distribution?
3. Suppose you read that the mean equals 10.0 and the standard deviation equals 6.0. Is the distribution skewed? (Assume that the lowest possible score is zero.) Explain.
4. Are statistically significant differences always large, substantive differences?
5. Should the Results section be an essay *or* should it be only a collection of statistics/tables?
6. According to this chapter, is it ever desirable to restate hypotheses that were originally stated in the introduction of a research report? Explain.
7. If statistical results are presented in a table, should all the entries in the table be discussed in the narrative? Explain.
8. Should 'descriptive statistics' *or* 'inferential tests' be reported first in Results sections?

Part B

Directions: Locate several quantitative research reports of interest to you in academic journals. Read them, and evaluate the descriptions of the results in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Select the one to which you gave the highest overall rating, and bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

Evaluating Analysis and Results Sections: Qualitative Research

Because human judgment is central in the analysis of qualitative data, there is much more subjectivity in the analysis of qualitative data than in the analysis of quantitative data. (See Chapter 10 for evaluation questions for quantitative Analysis and Results sections of research reports.) Consult Appendix A for additional information on the differences between qualitative and quantitative research.

___ **1. Were the Data Analyzed Independently by Two or More Individuals?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: As a general rule, the results of qualitative research are considered more dependable when the responses of participants are independently analyzed by two or more individuals (i.e., two or more individuals initially code and/or categorize the responses without consulting with each other). Then, they compare the results of their analyses and discuss any discrepancies in an effort to reach a consensus. Doing this assures consumers of research that the results represent more than just the impressions of one individual, which might be idiosyncratic. Examples 11.1.1 and 11.1.2 illustrate how this process might be described in a research report.

Example 11.1.1²

INDEPENDENT ANALYSIS BY TWO RESEARCHERS

Two independent research psychologists developed a list of *domains* or *topic areas* based on the content of the discussions and the focus group questions used to organize information into similar topics. Once each reviewer had independently identified their domains, the two reviewers compared their separate lists of domains until consensus was reached.

1 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”

2 Williams, J. K., Wyatt, G. E., Resell, J., Peterson, J., & Asuan-O’Brien, A. (2004). Psychosocial issues among gay- and non-gay-identifying HIV-seropositive African American and Latino MSM. *Cultural Diversity and Ethnic Minority Psychology, 10*(3), 268–286.

Example 11.1.2³

INDEPENDENT ANALYSIS BY TWO RESEARCHERS

Using a grounded theory approach, we used standard, qualitative procedures to code the data (Strauss & Corbin, 1998). Two coders, working independently, read a transcript of clients' unedited answers to each question and identified phenomena in the text that were deemed responsive to the question and thus, in the opinion of the coder, should be regarded as relevant data for inclusion in the analysis. Phenomena included all phrases or statements conveying meaningful ideas, events, objects, and actions. If both coders selected the same phrase or statement in the answer to a given question, then it was counted as an agreement. Overall, percent agreement between coders averaged 89% for this first step. Disagreements were resolved through discussion and consensus.

Notice that in the Example 11.1.2 above, the specific rate of agreement between the two coders (inter-rater reliability) is expressed as a percentage. This method of calculating agreement between independent coders' ratings or opinions is somewhat superior to a vague way of putting it as "the inter-rater agreement was high."

When giving your rating to this evaluation question, pay special attention to whether the coding process was first performed by the coders *independently*, to avoid any shared biases.

2. Did the Researchers Seek Feedback from Experienced Individuals and Auditors Before Finalizing the Results?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Seeking feedback helps to ensure the trustworthiness of the results. Example 11.2.1 is drawn from a report of research on incarcerated young men. The researchers had their preliminary results reviewed by two other individuals who had experienced incarceration (independent experienced individuals).

Example 11.2.1⁴

FEEDBACK FROM INDEPENDENT EXPERIENCED INDIVIDUALS

Finally, the data summary was reviewed by two individuals with a personal history of incarceration who were not involved in the data analytic process for critique of the face validity of the findings. Their feedback was incorporated into the discussion of our findings.

- 3 Beitel, M., Genova, M., Schuman-Olivier, Z., Arnold, R., Avants, S. K., & Margolin, A. (2007). Reflections by inner-city drug users on a Buddhist-based spirituality-focused therapy: A qualitative study. *American Journal of Orthopsychiatry*, 77(1), 1–9.
- 4 Seal, D. W., Belcher, L., Morrow, K., Eldridge, G., Binson, D., Kacanek, D., . . . Simms, R. (2004). A qualitative study of substance use and sexual behavior among 18- to 29-year-old men while incarcerated in the United States. *Health Education & Behavior*, 31(6), 775–789.

Often, researchers seek feedback on their preliminary results from outside experts who were not involved in conducting the research. The technical title for such a person in qualitative research is *auditor*. Example 11.2.2 describes the work of an auditor in a research project.

Example 11.2.2⁵

FEEDBACK FROM A CONTENT-AREA EXPERT (I.E., AUDITOR)

At three separate points . . . , the work of the analysis team was reviewed by an auditor. The first point came after domains had been agreed upon, the second point came after core ideas had been identified, and the third point came after the cross-analysis. In each case, the auditor made suggestions to the team regarding the names and ideas the team was working on. Adjustments were made after the team reached consensus on the feedback given by the auditor. Examples of feedback given by the auditor included suggestions on the wording of domain and category names and a request for an increased amount of specificity in the core ideas put forth by the team members. The auditor was a Caucasian female faculty member in the social psychology discipline whose research is focused in the area of domestic violence.

3. Did the Researchers Seek Feedback from the Participants (i.e., Use Member Checking) Before Finalizing the Results?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	<i>or</i> N/A	I/I
------------------------	---	---	---	---	---	----------------------	---------------	-----

Comment: As indicated in the discussion of Evaluation Question 2, seeking feedback helps to ensure the trustworthiness of the results. When researchers seek feedback on their preliminary results from the participants in the research, the process is called *member checking*. Using member checking is not always feasible, especially with very young participants and participants with limited cognitive abilities. The authors of Example 11.3.1 did use member checking.

Example 11.3.1⁶

FEEDBACK FROM “MEMBERS” (I.E., MEMBER CHECKING BY PARTICIPANTS)

To ensure methodological rigor, trustworthiness (Oktay, 2004; Straus & Corbin, 1998) of the data involved member (participant) checking to establish that the reconstructions were credible and that the findings were faithful to participants' experiences. Participants were

5 Wettersten, K. B. et al. (2004). Freedom through self-sufficiency: A qualitative examination of the impact of domestic violence on the working lives of women in shelter. *Journal of Counseling Psychology*, 51(4), 447–462.

6 Anderson, K. M., Danis, F. S., & Havig, K. (2011). Adult daughters of battered women: Recovery and posttraumatic growth following childhood adversity. *Families in Society: The Journal of Contemporary Social Services*, 92(2), 154–160.

provided written and oral summaries of their responses and given opportunities for correction, verification, and clarification through follow-up letters, telephone contacts, and interviews. For example, upon receiving their transcribed interviews, researchers told participants, “As you read through your transcript, you may want to make notes that would further clarify what was said or address an area that was not originally discussed.” And in the follow-up interview, participants were asked, “Are there any changes or additional comments that you would like to discuss in regard to the study’s findings?” Additionally, researchers conducted ongoing peer debriefing to review their audit trail regarding the research process.

4. Did the Researchers Name the Method of Analysis They Used and Provide a Reference for it?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Various methods for analyzing qualitative data have been suggested. Researchers should name the particular method they followed. Often, they name it and provide one or more references where additional information can be obtained. Example 11.4.1 illustrates this for two widely used methods for analyzing qualitative data: the grounded theory method and data triangulation⁷.

Example 11.4.1⁸

NAMING GROUNDED THEORY AND DATA TRIANGULATION AS THE METHODS OF ANALYSIS WITH REFERENCES FOR MORE INFORMATION ON EACH METHOD
(ITALICS ADDED FOR EMPHASIS)

According to Strauss and Corbin (1998), *grounded theory* is a “general methodology for developing theory that is grounded in data systematically gathered and analyzed” (p. 158). This approach uses “*data triangulation*” (Janesick, 1998) with multiple data sources (e.g., different families and family members, different groups and facilitators) and a “constant comparative method” (Glaser, 1967) by continually examining the analytic results with the raw data. The analysis proceeded in steps. First, a “start list” consisting of 42 descriptive codes was created on the basis of ongoing community immersion and fieldwork, as well as the perspectives of family beliefs (Weine, 2001b) and the prevention and access intervention framework used to develop the CAFES intervention (Weine, 1998). The codes addressed a variety of topics pertaining to refugee families suggested by prior

⁷ Notice that the method of triangulation used with qualitative data is very similar to the same method used with quantitative data – the gathering of data about the same phenomenon from several sources.

⁸ Weine, S., Feetham, S., Kulauzovic, Y., Knafl, K., Besic, S., Klebic, A., . . . Pavkovic, I. (2006). A family beliefs framework for socially and culturally specific preventative interventions with refugee youths and families. *American Journal of Orthopsychiatry*, 76(1), 1–9.

empirical and conceptual work. Texts were coded with only these codes, and they were supplemented with memos for any items of interest that did not match the code list. Out of the start list of 42 codes, 3 codes focused on adapting family beliefs.

5. Did the Researchers State *Specifically* How the Method of Analysis Was Applied?

Very unsatisfactory 1 2 3 4 5 Very satisfactory or N/A I/I

Comment: The previous evaluation question suggests that the particular method of analysis should be identified and that references for it should be provided. Sometimes, a method of analysis is so widely used and generally well known that it may not be necessary to add references about the method itself. However, it is still important that researchers explain in more detail, step by step, how they used it for the specific purposes of their study. Example 11.5.1 illustrates this for *content analysis* – one of the most common methods of analyzing text-based qualitative data in social sciences.

Example 11.5.1⁹

DESCRIBING THE STEPS USED IN THE ANALYSIS OF QUALITATIVE DATA USING THE METHOD OF CONTENT ANALYSIS

We conducted a content analysis to examine how rape is portrayed in print media. More specifically, we sought to answer the following research questions: (a) How pervasive is rape myth language in local newspaper reporting? and (b) Is the media using other indirect language that reinforces rape myths? To conduct this study, we used the Alliance for Audited Media (The New Audit Bureau of Circulations) to create a list of the top 100 circulated newspapers in the United States. We took out papers that had national readership accounting for their massive circulation, which included the New York Times, Wall Street Journal, and USA Today. Next, we grouped the newspapers by state and we further organized them into nine geographical regions, as designated by the Census Bureau. [. . .]

We utilized the database LexisNexis to conduct our search of articles containing the terms “rape” and/or “sexual assault” in the headline. Initially, we searched these terms in full in each circulation but our search yielded thousands of articles and many that were beyond the scope of the current research. Thus, we restricted our search of these terms to the headlines during the one-year period beginning on 1st January 2011, and ending on 1st January 2012, which provided us with a robust sample size for generalizability across the regions. In all, we found 386 articles. (See Table 1 for a breakdown per newspaper.) [in the original article]

9 Sacks, M., Ackerman, A. R., & Shlosberg, A. (2018). Rape myths in the media: A content analysis of local newspaper reporting in the United States. *Deviant Behavior*, 39(9), 1237–1246.

We borrowed a coding scheme from Turkewitz (2010), though we made modest changes to the coding instrument to include a few rape myths not contained in the original coding instrument. Our coding instrument was designed to provide as much detail about the media discourse on rape and sexual assault as possible. Therefore, we coded for various case characteristics, including details about the alleged victims, offenders, and incident details. We also sought to examine how the media used narratives to describe rape and sexual assault. However, for purposes of the current research, we specifically coded for the presence of commonly known rape myths in the newspaper coverage. More specifically, we coded for the following rape myths: (1) Victim Lying; (2) Victim Asked For It; (3) Victim Wanted To; (4) Victim Partially Responsible; (5) Perpetrator Couldn't Help It; (6) Not Traumatic/Big Deal; and (7) He's Not The Kind of Guy Who Would Do This. To ensure reliability in coding, two coders read and coded each article.

6. Did the Researchers Self-disclose their Backgrounds?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Sometimes, qualitative researchers disclose their own background characteristics as they relate to the variables under investigation. For instance, a researcher studying the social dynamics of men with HIV might reveal their own HIV status and that of their significant others. This is done in an effort to “clear the air” regarding any personal points of view and biases that might impact the researcher’s analysis of the data.

Example 11.6.1 shows a portion of such a disclosure in a study with a different context. The researchers included the statement in their Analysis section under the subheading Author Biases.

Example 11.6.1¹⁰

RESEARCHERS' SELF-DISCLOSURE

Mary Lee Nelson is a professor of counseling psychology. She came from a lower middle, working-class background, was the first in her family to pursue higher education, and had many of the experiences described by the research participants. This background provided her with important insights about the data. In addition, it might have biased her expectations about what participants’ experiences would be. She expected to hear stories of financial hardship, social confusion, loneliness, and challenges with personal and career identity development. Matt Englar-Carlson is a counseling psychologist and currently an associate professor of counselor education. He has a strong interest in new developments in social class theory. He comes from a middle-class, educated family background. He came to

10 Nelson, M. L., Englar-Carlson, M., Tierney, S. C., & Hau, J. M. (2006). Class jumping into academia: Multiple identities for counseling academics. *Journal of Counseling Psychology*, 53(1), 1–14.

the study with expectations that findings might conform to the social class worldview model, as developed by Liu (2001). Sandra C. Tierney is a recent graduate of a doctoral program in counseling psychology . . .

7. Are the Results of Qualitative Studies Adequately Supported with Examples of Quotations or Descriptions of Observations?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Qualitative researchers typically report few, if any, statistics in the Results section. Instead, they report on the themes and categories that emerged from the data, while looking for patterns that might have implications for theory development. Instead of statistics, quotations from participants or descriptions of observations of the participants' behaviors are used to support the general statements regarding results. This is illustrated in Example 11.7.1, in which a quotation is used to support a finding. As is typical in qualitative studies, researchers set off the quotation in block style (i.e., indented on left and right).

Example 11.7.1¹¹

RESULTS OF A QUALITATIVE STUDY SUPPORTED WITH QUOTATIONS

Although education was important to these men, there were barriers that they encountered in working toward their degrees, including expectations that they would fail and intrinsic pressure to succeed:

I am proud to be a Black man, and I am proud to have gotten where I am, but I'm real conscious of the fact that people are expecting less of me. There are days where I go at 150%, and there are days where I am tired and I can't go that hard; I can have great class presentations, and I can have a crappy presentation sometimes. When I am on a bad day or when I have a bad presentation—those stay with me longer than the good ones because of the fact that there are very few of us [in graduate school] and, thus, it's a burden that we've got to protect, we got to come tight with our game. And, not all the time I'm feeling that.

The use of extensive quotations is a technique used to produce what qualitative researchers refer to as *thick descriptions*. Not only do these descriptions help illustrate the point the researcher is making but they also allow the reader to feel the subjects' language and the emotional context of their situations, as well as assess if the researcher's interpretation bodes with the reader's own understanding. Example 11.7.2 illustrates how a quotation relays the research subject's view of his own offending and how he sees it within the context of being religious, in his own words.

11 Sánchez, F. J., Liu, W. M., Leathers, L., Goins, J., & Vilain, E. (2011). The subjective experience of social class and upward mobility among African American men in graduate school. *Psychology of Men & Masculinity*, 12(4), 368–382.

Example 11.7.2¹²**RESULTS OF A QUALITATIVE STUDY SUPPORTED WITH QUOTATIONS**

A similar self-serving interpretation of religious doctrine was evident in commentary from Cool, a 25-year-old male drug dealer:

The way it work is this. You go out and do some bad and then you ask for forgiveness and Jesus have to give it to you, and you know wipe the slate clean. So, I always do a quick little prayer right before and then I'm cool with Jesus. Also another thing is this; if you doing some wrong to another bad person, like if I go rob a dope dealer or a molester or something, then it don't count against me because it's like I'm giving punishment to them for Jesus. That's God's will. Oh you molested some kids? Well now I'm [God] sending Cool over your house to get your ass.

Such selective understanding of religious doctrine served offenders well in justifying their behavior, particularly when it came to considering the transcendental consequences of offending.

Consumers of qualitative research should make judgments as to how well the quotations illustrate and support the research findings, when giving a mark on this evaluation question.

___ **8. Are Appropriate Statistics Reported (Especially for Demographics)?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: The main description of the results of qualitative research is relatively free of statistics. However, statistical matters often arise when the results are written up. This could be as simple as reporting the numbers of cases (whose statistical symbol is *n*). For instance, instead of reporting that “*some students* were observed with their heads down on their desks,” it might be better to report that “*six students (or about a quarter of the class)* were observed with their heads down on their desks.” Too much emphasis on exact numbers, however, can be distracting in a qualitative research report. Hence, this evaluation question should be applied judiciously.

One of the most appropriate uses of statistics in qualitative research is to describe the demographics of the participants. When there are multiple demographic statistics reported, it is best to present them in a statistical table, which makes it easier for consumers of research to scan for relevant information. Example 11.8.1 shows a table of demographics presented in a qualitative research report.

12 Topalli, V., Brezina, T., & Bernhardt, M. (2013). With God on my side: The paradoxical relationship between religious belief and criminality among hardcore street offenders. *Theoretical Criminology*, 17(1), 49–69.

Example 11.8.1¹³DEMOGRAPHIC STATISTICS IN QUALITATIVE RESEARCH REPORTED
IN A TABLE**Table 1** Focus Group Participant Demographics (*N* = 28)

<i>Characteristics</i>		<i>n</i>	<i>%</i>
Gender	Men	21	75
	Women	7	25
Age	16–20	2	7
	21–30	7	25
	31–40	9	32
	41–50	8	29
	51–54	2	7
Marital status	Married	20	71
	Single	8	29
Income	<\$10,001	7	25
	\$10,001–\$20,000	8	29
	\$20,001–\$30,000	8	29
	\$30,001–\$40,000	4	14
	>\$40,000	1	3
Years of U.S. residence	2–5	8	29
	6–10	9	32
	11–15	4	14
	16–20	6	21
	21–25	1	4
Place of residence	Rural	20	71
	Urban	8	29
Employment	Construction worker	11	39
	Factory worker	5	18
	No outside employment	6	21
	Other ^a	6	21
Education	HS diploma/GED	17	61
	No HS diploma	7	25
	Some college	4	14

^a Examples include driver, caterer, baker, dry cleaner, nanny, and housecleaner.

- 13 Ames, N., Hancock, T. U., & Behnke, A. O. (2011). Latino church leaders and domestic violence: Attitudes and knowledge. *Families in Society: The Journal of Contemporary Social Services*, 92(2), 161–167.

Example 11.8.2 illustrates the reporting of demographic statistics in the narrative of a qualitative research report rather than in a table.¹⁴

Example 11.8.2¹⁵

DEMOGRAPHIC STATISTICS REPORTED IN QUALITATIVE RESEARCH

A purposive sample of 8 convicted child molesters, 7 European Americans and 1 Latino, aged 36 to 52 ($M = 44.0$, $SD = 6.4$), was recruited from an outpatient treatment facility for sex offenders in a northeastern urban community. Four men were single; the others were either separated ($n = 2$) or divorced ($n = 2$); 3 indicated being gay or bisexual. Participants' educational levels were GED ($n = 1$), high school graduate ($n = 2$), some college ($n = 3$), some graduate work ($n = 1$), and master's degree ($n = 1$). The median annual income was \$15,000–\$20,000.

___ **9. Overall, is the Results Section Clearly Organized?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The Results sections of qualitative research reports are often quite long. By using subheadings throughout the Results sections, researchers can help guide their readers through sometimes complex information. Example 11.9.1 shows the major headings (in bold) and subheadings (in italics) used to help readers through a long Results section of a qualitative research report.

Example 11.9.1¹⁶

MAJOR HEADINGS (IN BOLD) AND SUBHEADINGS (IN ITALICS) USED IN A LONG RESULTS SECTION OF A QUALITATIVE RESEARCH REPORT

Results

The Aboriginal Perspective: Cultural Factors That Serve As Barriers to Rehabilitation

The strength of the local and family hierarchy
Aboriginal fatalism

- 14 Demographic statistics are sometimes reported in the subsection on Participants in the Method section of a research report. Other times, they are reported in the Results section.
- 15 Schaefer, B. M., Friedlander, M. L., Blustein, D. L., & Maruna, S. (2004). The work lives of child molesters: A phenomenological perspective. *Journal of Counseling Psychology*, 51(2), 226–239.
- 16 Kendall, E., & Marshall, C. A. (2004). Factors that prevent equitable access to rehabilitation for Aboriginal Australians with disabilities: The need for culturally safe rehabilitation. *Rehabilitation Psychology*, 49(1), 5–13.

The Non-Aboriginal Perspective: Unhelpful Stereotypes

Fear of Aboriginal hostility

The self-sufficiency stereotype

Motivational stereotypes

The internal strife stereotype

___ 10. Overall, is the Presentation of the Results Adequate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and any additional considerations and concerns you may have. You may also want to consult the online resource for this chapter called Online Appendix: *Examining the Validity Structure of Qualitative Research*.

Chapter 11 Exercises

Part A

Directions: Answer the following questions.

1. When there are two or more individuals analyzing the data, what does *independently analyzed* mean?
2. What is the technical name of content-area experts who review preliminary research results for qualitative researchers?
3. What is the name of the process by which researchers seek feedback on their preliminary results from the participants in the research?
4. A researcher engages in self-disclosure in an effort to do what?
5. The results of qualitative studies should be supported with what type of material (instead of statistics)?
6. What is one of the most appropriate uses of statistics in qualitative research?
7. Because the Results sections of qualitative research reports are often quite long, what can researchers do to help guide readers?

Part B

Directions: Locate a qualitative research report of interest to you.¹⁷ Read it, and evaluate the description of the results in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Bring it to class for discussion, and be prepared to discuss both its strengths and weaknesses.

17 Researchers who conduct qualitative research often mention that it is qualitative, in the titles or abstracts of their reports. Thus, to locate examples of qualitative research using an electronic database, it is often advantageous to use *qualitative* as a search term.

Evaluating Analysis and Results Sections: Mixed Methods Research

Anne Li Kringen

This chapter discusses the evaluation of Analysis and Results sections in mixed methods research reports. Mixed methods research incorporates both qualitative and quantitative methods to address the same research topic. By incorporating both types of methods, mixed methods studies are ideally suited for rendering understanding of phenomena that can be difficult to understand using either a qualitative or a quantitative approach alone.

For example, researchers might want to understand how limited racial diversity in policing impacts new officers entering the profession.¹ A qualitative approach can shed light on how officers entering the profession feel about organizational culture and their individual experiences related to race, but it cannot address the question as to whether these experiences are representative of the experiences of officers entering the profession as a whole. In contrast, a quantitative approach can demonstrate how different levels of racial diversification relate to outcomes such as successful completion of the training academy and successful transition into the career, but it cannot effectively explain how individuals making these transitions feel about the experience.

Mixed methods allow researchers to include both methods, rendering an understanding of unique experience alongside a generalized understanding of trends and patterns. Given the inclusion of both qualitative and quantitative approaches, mixed methods research reports include descriptions of both quantitative and qualitative methods in Analysis sections and both qualitative and quantitative findings in Results sections.

The specific qualitative and quantitative methods used must be independently evaluated based on the relevant standards for each type of methodology. Likewise, presentation of the qualitative and quantitative results must be evaluated independently based on appropriate standards. The evaluation of Analysis and Results sections of quantitative research reports is covered in Chapter 10, and the evaluation of Analysis and Results sections of qualitative research reports in Chapter 11. Beyond specific evaluation of the qualitative and quantitative components, mixed methods research reports must also be evaluated for quality using a separate set of criteria

1 The research question is inspired by the author of the chapter's own research interests. Part of her research findings have been published here: Kringen, A. L. (2016). Examining the relationship between civil service commissions and municipal police diversity. *Criminal Justice Policy Review*, 27(5), 480–497.

unique to mixed methods research. These include aspects of design and implementation typically reported in Analysis sections as well as aspects of interpretation typically reported in Results sections.

1. Does the Methods Section Identify a Specific Mixed Methods Design?

Very unsatisfactory 1 2 3 4 5 Very satisfactory or N/A I/I²

Comment: While mixed methods studies involve the analysis of both qualitative and quantitative data, analyzing and reporting findings from both types of data do not necessarily reflect a true mixed methods study. Instead, mixed methods studies should incorporate a specific mixed methods design which dictates the logic used to integrate both the qualitative and quantitative analyses into a cohesive result.

Common mixed methods designs include the exploratory (or *sequential exploratory*) design where qualitative data are analyzed to guide a subsequent quantitative data collection and analysis, the explanatory (or *sequential explanatory*) design where quantitative data are analyzed to guide a subsequent qualitative data collection and analysis, and the *convergent design* where researchers use separate quantitative and qualitative analyses to triangulate results about a single topic or to merge findings from multiple data sources.

Given that each design is utilized for different types of projects to render insight into different types of questions, it is important that researchers clearly identify the specific mixed methods design utilized. Consider Examples 12.1.1 and 12.1.2 where such specific mixed methods designs are identified.

Example 12.1.1³

INDICATING AND EXPLAINING THE USE OF A MIXED METHODS CONVERGENT DESIGN

This study uses a mixed methods convergent design: a quantitative repeated measures design and qualitative methods consisting of a Grounded Theory design. The aim of a mixed methods design is to integrate quantitative and qualitative components to obtain additional knowledge (Boeije, Slagt, & Van Wesel, 2013; Creswell & Zhang, 2009). In this study, integration will be focused on interpreting how qualitative outcomes regarding patients' experiences with NET [narrative exposure therapy] enhance the understanding of the quantitative clinical outcomes.

- 2 Continuing with the same grading scheme as in the previous chapters, N/A stands for "Not applicable" and I/I stands for "Insufficient information to make a judgement."
- 3 Mauritz, M. W., Van Gall, B. G. I., Jongedijk, R. A., Schoonhoven, L., Nijhuis-van der Sanden, M. W. G., & Gossens, P. J. J. (2016). Narrative exposure therapy for posttraumatic stress disorder associated with repeated interpersonal trauma in patients with severe mental illness: A mixed methods design. *European Journal of Psychotraumatology*, 7(1), 32473.

Example 12.1.2⁴

EXPLAINING THE USE OF A SEQUENTIAL MIXED METHODS DESIGN

In this study, we used a sequential mixed methods design with a convergent mixed methods analysis (Teddle & Tashakkori, 2009) to generate new evidence about child perceptions of health. We first conducted a core qualitative study and when unexpected findings emerged, we generated new hypotheses that could not be fully understood using the existing data. We then turned to quantitative methods to aid in their interpretation and used generational theory as a lens to reflect upon both sets of data.

2. Does the Methods Section Link the Need for a Mixed Methods Approach to the Research Question(s)?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Given that mixed methods approaches are better suited to specific types of questions, it is important that research reports clearly link the choice to use mixed methods to the specific research question or questions that the researchers seek to address. For example, mixed methods are useful for understanding the patterns of larger trends while maintaining detail about individual cases. Consider Examples 12.2.1 and 12.2.2 where specific research questions are connected with the type of method used to investigate them.

Example 12.2.1⁵

INDICATING WHICH RESEARCH QUESTIONS ARE ANSWERED USING ONLY QUANTITATIVE METHODS, USING ONLY QUALITATIVE METHODS, AND THROUGH THE MIXED METHODS ANALYSIS

This article is guided by three research questions:

- RQ1 (Quantitative): What types of mixed methods designs are currently being conducted by military family researchers, and are they consistent with Creswell and Plano Clark (2011) or Greene and colleagues (1989)?
- RQ2 (Qualitative): In what ways is mixed methodology research being conducted in research on military families?
- RQ3 (Mixing): Using both the quantitative categories and the qualitative results, how much mixing is currently occurring in military family research, and what is the caliber of this mixed methodology research?

4 Michaelson, V., Pickett, W., Vandemeer, E., Taylor, B., & Davison, C. (2016). A mixed methods study of Canadian adolescents' perceptions of health. *International Journal of Qualitative Studies on Health and Well-being*, 11(1), 32891.

5 D'Aniello, C. & Moore, L. E. (2015). A mixed methods content analysis of military family literature. *Military Behavioral Health*, 3(3), 171–181. *Sexual Abuse*, 26(6), 657–676.

Example 12.2.2⁶**INDICATING WHICH QUESTIONS SPECIFICALLY RELY ON THE MIXED METHODS ANALYSIS**

Our study was designed to answer the following two research questions in the QUAN phase:

1. How do teachers' beliefs relate to their instructional technology practices?
2. How do factors other than beliefs relate to teachers' instructional technology practices?

Guided by these answers, we ultimately wanted to answer this question, which integrated the results of both methods, in the QUAL phase: Do teachers who work in technology schools and who are equipped to integrate technologies change their beliefs and consequently technology practices toward a student-centered paradigm?

____ **3. Does the Methods Section Clearly Explain Both the Quantitative and Qualitative Methods Utilized in the Study?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: While the importance of explaining the mixed methods design is paramount in a mixed methods study, explanation of the mixed methods design alone is insufficient without a detailed presentation of the specific qualitative and quantitative components incorporated in the mixed methods design.

While the mixed methods design determines how the two elements work together to address the research question, the logic of mixed methods designs assumes that the qualitative and quantitative methods employed are properly utilized. Research reports must clearly indicate the specifics of the qualitative and quantitative components so that readers can independently evaluate the quality of each component. Consider Examples 12.3.1, 12.3.2, and 12.3.3 that describe specific types of methodology.

Example 12.3.1⁷**DESCRIBING THE QUALITATIVE METHODOLOGY**

We used a modified grounded theory approach for analysis (Charmaz, 2003). In this approach, the investigators read through each transcript and identified key ideas that were present in the text and described experiences within one day of the index suicide attempt.

6 Palak, D., & Walls, R. T. (2009). Teachers' beliefs and technology practices: A mixed-methods approach. *Journal of Research on Technology in Education*, 41(4), 417–441.

7 Adler, A., Bush, A., Barg, F. K., Weissinger, G., Beck, A. T., & Brown, G. K. (2016). A mixed methods approach to identify cognitive warning signs for suicide attempts. *Archives of Suicide Research*, 20(4), 528–538.

The investigators discussed these key ideas and created a code for each key idea. Each code was defined and decision rules for when to apply the code to the text was entered into the NVivo software package. In addition, codes that described the key concepts we were looking to capture (e.g., state hopelessness) were added to the list of codes. Three coders (two master's-level research assistants and one PhD-level researcher) completed all of the coding. Practice coding was first conducted on four documents to establish initial reliability. Subsequently, 10% of transcripts were coded by all three coders who met bi-weekly to review coding and refine definitions. Previously coded transcripts were recoded when changes were made, such as when new codes were added or definitions revised. Inter-rater reliability was calculated within NVivo to ascertain consensus among coders until 100% agreement was reached. Coding was discussed until consensus was reached.

Example 12.3.2⁸

DESCRIBING THE QUANTITATIVE METHODOLOGY

This exploratory, mixed-methods study examines self-report data from a correctional subsample of 26 women and 25 men who are currently incarcerated for a sex offense against a child under the age of 13. Pen and paper surveys were administered in 2011 to participants in order to collect information on a range of demographic, victim, and offense characteristics. The instrument also included behavioral health measures to assess the presence of mental illness, substance use disorders, cognitive distortions, and sex addiction among participants. Due to the small sample size, data analysis is predominantly descriptive, although two regression models were used to further investigate bivariate findings.

Example 12.3.3⁹

DESCRIBING BOTH THE QUALITATIVE AND QUANTITATIVE METHODOLOGIES

Qualitative analysis of the interviews was accomplished using QSR NVivo 9 software. An inductive approach to thematic analysis was used to explore the data (Braun & Clarke, 2006). The transcripts were read and re-read and noteworthy aspects of the data were systematically coded. Then the coded text was organised into broad themes. Following this, the themes were reviewed, refined and named. Quantitative analyses were conducted using SPSS version 20(c) software. Data were screened and assumption violations dealt with using standard statistical practices (Tabachnick & Fidell, 2007). Multiple imputation was used to deal with missing data, as it has become the preferred method (Mackinnon, 2010; Sterne et al., 2009). Bivariate correlation analyses were performed to explore associations between parents' PA and the self-regulation variables. Where there were

8 Burgess-Proctor, A., Comartin, E. B., & Kubiak, S. P. (2017). Comparing female-and male-perpetrated child sexual abuse: A mixed-methods analysis. *Journal of Child Sexual Abuse*, 26(6), 657–676.

9 Butson, M. L., Borkoles, E., Hanlon, C., Morris, T., Romero, V., & Polman, R. (2014). Examining the role of parental self-regulation in family physical activity: A mixed-methods approach. *Psychology & Health*, 29(10), 1137–1155.

statistically significant correlations that were consistent with SCT and the TPB, multiple linear regression analyses were used to determine which self-regulation variables predicted PA measured by accelerometers and which self-regulation variables best predicted PA measured by self-report. The significance level was set at .05.

4. Are the Qualitative Results Presented in the Study Satisfactory According to Established Qualitative Standards?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: As noted in Chapter 11, there are several key issues related to presentation of qualitative results. These involve inclusion of expert and participant feedback, linking the specific application of the qualitative method to the analysis, disclosure of the researcher's background, and inclusion of quotes, descriptions, and observations. The results from the qualitative components of a mixed methods study require these factors within the mixed methods Results section. Consider Examples 12.4.1 and 12.4.2 illustrating how qualitative results can be described.

Example 12.4.1¹⁰

PRESENTING TRANSCRIPT EXCERPTS TO SUPPORT THE QUALITATIVE ANALYSIS

Given the perceived masculinity of drinking within this discourse, interviewees also expressed a belief that drinking excessively was more detrimental for a woman's perceived femininity than drinking per se (Table 1 [in the original article]). Although Sarah found it difficult to identify precisely her response to drunk women, it was clearly negative:

[3] Sarah (traditional)

It's more shocking to see someone, a woman who drinks like . . . much more than a man. So, I don't know. I guess yeah, it's much more shocking to see a woman getting drunk than a man.

What does "shocking" mean? Can you describe that more?

Mm . . . maybe not shocking but sort of . . . I don't know the word really but, if . . . if you see them and you are sort of . . . a bit, a bit repulsed, maybe . . .

And what happens when you see a man that binge drinks?

Well it's, um . . . it's the same, but in a . . . in a weird way, it's more accepted, I think.

[4] Jess (egalitarian)

I wouldn't think someone was less feminine for playing sport, playing football or something like that. But maybe if they're getting very drunk and being sick, then I don't think maybe, that isn't very feminine.

10 de Visser, R. O. & McDonnell, E. J. (2012). 'That's OK. He's a guy': A mixed-methods study of gender double-standards for alcohol use. *Psychology & Health*, 27(5), 618–639.

Example 12.4.2¹¹

INCLUDING QUOTATIONS WITHIN TEXT

The young people who participated in the qualitative component of our mixed methods study perceived health as “different for everyone.” The strength and consistency of this viewpoint was striking, and emerged between participants in individual groups and across focus groups. One participant emphasized the importance of this theme by identifying that “health is different for everyone” as the most important thing we had talked about in his focus group. Repeatedly, participants articulated that because each person is unique, each person has different needs, a different context, and different attitudes that fundamentally make their perception and experience of health customized.

One way that this theme emerged was in the way youth readily identified a diversity of behaviors, attitudes, and contexts that could be important to health in general. However, there was no consensus on what those aspects would be in a particular person. As one participant said, “Everyone has a different way of living” and so, “Different people need different things.”

5. Are the Quantitative Results Presented in the Study Satisfactory According to Established Quantitative Standards?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Chapter 10 clarifies several key issues related to the presentation of quantitative findings. These include referring to specific hypotheses or the research question, reporting appropriate statistics, supplementing tables with discussion in the narrative, providing descriptive statistics to inform inferential analyses, and overall cohesion in the presentation of quantitative results. The Results section of mixed methods research reports must adhere to these standards regarding the quantitative findings within the mixed methods design. Consider Examples 12.5.1 and 12.5.2 illustrating how quantitative results can be presented.

Example 12.5.1¹²

REFERRING BACK TO THE HYPOTHESIS WHEN DISCUSSING RESULTS

It was hypothesized that participants would show improvements between pre- and post-session measures of well-being and happiness. The study demonstrated statistically

11 Michaelson, V., Pickett, W., Vandemeer, E., Taylor, B., & Davison, C. (2016). A mixed methods study of Canadian adolescents’ perceptions of health. *International Journal of Qualitative Studies on Health and Well-being*, 11(1), 32891.

12 Paddon, H. L., Thomson, L. J. M., Menon, U., Lanceley, A. E., & Chatterjee, H. J. (2014). Mixed methods evaluation of well-being benefits derived from a heritage-in-health intervention with hospital patients. *Arts & Health*, 6(1), 24–58.

significant, overall enhancement of psychological well-being as determined by the PANAS measures, and subjective well-being and happiness as determined by the VAS measures. Positive PANAS, wellness and happiness VAS scores increased, and negative PANAS scores decreased in line with predictions, although there were no significant differences between the four patient groups. The average increase in positive mood was greater than the average decrease in negative mood supporting the view of Watson et al. (1988) that the two PANAS scales were independent and orthogonal.

Example 12.5.2¹³

USING A TABLE TO PRESENT RESULTS

Table 1 Treatment and Comparison Group Matching Characteristics Demographic Information

<i>Characteristic</i>	<i>DUI court</i>		<i>Comparison</i>		<i>Test statistic (p value)</i>
	<i>n</i>	<i>%</i>	<i>n</i>	<i>%</i>	
Sex					$\chi^2 = 0.12$
Male	390	81.3	384	81.5	(.91)
Female	90	18.8	87	18.5	
Race					$\chi^2 = 0.16$
White	437	91.0	431	91.5	(.93)
African American	24	5.0	21	4.5	
Other	19	4.0	19	4.0	
Age					$t = 1.00$
17–22	23	4.8	27	5.7	(.32)
23–38	91	19.0	115	24.4	
29–34	103	21.5	88	18.7	
35–40	78	16.3	68	14.4	
41–46	83	17.3	63	13.4	
47–52	58	12.1	67	14.2	
53+	44	9.2	43	9.1	
Risk score					$\chi^2 = 0.06$
Low	140	29.2	138	29.3	(.97)
Medium	318	66.3	313	66.5	
High	22	4.6	20	4.2	

13 Myer, A. J. & Makarios, M. D. (2017). Understanding the impact of a DUI court through treatment integrity: A mixed-methods approach. *Journal of Offender Rehabilitation*, 56(4), 252–276.

___ 6. Are the Findings of the Research Integrated/Mixed?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Mixed methods designs are conceived to render findings that extend beyond only qualitative or quantitative designs. Given the additional detail that mixed methods are designed to add, the findings of the research must likewise extend beyond the direct findings of the qualitative and quantitative components. Specifically, the results should be integrated where questions are answered through comparing and contrasting the qualitative and quantitative results. Examples 12.6.1 and 12.6.2 illustrate such integration of methods.

Example 12.6.1¹⁴

ILLUSTRATING HOW EACH METHOD HIGHLIGHTED DIFFERENT ASPECTS OF THE OVERALL FINDING

This study's TST [quantitative] and qualitative interview findings complement one another such that the TST analyses uncovered relationships between how participants tend to spontaneously describe themselves and self-stigma, while the qualitative interviews highlighted experiences of community stigma, how participants respond to these experiences, and how stigmas may influence each other. As such, each method illuminated a different aspect of stigma that may not have been captured without this approach. These TST findings may indicate that a tendency towards being self-reflective may protect against internalizing societal stigma, whereas the tendency to think of oneself in vague terms may increase risk of internalizing stigma. Thus, the tendency to be self-reflective may be a particularly important strength for individuals experiencing these three identities, which likely intersect in powerful ways to negatively impact recovery outcomes. Conversely, these three interacting stigmas may represent a particular barrier to developing more self-reflective styles of thinking, due to the negative impact of stigma on individuals' self-esteem and hopes for the future.

Example 12.6.2¹⁵

DEMONSTRATING CONSISTENCY BETWEEN QUALITATIVE AND QUANTITATIVE FINDINGS

The qualitative journal entries suggested that students experienced benefits from daily meditation such as feeling less overwhelmed, sleeping better, staying focused and feeling

14 West, M. L., Mulay, A. L., DeLuca, J. S., O'Donovan, K. & Yanos, P. T. (2018). Forensic psychiatric experiences, stigma, and self-concept: A mixed-methods study. *The Journal of Forensic Psychiatry & Psychology*, 29(4), 574–596.

15 Ramasubramanian, S. (2017). Mindfulness, stress coping and everyday resilience among emerging youth in a university setting: a mixed methods approach. *International Journal of Adolescence and Youth*, 22(3), 308–321.

happy or blissful. The emerging themes from the current analysis are consistent with prior research and applications of mindfulness (Amutio, Martinez-Taboada, Hermosilla, & Delgado, 2014; Grossman et al., 2004), giving the current data validity and indicating that across different settings, mindfulness training can achieve similar outcomes because of similar processes. Students repeatedly discussed how the mindfulness practice helped them relax, sleep better and be calmer about handling stressful situations such as upcoming exams, disappointing grades and work–life balance. These findings are reflected in the quantitative results as well.

7. Apart from Validity Issues Inherent in the Quantitative and Qualitative Components, Does the Researcher Address Validity Issues Specific to Mixed Methods Designs?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Several key validity issues emerge when using mixed methods. Among these are concerns that the data utilized in the qualitative and quantitative components might reflect different populations, that the conceptualizations of core ideas may differ between the qualitative and quantitative components, or that the mixed methods design utilized limits the types of conclusions that can be made. Consider Examples 12.7.1 and 12.7.2 discussing such limitations as mostly unique to mixed methods studies.

Example 12.7.1¹⁶

CONCLUSIONS LIMITED BY THE STUDY DESIGN

Lastly, our concurrent study design does not permit conclusions about the direction of effects between maternal and child characteristics and mothers' perspectives about the ease or difficulty of their and their child's transition. Constellations of different factors (including child, mother, nonfamilial caregiver, and situational factors) may combine to create ease or difficulty in the transition to child care. Our overall analysis suggests that in understanding the transition to nonfamilial care for infants and toddlers, it is important to consider maternal and child psychological characteristics as well as examine the social relationships and contextual factors that may converge to promote greater ease versus difficulty in the transition.

16 Swartz, R. A., Speirs, K. E., Encinger, A. J. & McElwain, N. L. (2016). A mixed methods investigation of maternal perspectives on transition experiences in early care and education. *Early Education and Development*, 27(2), 170–189.

Example 12.7.2¹⁷

DISCUSSING DISAGREEMENT ABOUT WHAT CONCLUSIONS CAN BE DRAWN FROM MIXED METHODS DESIGNS

We recognise that not all researchers will necessarily embrace the various meanings and reconciliations concerning mixed methods presented in the mixed methods literature and within our commentary. However, we hope that some of the strategies and reconciliations suggested throughout our commentary may push researchers towards expanding their thinking as to what qualitative inquiry can be (rather than what it should be) both apart from, and within, mixed methods genres of research. Indeed, in the researching and writing of this commentary, our own thinking and understanding concerning what mixed methods are and can be has expanded immeasurably. We hope to continue to grow in that respect and eventually begin to apply these new forms of knowledge in our own scholarship, teaching and mentoring. However, at the same time, we realise through researching and writing up the present commentary that we have barely scratched the surface of the myriad of issues and tensions that belie what some have termed a ‘third methodological movement’ (i.e. mixed methods) (Johnson et al. 2007, Teddlie and Tashakkori 2011) within the social sciences.

8. Do the Findings Include Consideration of Contradictory Data, Aberrant Cases or Surprising Results?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Due to inclusion of both qualitative and quantitative data and analysis, mixed methods approaches are able to uncover relationships between aggregate patterns while maintaining detail about individual units. Often, comparing between the qualitative and quantitative data uncovers surprising information. It may result from contradictions between the qualitative and quantitative data, or, in the case of consistent qualitative and quantitative findings overall, the mixed methods analysis may uncover data on aberrant cases that are inconsistent with major trends. As these data represent specific contradictory detail, it is important that they be included in the discussion of mixed methods results. Consider Example 12.8.1 which contrasts findings from the two different methods employed in a study of technology use by teachers and students.

17 McGannon, K. R. & Schweinbenz, A. N. (2011). Traversing the qualitative–quantitative divide using mixed methods: Some reflections and reconciliations for sport and exercise psychology. *Qualitative Research in Sport, Exercise and Health*, 3(3), 370–384.

Example 12.8.1¹⁸

CONTRADICTION BETWEEN QUALITATIVE AND QUANTITATIVE RESULTS

The qualitative analysis, which integrated the results of both methods, found that teachers' positive attitudes toward technology do not necessarily have the same influence on student technology use and instructional strategies that are compatible with the student-centered paradigm such as cooperative and project-based learning. These mixed methods results were contrary to those of the [quantitative] phase alone, where teachers' attitudes toward technology were found most significant for predicting student and teacher use of technology with a variety of instructional strategies. Although our survey items captured student use, teacher use, and instructional strategy use with technology, it was only through teachers' testimonies that we were able to describe how teachers had students use technology in the classroom.

9. Is the Use of Mixed Methods Justified?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Some research reports include the use of mixed methods when both qualitative and quantitative data are available. However, mixed methods studies are best when the choice of a mixed methods design reflects the needs of the study itself. Often, this choice can reflect the state of knowledge about a topic at the time the study was undertaken or nuances about the topic and measurement of its underlying key concepts that make the selection of a mixed methods approach ideal.

For example, a study on prisoner well-being might be ideally suited for a mixed methods design. Given that well-being might be difficult to measure using quantitative approaches, interviews with prisoners might render greater detail into their sense of well-being. However, inclusion of quantitative components addressing quantifiable aspects like health outcomes would provide additional detail while allowing the use of quantitative methods to discover patterns between prison features and/or conditions and prisoner well-being. The two approaches would inform each other and paint a clearer picture of prisoner well-being overall.

Given that the use of mixed methods approaches should be driven by the needs of the study itself, it is important that research reports clearly articulate the reasons that a mixed methods approach was used. Consider Examples 12.9.1 and 12.9.2 that discuss reasons for the use of mixed methods.

18 Palak, D., & Walls, R. T. (2009). Teachers' beliefs and technology practices: A mixed-methods approach. *Journal of Research on Technology in Education*, 41(4), 417–441.

Example 12.9.1¹⁹

LINKING THE METHOD TO THE ISSUE UNDER STUDY

Mixed methods studies facilitate a broader and deeper – and potentially more useful – understanding of issues by providing the benefits of different methods while compensating for some of their limitations (Tashakkori & Teddlie, 2003). Mixing methods can add experiential ‘flesh’ to statistical ‘bones’, and may be particularly useful for studying complex entities like gender which operate at both macro-social and micro-social levels. The mixed-methods approach adopted in this study was grounded in a critical realist epistemology (Bhaskar, 1989; Danermark, Ekstro, Jakobsen, & Karlson, 2002), and reflected an interest in addressing discourses and experiences via a discourse-dynamic approach to subjectivity (Willig, 2000).

Example 12.9.2²⁰

LINK THE CHOICE OF A MIXED METHODS DESIGN TO ISSUES RELATED TO MEASURING CONCEPTS

The explanatory mixed methods design (QUAN + QUAL) was followed by collecting quantitative and qualitative data sequentially across two phases (Creswell, 2002; Teddlie & Tashakkori, 2006). This mixed methods design was employed based on the empirical evidence in previous research on the relationship between teachers’ educational beliefs and their instructional technology practices: Teachers’ beliefs as a messy, ill-structured construct neither easily lends itself to empirical investigation nor entirely explains by itself how teachers are likely to use technology.

___ 10. Overall, is the Presentation of the Results Adequate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter and any additional considerations and concerns you may have. Be prepared to explain your evaluation.

-
- 19 de Visser, R. O. & McDonnell, E. J. (2012). ‘That’s OK. He’s a guy’: A mixed-methods study of gender double-standards for alcohol use. *Psychology & Health*, 27(5), 618-639.
- 20 Palak, D., & Walls, R. T. (2009). Teachers’ beliefs and technology practices: A mixed-methods approach. *Journal of Research on Technology in Education*, 41(4), 417-441.

Chapter 12 Exercises

Part A

Directions: Answer the following questions.

1. What does it mean that a study uses a *mixed methods* design?
2. How can researchers justify the use of mixed methods designs?
3. How should researchers link the *qualitative* and *quantitative* components of their mixed methods study to the research question?
4. What validity issues are unique to mixed methods designs?
5. What is the key concern when presenting results from a mixed methods study?
6. What should researchers do when the results of the qualitative component of a mixed methods study conflict with the results of the quantitative component?

Part B

Directions: Locate a mixed methods research report of interest to you.²¹ Read it, and evaluate the description of the results in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Bring it to class for discussion, and be prepared to discuss both its strengths and weaknesses.

21 Researchers who conduct this type of research often mention that it involves mixed methods in the titles or abstracts of their reports. Thus, to locate examples of mixed methods research using an electronic database, it is often advantageous to use *mixed methods* as a search term.

Evaluating Discussion Sections

The last section of a research article typically has the heading *Discussion*. However, expect to see variations such as Conclusion, Discussion and Conclusions, Discussion and Limitations, Conclusions and Implications, or Summary and Implications.

___ **1. In Long Articles, do the Researchers Briefly Summarize the Purpose and Results at the Beginning of the Discussion Section?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
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Comment: A summary at this point in a long research article reminds readers of the main focus of the research and its major findings. Often, such a summary begins by referring to the main research hypotheses, purposes, or questions addressed by the research. Example 13.1.1 shows the beginning of the first paragraph of a Discussion section that does this.

Example 13.1.1²

BEGINNING OF A DISCUSSION SECTION THAT REMINDS READERS OF THE PURPOSE OF THE RESEARCH

The aim of this study was to examine public opinion about primary schools in Turkey. According to the results of the study, the public image of these schools was below average. This result does not support the anticipated positive image of schools in Turkey. Because

1 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”

2 Ereş, F. (2011). Image of Turkish basic schools: A reflection from the province of Ankara. *The Journal of Educational Research*, 104(6), 431–441.

Turkey is a rapidly developing nation with the largest population of young people in Europe . . .

The Discussion section of a lengthy research article should also often reiterate the highlights of the findings of the study. Complex results should be summarized in order that readers be reminded of the most important findings. Example 13.1.2 shows the beginning of a Discussion section with such a summary of results. Note that specific statistics (previously reported in the Results sections of quantitative research reports) do not ordinarily need to be repeated in such a summary.

Example 13.1.2³

A SUMMARY OF FINDINGS AT THE BEGINNING OF THE DISCUSSION SECTION OF A RESEARCH ARTICLE

Our research demonstrates that racial microaggressions contribute to the race gap in adolescent offending. We show that African American middle-schoolers grapple with everyday racial microaggressions, reporting that they are called names, disrespected, and treated as intellectually inferior and dangerous on account of their race. Among our most notable findings is that one way racial microaggressions shape delinquency among Black adolescents in particular is by exacerbating the influence of general stresses on offending.

___ 2. Do the Researchers Acknowledge Specific Methodological Limitations?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Although the methodological limitations (i.e., weaknesses) may be discussed at any point in a research report, they are often discussed under the subheading Limitations within the Discussion section at the end of research reports.

The two most common types of limitations are weaknesses in measurement (i.e., observation or instrumentation) and weaknesses in sampling.

Examples 13.2.1, 13.2.2, and 13.2.3 show portions of descriptions of limitations that appeared in Discussion sections. Note that these limitations are important considerations in assessing the validity of the results of the studies.

3 De Coster, S., & Thompson, M. S. (2017). Race and general strain theory: Microaggressions as mundane extreme environmental stresses. *Justice Quarterly*, 34(5), 903–930.

Example 13.2.1⁴

ACKNOWLEDGMENT OF LIMITATIONS OF SAMPLING AND MEASURES IN A DISCUSSION SECTION

These survey findings, of course, have numerous limitations, the most important one being that the findings are based on one school in one community and, thus, are not representative of other rural communities. Moreover, despite the reassurance of confidentiality, students might not have felt secure enough to tell the truth about their drug use and therefore might have minimized their use. Finally, as indicated in the literature, young people who have a drug problem, such as the use of methamphetamines, are likely to drop out and not be found among the high school student population.

Example 13.2.2⁵

ACKNOWLEDGMENT OF LIMITATIONS OF SAMPLING AND MEASURES IN A DISCUSSION SECTION

Finally, the limitations of this study should be noted. First, the sample size in this study was small. Future studies should examine a larger sample in order to enhance the statistical power of the results. Second, we relied on self-reported scales to assess interpersonal stress . . . an alternative method, such as interviews, may yield a more objective assessment. Third, because the current study used a community sample of adolescents and did not examine clinically depressed adolescents, we must be cautious about generalizing the present findings to clinical samples.

Example 13.2.3⁶

ACKNOWLEDGMENT OF LIMITATIONS OF RESEARCH DESIGN IN A DISCUSSION SECTION

There are several limitations to the generalizability and validity of the conclusions that can be drawn from this study. First, other variables that were not included in the present models may be better predictors of mathematics growth or may explain the observed relationships among the included variables and mathematics growth. Most important, because this was a correlational study, it is impossible to draw causal inferences from the

- 4 Mitchell, J., & Schmidt, G. (2011). The importance of local research for policy and practice: A rural Canadian study. *Journal of Social Work Practice in the Addictions*, 11(2), 150–162.
- 5 Kuroda, Y., & Sakurai, S. (2011). Social goal orientations, interpersonal stress, and depressive symptoms among early adolescents in Japan: A test of the diathesis-stress model using the trichotomous framework of social goal orientations. *Journal of Early Adolescence*, 31(2), 300–322.
- 6 Judge, S., & Watson, S. M. R. (2011). Longitudinal outcomes for mathematics achievement for students with learning disabilities. *The Journal of Educational Research*, 104(3), 147–157.

results of the study. Therefore, any student effects reported in this study are correlational in nature, and manipulation of the variables used in this study may or may not produce similar results.

In Example 13.2.4, the researchers discuss the strengths of their study before discussing its limitations. This is especially appropriate when the study has special strengths to be pointed out to the readers.

Example 13.2.4⁷

LIMITATIONS DISCUSSED AFTER STRENGTHS ARE DESCRIBED

The study design is a strength. It utilized a national panel study with 2-year follow-ups spanning 8 years. With it we were able to examine report stability for use, age of onset, and logical consistency for the same youths. Furthermore, this is the first study to examine such measures of stability for marijuana use across nearly a decade of self-reported use. However, although marijuana use is illicit, the findings here would likely vary greatly from that of other illicit drug self-reports.

One limitation of this study is that the phrasing of the ever-use questions changed slightly during 1–2 survey years. These changes could have affected . . .

3. Are the Results Discussed in Terms of the Literature Cited in the Introduction?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The literature cited in the introduction sets the stage for the research. Thus, it is important to describe how the results of the current study relate to the literature cited at the beginning of the research report. The idea is that the current study adds to the existing literature and thus enriches and expands the state of knowledge on the topic.

Researchers might address issues such as the following:

- Are the results consistent with those previously reported in the literature? Or with only some of them? Or with none of them?
- Does the study fill a specific gap in the literature? What does it add?

These are important issues to consider when drawing conclusions from a particular study. For instance, if the results of a study being evaluated are inconsistent with the results of a

7 Shillington, A. M., Clapp, J. D., & Reed, M. B. (2011). The stability of self-reported marijuana use across eight years of the National Longitudinal Survey of Youth. *Journal of Child & Adolescent Substance Abuse*, 20(5), 407–420.

large number of other studies in the literature, the researcher should discuss this discrepancy and speculate on why his or her study is inconsistent with earlier ones. Examples 13.3.1 and 13.3.2 illustrate how some researchers refer to previously cited literature in their Discussion sections.

Example 13.3.1⁸

DISCUSSION IN TERMS OF LITERATURE MENTIONED IN THE INTRODUCTION

The present study provides results that are consistent with previous research. First, quizzes increased attendance (Azorlosa & Renner, 2006; Hovell et al., 1979; Wilder et al., 2001) and second, they increased self-reported studying (Azorlosa & Renner, 2006; Marchant, 2002; Ruscio, 2001; Wilder et al., 2001).

Example 13.3.2⁹

DISCUSSION IN TERMS OF LITERATURE MENTIONED IN THE INTRODUCTION

The univariate findings of the present study were consistent with those of researchers (Ackerman, Brown, & Izard, 2004) who have found that family instability (i.e., cohabiting with multiple partners over a 3-year period of time) is associated with poorer outcomes for children, compared with children whose mothers get married. I did not find, however, that cohabitation with multiple partners was significantly associated with child literacy in the multivariate analyses.

4. Have the Researchers Avoided Citing New References in the Discussion Section?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The relevant literature should be first cited in the introduction. The literature referred to in the Discussion section should be limited to that originally cited in the introduction.

Even though this is a good general rule to follow, there are exceptions. If a study finds something very unusual or unexpected, such findings could merit mentioning additional literature sources in the Discussion section. For example, sometimes the Results section includes a description of unexpected but interesting tangential findings that were clearly not a part of the original study hypotheses or goals, and it may be appropriate to include some elaboration and new citations in the Discussion to interpret these unexpected findings and put them within a proper framework.

8 Azorlosa, J. L. (2011). The effect of announced quizzes on exam performance: II. *Journal of Instructional Psychology*, 38(1), 3–7.

9 Fagan, J. (2011). Effect on preschoolers' literacy when never-married mothers get married. *Journal of Marriage and Family*, 73(5), 1001–1014.

Thus, interpret this evaluation question judiciously, taking into account whether there are good reasons for the new references.

___ 5. Are Specific Implications Discussed?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Research often has implications for practicing professionals. When this is the case, a statement of implications should specifically describe, whenever possible, what a person, group, or institution should do on the basis of the results of the current study. Consumers of research will want to know what the researchers (who are experts on the topic) think the implications are. Examples 13.5.1 and 13.5.2 illustrate how practical implications can be drawn from research results and presented in a Discussion section.

Example 13.5.1¹⁰

A STATEMENT ABOUT IMPLICATIONS FOLLOWING A GENERAL RE-STATEMENT OF STUDY'S FINDINGS

Overall, our study indicates that 1-year-old toddlers undergo a dramatic and painful transition when adapting to childcare. All the observed children demonstrated signs of distress, compatible with the phases of separation anxiety. Although the study is small, it points to a need to discuss how separation anxiety among toddlers in day care is handled. Longer and more flexible adaption time, shorter days and better staffing, especially in the early mornings and late afternoons, appear to be important measures to implement.

Example 13.5.2¹¹

A STATEMENT OF SPECIFIC IMPLICATIONS

The results of this study offer important implications for counselor education. We found that stereotypes related to race-ethnicity and gender do exist among individuals working toward licensure as a professional counselor. While it should be acknowledged that the existence of stereotypes does not automatically lead to discrimination against the stereotyped groups, if care is not exercised, then these stereotypes could easily guide someone's behavior and lead to discrimination. It is especially critical to avoid this in the counseling field, as clients require understanding and skillful counselors to help them when they are experiencing difficulties. Therefore, it is important that education about stereotypes and bias be consistently and thoroughly pursued in programs educating future counselors.

10 Klette, T., & Killén, K. (2018). Painful transitions: A study of 1-year-old toddlers' reactions to separation and reunion with their mothers after 1 month in childcare. *Early Child Development and Care* [online first].

11 Poyrazli, S., & Hand, D. B. (2011). Using drawings to facilitate multicultural competency development. *Journal of Instructional Psychology*, 38(2), 93–104.

Some studies have wider implications for policy and practice that are applicable at a local, national, and sometimes even international level. Examples 13.5.2 and 13.5.3 refer to such policy implications. (More information on systematic reviews and meta-analyses with implications for evidence-based practice and policy is provided in the next chapter – Chapter 14.)

Example 13.5.3¹²

A STATEMENT OF POLICY IMPLICATIONS

Our findings demonstrate that public transportation in an urban area serves as an efficient media vehicle by which alcohol advertisers can heavily expose school-aged youths and low-income groups. In light of the health risks associated with drinking among youths and low-income populations, as well as the established link between alcohol consumption among both youths and adults, the state of Massachusetts should consider eliminating alcohol advertising on its public transit system.

Other cities and states that allow alcohol advertising on their public transit systems should also consider eliminating this advertising to protect vulnerable populations, including underage students, from potentially extensive exposure.

Example 13.5.4¹³

A STATEMENT OF POLICY IMPLICATIONS

This study has important policy implications for interventions designed for adolescents with depressive symptomatology. In fact, interventions based on altering normative beliefs, which aim to correct erroneous perceptions about substance use, have shown success (see Hansen and Graham 1991). Specifically, our results indicate that adolescents with depressive symptomatology may be more likely to misuse alcohol (binge drink) because they misperceive how normative alcohol use is amongst their friends. Thus, normative beliefs-based interventions could be adapted specifically for adolescents with depressive symptomatology by taking into account the different attributional styles of depressed adolescents. If prevention programs specifically designed for adolescents with depression are able to correct misperceptions about alcohol usage and establish pro-social normative beliefs, this may be the key to preventing adolescents with depressive symptomatology from engaging in binge drinking.

12 Gentry, E., Poirier, K., Wilkinson, T., Nhean, S., Nyborn, J., & Siegel, M. (2011). Alcohol advertising at Boston subway stations: An assessment of exposure by race and socioeconomic status. *American Journal of Public Health, 101*(10), 1936–1941.

13 Harris, M. N., & Teasdale, B. (2017). The indirect effects of social network characteristics and normative beliefs in the association between adolescent depressive symptomatology and binge drinking. *Deviant Behavior, 38*(9), 1074–1088.

___ 6. Are the Results Discussed in Terms of any Relevant Theories?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: As indicated in earlier chapters, research that tests and/or develops theories is often important because theories provide the basis for numerous predictions and implications. If a study was introduced as theory-driven (or clearly based on certain theoretical considerations), it is appropriate to describe how the current results affect interpretation of the theory in the Discussion section at the end of the research article.

Example 13.6.1 is from the beginning of a discussion section in a study based on general strain theory.

Example 13.6.1¹⁴

DISCUSSION SECTION POINTING OUT RELATION TO THEORY

The results of this study partially support the more traditional viewpoints of general strain theory. On the one hand, while general strain theory predicts that stress, affective states, and coping will be significant predictors of deviance, these variables were not significant in our study. On the other hand, in line with general strain theory, we found that the removal of positive stimuli was a significant predictor of deviance. It is worth noting, however, this strain variable did not have the same power and influence as opportunity or peers. For this sample, the strongest predictor of criminal activity was respondents viewing crime as an opportunity and peer involvement in crime. Essentially, in the college environment respondents were more likely to commit acts of deviance when their friends implicitly supported the behavior and as opportunities presented themselves.

___ 7. Are Suggestions for Future Research Specific?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: It is uninformative for researchers to conclude with a simple phrase such as “more research is needed.” To be helpful, researchers should point to specific areas and research procedures that might be fruitful in future research. This is illustrated in Example 13.7.1.

14 Huck, J. L., Spraitz, J. D., Bowers Jr, J. H., & Morris, C. S. (2017). Connecting opportunity and strain to understand deviant behavior: A test of general strain theory. *Deviant Behavior*, 38(9), 1009–1026.

Example 13.7.1¹⁵

SPECIFIC SUGGESTIONS FOR FUTURE RESEARCH IN A DISCUSSION SECTION

[The] current study did not examine how different types of support (e.g., emotional and instrumental) may influence the relations between depression, peer victimization, and social support. Thus, future studies should examine how a combination of source and type of social support (e.g., emotional support from parents) may influence relations between stressors and outcomes.

Often, the suggestions for future research indicate how future studies can overcome the limitations in the current study. This is illustrated in Example 13.7.2.

Example 13.7.2¹⁶

SPECIFIC SUGGESTIONS FOR FUTURE RESEARCH IN VIEW OF CURRENT STUDY'S LIMITATIONS

There are several limitations to this study that also suggest directions for future research. First, all measures were completed by a single reporter, with no objective verification of sleep patterns and sleep disruptions. Future studies should include an objective measure of sleep patterns (e.g., actigraphy) and maternal functioning (e.g., missed days of work due to fatigue or sleepiness). Second, whereas this study highlights the relationship between child sleep disruptions and maternal sleep and functioning, future studies should include additional family focused variables, as disrupted child sleep likely affects all members of the family. For example, parents often disagree on how to handle child night wakings, which could negatively impact marital quality. Alternatively, a mother who is fatigued due to the disrupted sleep of one child may lack the energy to effectively parent other children. Finally, this study was limited by the relatively homogeneous sample, which favored educated Caucasian women. Future studies should continue to examine how children's sleep disturbances impact sleep and functioning in a more diverse sample, as well as include fathers and siblings.

8. Have the Researchers Distinguished between Speculation and Data-based Conclusions?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: It is acceptable for researchers to speculate in the Discussion section (e.g., what the results might have been if the methodology had been different). However, it is important that

15 Tanigawa, D., Furlong, M. J., Felix, E. D., & Sharkey, J. D. (2011). The protective role of perceived social support against the manifestation of depressive symptoms in peer victims. *Journal of School Violence, 10*(4), 393–412.

16 Meltzer, L. J., & Mindell, J. A. (2007). Relationship between child sleep disturbances and maternal sleep, mood, and parenting stress: A pilot study. *Journal of Family Psychology, 21*(1), 67–73.

researchers clearly distinguish between their speculation and the conclusions that can be justified by the data they have gathered. This can be done with some simple wording such as “It is interesting to speculate on the reasons for . . .”

9. Overall, is the Discussion Section Effective and Appropriate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	<i>or</i> N/A	I/I
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Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter, taking into account any additional considerations and concerns you may have.

Chapter 13 Exercises

Part A

Directions: Answer the following questions.

1. The methodological weaknesses of a study are sometimes discussed under what subheading?
2. What are the two most common types of limitations?
3. Is it ever appropriate to mention literature that was cited earlier in a research article *again* in the Discussion section at the end of a research article? Explain.
4. Suppose the entire statement of implications at the end of a research article is “Educators should pay more attention to students’ needs.” In your opinion, is this sufficiently specific? Explain.
5. Suppose this is the entire suggestion for future research stated at the end of a research article: “Due to the less-than-definitive nature of the current research, future research is needed on the effects of negative political campaign advertisements.” In your opinion, is this sufficiently specific? Explain.
6. Is it acceptable for researchers to speculate in the Discussion section of their research reports? Explain.

Part B

Directions: Locate several research reports of interest to you in academic journals. Read them, and evaluate the Discussion sections in light of the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Select the one to which you gave the highest overall rating, and bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

Evaluating Systematic Reviews and Meta-Analyses: Towards Evidence-Based Practice

Systematic reviews and meta-analyses are a distinct type of empirical studies – they use other, original empirical studies as their “sample,” to summarize their findings (i.e., evidence) related to a particular topic or intervention. The idea behind a systematic review is to make sure that an analysis of empirical literature on a specific topic is as comprehensive and unbiased as possible: it uses a deliberate and precise search strategy, includes all relevant studies meeting specific criteria, and takes their features and methods into account when summarizing their findings. For example, if we are interested in whether family therapy interventions for juvenile delinquents prevent further involvement in crime, a systematic review of all relevant empirical studies on such interventions would be very helpful, especially if it summarizes their results by giving more weight to the findings of more rigorous studies (the ones with random assignment to treatment and control groups,¹ larger samples, and longer follow-up periods for tracking recidivism outcomes).

Meta-analyses go a step further: besides including all relevant studies on a specific topic, researchers summarize the key results not just in a narrative fashion (this is what a systematic review does) but also by calculating an average size of the relationship between two variables (or an average difference in outcomes of an intervention) as a numerical result, often expressed as an *effect size*, across all studies included in the meta-analysis.² Other summary statistics besides effect size could be used³ but the attractiveness of the effect size estimate is its easy interpretation (it is often expressed similarly to a correction coefficient).

Using the same example about family therapy for troubled youths, we might want to know how much more effective family therapy is compared to other options, for example, compared to probation or community service in a control group (often called “treatment as usual” if it is

1 As you may recall from Chapter 9, random assignment to treatment and control groups is a key feature of a *true experiment*, which is also called a *randomized controlled trial*.

2 There is also a method of meta-synthesis, which is a counterpart to meta-analysis for summarizing the results of qualitative studies. But since its methods and procedures differ a lot from those employed in systematic reviews and meta-analyses and because the development of meta-synthesis as a type of research is still in its infancy, meta-synthesis is not covered in this text.

3 Besides effect sizes, other common summary statistics in meta-analyses include odds ratios (or hazard ratios, or relative risk ratios), as well as the mean difference or standardized mean difference (SMD).

a standard approach for this type of delinquent). In a meta-analysis, researchers would calculate the average difference in outcomes (in this example, recidivism) between the treatment and control groups, to help us understand not only how effective a specific intervention is (in this case, family therapy) but also how much *more* effective it is than the alternative approach. For example, if across all included studies with random assignment to treatment (family therapy) and control (probation) groups, 33% of juvenile offenders on average recidivate in the family therapy group and 55% of offenders recidivate while on probation within a year, the 22% difference would be the basis for expressing the effectiveness of family therapy numerically (the effect size can be calculated by taking into account the group sizes and standard deviations).

Thus, you can see how such systematic reviews and numerical summaries are especially suitable for providing comprehensive *evidence base* about interventions and practices.⁴ *Evidence-based practice* is a popular term but what makes a specific practice or intervention evidence-based is significant evidence of its effectiveness derived from systematic reviews and/or meta-analyses. This chapter outlines some important criteria for evaluating various components of a systematic review or meta-analysis in terms of their quality.

1. Have the Researchers Clearly Formulated Their Research Question or Hypothesis?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ⁵
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Comment: Just like with a research report of an original empirical study, it is very important that a researcher formulates a clear objective of the study, expressed as a research question or hypothesis (or a set of research questions and hypotheses). Often, the type of hypothesis would be related to “Does such-and-such intervention work?” but other research questions are possible as well⁶ (for example, estimates of prevalence for a certain condition). Example 14.1.1 illustrates some research questions that can be found in systematic reviews and meta-analyses across a range of social science disciplines.

- 4 Such interventions can refer to various treatments in medical and health sciences; teaching strategies and pedagogical tools in education; psychological interventions in psychology; policy changes or implementations in political science, sociology, and public health; crime/recidivism prevention programs and policing strategies in criminal justice; and so on. At the same time, other research questions can be addressed using systematic reviews and meta-analyses: for example, the evidence in support of a specific theory can be summarized or an average incidence of a specific condition in a population can be calculated from multiple studies.
- 5 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”
- 6 If you are interested in the types of research questions/topics that systematic reviews may address, a useful typology is provided in this article (mostly related to health sciences but still useful as a guide for other disciplines): Munn, Z., Stern, C., Aromataris, E., Lockwood, C., & Jordan, Z. (2018). What kind of systematic review should I conduct? A proposed typology and guidance for systematic reviewers in the medical and health sciences. *BMC Medical Research Methodology*, 18(1), 5.

Example 14.1.1

EXAMPLES OF STATEMENTS CLEARLY IDENTIFYING THE RESEARCH QUESTION IN A SYSTEMATIC REVIEW OR META-ANALYSIS

- (a) The primary question is whether counseling/psychotherapy is more effective in reducing symptoms of anxiety in school-age youth than control or comparison conditions.⁷
- (b) [I]t is the purpose of the current study to examine the overall positive and negative influences of violent video game playing in regards to aggression and visuospatial cognition in order to better understand the overall impact of these games on child and adolescent development.⁸
- (c) The purpose of this study was to systematically review the literature to examine the excess mortality rate of people with mental disorders, extending existing reviews of individual disorders. We sought to provide comprehensive estimates of individual- and population-level mortality rates related to mental disorders.⁹
- (d) In this systematic review and meta-analysis, we aimed to combine data from all published large-scale blood pressure lowering trials to quantify the effects of blood pressure reduction on cardiovascular outcomes and death across various baseline blood pressure levels, major comorbidities, and different pharmacological interventions.¹⁰
- (e) [O]ur primary objective in this article is to establish whether across the body of existing literature there is a substantively meaningful association between MCS [maternal cigarette smoking during pregnancy] and criminal/deviant behavior [of offspring].¹¹

2. Do the Researchers Explain in Detail How They Systematically Searched for Relevant Studies?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Since the most important feature of a systematic review or meta-analysis is considering all of the relevant studies for inclusion, it is especially critical that the researchers explain *how* and *where* they searched for these studies.

7 Erford, B., Kress, V., Giguere, M., Cieri, D., & Erford, B. (2015). Meta-analysis: Counseling outcomes for youth with anxiety disorders. *Journal of Mental Health Counseling*, 37(1), 63–94.

8 Ferguson, C. J. (2007). The good, the bad and the ugly: A meta-analytic review of positive and negative effects of violent video games. *Psychiatric Quarterly*, 78(4), 309–316.

9 Walker, E. R., McGee, R. E., & Druss, B. G. (2015). Mortality in mental disorders and global disease burden implications: A systematic review and meta-analysis. *JAMA Psychiatry*, 72(4), 334–341.

10 Ettehad, D., Emdin, C. A., Kiran, A., Anderson, S. G., Callender, T., Emberson, J., . . . & Rahimi, K. (2016). Blood pressure lowering for prevention of cardiovascular disease and death: A systematic review and meta-analysis. *The Lancet*, 387(10022), 957–967.

11 Pratt, T. C., McGloin, J. M., & Fearn, N. E. (2006). Maternal cigarette smoking during pregnancy and criminal/deviant behavior: A meta-analysis. *International Journal of Offender Therapy and Comparative Criminology*, 50(6), 672–690.

- Which databases did they comb through?
- Which keywords did they use in their searches?
- Did they look only for articles published within a certain time period?
- Did the search only include articles written in English or were other languages considered as well?
- Did the search only target studies published in peer-reviewed journals or were other sources included?¹²

Examples 14.2.1, 14.2.2, and 14.2.3 include descriptions of several different search strategies typical for systematic reviews.

Example 14.2.1¹³

A COMPREHENSIVE SEARCH STRATEGY INVOLVING MULTIPLE METHODS¹⁴

We used several strategies to perform an exhaustive search for literature fitting the eligibility criteria. First, a key word search was performed on an array of online abstract databases. Second, we reviewed the bibliographies of four past reviews of early family/parent training programs (Bernazzani et al. 2001; Farrington and Welsh 2007; Mrazek and Brown 1999; Tremblay et al. 1999). Third, we performed forward searches for works that had cited seminal studies in this area. Fourth, we performed hand searches of leading journals in the field. Fifth, we searched the publications of several research and professional agencies. Sixth, after finishing the searches and reviewing the studies as described later, we e-mailed the list to leading scholars knowledgeable in the specific area. These experts referred us to studies that we might have missed, particularly unpublished pieces such as dissertations. Finally, we consulted with an information specialist at the outset of our review and at points along the way to ensure that we had used appropriate search strategies.

Example 14.2.2¹⁵

A COMPREHENSIVE SEARCH STRATEGY INCLUDING ARTICLES PUBLISHED IN OTHER LANGUAGES

We identified publications estimating the prevalence of psychotic disorders (including psychosis, schizophrenia, schizophreniform disorders, manic episodes) and major depression

- 12 Typically, searches for sources other than peer-reviewed publications include what is called *grey literature* such as technical reports by agencies, government documents, and working papers. In addition, experts who are known to conduct relevant studies may be contacted to solicit information on unpublished works. For medical trials, researchers may also search trial registries like www.clinicaltrials.gov (maintained by the U.S. National Library of Medicine and containing over 250,000 ongoing and completed studies in over 200 countries, with new clinical trials being entered on a daily basis).
- 13 Piquero, A. R., Farrington, D. P., Welsh, B. C., Tremblay, R., & Jennings, W. G. (2009). Effects of early family/parent training programs on antisocial behavior and delinquency. *Journal of Experimental Criminology*, 5(2), 83–120.
- 14 The following excerpt originally includes multiple footnotes with the lists of specific databases searched and keywords used, as well as other details of the search. These footnotes have not been included here to save space.
- 15 Fazel, S., & Seewald, K. (2012). Severe mental illness in 33,588 prisoners worldwide: systematic review and meta-regression analysis. *The British Journal of Psychiatry*, 200(5), 364–373.

among prisoners that were published between 1 January 1966 and 31 December 2010. [...] we used the following databases: PsycINFO, Global Health, MEDLINE, Web of Science, PubMed, National Criminal Justice Reference Service, EMBASE, OpenSIGLE, SCOPUS, Google Scholar, scanned references and corresponded with experts in the field [...]. Key words used for the database search were the following: mental*, psych*, prevalence, disorder, prison*, inmate, jail, and also combinations of those. Non-English language articles were translated. We followed PRISMA¹⁶ [Preferred Reporting Items for Systematic Reviews and Meta-analyses] criteria.

Example 14.2.3¹⁷

A COMPREHENSIVE SEARCH STRATEGY SPECIFICALLY TARGETING THE INCLUSION OF UNPUBLISHED STUDIES

We conducted a comprehensive search for empirical research regarding the relationships between anger and aggressive driving. In order to do so, three recommended procedures were used to retrieve both published and unpublished studies on this focus. First, we conducted a computerised literature search of all relevant empirical articles published in journals indexed in the Psycinfo and ProQuest Dissertations & Theses databases using keywords such as: “trait anger”, “driving anger,” “aggressive driving”, “driving”, “aggressive drivers”, and “anger”. The search was limited to English language articles. Secondly, for all dissertation abstracts that were identified through the first search method, we attempted to obtain copies of the complete unpublished document. Thirdly, to gain access to additional unpublished studies, we directly contacted approximately 20 relevant researchers through email. In addition, we reviewed the references of all relevant manuscripts and we searched the table of contents of key journals in the field of transportation research to ensure that we had not missed other studies on this topic.

3. Have the Researchers Clearly Identified Their Criteria for Including or Excluding the Studies Produced by the Search?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: A comprehensive search to identify relevant studies is very important but it is just the first step. The second step is just as important: a well-developed and clear strategy for

16 PRISMA, or Preferred Reporting Items for Systematic Reviews and Meta-analyses, is a common acronym used in systematic reviews (especially in medical sciences) and refers to comprehensive reporting of the process and results of a systematic review and meta-analysis. A PRISMA-recommended flow diagram for the process of search and selection (inclusion/exclusion) of relevant studies is presented in Example 14.4.1. For more information about PRISMA, see Shamseer, L., Moher, D., Clarke, M., Gherzi, D., Liberati, A., Petticrew, M., . . . & Stewart, L. A. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: Elaboration and explanation. *BMJ: British Medical Journal (Online)*, 349, g7647.

17 Bogdan, S. R., Măirean, C., & Havarneanu, C. E. (2016). A meta-analysis of the association between anger and aggressive driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 42, 350–364.

deciding which among these studies should be included in the systematic review or meta-analysis and which ones should be excluded. A clearly described protocol for study selection should be provided by the researchers in the report and sometimes is registered by the researchers in advance, before the study takes place (to eliminate the possibility of changing it in response to how the search and selection shapes up).

Example 14.3.1 illustrates the list of criteria used for selecting which studies to include in a systematic review and meta-analysis of research literature evaluating whether people with schizophrenia have an increased risk for violence.

Example 14.3.1¹⁸

INCLUSION AND EXCLUSION CRITERIA ARE DESCRIBED CLEARLY

Our inclusion criteria included case-control studies (including cross-sectional surveys) and cohort studies, which allowed an estimation of the risk of violence in patients with schizophrenia and/or other psychoses compared with a general population comparison group.

Reports were excluded if: (i) Data were presented solely on all convictions not broken down for violence. (ii) There was no general population comparison data. Studies that used other psychiatric diagnoses as the comparator group were also excluded. (iii) Data were superseded by subsequent work and inclusion would involve duplication of data. [. . .] (iv) The cases included diagnoses of nonpsychotic illnesses such as personality disorder and major depression. However, we included one study where the proportion of psychoses was 95%.

We conducted a separate analysis of homicide only studies. For this analysis, studies were excluded if information on controls was taken from a different country and another time period or no data on controls were provided.

In Example 14.3.2, the authors are interested in whether mentoring programs reduce delinquency among at-risk youths. The researchers are very deliberate in describing the specific details of study methodology that would make a study either eligible or ineligible for inclusion in their meta-analysis.

Example 14.3.2¹⁹

STUDY INCLUSION/EXCLUSION CRITERIA ARE EXPLAINED IN DETAIL, WITH SPECIAL ATTENTION TO STUDY METHODOLOGY

Another criterion for inclusion in this review was that the study design involves a comparison that contrasted an intervention condition involving mentoring with a control

18 Fazel, S., Gulati, G., Linsell, L., Geddes, J. R., & Grann, M. (2009). Schizophrenia and violence: Systematic review and meta-analysis. *PLoS Medicine*, 6(8), e1000120.

19 Tolan, P. H., Henry, D. B., Schoeny, M. S., Lovegrove, P., & Nichols, E. (2014). Mentoring programs to affect delinquency and associated outcomes of youth at risk: A comprehensive meta-analytic review. *Journal of Experimental Criminology*, 10(2), 179–206.

condition. Control conditions could be “no treatment,” “waiting list,” “treatment as usual,” or “placebo treatment.” To ensure comparability across studies, we made an a priori rule to not include comparisons to another experimental or actively applied intervention beyond treatment as usual. However, there were no such cases among the studies otherwise meeting criteria for inclusion.

We coded studies according to whether they were experimental or quasi-experimental designs. To qualify as experimental or quasi-experimental for the purposes of this review, we required each study to meet at least one of three criteria: (1) Random assignment of subjects to treatment and control conditions or assignment by a procedure plausibly equivalent to randomization; (2) individual subjects in the treatment and control conditions were prospectively matched on pretest variables and/or other relevant personal and demographic characteristics; and (3) use of a comparison group with demonstrated retrospective pretest equivalence on the outcome variables and demographic characteristics as described below.

Randomized controlled trials that met the above conditions were clearly eligible for inclusion in the review. Single-group pretest-post-test designs (studies in which the effects of treatment are examined by comparing measures taken before treatment to measures taken after treatment on a single subject sample) were never eligible. A few nonequivalent comparison group designs (studies in which treatment and control groups were compared even though the research subjects were not randomly assigned to those groups) were included. Such studies were only included if they matched treatment and control groups prior to treatment on at least one recognized risk variable for delinquency, had pretest measures for outcomes on which the treatment and control groups were compared and had no evidence of group non-equivalence. We required that non-randomized quasi-experimental studies employed pre-treatment measures of delinquent, criminal, or antisocial behavior, or significant risk factors for such behavior, that were reported in a form that permitted assessment of the initial equivalence of the treatment and control groups on those variables.

Notice that if specific criteria for study inclusion or exclusion from the analysis are not clearly listed or outlined in the article, then you should give a low mark on this evaluation question.

___ **4. Are There Enough Studies Included in the Final Sample for Analysis?**

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Since the most important part of a systematic review is a highly structured and explicit search and selection process, often requiring the inclusion of only those studies that have a rigorous methodology (for example, only randomized controlled trials would be considered for inclusion), it should be no surprise that some systematic reviews end up with very few studies meeting the researchers’ criteria. In the famous Cochrane Library²⁰ – one of the most

20 See more information about the Cochrane Library and relevant links in the online resources for the chapter.

comprehensive online collections of rigorous systematic reviews on health care and medical interventions – there are thousands of reviews with just 2 or 3 studies included, and even hundreds of reviews with zero included studies²¹ (no studies have apparently met the criteria for inclusion)!

At the same time, it is clear that making any sort of generalizations based on just a handful of studies is less convincing than gathering evidence from dozens of well-done empirical studies. This is especially important for meta-analyses since compiling numerical averages for just a few studies does not make much sense.

Thus, when answering this evaluation question, give higher marks to reviews and meta-analyses that include at least 10 studies, and highest marks to reviews that include over 20 studies²². Such reviews clearly provide a more solid evidence base, especially if the included studies are scientifically rigorous and have larger samples.²³

Example 14.4.1 presents a brief description and a flow diagram with explanations for how the final selection of studies is arrived at, after the inclusion and exclusion criteria have been applied. The researchers have set out to summarize the results of sex education and HIV prevention across a range of developing countries.

Example 14.4.1²⁴

AN EXAMPLE OF A SYSTEMATIC REVIEW AND META-ANALYSIS THAT INCLUDES A VERY RESPECTABLE NUMBER OF STUDIES, WITH THE STUDY SELECTION PROCESS MAPPED ON A FLOW DIAGRAM

Of 6191 studies initially identified, 64 studies in 63 articles met the inclusion criteria for this review (Figure 1). In five cases, more than one article presented data from the same study. If articles from the same study presented different outcomes or follow-up times, both articles were retained and included in the review as one study. If both articles presented similar data, such as by providing an update with longer follow-up, the most recent article or the article with the largest sample size was chosen for inclusion.

[See Figure 14.4.1, p. 172.]

- 21 These are often referred to as *zombie reviews* or *empty reviews*. For more information, see this article: Yaffe, J., Montgomery, P., Hopewell, S., & Shepard, L. D. (2012). Empty reviews: A description and consideration of Cochrane systematic reviews with no included studies. *PLoS One*, 7(5), e36626.
- 22 This guideline is a rule of thumb that has been developed by the second author of this textbook (Maria Tcherni-Buzzeo) based on her subjective interpretation of research literature that emerged from carefully reading hundreds of systematic reviews and meta-analyses. No specific guidelines in research literature have been found on what number of studies included into a systematic review can be considered either *sufficient* or *substantial*.
- 23 At the same time, researchers often have to make trade-offs between the number of studies and their quality when deciding which studies to include: methodologically weaker studies are more numerous but evidence based on such studies is less convincing.
- 24 Fonner, V. A., Armstrong, K. S., Kennedy, C. E., O'Reilly, K. R., & Sweat, M. D. (2014). School based sex education and HIV prevention in low- and middle-income countries: A systematic review and meta-analysis. *PloS One*, 9(3), e89692.

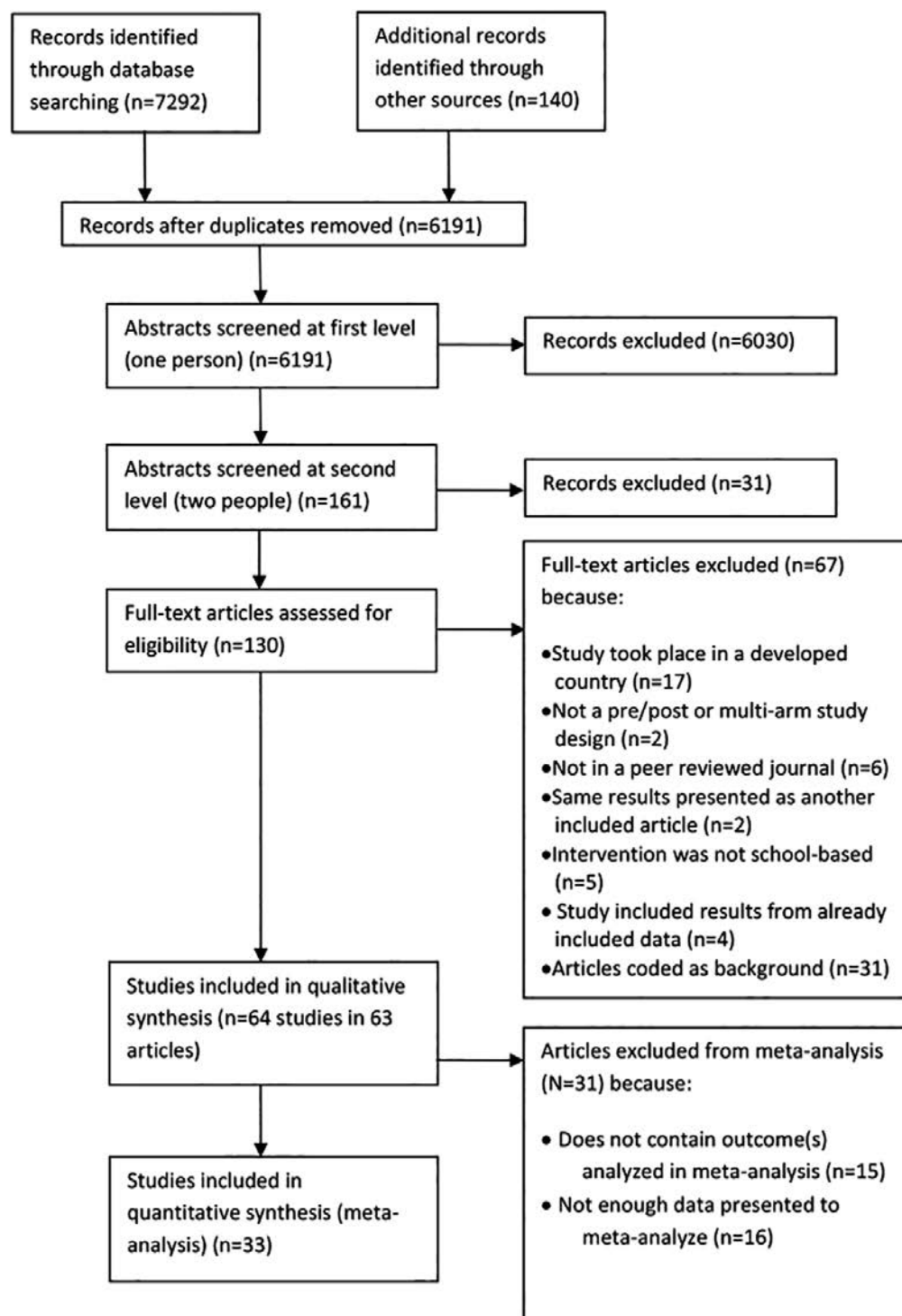


Figure 14.4.1²⁵ Disposition of Citations During the Search and Screening Process.

Source: Fonner et al., 2014 (doi:10.1371/journal.pone.0089692.g001)

25 Source: Figure 1 in Fonner et al., 2014 (doi:10.1371/journal.pone.0089692.g001).

5. Have the Researchers Addressed the Issue of Heterogeneity among the Included Studies?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: The previous evaluation question asked about the number of studies included in a systematic review or meta-analysis. However, it is not enough for the researchers to include enough studies in their analysis. It is also essential that the researchers evaluate the included studies and classify them on any important dimensions related to study design and other characteristics.

For example, a systematic review of interventions for sex offenders may include a wide range of studies with different types of treatments, administered in different settings (some in hospitals, some in prisons, and other ones – in the community), to different types of sex offenders (some study samples included only rapists, others – only child molesters, still others – both groups), using different types of design (some studies included random assignment to treatment and control groups, others – before-and-after comparisons of non-randomly-assigned groups)²⁶. Such variability among the included studies is referred to as *heterogeneity*, which can lead to “comparing apples to oranges.”

Heterogeneity in the included studies is often specifically measured in a meta-analysis. A high heterogeneity may mean that studies need to be subdivided into groups and can only be meaningfully compared and summarized within those groups. Example 14.5.1 discusses some standard ways of calculating heterogeneity.

Example 14.5.1²⁷

THE REASONING BEHIND TYPICAL TESTS FOR HETEROGENEITY IN META-ANALYSES EXPLAINED

A fundamental concern in meta-analysis is the pooling together of commensurate studies (avoiding an “apples and oranges” comparison; Lipsey & Wilson, 2001, p. 2). We test for the presence of heterogeneity in the effect size distributions using a Cochran’s Q statistic and an I^2 test. The Q statistic tests whether differences between study effect sizes are the result of random subject-level sampling error (i.e., whether samples for each of the studies were drawn from the same population; Lipsey & Wilson, 2001). The I^2 test ranges from 0 to 100%, and estimates the percent of total variation across the effect sizes that is due to the true effect of the treatment rather than to sampling variation (Higgins, Thompson, Deeks, & Altman, 2003).

26 This example is roughly based on Hanson, R. K., Bourgon, G., Helmus, L., & Hodgson, S. (2009). *A meta-analysis of the effectiveness of treatment for sexual offenders: Risk, need, and responsivity*. Public Safety Canada.

27 Wong, J. S., Bouchard, J., Gravel, J., Bouchard, M., & Morselli, C. (2016). Can at-risk youth be diverted from crime? A meta-analysis of restorative diversion programs. *Criminal Justice and Behavior*, 43(10), 1310–1329.

6. Have the Researchers Addressed the Possibility of Bias among the Included Studies?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Another important consideration is to assess a possible risk of bias in the included studies. For example, some common biases include: *attrition bias* (participants dropping out of treatment before it is completed or refusing to continue participating in a study), *selective reporting bias* (statistically significant results are more likely to be reported within the study than null findings), and *publication bias* (studies with statistically significant findings are more likely to be published).²⁸ If these biases are not taken into account when researchers analyze the findings of studies on a specific intervention, it can erroneously lead to overly optimistic conclusions about the effectiveness of the assessed intervention.

Examples 14.6.1 and 14.6.2 present some options for how *publication bias* (sometimes also called a *file-drawer problem*) can be reasonably addressed in meta-analyses.

Example 14.6.1²⁹

A RISK OF PUBLICATION BIAS AMONG THE INCLUDED STUDIES
COMPREHENSIVELY ADDRESSED

A common problem in conducting meta-analysis is that many studies remain unpublished because of non-significant findings. The studies included in a meta-analysis may therefore not be a random sample of all studies that were conducted. To examine whether such publication bias or “file drawer problem” exists we computed fail safe numbers using Orwin’s formula (Lipsey & Wilson, 2001). It calculates the number of additional studies needed to reduce an observed mean effect size to a desired minimal effect size (Orwin, 1983). Meta-analytic findings are considered to be robust if the fail-safe number exceeds the critical value obtained with Rosenthal (1995) formula $5 * k + 10$, in which k is the number of studies used in the meta-analysis. If the fail-safe number falls below this critical value, a publication bias or file drawer problem may exist (see Results section [*in the original article*]).

Another method to examine file drawer bias is by funnel plot examination. This method examines the distribution of each individual study’s effect size on the horizontal axis against its sample size, standard error, or precision (the reciprocal of the standard error on the vertical axis). If no file-drawer bias is present, the distribution of effect sizes should be shaped as a funnel. Violation of funnel plot symmetry therefore reflects file-drawer bias (Sutton, 2009). Furthermore, the missing effect sizes can be substituted (“filled”) to

28 Publication bias may be a concern for some topics more than others. See a good discussion of this issue geared towards social sciences in Pratt, T. C. (2010). Meta-analysis in criminal justice and criminology: What it is, when it’s useful, and what to watch out for. *Journal of Criminal Justice Education*, 21(2), 152–168.

29 van Langen, M. A., Wissink, I. B., Van Vugt, E. S., Van der Stouwe, T., & Stams, G. J. J. M. (2014). The relation between empathy and offending: A meta-analysis. *Aggression and Violent Behavior*, 19(2), 179–189.

calculate overall effects corrected for file drawer bias. Selectivity bias according to the funnel plot was examined using MIX 2.0 (Bax, 2011).

Example 14.6.2³⁰

A POSSIBILITY OF PUBLICATION BIAS ADEQUATELY ADDRESSED

Using only published work in a meta-analysis is potentially controversial over the inferential errors that could be made concerning “publication bias” (see Egger and Smith, 1998; Rosenthal, 1979). In particular, the effect sizes may be inflated and the range of values restricted because studies revealing nonsignificant relationships may be more likely either to be rejected for publication or to remain unsubmitted to journals by authors (see also the discussion by Cooper, DeNeve, and Charleton, 1997; Lipsey and Wilson, 2001; Olson et al., 2002). Nevertheless, the effect sizes in our data ranged from $-.445$ to $.620$ (with a standard deviation of $.130$), which indicates that considerable variation in effect sizes exists – something that would be unlikely if publication bias were present. Subsequent analyses also reveal no significant problems with outliers or truncation in the distribution of effect sizes or the empirical Bayes residuals. Thus, the probability that our results are an artifact of publication bias is exceptionally low.

A bias may also result from another area: a study funding source. For example, if a study finds that drinking coffee is hugely beneficial for one’s health (the more coffee people consume, the healthier they are), it is important to check whether the study was funded by a United Coffee Association of America (which is a made-up name, but we are sure you get the gist).

7. For Meta-analysis, are the Procedures for Data Extraction and Coding Described Clearly?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: It is important that the researchers who conducted the meta-analysis meticulously describe the procedures for how the data were extracted from the included studies and coded for analysis. This allows the reader to evaluate the study more meaningfully and allows other researchers to replicate the meta-analysis in a few years, after more original studies on the topic get published. If the same procedures can be followed, it would make it easier to compare the results of the new meta-analysis to the previous one and see if things have changed over time.

For example, if a researcher suspects that the rate of mental illness among prisoners has been increasing over the recent decades, a new meta-analysis conducted using the same data-coding procedures as the previous one on the topic can help answer this question. If the data extraction and coding cannot be replicated, it would be hard to say whether the rates

30 Pratt, T. C., Turanovic, J. J., Fox, K. A., & Wright, K. A. (2014). Self-control and victimization: A meta-analysis. *Criminology*, 52(1), 87–116.

of mental illness among prisoners have changed or if it is simply the new coding procedures that have affected the results (or the newly published studies using a different way of measuring mental illness).

The specific ways of coding information extracted from each study included into a meta-analysis depend on the academic field and the research question the analysis is supposed to answer. Generally, the following important components of each study are coded in meta-analyses:

- study sample characteristics (size, type of subjects)
- the type of intervention
- comparability of comparison group
- the way outcomes were assessed
- the type of study design (true experiment, quasi-experiment, etc.).

Example 14.7.1 is an excerpt from a meta-analysis of so-called “hot spots” policing interventions and their impact on crime, and lists the variables on which the researchers coded the included studies (a very reasonable set of variables for the research question).

Example 14.7.1³¹

A CLEAR ACCOUNT OF THE CODING VARIABLES FOR THE STUDIES INCLUDED IN META-ANALYSIS

The eligible studies were coded on the following criteria:

- study identifiers (title, author, year, publication type)
- location of intervention (Country, Region, State, City)
- size of intervention, control and catchment areas (e.g., km², number of residents, number of households)
- research design (randomized control trial, pre-post w/catchment and control, etc.)
- nature (type) of focused policing intervention. This was divided into the categories mentioned in the criteria section above [*in the original article*]
- crime type targeted
- length of pre-assessment, intervention and follow-up period
- unit of analysis/sample size. This depended on the study design. For example, some evaluations considered changes in only one treatment, catchment (for a definition, see below [*in the original article*]) and control area whereas others examined changes in many
- pre- and post-outcome measure statistics
 - in intervention area(s)
 - in catchment area(s)
 - in control area(s)

31 Bowers, K. J., Johnson, S. D., Guerette, R. T., Summers, L., & Poynton, S. (2011). Spatial displacement and diffusion of benefits among geographically focused policing initiatives: A meta-analytical review. *Journal of Experimental Criminology*, 7(4), 347–374.

- measures of effect size and inferential statistical tests employed. The types of test used varied according to the study design employed (see above). For example, some studies employed time-series analyses, others used difference in difference statistics, others reported F tests, while others reported descriptive statistics alone
- effect sizes for the treatment area and the catchment area(s).

On the other hand, sometimes the variables that the included studies were coded on are described vaguely or some variables are used that are inconsequential for the research question (for example, whether the study results were presented on a graph). In such cases, you can give lower marks on this evaluation question.

8. For Meta-analysis, are the Numerical Results Explained in a Way That is Understandable to a Non-specialist?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: All systematic reviews and meta-analyses included into the Cochrane Library (which is mentioned above, in Evaluation Question 4) have a wonderful feature: after the abstract, the article must include a Plain Language Summary that explains in easy-to-understand terms why this review is important, which questions it tries to answer, which studies were included in the review, and what the evidence from the review tells us.

It would be great to have the same requirement – a summary written in plain language – for each original research article, systematic review, and meta-analysis published elsewhere. In the absence of such a convenient feature, it is important to look for explanations and interpretations of the meta-analysis results, especially numerical results, in the text of the meta-analysis itself (usually, in the Results or Discussion section).

Examples 14.8.1 and 14.8.2 provide illustrations for how such easy-to-understand explanations can be accomplished using comparison and application.

Example 14.8.1³²

COMPARISONS ARE PROVIDED TO HELP PUT THE OBTAINED RESULTS INTO PERSPECTIVE

The excess mortality associated with considerable social exclusion is extreme. We found all cause mortality SMRs [standardized mortality ratios] of 7.9 in male individuals and 11.9 in female individuals. By comparison, mortality rates for individuals aged 15–64 years in the most deprived areas of England and Wales are 2.8 times higher than those in the least deprived areas for male individuals and 2.1 times higher for female individuals.

32 Aldridge, R. W., Story, A., Hwang, S. W., Nordentoft, M., Luchenski, S. A., Hartwell, G., . . . & Hayward, A. C. (2018). Morbidity and mortality in homeless individuals, prisoners, sex workers, and individuals with substance use disorders in high-income countries: A systematic review and meta-analysis. *The Lancet*, 391(10117), 241–250.

Example 14.8.2³³

PLACING THE OBTAINED NUMERICAL RESULT INTO CONTEXT AND PROVIDING AN
EXAMPLE TO BETTER ILLUSTRATE ITS IMPLICATIONS

[From the *Results* Section]:

Results showed a significant female advantage on school marks, reflecting an overall estimated *d* of 0.225 (95% CI [0.201, 0.249]). As the confidence interval did not include zero, the overall effect size is significant with $p < .05$.

[From the *Discussion* Section]:

The most important finding observed here is that our analysis of 502 effect sizes drawn from 369 samples revealed a consistent female advantage in school marks for all course content areas. In contrast, meta-analyses of performance on standardized tests have reported gender differences in favor of males in mathematics (e.g., Else-Quest et al., 2010; Hyde et al., 1990; but see Lindberg et al., 2010) and science achievement (Hedges & Nowell, 1995), whereas they have shown a female advantage in reading comprehension (e.g., Hedges & Nowell, 1995). This contrast in findings makes it clear that the generalized nature of the female advantage in school marks contradicts the popular stereotypes that females excel in language whereas males excel in math and science (e.g., Halpern, Straight, & Stephenson, 2011). Yet the fact that females generally perform better than their male counterparts throughout what is essentially mandatory schooling in most countries seems to be a well-kept secret considering how little attention it has received as a global phenomenon. [. . .]

To put the present findings in perspective, an effect size of 0.225 would reflect approximately a 16% nonoverlap between distributions of males and females (Cohen, 1988). Thus, a crude way to interpret this finding is to say that, in a class of 50 female and 50 male students, there could be eight males who are forming the lower tail of the class marks distribution. These males would be likely to slow down the class, for example, and this could have cumulative effects on their school marks. Of course, this is not a completely accurate way to interpret the nonoverlap, but it should serve to illustrate the importance of this finding.

9. Have the Researchers Explained the Limitations of their Analysis?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Every research study has limitations, and systematic reviews and meta-analyses are no exception. Most limitations in meta-analyses have to do with the original studies included:

33 Voyer, D., & Voyer, S. D. (2014). Gender differences in scholastic achievement: A meta-analysis. *Psychological Bulletin*, 140(4), 1174–1204.

the lower the scientific quality of included studies and the lower the number of studies included, the more limited the results of the meta-analysis are. Limitations of systematic reviews and meta-analyses may also have a lot to do with the study search and selection procedures. In any case, if the authors do not list any limitations or if the only stated limitation of their review is that they omitted non-English-language studies, you can give a low mark on this evaluation question.

Example 14.9.1 illustrates a reasonable set of limitations in a systematic review of interventions aiming to help people quit smoking, and Example 14.9.2 discusses limitations along with the strengths of a systematic review of mother–infant separations in prison.

Example 14.9.1³⁴

LIMITATIONS OF THE CONDUCTED SYSTEMATIC REVIEW DETAILED

This review has several limitations. First, our literature search was conducted using key words to identify appropriate studies and may have missed some relevant articles that were not picked up from database searches. Second, our analysis was limited to economic studies assessing specific pharmacotherapies and brief counseling for smoking cessation and does not include other programs. Third, considerable heterogeneity among study methods, interventions, outcome variables, and cost components limits our ability to compare studies directly and determine specific policy recommendations.

Example 14.9.2³⁵

STRENGTHS AND WEAKNESSES OF THE CONDUCTED SYSTEMATIC REVIEW ADEQUATELY ADDRESSED

Given the date range, some of the key work in the area was excluded (e.g. Edge, 2006), however, these particular works were referred to in the more recent documents. Involvement from a prisoner or prison worker would have added critical reflections on the literature (e.g. Sweeney, Beresford, Faulkner, Nettle, & Rose, 2009). However, there were direct quotations from women who had been separated from their infants which added more detail to the impact of the experience of separation. Whilst the focus on the UK kept the review directly relevant to the policy, a review of international literature might have added some further insights around the use of attachment theory in prison policy and practice.

34 Ruger, J. P., & Lazar, C. M. (2012). Economic evaluation of pharmaco- and behavioral therapies for smoking cessation: A critical and systematic review of empirical research. *Annual Review of Public Health*, 33, 279–305.

35 Powell, C., Ciclitira, K., & Marzano, L. (2017). Mother–infant separations in prison. A systematic attachment-focused review of the academic and grey literature. *The Journal of Forensic Psychiatry & Psychology*, 28(6), 790–810.

10. Have the Researchers Interpreted the Results of Their Analysis to Draw Specific Implications for Practice?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
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Comment: Most meta-analyses and systematic reviews inform evidence-based policies and practice. If the research question investigated in such a review has specific implications for practice, the researchers must make it clear what these implications are.

Even if a systematic review or meta-analysis did not arrive at any conclusive results or strong findings, it is important for the researchers to state that implications for practice cannot be drawn and explain the reasons for that (rather than have the readers guessing).

In Example 14.10.1, the researchers make specific implications of their meta-analysis very clear in terms of suggested best policies and laws regarding sex offenders.

Example 14.10.1³⁶

POLICY IMPLICATIONS OF THE RESULTS OF META-ANALYSIS CLEARLY DRAWN

There is strong evidence that (a) there is wide variability in recidivism risk for individuals with a history of sexual crime; (b) risk predictably declines over time; and (c) risk can be very low – so low, in fact, that it becomes indistinguishable from the rate of spontaneous sexual offenses for individuals with no history of sexual crime but who have a history of nonsexual crime. These findings have clear implications for constructing effective public protection policies for sexual offenders.

First, the most efficient public protection policies will vary their responses according to the level of risk presented. Uniform policies that apply the same strategies to all individuals with a history of sexual crime are likely insufficient to manage the risk of the highest risk offenders, while over-managing and wasting resources on individuals whose risk is very low. [. . .]

The second implication is that efficient public policy responses need to include a process for reassessment. We cannot assume that our initial risk assessment is accurate and true for life. All systems that classify sexual offenders according to risk level also need a mechanism to reclassify individuals: the individuals who do well should be reassigned to lower risk levels, and individuals who do poorly should be reassigned to higher risk levels. The results of the current study, in particular, justify automatically lowering risk based on the number of years sexual offense-free in the community. [. . .]

The third implication is that there should be an upper limit to the absolute duration of public protection measures. In the current study, there were few individuals who presented more than a negligible risk after 15 years, and none after 20 years. [. . .]

36 Hanson, R. K., Harris, A. J., Letourneau, E., Helmus, L. M., & Thornton, D. (2018). Reductions in risk based on time offense-free in the community: Once a sexual offender, not always a sexual offender. *Psychology, Public Policy, and Law*, 24(1), 48–63.

Critics may argue that we cannot be too safe when it comes to the risk of sexual offenses. Although the harm caused by sexual offenses is serious, there are, however, finite resources that can be accorded to the problem of sexual victimization. From a public protection perspective, it is hard to justify spending these resources on individuals whose objective risk is already very low prior to intervention. Consequently, resources would be better spent on activities more likely to reduce the public health burden of sexual victimization . . .

____ 11. Overall, is the Systematic Review or Meta-analysis Adequate?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Rate this evaluation question after considering your answers to the earlier ones in this chapter, taking into account any additional considerations and concerns you may have. Make sure to put more weight on whether the systematic review or meta-analysis has been conducted properly rather than on whether it has produced interesting results.

Chapter 14 Exercises

Part A

Directions: Answer the following questions.

1. What is the main difference between a literature review and a systematic review?
2. How is a meta-analysis different from a systematic review?
3. Why are systematic reviews and meta-analyses especially suitable for providing a comprehensive evidence base about interventions and practices?
4. Which aspects of a systematic search for relevant studies should be documented in a systematic review or meta-analysis?
5. Often, researchers would publish their protocol for study selection ahead of conducting their systematic review or meta-analysis. Why is it important?
6. Can you explain what *heterogeneity* among included studies means?
7. What is publication bias? How can it affect the results of meta-analyses?
8. What are some important components of a study typically coded in a meta-analysis? Is there anything else important you think should be added to this list?

Part B

Directions: Search for meta-analyses and systematic reviews on a topic of interest to you in academic journals. Read them, and evaluate them using the evaluation questions in this chapter, taking into account any other considerations and concerns you may have. Select the one to which you gave the highest overall rating, and bring it to class for discussion. Be prepared to discuss its strengths and weaknesses.

Putting It All Together

As a final step, a consumer of research should make an overall judgment on the quality of a research report by considering the report as a whole. The following evaluation questions are designed to help in this activity.

___ 1. In Your Judgment, Has the Researcher Selected an Important Problem?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I ¹
------------------------	---	---	---	---	---	----------------------	--------	------------------

Comment: Evaluation Question 2 in Chapter 4 asks whether the researcher has established the importance of the problem area. The evaluation question being considered here is somewhat different from the previous one because this question asks whether *the evaluator* judges the problem to be important² – even if the researcher has failed to make a strong case for its importance. In such a case, a consumer of research would give the research report a high rating on this evaluation question but a low rating on Evaluation Question 2 in Chapter 4.

Note that a methodologically strong study on a trivial problem is a flaw that cannot be compensated for even with the best research methodology and report writing. On the other hand, a methodologically weak and poorly written study on an important topic may be judged to make a contribution – especially if there are no stronger studies available on the same topic.

___ 2. Were the Researchers Reflective?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

- 1 Continuing with the same grading scheme as in the previous chapters, N/A stands for “Not applicable” and I/I stands for “Insufficient information to make a judgement.”
- 2 For some amusing examples of studies that focus on seemingly trivial research problems, see the links to Ig Nobel Prize Winners in the online resources for this chapter.

Comment: Researchers should reflect on their methodological decisions and share these reflections with their readers. This shows that careful thinking underlies their work. For instance, do they reflect on why they worked with one kind of sample rather than another? Do they discuss their reasons for selecting one measure over another for use in their research? Do they discuss their rationale for other procedural decisions made in designing and conducting their research?

Researchers also should reflect on their interpretations of the data. Are there other ways to interpret the data? Are the various possible interpretations described and evaluated? Do they make clear why they favor one interpretation over another? Do they consider alternative explanations for the study results?

Such reflections can appear throughout research reports and often are repeated in the Discussion section at the end.

___ 3. Is the Report Cohesive?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Do the researchers make clear the heart of the matter (usually the research hypotheses, purposes, or questions) and write a report that revolves around it? Is the report cohesive (i.e., does it flow logically from one section to another)? Note that a scattered, incoherent report has little chance of making an important contribution to the understanding of a topic.

___ 4. Does the Report Extend the Boundaries of the Knowledge on a Topic, Especially for Understanding Relevant Theories?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: By introducing new variables or improved methods, researchers are often able to expand understanding of a problem. It is especially helpful when their findings provide insights into various theories or provide data that may be used for theory development. When researchers believe their data clearly extend the boundaries of what is known about a research problem, they should state their reasons for this belief.

Example 15.4.1 is from the introduction to a research report. The researchers state that their research has the potential to extend the boundaries of knowledge by filling in gaps in knowledge of a topic.

Example 15.4.1³

RESEARCHERS STATE IN THE INTRODUCTION THAT THEIR STUDY WILL EXTEND KNOWLEDGE BY FILLING GAPS (ITALICS ADDED FOR EMPHASIS)

Close relationships are the setting in which some of life's most tumultuous emotions are experienced. Echoing this viewpoint, Berscheid and Reis (1998) have argued that identifying both the origins and the profile of emotions that are experienced in a relationship is essential if one wants to understand the core defining features of a relationship. Against this backdrop, one might expect that a great deal would be known about emotions in relationships, especially how significant relationship experiences at critical stages of social development forecast the type and intensity of emotions experienced in adult attachment relationships. Surprisingly little is known about these issues, however (see Berscheid & Regan, 2004; Shaver, Morgan, & Wu, 1996). *Using attachment theory (Bowlby, 1969, 1973, 1980) as an organizing framework, we designed the current longitudinal study to fill these crucial conceptual and empirical gaps in our knowledge.*

Example 15.4.2 is excerpted from the Discussion section of a research report in which the researchers explicitly state that their findings replicate and extend what is known about an issue.

Example 15.4.2⁴

RESEARCHERS STATE IN THE DISCUSSION SECTION THAT THEIR STUDY EXTENDED KNOWLEDGE OF THE TOPIC (ITALICS ADDED FOR EMPHASIS)

The present study extends beyond prior descriptions of interventions for homeless families by providing detailed information about a comprehensive health center-based intervention. Findings demonstrate that it is feasible to integrate services that address the physical and behavioral health and support needs of homeless families in a primary health care setting. Detailed descriptive data presented about staff roles and activities begin to establish parameters for fidelity assessment, an essential first step to ensure adequate replication and rigorous testing of the HFP model in other settings.

Example 15.4.3 is excerpted from the Discussion section of a research report in which the researchers note that their results provide support for a theory.

- 3 Simpson, J. A., Collins, W. A., Tran, S., & Haydon, K. C. (2007). Attachment and the experience and expression of emotions in romantic relationships: A developmental perspective. *Journal of Personality and Social Psychology*, 92(2), 355–367.
- 4 Weinreb, L., Nicholson, J., Williams, V., & Anthes, F. (2007). Integrating behavioral health services for homeless mothers and children in primary care. *American Journal of Orthopsychiatry*, 77, 142–152.

Example 15.4.3⁵

RESEARCHERS STATE IN THE DISCUSSION SECTION THAT THEIR STUDY HELPS TO SUPPORT A THEORY (ITALICS ADDED FOR EMPHASIS):

Study 1 provided evidence in support of the first proposition of a new dialect theory of communicating emotion. As in previous studies of spontaneous expressions (Camras, Oster, Campos, Miyake, & Bradshaw, 1997; Ekman, 1972), posed emotional expressions converged greatly across cultural groups, in support of basic universality. However, reliable cultural differences also emerged. Thus, the study provided direct empirical support for a central proposition of dialect theory, to date supported only by indirect evidence from emotion recognition studies (e.g., Elfenbein & Ambady, 2002b). Differences were not merely idiosyncratic.

5. Are any Major Methodological Flaws Unavoidable or Forgivable?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: No study is perfect, but some are more seriously flawed than others. When serious flaws are encountered, consider whether they were unavoidable. For instance, obtaining a random sample of street prostitutes for a study on AIDS transmission is probably impossible. However, if the researchers went to considerable effort to contact potential participants at different times of the day in various locations (not just the safer parts of a city) and obtained a high rate of participation from those who were contacted, the failure to obtain a random sample would be forgivable because the flaw was unavoidable and considerable effort was made to overcome the flaw.

Contrast the preceding example with a study in which researchers want to generalize from a sample of fourth graders to a larger population but simply settle for a classroom of students who are readily accessible because they attend the university’s demonstration school on the university campus. The failure to use random sampling, or at least to use a more diverse sample from various classrooms, is not unavoidable and should be counted as a flaw.

Unless some flaws under some circumstances are tolerated, the vast majority of research in the social and behavioral sciences would need to be summarily rejected. Instead, as a practical matter, consumers of research tolerate certain flaws but interpret the findings from seriously flawed studies with considerable caution.

6. Is the Research Likely to Inspire Additional Research?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Even if a study is seriously flawed, it can receive a high evaluation on this question if it is likely to inspire others to study the problem. Seriously flawed research is most likely to

5 Elfenbein, H. A., Beaupré, M., Lévesque, M., & Hess, U. (2007). Toward a dialect theory: Cultural differences in the expression and recognition of posed facial expressions. *Emotion*, 7(1), 131–146.

get high ratings on this evaluation question if it employs novel research methods, has surprising findings, or helps to advance the development of a theory. Keep in mind that science is an incremental enterprise, with each study contributing to the base of knowledge about a topic. A study that stimulates the process and moves it forward is worthy of attention – even if it is seriously flawed or is only a pilot study.

___ 7. Is the Research Likely to Help in Decision Making?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Even seriously flawed research sometimes can help decision makers. Suppose a researcher conducted an experiment on a new drug-resistance educational program with no control group (usually considered a serious flaw) and found that students' illicit drug usage actually went up from pretest to post-test. Such a finding might lead to the decision to abandon the educational program, especially if other studies with different types of flaws produced results consistent with this one.

When applying this evaluation question, consider the following: In the absence of any other studies on the same topic, would this study help decision makers arrive at more informed decisions than they would if the study did not exist?

___ 8. All Things Considered, is the Report Worthy of Publication in an Academic Journal?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: Given that space is limited in academic journals, with some journals rejecting more than 90% of the research reports submitted, is the report being evaluated worthy of publication?

___ 9. Would You be Proud to Have Your Name on the Research Article as a Co-author?

Very unsatisfactory	1	2	3	4	5	Very satisfactory	or N/A	I/I
------------------------	---	---	---	---	---	----------------------	--------	-----

Comment: This is the most subjective evaluation question in this book, and it is fitting that it is last. Would you want to be personally associated with the research you are evaluating?

Concluding Comment

We hope that as a result of reading and working through this book, you have become a critical consumer of research while recognizing that conducting solid research in the social and behavioral sciences is often difficult (and conducting “perfect research” is impossible).

Note that the typical research methods textbook attempts to show *what should be done in the ideal*. Textbook authors do this because their usual purpose is to train students in how to conduct research. Unless a student knows what the ideal standards for research are, he or she is likely to fall unintentionally into many traps.

However, when evaluating reports of research in academic journals, it is unreasonable to hold each research article up to ideal “textbook standards.” Researchers conduct research under less-than-ideal conditions, usually with limited resources. In addition, they typically are forced to make many compromises (especially in measurement and sampling) given the practical realities of typical research settings. A fair and meaningful evaluation of a research article takes these practical matters into consideration.

Quantitative, Qualitative, and Mixed Methods Research: An Overview¹

Because *quantitative* researchers reduce information to statistics such as averages, percentages, and so on, their research reports are easy to spot. If a report has a Results section devoted mainly to the presentation of statistical data, it is a report of quantitative research. This approach to research dominated the social and behavioral sciences throughout most of the 1900s and still represents the majority of published research in the 2000s. Thus, for most topics, you are likely to locate many more articles reporting quantitative research than qualitative research.

Ideally, those who conduct *quantitative research* should do the following:

1. Start with one or more very specific, explicitly stated research hypotheses, purposes, or questions, ideally derived from theory and/or previous research. Make research plans that focus narrowly on the stated hypotheses, purposes, or questions (as opposed to being wide-ranging and exploratory).
2. Select a random sample (like drawing names out of a hat) from a population so that the sample is representative of the population from which it was drawn.²
3. Use a relatively large sample of participants, sometimes as many as 1,500 for a national survey. Some quantitative researchers use even larger samples, but many use much smaller ones because of limited resources. A study with a large sample is usually a quantitative one.
4. Make observations with measures that can be scored objectively, such as multiple-choice achievement tests and attitude scales in which participants mark choices such as “strongly agree” and “strongly disagree.”
5. Describe results using statistics, and make inferences to the population from which the sample was drawn (i.e., inferring that what the researcher found by studying a sample is similar to what he or she would have found by studying the entire population from which the sample was drawn).

1 This appendix is based in part on material drawn with permission from Galvan, J. L. (2009). *Writing literature reviews: A guide for students of the social and behavioral sciences* (4th ed.). Glendale, CA: Pyrczak Publishing. Copyright © 2009 by Pyrczak Publishing. All rights reserved.

2 It is representative except for the effects of random errors, which can be assessed with inferential statistics. Chapter 7 points out that researchers do not always sample or need random samples.

In addition, quantitative research is characterized by “distance” between researchers and their participants. That is, quantitative researchers typically have limited contact with their participants. In fact, it is not uncommon for the researcher to have no direct contact with them. For instance, a quantitative researcher might have teachers administer tests to students without ever seeing or talking with the students. Even if the researcher is physically present in the research setting, he or she usually follows a prearranged script for the study and avoids unplanned personal interactions.

In great many quantitative studies, the researchers do not know their participants at all because the researchers use *secondary data*, i.e., data that have been collected previously by other researchers. Secondary data are available, for example, through governmental agencies like the U.S. Census Bureau³ and the Centers for Disease Control and Prevention (CDC),⁴ as well as through survey initiatives like the National Longitudinal Study of Adolescent to Adult Health (Add Health)⁵ and the Monitoring the Future (MTF)⁶ project.

Qualitative research also has a long tradition in the social and behavioral sciences, but it has gained a large following in many applied fields only in recent decades. It is also often easy to identify because the titles of the articles frequently contain the word *qualitative*. In addition, qualitative researchers usually identify their research as qualitative in their Introductions as well as in other parts of their reports.⁷ You can also identify qualitative research because the Results section will be presented in a narrative describing themes and trends, which are very often illustrated with quotations from the participants.

In the ideal case, those who conduct *qualitative research* should do the following:

1. Start with a general research question or problem, and *not* formulate hypotheses derived from previously published literature or theories. Although qualitative researchers avoid starting with hypotheses and theories, they may emerge while the research is being conducted (i.e., a qualitative researcher may formulate hypotheses or theories that explain his or her observations). Such hypotheses and theories are subject to change as additional data are collected during the study. Thus, there is a fluid interaction between the data collection, data analysis, and any hypotheses or theories that may emerge.
2. Select a purposive sample – not a random one. A purposive sample is one in which the researcher has some special research interest and is not necessarily representative of a larger population. In other words, the researcher intentionally draws what he or she believes to be an appropriate sample for the research problem, without regard to random selection.
3. Use a relatively small sample – sometimes as small as one exemplary case, but more often small groups of people or aggregate units such as classrooms, churches, and so on.
4. Observe with relatively unstructured measures such as semi-structured interviews, unstructured direct observations, and so on.

3 www.census.gov/

4 www.cdc.gov/

5 www.cpc.unc.edu/projects/addhealth

6 www.monitoringthefuture.org/

7 Note that quantitative researchers rarely explicitly state that their research is quantitative. Because the overwhelming majority of research reports in journals are quantitative, readers will assume that it is quantitative unless told otherwise.

5. Observe intensively (e.g., spending extended periods of time with the participants to gain in-depth insights into the phenomena of interest).
6. Present results mainly or exclusively in words, with an emphasis on understanding the particular purposive sample studied and a de-emphasis on making generalizations to larger populations.

In addition, qualitative research is characterized by the researchers' awareness of their own orientations, biases, and experiences that might affect their collection and interpretation of data. It is not uncommon for qualitative researchers to include in their research reports a statement on these issues and what steps they took to see beyond their own subjective experiences in order to understand their research problems from the participants' points of view. Thus, there is a tendency for qualitative research to be personal and interactive. This is in contrast to quantitative research, in which researchers attempt to be objective and distant.

On the other hand, the personal nature of interactions between the qualitative researcher and her participants can create a unique sort of ethical dilemmas the researcher must navigate: from the issue of possible deception involved in gaining access to or trust from the persons of interest, to maintaining confidentiality when the knowledge gained has to be carefully guarded and participants' identities protected, to guilty knowledge when the researcher accidentally learns about some dangerous or even criminal activities being planned, to maintaining some distance in situations where the researcher is compelled to significantly intervene or provide substantial assistance.

As can be seen in this appendix, the fact that the two research traditions are quite distinct must be taken into account when research reports are being evaluated. Those who are just beginning to learn about qualitative research are urged to read the online resource provided for Chapter 11 of this book, *Examining the Validity Structure of Qualitative Research*, which discusses some important issues related to its evaluation.

Besides quantitative and qualitative, the third type of studies combining the first two – *mixed methods* research – has been gaining momentum in social sciences in the last 15–20 years. The advantage of mixed methods is to use the strengths of both quantitative and qualitative research while compensating for the weaknesses of each of the two approaches.

To begin, qualitative information such as words, pictures, and narratives can add meaning and depth to quantitative data. Likewise, quantitative data have the ability of enhancing clarity and precision to collected words, pictures, and narratives. Second, employing a mixed methods approach unbinds a researcher from a mono-method approach, thus, increasing their ability to accurately answer a wider range of research questions. Third, it can increase the specificity and generalizability of results by drawing from both methodological approaches. Mixing qualitative and quantitative techniques also has the potential to enhance validity and reliability, resulting in stronger evidence through convergence of collected data and findings. Lastly, examining an object of study by triangulating research methods allows for more complete knowledge – uncovering significant insights that mono-method research could overlook or miss completely (see Jick 1979).⁸

8 Brent, J. J., & Kraska, P. B. (2010). Moving beyond our methodological default: A case for mixed methods. *Journal of Criminal Justice Education*, 21(4), 412–430.

Ideally, those who conduct *mixed methods research* should do the following:

1. Determine the type of mixed methods design that would best serve the goal of answering the research questions. Should quantitative data be analyzed first and then a qualitative approach employed to clarify the specific subjective experiences and important details? Or should the project start with the qualitative data collection stage and then complement these data with the big-picture trends and patterns gleaned from the analysis of quantitative data?
2. Continue with the steps outlined above for quantitative and qualitative data collection, respectively.
3. Integrate the results from both methods and analyze whether both sets of results lead to the same conclusions and whether there are some important discrepancies or aberrations stemming from the comparison of data gathered using qualitative versus quantitative methods.
4. Draw conclusions and generalize the results taking into account the differences in samples and approaches between the two methods.

A Special Case of Program or Policy Evaluation

What is evaluation research? Evaluation research tests the effects of programs or policies. It helps determine which programs and policies are effective and how well they are working (or why they are not working). It also helps determine the financial side of things through *cost-effectiveness* analysis (how much return on investment the approach will bring) and *cost-benefit* analyses (comparing the costs and benefits of different approaches).

Often, evaluation studies form the basis of evidence (as in: *evidence-based* policies and practices). The importance of these studies cannot be overstated: local and federal governments, non-profit organizations, foundations, and treatment providers want to know which initiatives are worth spending their money on (in terms of program effectiveness and its cost-effectiveness) and thus, which ones should be implemented as their practices. For example, if a state government wants to reduce the rate of opioid overdose deaths, what is the best policy or program to invest in? Should the government fund more drug treatment programs or distribute antidotes like naloxone that reverse opioid overdoses? How much would each approach cost? Which one is more effective? Evaluation research helps answer these types of questions.

How are the effects of programs and policies assessed?

There are two main approaches to evaluating program effectiveness:

- the intended program outcomes are tracked and measured (called *impact assessment*)
- the implementation of the program is carefully examined (called *process evaluation*, or *process analysis*).

For example, how would we assess the *impact* of a drug treatment program? The most obvious answer is: we would need to measure the drug use among program participants/graduates before and after the program completion. If their drug use has declined, the program is effective, right? Unfortunately, it is not that simple How would we know if it is the program impact or some other impact (for example, the fact that the participants were arrested for a drug crime before starting the program) that has caused the outcomes (the reduction in drug use)?

As was explained in Chapter 9, the best method of determining causality (whether X caused Y, where X is participation in the program and Y is the outcome) is to conduct a true experiment,

with random assignment of participants to the program (a randomly assigned half of the study participants would undergo treatment X and the other half would serve as a control group).

Let us consider a situation where the program impact was assessed, and the researchers have found that they cannot reject the null hypothesis: that is, the difference between the treatment and control group participants' drug use (after program completion) is close to zero, which means that the level of drug use among those who completed the treatment program is similar to the level of drug use among those who did not go through the program. Is it because the program does not work (not effective)? Or is it because the program has been poorly *implemented* (for example, the counselors' training is not adequate or there are not enough resources to fully administer the program)? To answer this type of research question, a *process evaluation* needs to be conducted.

Often, the process is analyzed using observations and interviews with program participants and program administrators (qualitative approach), whereas the impact is assessed using numerical data analyses on program outcomes (quantitative approach). In an ideal program evaluation, a mixed methods approach would be used, combining the qualitative analysis of the program process and the quantitative assessment of its outcomes.

What about the program/policy costs?

Ideally, after the outcomes have been assessed and the program or policy is found effective, the cost analyses should be conducted to figure out whether the effective program is also *efficient* (delivers the results at a reasonable cost) or if there is another effective alternative that costs less. For example, if probation has been found to reduce recidivism (the rate of reoffending) just as much as incarceration does, we would want to compare how much it costs to supervise an offender on probation versus keeping him or her behind bars. If it costs about \$3,500 per year on average to supervise an offender in the community and about \$30,000 per year on average to keep a person in prison,¹ the analysis would be very helpful for the government in deciding the best course of action in crafting sentencing laws for offenders who committed minor crimes. *Cost-effectiveness* and *cost-benefit* analyses help answer these types of research questions.

How difficult is it to evaluate programs/policies?

Finally, just a few remarks on the complexity of program evaluation research. Obviously, there are many important details, considerations, and planning that go into developing a high-quality program evaluation study. Here are some examples of such important aspects:

- assessing needs (what the program has to remedy) and objectives (what the program is intended to achieve)
- determining who the intended program participants are and what the mechanism is for their selection/enrollment into the program (the feasibility of using random assignment)

1 www.uscourts.gov/news/2013/07/18/supervision-costs-significantly-less-incarceration-federal-system.

- assessing the logic of program theory (how the program components and activities are supposed to contribute to its intended outcomes)
- translating it into the timeline for assessment (for example, how long after the completion of the program its outcomes are supposed to last, i.e., whether only the immediate outcomes are assessed or more distant ones as well)
- coordinating between program providers and evaluators (e.g., who would ensure the collection of necessary data and its delivery to the researchers)
- considering ethical issues involved in program evaluation (for example, if the program is found to have no significant positive effects, how to deliver the news to program providers)

Almost all federal grants in the United States that fund programs and interventions now come with a mandatory requirement that a certain percentage of the grant funds must be spent on program evaluation. Program evaluation studies are the first step in building the *evidence base* for policies and practices (the next step is to compile the results from multiple evaluation studies and replications and summarize them in systematic reviews and meta-analyses, as explained in Chapter 14).

The Limitations of Significance Testing

Most of the quantitative research you evaluate will contain significance tests. They are important tools for quantitative researchers but have two major limitations. Before discussing the limitations, consider the purpose of significance testing and the types of information it provides.

The Function of Significance Testing

The function of significance testing is to help researchers evaluate the role of chance errors due to sampling. Statisticians refer to these chance errors as *sampling errors*. As you will see later in this appendix, it is very important to note that the term *sampling errors* is a statistical term that refers only to *chance* errors. Where do these sampling errors come from? They result from random sampling. Random sampling (e.g., drawing names out of a hat) gives everyone in a population an equal chance of being selected. Random sampling also produces random errors (once again, known as *sampling errors*). Consider Examples C1 and C2 to get a better understanding of this problem. Note in Example C1 that when whole populations are tested, there are no sampling errors and, hence, significance tests are not needed. It is also important to note that *a real difference can be a small difference* (in this example, less than a full point on a 30 item test).

Example C1

EXAMPLE WITH NO SAMPLING ERRORS BECAUSE A WHOLE POPULATION OF TENTH GRADERS WAS TESTED

A team of researchers tested all 500 tenth graders in a school district with a highly reliable and valid current events test consisting of 30 multiple-choice items. The team obtained a mean (the most popular average) of 15.9 for the girls and a mean of 15.1 for the boys. In this case, the 0.8-point difference in favor of the girls is “real” because *all* boys and girls were tested. The research team did not need to conduct a significance test to help them determine whether the 0.8-point difference was due to studying just a random sample of girls, which might not be representative of all girls, and a random sample of boys, which might not be representative of all boys. (Remember that the function of significance testing

is to help researchers evaluate the role of chance errors due to sampling when they want to generalize the results obtained on a sample to a population.)

Example C2

EXAMPLE OF SAMPLING ERRORS WHEN SAMPLES ARE USED

A different team of researchers conducted the same study with the same test at about the same time as the research team in Example C1. (They did not know the other team was conducting a population study.) This second team drew a random sample of 30 tenth-grade girls and 30 tenth-grade boys and obtained a mean of 16.2 for the girls and a mean of 14.9 for the boys. Why didn't they obtain the same values as the first research team? Obviously, it is because this research team sampled. Hence, the difference in results between the two studies is due to the *sampling errors* in this study.

In practice, typically only one study is conducted using random samples. If researchers are comparing the means for two groups, there will almost always be at least a small difference (and sometimes a large difference). In either case, it is conventional for quantitative researchers to conduct a significance test, which yields a probability that the difference between the means is due to sampling errors (and thus, no real difference exists between the two groups in the population). If there is a low probability that sampling errors created the difference (such as less than 5 out of 100, or $p < 0.05$), then the researchers will conclude that the difference is due to something other than chance. Such a difference is called a *statistically significant difference*.

The Limitations of Significance Testing

There are three major limitations to significance testing. Without knowing them, those who conduct and evaluate the results of quantitative research are likely to be misled.

First, *a significant difference can be large or small*. While it is true that larger differences tend to be statistically significant, significance tests are built on a combination of factors that can offset each other.¹ Under certain common circumstances, small differences are statistically significant. Therefore, the first limitation of significance testing is that it does not tell us whether a difference (or relationship) is large or small. (Remember that small differences can be “real” [see Example C1], and these can be detected by significance tests.) The obvious implication for those who are evaluating research reports is that they need to consider the magnitude of any significant differences that are reported (so-called *substantive significance*). For instance, for the difference between two means, ask “By *how many points* do the two groups differ?” and “Is this a large difference?”

1 If the difference between two means is being tested for statistical significance, three factors are combined mathematically to determine the probability: the size of the difference, the size of the sample, and the amount of variation within each group. One or two of these factors can offset the other(s). For this reason, sometimes small differences are statistically significant, and sometimes large differences are *not* statistically significant.

The second limitation of significance testing is that a significance test does not indicate whether the result is of practical significance. For instance, a school district might have to spend millions of dollars to purchase computer-assisted instructional software to get a statistically significant improvement (which might be indicated by a research report). If there are tight budgetary limits, the results of the research would be of no practical significance to the district. When considering practical significance, the most common criteria are as follows: (a) cost in relation to benefit of a statistically significant improvement (e.g., how many points of improvement in mathematics achievement can we expect for each dollar spent?); (b) the political acceptability of an action based on a statistically significant research result (e.g., will local politicians and groups that influence them, such as parents, approve of the action?); and (c) the ethical and legal status of any action that might be suggested by statistically significant results.

The third limitation is that statistical significance tests are designed to assess only sampling error (errors due to random sampling). More often than not, research published in academic journals is based on samples that are clearly not drawn at random (e.g., using students in a professor's class as research participants or using volunteers). Strictly speaking, there are no significance tests appropriate for testing differences when nonrandom samples are used. Nevertheless, quantitative researchers routinely apply significance tests to such samples. As a consequence, consumers of research should consider the results of such tests as providing only tenuous information.

A Crisis of Significance Testing and Research Ethics

In recent years, there has been essentially a widening crisis of significance testing that has started out in Psychology and spread to other sciences. What some researchers had published earlier as statistically significant results supporting novel theories turned out to be essentially a fluke of statistical significance testing (the use of small samples and removal of outliers could help reach statistical significance).

For example, a study conducted by Open Science Collaboration² has replicated 100 experimental and correlational studies published in three well-regarded Psychology journals. Only about a third of carefully conducted, high-quality replications (using large samples) have reached statistically significant results. This put many other famous findings from psychological studies in doubt.³ A movement to replicate classical psychological experiments, as well as important studies in other fields of sciences, is now under way.

2 Open Science Collaboration (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. Available at: <http://science.sciencemag.org/content/sci/349/6251/aac4716.full.pdf>

3 For example, a TED Talk by Amy Cuddy about how “power poses” can make you more confident and bring about success in life was for a few years one of the most watched TED Talks: www.ted.com/talks/amy_cuddy_your_body_language_shapes_who_you_are. More recently, multiple high-quality replications of the “power pose” impact have found essentially no statistically significant effects: Jonas, K. J., Cesario, J., Alger, M., Bailey, A. H., Bombari, D., Carney, D., . . . & Jackson, B. (2017). Power poses – where do we stand? *Comprehensive Results in Social Psychology*, 2(1), 139–141. Available at: www.tandfonline.com/doi/abs/10.1080/23743603.2017.1342447

So, why do studies with statistically significant results that get published in respectable journals fail to replicate? One likely reason for this is a pressure to publish being so strong for most researchers (so called “publish or perish” pressure⁴) that they would go to great lengths to produce a coveted publication. Sometimes, it also means they would use unethical practices like *data dredging* (data mining to uncover some patterns in the data without having any specific hypotheses), “massaging” the data (for example, removing some outliers to reach statistical significance or using questionable imputation methods for missing data), and “shopping” for a statistical model that would produce statistically significant results.

Such unethical research practices are not easy to detect for journal editors and peer reviewers because they are not always evident in research reports. The situation is dire enough that there are calls in the research community to get rid of significance testing all together⁵ since it underlies (and thus tacitly supports) these unethical practices. At the same time, there are no easy alternatives to statistical significance testing so a more reasonable suggestion seems to be to lower the standard threshold for statistical significance from $p < 0.05$ to < 0.005 , to make it more difficult for statistical flukes to lead to publications.⁶

Concluding Comments

Thus, it seems that statistical significance testing is unlikely to be discarded as a method any time soon. It plays an important role in quantitative research when differences are being assessed in light of sampling error (i.e., chance error). If researchers are trying to show that there is a real difference (when using random samples), their first hurdle is to use statistics (including the laws of probability) to show that the difference is statistically significant. If they pass this hurdle, they should then consider how large the difference is in absolute terms (e.g., 100 points on College Boards versus 10 points on College Boards).⁷ Then, they should evaluate the practical significance of the result. If they used nonrandom samples, any conclusions regarding significance (the first hurdle) should be considered highly tenuous.

Because many researchers are better trained in their content areas than in statistical methods, it is not surprising that some make the mistake of assuming that when they have statistically significant results, by definition they have important results and discuss their results accordingly. As a savvy consumer of research, you will know to consider the absolute size (substantive significance) of any differences, as well as the practical significance of the results when evaluating their research.

- 4 You can find an excellent explanation of this pressure and of the origins of the “publish or perish” phrase, as well as some examples of unethical research practices stemming from this pressure, in Rawat, S., & Meena, S. (2014). Publish or perish: Where are we heading? *Journal of Research in Medical Sciences: The Official Journal of Isfahan University of Medical Sciences*, 19(2), 87–89. Available at: www.ncbi.nlm.nih.gov/pmc/articles/PMC3999612/
- 5 Trafimow, D., & Marks, M. (2015). Editorial. *Basic and Applied Social Psychology*, 37(1), 1–2.
- 6 Ioannidis, J. P. (2018). The proposal to lower p value thresholds to .005. *JAMA*, 319(14), 1429–1430.
- 7 Sometimes also indicated by an *effect size*, which is basically a way to quantify the difference in outcomes between two groups beyond calling it statistically significant (see more information about it in Chapter 14).

Checklist of Evaluation Questions

Following are the evaluation questions presented in Chapter 2 through Chapter 15 of this book. You may find it helpful to go back to the relevant chapter and look for explanations and examples for any questions that are unclear to you. Keep in mind that your professor may require you to justify each of your responses.

Chapter 2 Evaluating Titles

- ☐ 1. Is the title sufficiently specific?
- ☐ 2. Is the title reasonably concise?
- ☐ 3. Are the primary variables mentioned in the title?
- ☐ 4. When there are many variables, are the *types* of variables referred to?
- ☐ 5. Does the title identify the types of individuals who participated or the types of aggregate units in the sample?
- ☐ 6. If a study is strongly tied to a theory, is the name of the specific theory mentioned in the title?
- ☐ 7. Has the author avoided describing results in the title?
- ☐ 8. Has the author avoided using a 'yes–no' question as a title?
- ☐ 9. If there are a main title and a subtitle, do both provide important information about the research?
- ☐ 10. If the title implies causality, does the method of research justify it?
- ☐ 11. Is the title free of jargon and acronyms that might be unknown to the audience for the research report?
- ☐ 12. Are any highly unique or very important characteristics of the study referred to in the title?
- ☐ 13. Overall, is the title effective and appropriate?

Chapter 3 Evaluating Abstracts

- ☐ 1. Is the purpose of the study referred to or at least clearly implied?
- ☐ 2. Does the abstract mention highlights of the research methodology?

- ___ 3. Has the researcher omitted the titles of measures (except when these are the focus of the research)?
- ___ 4. Are the highlights of the results described?
- ___ 5. If the study is strongly tied to a theory, is the theory mentioned in the abstract?
- ___ 6. Has the researcher avoided making vague references to implications and future research directions?
- ___ 7. Does the abstract include purpose/objectives, methods, and results of the study?
- ___ 8. Overall, is the abstract effective and appropriate?

Chapter 4 Evaluating Introductions and Literature Reviews

- ___ 1. Does the researcher begin by identifying a specific problem area?
- ___ 2. Does the researcher establish the importance of the problem area?
- ___ 3. Are any underlying theories adequately described?
- ___ 4. Does the introduction move from topic to topic instead of from citation to citation?
- ___ 5. Are very long introductions broken into subsections, each with its own sub-heading?
- ___ 6. Has the researcher provided adequate conceptual definitions of key terms?
- ___ 7. Has the researcher cited sources for “factual” statements?
- ___ 8. Do the specific research purposes, questions, or hypotheses logically flow from the introductory material?
- ___ 9. Overall, is the introduction effective and appropriate?

Chapter 5 A Closer Look at Evaluating Literature Reviews

- ___ 1. Has the researcher avoided citing a large number of sources for a single point?
- ___ 2. Is the literature review critical?
- ___ 3. Is current research cited?
- ___ 4. Has the author cited any contradictory research findings?
- ___ 5. Has the researcher distinguished between opinions and research findings?
- ___ 6. Has the researcher noted any gaps in the literature?
- ___ 7. Has the researcher interpreted research literature in light of the inherent limits of empirical research?
- ___ 8. Has the researcher avoided overuse of direct quotations from the literature?
- ___ 9. After reading the literature review, does a clear picture emerge of what the previous research has accomplished and which questions still remain unresolved?
- ___ 10. Overall, is the literature review portion of the introduction appropriate and effective?

Chapter 6 Evaluating Samples when Researchers Generalize

- ___ 1. Was random sampling used?
- ___ 2. If random sampling was used, was it stratified?

- ☐ 3. If some potential participants refused to participate, was the rate of participation reasonably high?
- ☐ 4. If the response rate was low, did the researcher make multiple attempts to contact potential participants?
- ☐ 5. Is there reason to believe that the participants and nonparticipants are similar on relevant variables?
- ☐ 6. If a sample is not random, was it at least drawn from the target group for the generalization?
- ☐ 7. If a sample is not random, was it drawn from diverse sources?
- ☐ 8. If a sample is not random, does the researcher explicitly discuss this limitation and how it may have affected the generalizability of the study findings?
- ☐ 9. Has the author described relevant characteristics (demographics) of the sample?
- ☐ 10. Is the overall size of the sample adequate?
- ☐ 11. Is the number of participants in each subgroup sufficiently large?
- ☐ 12. Has informed consent been obtained?
- ☐ 13. Has the study been approved by an ethics review board (Institutional Review Board (IRB) if in the United States or a similar agency if in another country)?
- ☐ 14. Overall, is the sample appropriate for generalizing?

Chapter 7 Evaluating Samples when Researchers Do *Not* Generalize

- ☐ 1. Has the researcher described the sample/population in sufficient detail?
- ☐ 2. For a pilot study or developmental test of a theory, has the researcher used a sample with relevant demographics?
- ☐ 3. Even if the purpose is not to generalize to a population, has the researcher used a sample of adequate size?
- ☐ 4. Is the sample size adequate in terms of its orientation (quantitative versus qualitative)?
- ☐ 5. If a purposive sample has been used, has the researcher indicated the basis for selecting participants?
- ☐ 6. If a population has been studied, has it been clearly identified and described?
- ☐ 7. Has informed consent been obtained?
- ☐ 8. Has the study been approved by an ethics review committee?
- ☐ 9. Overall, is the description of the sample adequate?

Chapter 8 Evaluating Measures

- ☐ 1. Have the actual items and questions (or at least a sample of them) been provided?
- ☐ 2. Are any specialized response formats, settings, and/or restrictions described in detail?

- ___ 3. When appropriate, were multiple methods used to collect data/information on key variables?
- ___ 4. For published measures, have sources been cited where additional information can be obtained?
- ___ 5. When delving into sensitive matters, is there reason to believe that accurate data were obtained?
- ___ 6. Have steps been taken to keep the measures from influencing any overt behaviors that were observed?
- ___ 7. If the collection and coding of observations involves subjectivity, is there evidence of inter-rater (or inter-observer) reliability?
- ___ 8. If a measure is designed to measure a single unitary trait, does it have adequate internal consistency?
- ___ 9. For stable traits, is there evidence of temporal stability?
- ___ 10. When appropriate, is there evidence of content validity?
- ___ 11. When appropriate, is there evidence of empirical validity?
- ___ 12. Do the researchers discuss obvious limitations of their measures?
- ___ 13. Overall, are the measures adequate?

Chapter 9 Evaluating Experimental Procedures

- ___ 1. If two or more groups were compared, were the participants assigned at random to the groups?
- ___ 2. If two or more groups were compared, were there enough participants (or aggregate units) per group?
- ___ 3. If two or more comparison groups were not formed at random, is there evidence that they were initially equal in important ways?
- ___ 4. If only a single participant or a single group is used, have the treatments been alternated?
- ___ 5. Are the treatments described in sufficient detail?
- ___ 6. If the treatments were administered by individuals other than the researcher, were those individuals properly trained and monitored?
- ___ 7. If each treatment group had a different person administering a treatment, did the researcher try to eliminate the personal effect?
- ___ 8. If treatments were self-administered, did the researcher check on treatment compliance?
- ___ 9. Except for differences in the treatments, were all other conditions the same in the experimental and control groups?
- ___ 10. Were the effects or outcomes of treatment evaluated by individuals who were *not* aware of the group assignment status?
- ___ 11. When appropriate, have the researchers considered possible demand characteristics?
- ___ 12. Is the setting for the experiment natural?

- ☐ 13. Has the researcher distinguished between random selection and random assignment?
- ☐ 14. Has the researcher considered attrition?
- ☐ 15. Has the researcher used ethical and politically acceptable treatments?
- ☐ 16. Overall, was the experiment properly conducted?

Chapter 10 Evaluating Analysis and Results Sections: Quantitative Research

- ☐ 1. When percentages are reported, are the underlying numbers of cases also reported?
- ☐ 2. Are means reported only for approximately symmetrical distributions?
- ☐ 3. If any differences are statistically significant and small, have the researchers noted that they are small?
- ☐ 4. Is the Results section a cohesive essay?
- ☐ 5. Does the researcher refer back to the research hypotheses, purposes, or questions originally stated in the introduction?
- ☐ 6. When there are a number of related statistics, have they been presented in a table?
- ☐ 7. If there are tables, are their highlights discussed in the narrative of the Results section?
- ☐ 8. Have the researchers presented descriptive statistics before presenting the results of inferential tests?
- ☐ 9. Overall, is the presentation of the results comprehensible?
- ☐ 10. Overall, is the presentation of the results adequate?

Chapter 11 Evaluating Analysis and Results Sections: Qualitative Research

- ☐ 1. Were the data analyzed independently by two or more individuals?
- ☐ 2. Did the researchers seek feedback from experienced individuals and auditors before finalizing the results?
- ☐ 3. Did the researchers seek feedback from the participants (i.e., use member checking) before finalizing the results?
- ☐ 4. Did the researchers name the method of analysis they used and provide a reference for it?
- ☐ 5. Did the researchers state *specifically* how the method of analysis was applied?
- ☐ 6. Did the researchers self-disclose their backgrounds?
- ☐ 7. Are the results of qualitative studies adequately supported with examples of quotations or descriptions of observations?
- ☐ 8. Are appropriate statistics reported (especially for demographics)?
- ☐ 9. Overall, is the Results section clearly organized?
- ☐ 10. Overall, is the presentation of the results adequate?

Chapter 12 Evaluating Analysis and Results Sections: Mixed Methods Research

- ☐ 1. Does the Methods section identify a specific mixed methods design?
- ☐ 2. Does the Methods section link the need for a mixed methods approach to the research question(s)?
- ☐ 3. Does the Methods section clearly explain both the quantitative and qualitative methods utilized in the study?
- ☐ 4. Are the qualitative results presented in the study satisfactory according to established qualitative standards?
- ☐ 5. Are the quantitative results presented in the study satisfactory according to established quantitative standards?
- ☐ 6. Are the findings of the research integrated/mixed?
- ☐ 7. Apart from validity issues inherent in the quantitative and qualitative components, does the researcher address validity issues specific to mixed methods designs?
- ☐ 8. Do the findings include consideration of contradictory data, aberrant cases or surprising results?
- ☐ 9. Is the use of mixed methods justified?
- ☐ 10. Overall, is the presentation of the results adequate?

Chapter 13 Evaluating Discussion Sections

- ☐ 1. In long articles, do the researchers briefly summarize the purpose and results at the beginning of the Discussion section?
- ☐ 2. Do the researchers acknowledge specific methodological limitations?
- ☐ 3. Are the results discussed in terms of the literature cited in the introduction?
- ☐ 4. Have the researchers avoided citing new references in the Discussion section?
- ☐ 5. Are specific implications discussed?
- ☐ 6. Are the results discussed in terms of any relevant theories?
- ☐ 7. Are suggestions for future research specific?
- ☐ 8. Have the researchers distinguished between speculation and data-based conclusions?
- ☐ 9. Overall, is the Discussion section effective and appropriate?

Chapter 14 Evaluating Systematic Reviews and Meta-Analyses: Towards Evidence-Based Practice

- ☐ 1. Have the researchers clearly formulated their research question or hypothesis?
- ☐ 2. Do the researchers explain in detail how they systematically searched for relevant studies?
- ☐ 3. Have the researchers clearly identified their criteria for including or excluding the studies produced by the search?

Appendix D: Checklist of Evaluation Questions

- ___ 4. Are there enough studies included in the final sample for analysis?
- ___ 5. Have the researchers addressed the issue of heterogeneity among the included studies?
- ___ 6. Have the researchers addressed the possibility of bias among the included studies?
- ___ 7. For meta-analysis, are the procedures for data extraction and coding described clearly?
- ___ 8. For meta-analysis, are the numerical results explained in a way that is understandable to a non-specialist?
- ___ 9. Have the researchers explained the limitations of their analysis?
- ___ 10. Have the researchers interpreted the results of their analysis to draw specific implications for practice?
- ___ 11. Overall, is the systematic review or meta-analysis adequate?

Chapter 15 Putting It All Together

- ___ 1. In your judgment, has the researcher selected an important problem?
- ___ 2. Were the researchers reflective?
- ___ 3. Is the report cohesive?
- ___ 4. Does the report extend the boundaries of the knowledge on a topic, especially for understanding relevant theories?
- ___ 5. Are any major methodological flaws unavoidable or forgivable?
- ___ 6. Is the research likely to inspire additional research?
- ___ 7. Is the research likely to help in decision making?
- ___ 8. All things considered, is the report worthy of publication in an academic journal?
- ___ 9. Would you be proud to have your name on the research article as a co-author?

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